```
# ==== run.py ====
# -*- coding: utf-8 -*-
"""系统启动入口。
运行：
    python run.py                启动并自动打开浏览器
    python run.py --no-browser   只启动服务，不打开浏览器
    python run.py --dev          开发模式（开启代码热重载与调试器）
    python run.py --port 5001    指定端口
默认以 127.0.0.1:5000 启动，就绪后自动打开浏览器。
"""
import os
import sys
import threading
import time
import webbrowser
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")   # 控制台中文正常显示
from app import create_app
from app.models import db
PORT = 5000
for i, a in enumerate(sys.argv):
    if a == "--port" and i + 1 < len(sys.argv):
        PORT = int(sys.argv[i + 1])
URL = "http://127.0.0.1:%d" % PORT
NO_BROWSER = "--no-browser" in sys.argv
DEV = "--dev" in sys.argv
app = create_app()
def open_browser_later(delay=1.2):
    """等服务工作起来再打开浏览器，避免打开时页面还没就绪。
    放在后台线程里，不用 sleep 阻塞服务启动；线程设为守护线程，
    用户 Ctrl+C 停止服务时不会残留。
    """
    def run():
        time.sleep(delay)
        try:
            webbrowser.open(URL)
        except Exception:
            pass            # 打开失败不影响服务本身
    t = threading.Thread(target=run, daemon=True)
    t.start()
if __name__ == "__main__":
    with app.app_context():
        db.create_all()     # 首次运行时自动建表
        from app.utils.schema import ensure_schema
        added = ensure_schema(db)     # 旧库补齐新增字段
        if added:
            print("已补充字段：%s" % "、".join(added))
    print("=" * 58)
    print("   历史遗留废弃矿山自然恢复适宜性评价系统  Ｖ１．０")
    print("=" * 58)
```

```
    print("   访问地址：%s" % URL)
    print("   演示账号：admin      密码：admin123")
    print("   停止运行：按 Ctrl+C，或直接关闭本窗口")
    print("=" * 58)
    if not NO_BROWSER and not DEV:
        print("   正在打开浏览器……")
    print()
    # 开发模式才启用热重载（重载会二次启动进程，故此时不自动开浏览器）
    if not NO_BROWSER and not DEV:
        open_browser_later()
    app.run(host="127.0.0.1", port=PORT, debug=DEV,
    use_reloader=DEV)
# ==== app/__init__.py ====
# -*- coding: utf-8 -*-
"""Flask 应用工厂。"""
from datetime import date
from flask import Flask
from flask_login import LoginManager
from flask_wtf.csrf import CSRFProtect, CSRFError
from app.config import Config
from app.models import db, User
from app.utils.constants import ROLE_LABELS
login_manager = LoginManager()
login_manager.login_view = "auth.login"
login_manager.login_message = "请先登录"
csrf = CSRFProtect()
@login_manager.user_loader
def load_user(user_id):
    return db.session.get(User, int(user_id))
def create_app(config_class=Config):
    app = Flask(__name__)
    app.config.from_object(config_class)
    db.init_app(app)
    login_manager.init_app(app)
    # CSRF 防护：WTF_CSRF_ENABLED 只对 FlaskForm 生效，页面上那些手写 POST
    表单
    # （账号增删改、项目 / 图斑删除等）以前是没校验的；同时模板里的 csrf_token()
    # 也需要它才会存在——少了这一行，保存过一次 AHP 权重后 /ahp/ 直接 500。
    csrf.init_app(app)
    # 模板里展示时间统一走 localdt：库里存 UTC，展示成本地时间
    from app.utils.timefmt import to_local
    app.jinja_env.filters["localdt"] = to_local
    # 注册视图（各模块路由）
    from app.views import auth, project, unit, standard,
    constraint, \
        ahp, indicator, evaluate, report, search
    app.register_blueprint(auth.bp)
    app.register_blueprint(project.bp)
    app.register_blueprint(unit.bp)
    app.register_blueprint(standard.bp)
```

```
app.register_blueprint(constraint.bp)
app.register_blueprint(ahp.bp)
app.register_blueprint(indicator.bp)
app.register_blueprint(evaluate.bp)
app.register_blueprint(report.bp)
app.register_blueprint(search.bp)
#  全局登录保护：除登录页与静态资源外，均需登录
from flask import request, redirect, url_for,
send_from_directory, flash
from flask_login import current_user, login_user,
logout_user
PUBLIC_ENDPOINTS = {"auth.login", "auth.register",
"static", "favicon"}
#  浏览器默认会请求根路径下的 /favicon.ico。
#  页面里虽已用 <link rel="icon"> 指向 static 目录，但出错页等场景仍会
#  回退到这个路径，补一条路由免得日志里一直刷 404。
@app.route("/favicon.ico")
def favicon():
    return send_from_directory(app.static_folder,
"favicon.ico",
                               mimetype="image/vnd.micro
soft.icon")
@app.before_request
def require_login():
    #  截图模式（仅用于生成说明书配图）：自动以管理员身份登录
    if app.config.get("SCREENSHOT_MODE"):
        if (request.endpoint not in ("auth.login",
"auth.register", "static")
                and not current_user.is_authenticated):
            admin = User.query.filter_by(role="admin").f
irst()
            if admin:
                login_user(admin)
        return None
    if request.endpoint in PUBLIC_ENDPOINTS or
request.endpoint is None:
        return None
    if not current_user.is_authenticated:
        return redirect(url_for("auth.login"))
    #  账号被停用后立即失效：不然已经登录的会话还能继续用下去
    if not current_user.active:
        logout_user()
        flash("该账号已停用，请联系管理员", "toast-error")
        return redirect(url_for("auth.login"))
    return None
#  侧边栏导航：按评价流程的顺序编排，分为「评价流程」与「系统管理」两组。
#  不使用模块编号，避免出现与实际流程不一致的代号。
#  每项带图标名（内联 SVG，见 templates/_macros.html 的 ICONS）
#  与可选的待办角标（该步骤还没做完的图斑数）。
FLOW_LINKS = [
```

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        ("项目与矿山档案", "/project/", "folder"),
        ("政策与标准知识库", "/standard/", "book"),
        ("限制性指标判定", "/constraint/", "shield"),
        ("微地质单元", "/unit/", "layers"),
        ("指标体系与权重", "/ahp/", "sliders"),
        ("指标数据采集", "/indicator/", "database"),
        ("适宜性定量评价", "/evaluate/", "target"),
        ("报告与结果输出", "/report/", "file"),
    ]
    #  「用户与操作日志」已并入「个人设置」，不再是独立菜单，
    #  因此系统管理组只留一个入口。
    ADMIN_LINKS = [
        ("个人设置", "/profile", "user"),
    ]
    @app.context_processor
    def inject_nav():
        path = request.path or "/"
        def hit(prefix):
            """当前路径是否属于某个入口（/unit/ 也匹配 /unit/3）。"""
            return path.startswith(prefix.rstrip("/"))
        badges = {}
        if current_user.is_authenticated:
            from app.services import dashboard as dash
            badges = dash.nav_badges()
        def build(links):
            return [{"label": label, "href": href, "icon":
    icon,
                     "active": hit(href), "badge":
    badges.get(href, 0)}
                    for label, href, icon in links]
        groups = [
            {"title": "评价流程", "links": build(FLOW_LINKS)},
            {"title": "系统管理", "links": build(ADMIN_LINKS)},
        ]
        #  面包屑用的当前页名：按路径命中的入口取，未命中时给个通用名
        if path == "/":
            page_title = "工作台"
        elif path.startswith("/profile"):
            page_title = "个人设置"
        elif path.startswith(("/users", "/logs")):
            page_title = "账号与日志"
        elif path.startswith("/search"):
            page_title = "搜索结果"
        else:
            page_title = "工作台"
            for label, href, _icon in FLOW_LINKS:
                if hit(href):
                    page_title = label
                    break
        #  层级引导用：项目是归档单位，图斑才是评价对象。
```

```
    #  下游各页面按图斑组织数据，因此需要提示「哪些项目下还没有图斑」。
    from app.models import Project, MinePatch
    projects = Project.query.order_by(Project.id).all()
    return {
        "nav_groups": groups,
        "home_active": path == "/",
        "page_title": page_title,
        "today": date.today().strftime("%Y-%m-%d"),
        "search_q": request.args.get("q", ""),
        "role_labels": ROLE_LABELS,
        "project_count": len(projects),
        "patch_count": MinePatch.query.count(),
        "projects_without_patch": [p for p in projects
                                   if p.patches.count()
== 0],
    }
#  操作审计：统一在响应阶段记录成功的写操作（操作日志）
from app.services import audit
@app.after_request
def audit_writes(response):
    if request.method in ("POST", "PUT", "PATCH",
"DELETE"):
        audit.record(request.endpoint,
response.status_code)
    return response
#  错误页：给出可读提示，而不是默认的英文错误页
from flask import render_template
@app.errorhandler(403)
def forbidden(e):
    return render_template("error.html", code=403,
                           message="没有权限执行该操作"), 403
@app.errorhandler(404)
def not_found(e):
    return render_template("error.html", code=404,
                           message="请求的页面不存在"), 404
@app.errorhandler(413)
def too_large(e):
    """上传超过体积上限时给中文页；默认是英文的 Request Entity Too Large。"""
    limit = app.config.get("MAX_CONTENT_LENGTH") or 0
    return render_template(
        "error.html", code=413,
        message="上传的文件太大（上限 %d MB），请压缩或拆分后重试"
                % (limit // 1024 // 1024)), 413
@app.errorhandler(500)
def server_error(e):
    """未预料的错误也给中文页；生产模式下 Flask 不会带出堆栈。"""
    return render_template("error.html", code=500,
                           message="系统处理时出现异常，本次操作未生效，"
                                   "请返回重试或联系管理员"), 500
@app.errorhandler(CSRFError)
```

```
    def csrf_error(e):
        """令牌过期或来源不符时给一句人话，并把人送回上一页。
        原来校验一失败就是默认英文 400 页，或者（对手写表单）干脆什么都不发生，
        用户只会觉得「点了没反应」。回跳目标只取本站地址，避免开放重定向。
        """
        from flask import flash, redirect, url_for
        flash("页面停留时间过久或提交来源校验未通过，请重新提交一次", "error")
        referrer = request.referrer or ""
        if referrer.startswith(request.host_url):
            return redirect(referrer)
        return redirect(url_for("auth.index"))
    return app
# ==== app/config.py ====
# -*- coding: utf-8 -*-
"""应用配置。"""
import os
import secrets
from datetime import timedelta
BASE_DIR = os.path.abspath(os.path.dirname(__file__))
def _secret_key():
    """会话签名密钥：优先取环境变量，否则首次运行随机生成并存到本地文件。
    原来是一个写死在代码里的默认值，任何拿到源码的人都能自签会话 cookie。
    单机本地跑风险有限，但改成本机随机值几乎零成本。
    生成的文件已加进 .gitignore，不会进版本库。
    """
    env = os.environ.get("SECRET_KEY")
    if env:
        return env
    path = os.path.join(BASE_DIR, ".secret_key")
    try:
        if os.path.exists(path):
            with open(path, encoding="utf-8") as f:
                saved = f.read().strip()
            if saved:
                return saved
        key = secrets.token_hex(32)
        with open(path, "w", encoding="utf-8") as f:
            f.write(key)
        return key
    except OSError:
        # 目录不可写时退回固定值，保证系统仍能起来（此时会退化为原来的行为）
        return "mine-restoration-dev-key"
class Config:
    SECRET_KEY = _secret_key()
    # 单机部署，SQLite 数据库文件放在应用目录下
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE_DIR, "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    # 禁止跨站请求伪造（CSRF），登录与表单提交依赖此项
    WTF_CSRF_ENABLED = True
```

```
    # 会话 cookie：不交给脚本读取，跨站请求不带
    SESSION_COOKIE_HTTPONLY = True
    SESSION_COOKIE_SAMESITE = "Lax"
    # 模板改动立即生效，不必重启服务。
    # Flask 在非 debug 下默认把模板缓存到进程里，改完模板刷新浏览器还是旧页面，
    # 容易被误当成「改了没生效」。本系统只在本机运行，这点开销可以忽略。
    TEMPLATES_AUTO_RELOAD = True
    # 「记住我」：登录时勾选则下发持久 cookie，关闭浏览器后仍可自动登录
    REMEMBER_COOKIE_DURATION = timedelta(days=30)
    REMEMBER_COOKIE_HTTPONLY = True
    REMEMBER_COOKIE_SAMESITE = "Lax"
    # 影像与数据文件上传（遥感影像挂接、批量导入）
    UPLOAD_FOLDER = os.path.join(BASE_DIR, "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024          # 单次上传上限
    32 MB
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
    ALLOWED_DATA_EXT = {"csv", "xlsx", "xls"}
    # 头像只用浏览器能直接显示的格式（不含 bmp / tif），并单独限制体积
    ALLOWED_AVATAR_EXT = {"png", "jpg", "jpeg", "gif",
    "webp"}
    MAX_AVATAR_LENGTH = 4 * 1024 * 1024            # 4 MB
# ==== app/models.py ====
# -*- coding: utf-8 -*-
"""数据模型：系统全部数据表。
本文件由甲在阶段 0 一次性写完，作为两人独立开发的地基。
约定：两人只读此文件；确需改表时，由乙提需求、甲统一修改。
"""
import json
from flask_login import UserMixin
from flask_sqlalchemy import SQLAlchemy
from app.utils.timefmt import utcnow    # 替代已弃用的
    datetime.utcnow()
db = SQLAlchemy()
class User(UserMixin, db.Model):
    """用户（账户与权限）。"""
    __tablename__ = "users"
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(64), unique=True,
    nullable=False, index=True)
    password_hash = db.Column(db.String(256),
    nullable=False)
    # 职务只有两种：管理员 admin（可管理账号与他人的职务） / 操作员 operator
    role = db.Column(db.String(20), nullable=False,
    default="operator")
    active = db.Column(db.Boolean, nullable=False,
    default=True)
    created_at = db.Column(db.DateTime, default=utcnow)
    # ---- 个人资料（用户可在「个人设置」里自行维护）----
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    display_name = db.Column(db.String(64), nullable=True)
    # 昵称
    email = db.Column(db.String(128), nullable=True)
    phone = db.Column(db.String(32), nullable=True)
    gender = db.Column(db.String(8), nullable=True)
    # 男 / 女 / 保密
    avatar = db.Column(db.String(256), nullable=True)
    # 头像，相对 static 的路径
    @property
    def is_admin(self):
        return self.role == "admin"
    @property
    def shown_name(self):
        """界面上显示的名字：优先用昵称，没填就用账号名。"""
        return self.display_name or self.username
    @property
    def initial(self):
        """没有头像时用首字占位。"""
        return (self.shown_name or "?")[0].upper()
    def to_dict(self):
        return {"id": self.id, "username": self.username,
                "display_name": self.display_name,
    "gender": self.gender,
                "role": self.role, "active": self.active}
class OperationLog(db.Model):
    """操作日志。"""
    __tablename__ = "operation_logs"
    id = db.Column(db.Integer, primary_key=True)
    user_id = db.Column(db.Integer, db.ForeignKey("users.id"
    ), nullable=True)
    action = db.Column(db.String(128), nullable=False)
    # 操作类型
    target_type = db.Column(db.String(64), nullable=True)
    # 对象类型
    target_id = db.Column(db.Integer, nullable=True)
    # 对象主键
    detail = db.Column(db.Text, nullable=True)
    # 操作详情
    created_at = db.Column(db.DateTime, default=utcnow,
    index=True)
class Project(db.Model):
    """评价项目（项目与矿山档案）。"""
    __tablename__ = "projects"
    id = db.Column(db.Integer, primary_key=True)
    name = db.Column(db.String(128), nullable=False)
    code = db.Column(db.String(64), nullable=True)
    # 项目编号
    description = db.Column(db.Text, nullable=True)
    status = db.Column(db.String(20), default="进行中")
    # 进行中 / 已完成 / 已归档
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    created_by = db.Column(db.Integer,                                  401
    db.ForeignKey("users.id"), nullable=True)                           402
    created_at = db.Column(db.DateTime, default=utcnow)                 403
    patches = db.relationship("MinePatch",                              404
    backref="project", lazy="dynamic",                                  405
                                    cascade="all, delete-orphan")       406
class MinePatch(db.Model):                                              407
    """矿山图斑。"""                                                    408
    __tablename__ = "mine_patches"                                      409
    id = db.Column(db.Integer, primary_key=True)                        410
    project_id = db.Column(db.Integer,                                  411
    db.ForeignKey("projects.id"), nullable=False)                       412
    name = db.Column(db.String(128), nullable=False)                    413
    admin_region = db.Column(db.String(128), nullable=True)             414
    # 所在行政区                                                        415
    center_x = db.Column(db.Float, nullable=True)                       416
    # 中心经度 / 横坐标                                                 417
    center_y = db.Column(db.Float, nullable=True)                       418
    # 中心纬度 / 纵坐标                                                 419
    area_ha = db.Column(db.Float, nullable=True)                        420
    # 图斑面积（公顷）                                                  421
    mineral_type = db.Column(db.String(64), nullable=True)              422
    # 开采矿种                                                          423
    mining_method = db.Column(db.String(64), nullable=True)             424
    # 开采方式                                                          425
    stop_years = db.Column(db.Float, nullable=True)                     426
    # 停产年限                                                          427
    responsible_entity_missing = db.Column(db.Boolean,                  428
    default=True)   # 责任主体是否缺失                                 429
    # 遥感底图 / 无人机正射影像挂接                                     430
    image_path = db.Column(db.String(256), nullable=True)               431
    # 相对 static 的路径                                                432
    image_note = db.Column(db.String(256), nullable=True)               433
    # 影像说明（如来源、时相）                                          434
    image_uploaded_at = db.Column(db.DateTime,                          435
    nullable=True)                                                      436
    created_at = db.Column(db.DateTime, default=utcnow)                 437
    units = db.relationship("MicroUnit", backref="patch",               438
    lazy="dynamic",                                                     439
                                    cascade="all, delete-orphan")       440
class ReferenceEcosystem(db.Model):                                     441
    """参照生态系统：标准化基准值，与图斑绑定。"""                     442
    __tablename__ = "reference_ecosystems"                              443
    id = db.Column(db.Integer, primary_key=True)                        444
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat            445
    ches.id"), nullable=False)                                          446
    coverage_benchmark = db.Column(db.Float, default=75.0)              447
    # 植被覆盖度基准 %                                                  448
    shannon_benchmark = db.Column(db.Float, default=0.7)                449
    # Shannon-Wiener 基准                                               450
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    dominant_species = db.Column(db.String(256),                    451
    nullable=True)   # 建群种                                        452
    soil_benchmark = db.Column(db.String(256),                      453
    nullable=True)     # 土壤基准                                   454
class Standard(db.Model):                                           455
    """地方标准 / 政策文件（政策与标准知识库）。"""                    456
    __tablename__ = "standards"                                     457
    id = db.Column(db.Integer, primary_key=True)                    458
    province = db.Column(db.String(32), nullable=False)             459
    # 省份                                                          460
    name = db.Column(db.String(256), nullable=False)                461
    # 名称                                                          462
    doc_number = db.Column(db.String(128), nullable=True)           463
    # 标准编号 / 文号                                               464
    publish_date = db.Column(db.String(32), nullable=True)          465
    implement_date = db.Column(db.String(32), nullable=True)        466
    # 类别：地方标准 / 政策文件                                     467
    category = db.Column(db.String(32), nullable=False,             468
    default="地方标准")                                             469
    summary = db.Column(db.Text, nullable=True)                     470
    # 特点概述                                                      471
    rules = db.relationship("StandardRule",                         472
    backref="standard",                                             473
                                cascade="all, delete-orphan")       474
class StandardRule(db.Model):                                       475
    """结构化规则：将标准条文拆解为可执行规则。"""                    476
    __tablename__ = "standard_rules"                                477
    id = db.Column(db.Integer, primary_key=True)                    478
    standard_id = db.Column(db.Integer,                             479
    db.ForeignKey("standards.id"), nullable=False)                  480
    item = db.Column(db.String(128), nullable=False)                481
    # 判定项                                                        482
    operator = db.Column(db.String(8), nullable=False)              483
    # 比较符：> >= < <= ==                                          484
    threshold = db.Column(db.String(64), nullable=True)             485
    # 阈值                                                          486
    condition_type = db.Column(db.String(32),                       487
    default="必要条件")   # 必要条件 / 充分条件                     488
    region = db.Column(db.String(64), nullable=True)                489
    # 适用分区（宁夏分区差异化）                                    490
    note = db.Column(db.Text, nullable=True)                        491
class ConstraintCheck(db.Model):                                    492
    """限制性判定记录。"""                                          493
    __tablename__ = "constraint_checks"                             494
    id = db.Column(db.Integer, primary_key=True)                    495
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat        496
    ches.id"), nullable=False)                                      497
    policy_pass = db.Column(db.Boolean, nullable=True)              498
    # 政策约束                                                      499
    location_pass = db.Column(db.Boolean, nullable=True)            500
```

```
    # 区位约束
    pollution_pass = db.Column(db.Boolean, nullable=True)
    # 土壤污染
    geo_pass = db.Column(db.Boolean, nullable=True)
    # 地质灾害
    farmland_pass = db.Column(db.Boolean, nullable=True)
    # 耕地属性
    passed = db.Column(db.Boolean, nullable=True)
    # 总体是否通过
    reason = db.Column(db.Text, nullable=True)
    # 不通过原因
    checked_at = db.Column(db.DateTime, default=utcnow)
class MicroUnit(db.Model):
    """微地质单元：岩质边坡 / 碎石边坡 / 平台。"""
    __tablename__ = "micro_units"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    unit_type = db.Column(db.String(32), nullable=False)
    # 单元类型
    area = db.Column(db.Float, nullable=True)
    # 面积
    ratio = db.Column(db.Float, nullable=True)
    # 面积占比（自动计算）
    description = db.Column(db.Text, nullable=True)
    # 现场描述
class Indicator(db.Model):
    """指标定义：指标树，含角色与启用单元类型。"""
    __tablename__ = "indicators"
    id = db.Column(db.Integer, primary_key=True)
    code = db.Column(db.String(32), unique=True,
    nullable=False)  # 指标编码
    name = db.Column(db.String(64), nullable=False)
    # 指标名称
    level = db.Column(db.Integer, nullable=False)
    # 1/2/3 级
    parent_code = db.Column(db.String(32), nullable=True)
    # 父指标编码
    # 角色：土壤污染指标同时作为限制性指标与评价指标
    is_constraint = db.Column(db.Boolean, default=False)
    is_evaluative = db.Column(db.Boolean, default=True)
    unit = db.Column(db.String(32), nullable=True)
    # 单位
    ideal_value = db.Column(db.String(64), nullable=True)
    # 理想值（可配置）
    ideal_type = db.Column(db.String(32), nullable=True)
    # benefit/cost/mid/discrete
    # 启用该指标的微地质单元（逗号分隔）。岩质边坡不启用土壤指标、平台不启用地形指标
    enabled_units = db.Column(db.String(128), nullable=True)
    description = db.Column(db.Text, nullable=True)
```

```
    # 取值依据说明
    def units_list(self):
        return [s for s in (self.enabled_units or
    "").split(",") if s]
class WeightScheme(db.Model):
    """权重方案：按微地质单元类型分别保存。"""
    __tablename__ = "weight_schemes"
    id = db.Column(db.Integer, primary_key=True)
    unit_type = db.Column(db.String(32), nullable=False)
    # 所属单元类型
    name = db.Column(db.String(64), nullable=True)
    # 方案名称
    level = db.Column(db.Integer, nullable=False,
    default=1)          # 1=一级指标 2=二级指标
    is_default = db.Column(db.Boolean, default=False)
    # 是否默认（系统默认）
    created_at = db.Column(db.DateTime, default=utcnow)
class WeightItem(db.Model):
    """权重明细：方案内的各项权重。"""
    __tablename__ = "weight_items"
    id = db.Column(db.Integer, primary_key=True)
    scheme_id = db.Column(db.Integer,
    db.ForeignKey("weight_schemes.id"), nullable=False)
    indicator_code = db.Column(db.String(32),
    nullable=False)        # 指标编码
    weight = db.Column(db.Float, nullable=False)
    # 权重值
class ExpertMatrix(db.Model):
    """专家判断矩阵：多专家打分的原始矩阵。"""
    __tablename__ = "expert_matrices"
    id = db.Column(db.Integer, primary_key=True)
    scheme_id = db.Column(db.Integer,
    db.ForeignKey("weight_schemes.id"), nullable=True)
    # 直接标注适用对象，便于无需先建权重方案即可录入专家矩阵
    unit_type = db.Column(db.String(32), nullable=True)
    # 微地质单元类型
    l1_name = db.Column(db.String(64), nullable=True)
    # 所属一级指标（二级矩阵用）
    expert_name = db.Column(db.String(64), nullable=True)
    # 专家姓名
    specialty = db.Column(db.String(64), nullable=True)
    # 专家专业
    level = db.Column(db.Integer, nullable=False,
    default=1)          # 矩阵对应指标层级
    matrix_json = db.Column(db.Text, nullable=False)
    # 判断矩阵（JSON 二维数组）
    cr = db.Column(db.Float, nullable=True)
    # 该专家矩阵的一致性比例
    created_at = db.Column(db.DateTime, default=utcnow)
class IndicatorData(db.Model):
```

```
    """指标实测数据：按微地质单元组织。"""
    __tablename__ = "indicator_data"
    id = db.Column(db.Integer, primary_key=True)
    unit_id = db.Column(db.Integer, db.ForeignKey("micro_uni
    ts.id"), nullable=False, unique=True)
    soil_ratio = db.Column(db.Float, nullable=True)
    # 土石比
    thickness = db.Column(db.Float, nullable=True)
    # 土层厚度 cm
    ph = db.Column(db.Float, nullable=True)
    organic_matter = db.Column(db.Float, nullable=True)
    # 有机质 g/kg
    # 重金属含量，JSON 字符串：{"Hg": 0.05, ...}，未检测项缺失或为 null
    heavy_metals = db.Column(db.Text, nullable=True)
    coverage = db.Column(db.Float, nullable=True)
    # 植被覆盖度 %
    shannon = db.Column(db.Float, nullable=True)
    # Shannon-Wiener 多样性指数
    slope = db.Column(db.Float, nullable=True)
    # 坡度 °
    aspect = db.Column(db.String(16), nullable=True)
    # 坡向：阳坡/阴阳坡/阴坡
    def heavy_metals_dict(self):
        """把重金属 JSON 解析成字典。
        字段是手工可改的文本，被脏写或导入半个 JSON 时不能让整页 500：
        解析失败按「未检测」处理（空字典），并把原文留在库里供排查。
        """
        if not self.heavy_metals:
            return {}
        try:
            data = json.loads(self.heavy_metals)
        except (ValueError, TypeError):
            return {}
        return data if isinstance(data, dict) else {}
class EvaluationResult(db.Model):
    """评价结果。"""
    __tablename__ = "evaluation_results"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    final_score = db.Column(db.Float, nullable=True)
    # 最终评价因子 F
    grade = db.Column(db.String(32), nullable=True)
    # 适宜性等级
    measures = db.Column(db.Text, nullable=True)
    # 推荐措施
    unit_scores = db.Column(db.Text, nullable=True)
    # 各单元得分 JSON
    # 计算参数快照：本次用的权重来源、参照生态系统基准、各指标理想取值。
    # 参数现在可以在界面上改，不留快照就无法复现这一次的结果。
```

```
    params_json = db.Column(db.Text, nullable=True)
    created_by = db.Column(db.Integer,
    db.ForeignKey("users.id"), nullable=True)
    created_at = db.Column(db.DateTime, default=utcnow)
class ScoreDetail(db.Model):
    """得分明细：每一层指标的权重、标准化值、加权贡献，供追溯与复核。"""
    __tablename__ = "score_details"
    id = db.Column(db.Integer, primary_key=True)
    result_id = db.Column(db.Integer,
    db.ForeignKey("evaluation_results.id"), nullable=False)
    unit_id = db.Column(db.Integer, nullable=True)
    l1_name = db.Column(db.String(64), nullable=True)
    # 一级指标
    l2_name = db.Column(db.String(64), nullable=True)
    # 二级指标
    raw_value = db.Column(db.String(64), nullable=True)
    # 实测值
    std_value = db.Column(db.Float, nullable=True)
    # 标准化值
    weight = db.Column(db.Float, nullable=True)
    # 权重
    contribution = db.Column(db.Float, nullable=True)
    # 加权贡献
# ==== app/forms.py ====
# -*- coding: utf-8 -*-
"""表单定义（共享，由甲统一维护）。
各模块的 Flask-WTF 表单集中于此。
"""
import re
from flask_wtf import FlaskForm
from wtforms import SelectField, SubmitField, StringField,
    PasswordField, BooleanField
from wtforms.validators import (DataRequired, Length,
    Optional, EqualTo,
                                ValidationError)
from app.utils.constants import UNIT_TYPES, ASPECT_TYPES,
    GENDERS
# 邮箱格式：够用的轻量校验。
# WTForms 自带的 Email() 需要额外安装 email_validator,
# 为一个字段引入依赖不划算，这里自己用正则判断。
EMAIL_RE = re.compile(r"^[^@\s]+@[^@\s]+\.[A-Za-z]{2,}$")
def email_format(form, field):
    v = (field.data or "").strip()
    if v and not EMAIL_RE.match(v):
        raise ValidationError("邮箱格式不正确")
def email_ok(value):
    """邮箱格式判断，供不用 WTForms 的表单复用（如管理员编辑用户）。"""
    v = (value or "").strip()
    return (not v) or bool(EMAIL_RE.match(v))
class AHPSelectForm(FlaskForm):
```

```
    """选择微地质单元类型与指标层级，以确定判断矩阵规模。"""
    unit_type = SelectField("微地质单元类型",
    validators=[DataRequired()])
    level = SelectField("指标层级",
                        choices=[("1", "一级指标"), ("2",
    "二级指标")],
                        validators=[DataRequired()])
    l1 = SelectField("一级指标", validators=[],
    validate_choice=False)  # 动态填充，层级=2 时使用
    submit = SubmitField("生成判断矩阵")
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        self.unit_type.choices = [(t, t) for t in
    UNIT_TYPES]
class EmptyForm(FlaskForm):
    """仅用于承载 CSRF 令牌的空表单（矩阵录入等自定义表单使用）。"""
    pass
class LoginForm(FlaskForm):
    """登录表单。支持「记住我」以在关闭浏览器后保持登录状态。
    校验消息用中文写明：登录框带 novalidate，浏览器不会代填提示，
    字段为空时若不自己给消息，用户点了「登录」会以为没反应。
    """
    username = StringField("账户名称",
                           validators=[DataRequired(message=
    "请输入账户名称"),
                                       Length(max=64,

    message="账户名称不能超过 64 个字符")])
    password = PasswordField("密码",
                             validators=[DataRequired(messag
    e="请输入密码")])
    remember = BooleanField("记住我")
    submit = SubmitField("登录")
GENDER_CHOICES = [("", "不填")] + [(g, g) for g in GENDERS]
class ProfileForm(FlaskForm):
    """个人资料：昵称、性别与联系方式。字段都可留空。
    职务不在这里 —— 只能由管理员在「个人设置 · 账号」里改。
    """
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format(),
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_info = SubmitField("保存资料")
class RegisterForm(FlaskForm):
    """注册：账号与密码必填，其余选填。注册后默认是操作员。"""
```

```
    username = StringField("账号名称",                                    751
                           validators=[DataRequired("请填写账号名称      752
    "),                                                                   753
                                       Length(min=2, max=64,              754
                                              message="账号名称           755
    2-64 个字符")])                                                       756
    password = PasswordField("密码",                                      757
                             validators=[DataRequired("请设置密码         758
    "),                                                                   759
                                         Length(min=6,                    760
    max=64,                                                               761
                                                                          762
    message="密码至少 6 位")])                                            763
    confirm = PasswordField("确认密码",                                   764
                            validators=[DataRequired(),                   765
                                        EqualTo("password",               766
                                                                          767
    message="两次输入的密码不一致")])                                     768
    display_name = StringField("昵称", validators=[Optional(),            769
    Length(max=64)])                                                      770
    gender = SelectField("性别", choices=GENDER_CHOICES,                  771
    validators=[Optional()])                                              772
    email = StringField("邮箱", validators=[Optional(),                   773
    email_format,                                                         774
                                        Length(max=128)])                 775
    phone = StringField("手机号", validators=[Optional(),                 776
    Length(max=32)])                                                      777
    submit_reg = SubmitField("注册并登录")                                778
class PasswordForm(FlaskForm):                                            779
    """修改密码：需要先验证当前密码。"""                                  780
    old_password = PasswordField("当前密码",                              781
    validators=[DataRequired()])                                          782
    new_password = PasswordField("新密码",                                783
                                 validators=[DataRequired(),              784
    Length(min=6, max=64,                                                 785
                                                                          786
    message="新密码至少 6 位")])                                          787
    confirm = PasswordField("确认新密码",                                 788
                            validators=[DataRequired(),                   789
                                                                          790
    EqualTo("new_password",                                               791
                                                                          792
    message="两次输入的新密码不一致")])                                   793
    submit_pwd = SubmitField("修改密码")                                  794
# ==== app/utils/constants.py ====                                        795
# -*- coding: utf-8 -*-                                                   796
"""评价指标常量表。                                                       797
全部数值为系统内置值，作为计算的唯一基准：                                798
- 取值规定：指标标准化理想极值与取值依据                                  799
- 等级划分规定：适宜性等级划分                                            800
```

```
- 系统默认权重：按微地质单元分别确定的一级、二级指标权重
- 权重计算方法：平均随机一致性指标 ＲＩ
"""
# ----------------------------------------------------------------
    ----------
# 一、微地质单元类型
# ----------------------------------------------------------------
    ----------
# 流程约定："按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡
# 三种地质微单元类型"
ＵＮＩＴ＿ＴＹＰＥＳ ＝ （"岩质边坡"， "碎石边坡"， "平台"）
# 每个单元类型启用的指标类别（系统约定：岩质边坡无土壤指标，平台无地形指标）
# 供数据录入界面与评价引擎判断"该单元是否需要某类指标"
ＵＮＩＴ＿ＥＮＡＢＬＥＤ＿ＩＮＤＩＣＡＴＯＲＳ ＝ ｛
    "岩质边坡"： （"植被指标"， "地形指标"）， # 不设土壤指标
    "碎石边坡"： （"土壤指标"， "植被指标"， "地形指标"），
    "平台"： （"土壤指标"， "植被指标"），  # 不设地形指标
｝
# ----------------------------------------------------------------
    ----------
# 二、指标标准化理想极值（指标取值规定）
# ----------------------------------------------------------------
    ----------
ＩＤＥＡＬ＿ＳＯＩＬ＿ＲＡＴＩＯ ＝ ０．８ # 土石比理想值
ＩＤＥＡＬ＿ＴＨＩＣＫＮＥＳＳ ＝ ２０．０ # 土层厚度理想值 （ｃｍ）
ＩＤＥＡＬ＿ＰＨ＿ＲＡＮＧＥ ＝ （５．０， ９．０） # ｐＨ 理想区间（适中型指标）
ＩＤＥＡＬ＿ＯＲＧＡＮＩＣ ＝ ３０．０ # 有机质理想值
    （ｇ／ｋｇ），全国第二次土壤普查养分等级划分标准二等最低值
ＩＤＥＡＬ＿ＣＯＶＥＲＡＧＥ ＝ ７５．０ # 植被覆盖度参照生态系统 （％）
ＩＤＥＡＬ＿ＳＨＡＮＮＯＮ ＝ ０．７ # Ｓｈａｎｎｏｎ-Ｗｉｅｎｅｒ 多样性指数参照生态系统
# 重金属极值 （ｍｇ／ｋｇ），《绿化种植土壤》ＣＪ／Ｔ ３４０—２０１６ 土壤重金属含量 ＩＶ 级最高值
# 取值依据：镉、汞、铅、铬、砷、镍、铜、锌分别为 2、2、５３０、４００、４５、２２０、６００、８００
ＨＥＡＶＹ＿ＭＥＴＡＬＳ ＝ （"Ｃｄ"， "Ｈｇ"， "Ｐｂ"， "Ｃｒ"， "Ａｓ"， "Ｎｉ"， "Ｃｕ"，
    "Ｚｎ"）
ＨＥＡＶＹ＿ＭＥＴＡＬ＿ＬＩＭＩＴ ＝ ｛
    "Ｃｄ"： ２．０， "Ｈｇ"： ２．０， "Ｐｂ"： ５３０．０， "Ｃｒ"： ４００．０，
    "Ａｓ"： ４５．０， "Ｎｉ"： ２２０．０， "Ｃｕ"： ６００．０， "Ｚｎ"： ８００．０，
｝
# 污染指标下 8 种重金属"重要性相当，权重平均分配"（权重计算方法）
ＨＥＡＶＹ＿ＭＥＴＡＬ＿ＥＱＵＡＬ＿ＷＥＩＧＨＴ ＝ １．０ ／ ８．０
# 坡度分界，取值规定引《新一轮退耕还林还草总体方案》发改西部〔２０１４〕１７７２ 号
ＳＬＯＰＥ＿ＢＲＥＡＫ ＝ ２５．０
# 坡向离散赋值，取值依据："阳坡、阴阳坡、阴坡分别赋值 0．2、０．６、１"
ＡＳＰＥＣＴ＿ＳＣＯＲＥＳ ＝ ｛"阳坡"： ０．２， "阴阳坡"： ０．６， "阴坡"： １．０｝
ＡＳＰＥＣＴ＿ＴＹＰＥＳ ＝ （"阳坡"， "阴阳坡"， "阴坡"）
# ｐＨ 适中型标准化的容差边界（区间外线性衰减，避免得分突变）
ＰＨ＿ＴＯＬ＿ＬＯＷ ＝ ４．０
ＰＨ＿ＴＯＬ＿ＨＩＧＨ ＝ １０．０
# ----------------------------------------------------------------
    ----------
```

```
# 三、适宜性等级划分
# ------------------------------------------------------------
    ----------------
GRADE_LEVELS = (
    {"grade": "不适宜", "min": None, "max": 0.3,
     "meaning": "存在限制性条件或区域生态本底较差，自然恢复影响因子严重制约自然恢复",
     "measure": "生态重建、人工辅助"},
    {"grade": "基本适宜", "min": 0.3, "max": 0.7,
     "meaning": "生态本底良好，核心自然恢复影响因子基本满足自然恢复需求",
     "measure": "自然恢复为主，可辅以封闭场地和少量补种、补植、飞播等"},
    {"grade": "适宜", "min": 0.7, "max": None,
     "meaning": "各自然恢复影响因子均能满足自然恢复需求，自然演替速度快",
     "measure": "自然恢复"},
)
def grade_of(score):
    """按适宜性等级划分表判定等级。"""
    for g in GRADE_LEVELS:
        lo, hi = g["min"], g["max"]
        if lo is None and score < hi:
            return g
        if hi is None and score >= lo:
            return g
        if lo is not None and hi is not None and lo <=
    score < hi:
            return g
    # score 恰好等于 0.7 时落入"适宜"
    return GRADE_LEVELS[-1]
# ------------------------------------------------------------
    ----------------
# 四、平均随机一致性指标 RI（权重计算方法给出 n=2，3，4，5，其余为常用扩展值）
# ------------------------------------------------------------
    ----------------
RI_TABLE = {1: 0.0, 2: 0.0, 3: 0.58, 4: 0.90, 5: 1.12, 6:
    1.24,
            7: 1.32, 8: 1.41, 9: 1.45, 10: 1.49}
# 一致性比例阈值（方法要求：当 CR<0.1 时，判断矩阵满足一致性要求）
CR_THRESHOLD = 0.1
# ------------------------------------------------------------
    ----------------
# 五、默认权重方案（系统内置）
# ------------------------------------------------------------
    ----------------
# 指标取值规定："划分岩质边坡、碎石土边坡、平台三类微地质单元开展独立评价，
# 分别计算评价指标权重"（方法概述与流程均作"碎石边坡"，
# 系统统一采用"碎石边坡"）。权重由 10 位专家打分后经 AHP 方法确定。
# 下表为系统内置的归一化权重，作为系统默认方案。
DEFAULT_WEIGHTS = {
    "岩质边坡": {
        "l1": {"植被指标": 0.45, "地形指标": 0.55},
        "l2": {
```

```
                    "植被指标"： {"植被覆盖度"： 0.67， "生物多样性"： 0.33}，
                    "地形指标"： {"坡度"： 0.64， "坡向"： 0.36}，
                },
            },
            "碎石边坡"： {
                "l1"： {"土壤指标"： 0.44， "植被指标"： 0.31， "地形指标"： 0.25}，
                "l2"： {
                    "土壤指标"： {"土石比"： 0.25， "土层厚度"： 0.25， "pH"： 0.17，
                                "有机质"： 0.17， "污染指标"： 0.17}，
                    "植被指标"： {"植被覆盖度"： 0.64， "生物多样性"： 0.36}，
                    "地形指标"： {"坡度"： 0.67， "坡向"： 0.33}，
                },
            },
            "平台"： {
                "l1"： {"土壤指标"： 0.54， "植被指标"： 0.46}，
                "l2"： {
                    "土壤指标"： {"土石比"： 0.25， "土层厚度"： 0.25， "pH"： 0.17，
                                "有机质"： 0.17， "污染指标"： 0.17}，
                    "植被指标"： {"植被覆盖度"： 0.61， "生物多样性"： 0.39}，
                },
            },
        }
# ------------------------------------------------------------------------
    ------------------
# 六、限制性指标（指标清单与评价流程）
# ------------------------------------------------------------------------
    ------------------
# 方法概述与评价流程：四项限制性指标；流程段另明确"是否为耕地"为约束条件
CONSTRAINT_ITEMS = ("政策"， "区位约束"， "土壤污染"， "地质灾害"， "耕地属性")
# ------------------------------------------------------------------------
    ------------------
# 七、账号职务
# ------------------------------------------------------------------------
    ------------------
# 只分两种：管理员可管理账号、维护基础数据并查看操作日志，操作员用业务功能。
ROLE_LABELS = {"admin": "管理员"， "operator": "操作员"}
ROLE_CODES = ("admin"， "operator")
DEFAULT_ROLE = "operator"                # 新注册账号的默认职务
ROLE_ORDER = ("admin"， "operator")   # 列表排序用：管理员在前
# 性别（注册与个人资料中选填）
GENDERS = ("男"， "女"， "保密")
# ==== app/utils/schema.py ====
# -*- coding: utf-8 -*-
"""轻量结构迁移。
SQLite 下 `db.create_all()` 只建缺失的表，不会给已有表补列。
本模块在启动时检查目标列是否存在，缺失则 ALTER TABLE 补上，
使旧数据库无需重建即可用上新字段。
"""
from sqlalchemy import text
# （表名， 列名， 列定义）
```

```
REQUIRED_COLUMNS = [
    # 遥感影像挂接
    ("mine_patches", "image_path", "VARCHAR(256)"),
    ("mine_patches", "image_note", "VARCHAR(256)"),
    ("mine_patches", "image_uploaded_at", "DATETIME"),
    # 指标维护
    ("indicators", "enabled_units", "VARCHAR(128)"),
    ("indicators", "description", "TEXT"),
    # 多专家矩阵
    ("expert_matrices", "unit_type", "VARCHAR(32)"),
    ("expert_matrices", "l1_name", "VARCHAR(64)"),
    ("expert_matrices", "specialty", "VARCHAR(64)"),
    ("expert_matrices", "cr", "FLOAT"),
    # 用户个人资料（账户页）
    ("users", "display_name", "VARCHAR(64)"),
    ("users", "email", "VARCHAR(128)"),
    ("users", "phone", "VARCHAR(32)"),
    ("users", "gender", "VARCHAR(8)"),
    ("users", "avatar", "VARCHAR(256)"),
    # 评价结果的可追溯信息：计算参数快照与计算人
    ("evaluation_results", "params_json", "TEXT"),
    ("evaluation_results", "created_by", "INTEGER"),
]
# 职务体系由「管理员 / 评价员 / 查看者」三种收敛为「管理员 / 操作员」两种，
# 旧数据里的后两种统一归为操作员。
ROLE_MIGRATION = {"evaluator": "operator", "viewer":
    "operator"}
def _add_columns(db):
    added = []
    with db.engine.connect() as conn:
        for table, column, ddl in REQUIRED_COLUMNS:
            rows = conn.execute(text("PRAGMA
    table_info(%s)" % table)).fetchall()
            if not rows:
                continue                          #
    表还不存在，create_all 会建
            existing = {r[1] for r in rows}
            if column in existing:
                continue
            conn.execute(text(
                "ALTER TABLE %s ADD COLUMN %s %s" % (table,
    column, ddl)))
            conn.commit()
            added.append("%s.%s" % (table, column))
    return added
def _migrate_roles(db):
    """把旧角色名统一为新的两种，返回改动的行数。"""
    changed = 0
    with db.engine.connect() as conn:
        rows = conn.execute(text("PRAGMA
```

```
    table_info(users)")).fetchall()
        if not rows:
            return 0
        for old, new in ROLE_MIGRATION.items():
            r = conn.execute(text("UPDATE users SET role =
    :new "
                                  "WHERE role = :old"),
                                {"new": new, "old": old})
            changed += r.rowcount or 0
        conn.commit()
    return changed
def ensure_schema(db):
    """为已有表补齐缺失列，并做必要的数据迁移。返回变更清单。"""
    try:
        changed = _add_columns(db)
        n = _migrate_roles(db)
        if n:
            changed.append("用户角色归一为管理员/操作员 %d 条" % n)
        return changed
    except Exception as e:                                  #
    迁移失败不应阻断启动
        print("[schema] 结构检查跳过：%s" % e)
        return []
# ==== app/utils/uploads.py ====
# -*- coding: utf-8 -*-
"""上传文件处理（遥感影像、批量导入）。"""
import os
import re
import uuid
from datetime import datetime
from werkzeug.utils import secure_filename
def _safe_ext(filename):
    return os.path.splitext(filename)[1].lower().lstrip(".")
def ext_allowed(filename, allowed):
    return _safe_ext(filename) in allowed
def save_upload(file_storage, subdir, allowed_ext,
    upload_root):
    """保存上传文件，返回相对 static 的路径（如 uploads/patches/xxx.png）。
    文件名重命名为「时间戳_随机串.扩展名」，避免中文名与重名问题。
    """
    if file_storage is None or not file_storage.filename:
        return None, "未选择文件"
    ext = _safe_ext(file_storage.filename)
    if ext not in allowed_ext:
        return None, "不支持的文件类型：.%s" % ext
    target_dir = os.path.join(upload_root, subdir)
    os.makedirs(target_dir, exist_ok=True)
    # 仅保留安全字符，并加时间戳与随机串防重名
    base = secure_filename(os.path.splitext(file_storage.fil
    ename)[0]) or "file"
```

```
    base = re.sub(r"[^A-Za-z0-9_\-]", "", base)[:40] or "file"
    name = "%s_%s.%s" % (datetime.now().strftime("%Y%m%d%H%M%S"),
                         uuid.uuid4().hex[:6], ext)
    abs_path = os.path.join(target_dir, name)
    file_storage.save(abs_path)
    rel = os.path.join("uploads", subdir, name).replace("\\", "/")
    return rel, None
def remove_upload(rel_path, upload_root):
    """删除已保存的上传文件（相对 static 的路径）。
    路径必须落在上传根目录之内：rel_path 来自数据库，正常都是 save_upload
    生成的，但万一被改脏（手工改库、导入异常），`../` 这种值会删掉上传根目录
    之外的文件。这里用 realpath 做一次围栏校验。
    """
    if not rel_path:
        return
    root = os.path.realpath(upload_root)
    abs_path = os.path.realpath(
        os.path.join(root, rel_path.replace("uploads/", "", 1)))
    if abs_path != root and not abs_path.startswith(root + os.sep):
        return
    try:
        if os.path.isfile(abs_path):
            os.remove(abs_path)
    except OSError:
        pass
# ==== app/utils/charts.py ====
# -*- coding: utf-8 -*-
"""轻量图表。
用纯 SVG 输出，不依赖任何绘图库或前端脚本：
- 服务端渲染，页面与打印（PDF）中都能稳定显示
- 每个图元都带数值标签，黑白打印时仍可读（不单靠颜色区分）
- 单色方案，符合界面「一个视图一个强调色」的约定
"""
from html import escape
ACCENT = "#559FCF"
ACCENT_SOFT = "#64BBE0"
INK = "#23262b"
MUTED = "#6b7280"
LINE = "#dfe3e8"
# 科研顶刊配色
GRAYS = ["#6FAE47", "#FCE5D6", "#FFF2CC", "#DAE3F2",
         "#E9C9C9", "#559FCF"]
def _fmt(v, digits=4):
    try:
        return ("%%.%df" % digits) % float(v)
```

```
    except (TypeError, ValueError):                                    1101
        return str(v)                                                  1102
def hbars(items, width=620, bar_h=22, gap=12, label_w=150,             1103
    value_w=76):                                                       1104
    """水平条形图。items: [(标签, 数值, 显示文本)]"""                    1105
    if not items:                                                      1106
        return '<p class="muted">暂无数据</p>'                          1107
    n = len(items)                                                     1108
    height = n * (bar_h + gap) + 10                                    1109
    max_v = max((abs(float(v)) for _, v, _ in items),                  1110
    default=1) or 1                                                    1111
    body_w = width - label_w - value_w                                 1112
    parts = ['<svg viewBox="0 0 %d %d" width="100%%"                   1113
    height="%d" '                                                      1114
             'role="img" aria-label="条形图" '                         1115
             'style="max-width:%dpx;font-family:inherit;">'            1116
    % (width, height, height, width)]                                  1117
    for i, (label, value, disp) in enumerate(items):                   1118
        y = i * (bar_h + gap) + 5                                      1119
        frac = (abs(float(value)) / max_v) if max_v else 0             1120
        w = max(2.0, body_w * frac)                                    1121
        fill = GRAYS[i % len(GRAYS)]                                   1122
        parts.append(                                                  1123
            '<text x="%d" y="%d" text-anchor="end"                     1124
    font-size="12" fill="%s">%s</text>'                                1125
            % (label_w - 10, y + bar_h * 0.72, INK,                    1126
    escape(str(label))))                                               1127
        parts.append(                                                  1128
            '<rect x="%d" y="%d" width="%.1f" height="%d"              1129
    rx="3" fill="%s"/>'                                                1130
            % (label_w, y, w, bar_h, fill))                            1131
        parts.append(                                                  1132
            '<text x="%d" y="%d" font-size="12" fill="%s" '            1133
            'style="font-variant-numeric:tabular-nums;">%s</           1134
    text>'                                                             1135
            % (label_w + w + 8, y + bar_h * 0.72, MUTED,               1136
    escape(str(disp))))                                                1137
    parts.append('</svg>')                                             1138
    return "".join(parts)                                              1139
def donut(items, size=210, thickness=34):                              1140
    """环形图（权重分布）。items: [(标签, 数值, 显示文本)]              1141
    图例中给出数值与占比，黑白打印时靠图例与灰度深浅区分。              1142
    """                                                                1143
    vals = [(l, float(v), d) for l, v, d in items if                   1144
    float(v) > 0]                                                      1145
    if not vals:                                                       1146
        return '<p class="muted">暂无数据</p>'                          1147
    total = sum(v for _, v, _ in vals) or 1                            1148
    r = (size - thickness) / 2                                         1149
    cx = cy = size / 2                                                 1150
```

```
import math
parts = ['<svg viewBox="0 0 %d %d" width="%d"
height="%d" role="img" '
         'aria-label="权重分布环形图">' % (size, size, size,
size)]
angle = -math.pi / 2
for i, (label, v, disp) in enumerate(vals):
    sweep = 2 * math.pi * (v / total)
    a0, a1 = angle, angle + sweep
    x0, y0 = cx + r * math.cos(a0), cy + r *
math.sin(a0)
    x1, y1 = cx + r * math.cos(a1), cy + r *
math.sin(a1)
    large = 1 if sweep > math.pi else 0
    parts.append(
        '<path d="M %.2f %.2f A %.2f %.2f 0 %d 1 %.2f
%.2f" '
        'fill="none" stroke="%s" stroke-width="%d"/>'
        % (x0, y0, r, r, large, x1, y1, GRAYS[i %
len(GRAYS)], thickness))
    angle = a1
parts.append('<text x="%d" y="%d" text-anchor="middle"
font-size="12" '
             'fill="%s">合计</text>' % (cx, cy - 2,
MUTED))
parts.append('<text x="%d" y="%d" text-anchor="middle"
font-size="15" '
             'fill="%s" style="font-variant-numeric:tabu
lar-nums;">%s</text>'
             % (cx, cy + 18, INK, _fmt(total, 4)))
parts.append('</svg>')
legend = ['<table style="width:auto;margin-top:10px;"><t
head><tr>'
          '<th scope="col">项目</th><th
scope="col">数值</th>'
          '<th scope="col">占比</th></tr></thead><tbody>']
for i, (label, v, disp) in enumerate(vals):
    legend.append(
        '<tr><td style="text-align:left;">'
        '<span style="display:inline-block;width:10px;he
ight:10px;'
        'border-radius:2px;background:%s;margin-right:6p
x;"></span>%s</td>'
        '<td class="mono">%s</td><td
class="mono">%.1f%%</td></tr>'
        % (GRAYS[i % len(GRAYS)], escape(str(label)),
escape(str(disp)),
           100.0 * v / total))
legend.append('</tbody></table>')
return ('<div style="display:flex;flex-wrap:wrap;gap:20p
```

```
    x;align-items:center;">'                                                1201
            + "".join(parts) + "".join(legend) + '</div>')                  1202
# ==== app/utils/browser.py ====                                            1203
# -*- coding: utf-8 -*-                                                     1204
"""HTML 转 PDF（报告导出）。                                                1205
复用本机已安装的 Chrome/Edge 无头模式打印，避免引入 reportlab、             1206
weasyprint 等额外依赖。生成过程使用系统临时目录中的 ASCII 文件名，          1207
避免中文路径在 file:// URL 下出问题。                                       1208
"""                                                                         1209
import os                                                                   1210
import shutil                                                               1211
import subprocess                                                           1212
import tempfile                                                             1213
import uuid                                                                 1214
CANDIDATES = [                                                              1215
    r"C:\Program Files\Google\Chrome\Application\chrome.exe"                 1216
    ,                                                                       1217
    r"C:\Program Files (x86)\Google\Chrome\Application\chrom                 1218
    e.exe",                                                                 1219
    r"C:\Program Files (x86)\Microsoft\Edge\Application\msed                 1220
    ge.exe",                                                                1221
    r"C:\Program Files\Microsoft\Edge\Application\msedge.exe                 1222
    ",                                                                      1223
]                                                                           1224
# 允许通过环境变量覆盖，便于在其他机器上指定浏览器                          1225
ENV_KEY = "DSH_BROWSER_PATH"                                                1226
def browser_path():                                                         1227
    """定位可用的浏览器可执行文件，找不到返回 None。"""                     1228
    override = os.environ.get(ENV_KEY)                                      1229
    if override and os.path.isfile(override):                               1230
        return override                                                     1231
    for p in CANDIDATES:                                                    1232
        if os.path.isfile(p):                                               1233
            return p                                                        1234
    for name in ("chrome", "msedge", "chromium"):                           1235
        found = shutil.which(name)                                          1236
        if found:                                                           1237
            return found                                                    1238
    return None                                                             1239
def html_to_pdf(html, timeout=90):                                          1240
    """把 HTML 字符串渲染为 PDF，返回字节；失败返回 (None, 原因)。"""       1241
    exe = browser_path()                                                    1242
    if exe is None:                                                         1243
        return None, ("未找到可用的浏览器（Chrome/Edge），无法生成 PDF。"    1244
                      "可设置环境变量 %s 指向浏览器可执行文件。" % ENV_KEY)  1245
    tmpdir = tempfile.mkdtemp(prefix="dshpdf_")                             1246
    tag = uuid.uuid4().hex[:8]                                              1247
    html_path = os.path.join(tmpdir, "r_%s.html" % tag)                     1248
    pdf_path = os.path.join(tmpdir, "r_%s.pdf" % tag)                       1249
    try:                                                                    1250
```

```
            with open(html_path, "w", encoding="utf-8") as f:
                f.write(html)
            cmd = [
                exe, "--headless=new", "--disable-gpu",
        "--no-sandbox",
                "--no-first-run", "--no-default-browser-check",
                "--no-pdf-header-footer",
                "--virtual-time-budget=4000",
                "--print-to-pdf=%s" % pdf_path,
                "file:///" + html_path.replace("\\", "/"),
            ]
            p = subprocess.run(cmd, capture_output=True,
        text=True,
                               timeout=timeout,
        encoding="utf-8", errors="replace")
            if not os.path.isfile(pdf_path) or
        os.path.getsize(pdf_path) < 800:
                return None, "浏览器未能生成 PDF（%s）" % (p.stderr or
        "").strip()[:200]
            with open(pdf_path, "rb") as f:
                return f.read(), None
        except subprocess.TimeoutExpired:
            return None, "生成 PDF 超时，请稍后重试"
        except Exception as e:
            return None, "生成 PDF 失败：%s" % e
        finally:
            shutil.rmtree(tmpdir, ignore_errors=True)
# ==== app/services/ahp_service.py ====
# -*- coding: utf-8 -*-
"""AHP 权重引擎。
按权重计算方法实现：
- 式（２）（３）（４）：特征根法求权重（行元素乘积 → n 次方根 → 归一化）
- 式（５）（６）（７）：一致性检验（λmax、CI、CR）
- 方法要求 CR < 0.1；RI 按阶数查表
"""
import numpy as np
from app.utils.constants import RI_TABLE, CR_THRESHOLD
def matrix_from_upper(upper):
    """由判断矩阵上三角构造完整互反矩阵。
    参数
    ----
    upper : list[list[float]] 或 np.ndarray，上三角元素（不含对角线）。
        例：3 阶时传入 [[a12, a13], [a23]]。
        对角线为 1，下三角取倒数（a_ji = 1/a_ij）。
    """
    upper = np.asarray(upper, dtype=float)
    k = upper.shape[0]
    n = int(round((1 + (1 + 8 * k) ** 0.5) / 2))
    A = np.ones((n, n), dtype=float)
    idx = 0
```

```
    for i in range(n):
        for j in range(i + 1, n):
            A[i, j] = upper[idx]
            A[j, i] = 1.0 / upper[idx]
            idx += 1
    return A
def weights_and_consistency(A):
    """由判断矩阵 A 计算权重与一致性指标。
    返回
    ----
    dict：{weights: np.ndarray, lambda_max, ci, cr, n,
    consistent: bool}
    """
    A = np.asarray(A, dtype=float)
    n = A.shape[0]
    # 式(2) 行元素乘积
    row_prod = np.prod(A, axis=1)
    # 式(3) 开 n 次方根
    root = row_prod ** (1.0 / n)
    # 式(4) 归一化
    weights = root / root.sum()
    # 式(5) 最大特征值 λmax = (1/n) Σ (A·W)_i / W_i
    aw = A.dot(weights)
    lambda_max = float(np.sum(aw / weights) / n)
    # 式(6) 一致性指标 CI
    ci = (lambda_max - n) / (n - 1) if n > 1 else 0.0
    # 式(7) 一致性比例 CR = CI / RI
    ri = RI_TABLE.get(n, 1.49)
    cr = ci / ri if ri else 0.0
    return {
        "weights": weights,
        "lambda_max": lambda_max,
        "ci": ci,
        "cr": cr,
        "n": n,
        "consistent": cr < CR_THRESHOLD,
    }
def find_most_inconsistent(A):
    """定位判断矩阵中最不一致的一对比较。
    依据：完全一致矩阵应满足 a_ij = w_i / w_j。
    计算每个上三角元素与其理想值的相对偏差，返回偏差最大者。
    返回
    ----
    dict：{i, j, value, expected, deviation}，偏差最大的 (i, j)
    及对应取值。
    """
    A = np.asarray(A, dtype=float)
    r = weights_and_consistency(A)
    w = r["weights"]
    n = A.shape[0]
```

```
    best = None                                                    1351
    for i in range(n):                                             1352
        for j in range(i + 1, n):                                  1353
            expected = w[i] / w[j]                                 1354
            if expected == 0:                                      1355
                continue                                           1356
            deviation = abs(A[i, j] - expected) / expected         1357
            if best is None or deviation >                         1358
    best["deviation"]:                                             1359
                best = {"i": i, "j": j, "value": A[i, j],          1360
                        "expected": expected, "deviation":         1361
    deviation}                                                     1362
    return best                                                    1363
def aggregate_matrices(matrices):                                  1364
    """多专家判断矩阵聚合：几何平均（保持互反性）。                1365
    参数                                                           1366
    ----                                                           1367
    matrices : list[np.ndarray]，多个同阶判断矩阵。                1368
    """                                                            1369
    stack = np.stack([np.asarray(m, dtype=float) for m in          1370
    matrices])                                                     1371
    return np.prod(stack, axis=0) ** (1.0 / len(stack))            1372
def judge_from_weights(weights):                                   1373
    """由权重向量构造完全一致判断矩阵并检验。                      1374
    用途：验证给定权重向量能否作为某个完全一致判断矩阵的主特征向量 1375
    （A_ij = w_i / w_j 时 λmax = n、CR = 0）。                      1376
    """                                                            1377
    w = np.asarray(weights, dtype=float)                           1378
    n = w.shape[0]                                                 1379
    A = np.outer(w, 1.0 / w)                                       1380
    r = weights_and_consistency(A)                                 1381
    r["matrix"] = A                                                1382
    return r                                                       1383
# ==== app/services/standardization.py ====                        1384
# -*- coding: utf-8 -*-                                            1385
"""指标标准化（的一部分）。                                        1386
按标准化公式实现：                                                 1387
    x' = (x - x_min) / (x_max - x_min)                             1388
按指标性质分为四类：越大越优 / 越小越优 / 适中 / 离散赋值。        1389
所有理想极值取自 app.utils.constants，可配置。                     1390
"""                                                                1391
from app.utils.constants import (                                  1392
    IDEAL_SOIL_RATIO, IDEAL_THICKNESS, IDEAL_PH_RANGE,             1393
    IDEAL_ORGANIC,                                                 1394
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,              1395
    HEAVY_METAL_EQUAL_WEIGHT, SLOPE_BREAK, ASPECT_SCORES,          1396
    PH_TOL_LOW, PH_TOL_HIGH,                                       1397
)                                                                  1398
def std_benefit(value, ideal):                                     1399
    """越大越优指标标准化：以理想值为上限，按比例给分，饱和于 [0，1]。"""  1400
```

```
    if value is None:                                                    1401
        return None                                                      1402
    return max(0.0, min(1.0, value / ideal))                             1403
def std_cost(value, extreme):                                            1404
    """越小越优指标标准化：以极值为参考，线性折减至 [0,1]。"""                1405
    if value is None:                                                    1406
        return None                                                      1407
    return max(0.0, min(1.0, 1.0 - value / extreme))                     1408
def std_mid_ph(value, lo, hi, tol_lo=PH_TOL_LOW,                         1409
    tol_hi=PH_TOL_HIGH):                                                 1410
    """适中型指标标准化（pH）：理想区间内取 1，区间外线性衰减。"""           1411
    if value is None:                                                    1412
        return None                                                      1413
    if lo <= value <= hi:                                                1414
        return 1.0                                                       1415
    if value > hi:                                                       1416
        return max(0.0, (tol_hi - value) / (tol_hi - hi))                1417
    return max(0.0, (value - tol_lo) / (lo - tol_lo))                    1418
def std_slope(value, break_point=SLOPE_BREAK, upper=90.0):               1419
    """坡度标准化。                                                      1420
    取值规定：以 25° 为陡坡重要分界。25° 及以下记满分 1，                1421
    超过 25° 线性折减，90° 记 0。                                        1422
    """                                                                  1423
    if value is None:                                                    1424
        return None                                                      1425
    if value <= break_point:                                             1426
        return 1.0                                                       1427
    return max(0.0, (upper - value) / (upper - break_point))             1428
def aspect_score(aspect):                                                1429
    """坡向离散赋值：阳坡 0.2 / 阴阳坡 0.6 / 阴坡 1.0（取值规定）。"""     1430
    if aspect is None:                                                   1431
        return None                                                      1432
    return ASPECT_SCORES.get(aspect)                                     1433
def pollution_score(heavy_metals, limits=None):                          1434
    """污染指标：8 种重金属等权（各 0.125）合成。                        1435
    方法约定：未检测 Cr、Zn 的，从矿山类型看未受污染，按无污染（=1）处理。  1436
    heavy_metals : dict，键为金属名，值为含量 mg/kg；缺失键视为未检测。    1437
    limits : 可选，金属 → 极值；不传则用代码常量（界面改过就传库里的值）。  1438
    """                                                                  1439
    table = limits or HEAVY_METAL_LIMIT                                  1440
    total = 0.0                                                          1441
    for metal, extreme in table.items():                                 1442
        value = heavy_metals.get(metal)                                  1443
        total += std_cost(value, extreme) if value is not                1444
    None else 1.0                                                        1445
    return total / len(table)                                            1446
def standardize_unit(unit_type, data, weights=None,                      1447
    settings=None):                                                      1448
    """对单个微地质单元的全部指标做标准化。                              1449
    参数                                                                 1450
```

```
    ----                                                                1451
    unit_type : str，岩质边坡 / 碎石边坡 / 平台                          1452
    data : dict，实测数据，字段见接口契约：                              1453
        soil_ratio, thickness, ph, organic_matter,                      1454
    heavy_metals(dict),                                                 1455
        coverage, shannon, slope, aspect                                1456
    weights : dict，该单元类型的权重（app.utils.constants.DEFAULT_WEIG  1457
    HTS 结构），                                                        1458
                未提供时用默认值。                                      1459
    settings : app.services.unit_config.UnitSettings，图斑级的基准值   1460
    与理想取值。                                                        1461
                不传则全部按代码常量 —— 历史数据与验证算例走这条路径，结果不变。  1462
    返回                                                                1463
    ----                                                                1464
    dict：{ 一级指标： { 二级指标： （实测值， 标准化值， 权重， 贡献） } }  1465
    标准化值可能为 None，表示该指标不适用或方法待定。                   1466
    """                                                                 1467
    from app.utils.constants import DEFAULT_WEIGHTS                     1468
    from app.services import unit_config as uc                          1469
    if weights is None:                                                 1470
        weights = DEFAULT_WEIGHTS[unit_type]                            1471
    if settings is None:                                                1472
        settings = uc.UnitSettings()                                    1473
    # 理想取值：库里有数字就用库里的，否则用常量                        1474
    def ideal(code, fallback):                                          1475
        return settings.ideal(code, fallback)                           1476
    l1_enabled = weights["l1"].keys()                                   1477
    l2w = weights["l2"]                                                 1478
    result = {}                                                         1479
    # 一级指标是否启用：库里配过就按库里的（界面上的「启用单元」），没配按常量  1480
    def group_on(l1_name, l1_code):                                     1481
        return settings.group_enabled(unit_type, l1_name,               1482
    l1_code) is not False                                               1483
    # ---- 土壤指标 ----                                               1484
    if "土壤指标" in l1_enabled and group_on("土壤指标", "soil"):       1485
        result["土壤指标"] = {                                          1486
            "土石比": (data.get("soil_ratio"),                          1487
                       std_benefit(data.get("soil_ratio"),              1488
                                   ideal("soil_ratio",                  1489
    IDEAL_SOIL_RATIO)),                                                 1490
                       l2w["土壤指标"]["土石比"]),                      1491
            "土层厚度": (data.get("thickness"),                         1492
                         std_benefit(data.get("thickness"),             1493
                                     ideal("thickness",                 1494
    IDEAL_THICKNESS)),                                                  1495
                         l2w["土壤指标"]["土层厚度"]),                  1496
            "pH": (data.get("ph"),                                      1497
                   std_mid_ph(data.get("ph"),                           1498
    *IDEAL_PH_RANGE),                                                   1499
                   l2w["土壤指标"]["pH"]),                              1500
```

```
                "有机质": (data.get("organic_matter"),                  1501
                          std_benefit(data.get("organic_matter")        1502
    ,                                                                   1503
                                      ideal("organic_matter",           1504
    IDEAL_ORGANIC)),                                                    1505
                          l2w["土壤指标"]["有机质"]),                    1506
                "污染指标": (data.get("heavy_metals") or {},            1507
                             pollution_score(data.get("heavy_meta       1508
    ls") or {},                                                         1509
                                                                        1510
    uc.metal_limits(settings)),                                         1511
                          l2w["土壤指标"]["污染指标"]),                  1512
            }                                                           1513
        # ---- 植被指标 ----                                           1514
        if "植被指标" in l1_enabled and group_on("植被指标", "veg"):    1515
            result["植被指标"] = {                                      1516
                "植被覆盖度": (data.get("coverage"),                    1517
                             std_benefit(data.get("coverage"),          1518
                                         ideal("coverage",              1519
    IDEAL_COVERAGE)),                                                   1520
                             l2w["植被指标"]["植被覆盖度"]),             1521
                "生物多样性": (data.get("shannon"),                     1522
                             std_benefit(data.get("shannon"),           1523
                                         ideal("shannon",               1524
    IDEAL_SHANNON)),                                                    1525
                             l2w["植被指标"]["生物多样性"]),             1526
            }                                                           1527
        # ---- 地形指标 ----                                           1528
        if "地形指标" in l1_enabled and group_on("地形指标", "terrain"):1529
            result["地形指标"] = {                                      1530
                "坡度": (data.get("slope"),                             1531
                         std_slope(data.get("slope")),                  1532
                         l2w["地形指标"]["坡度"]),                      1533
                "坡向": (data.get("aspect"),                            1534
                         aspect_score(data.get("aspect")),              1535
                         l2w["地形指标"]["坡向"]),                      1536
            }                                                           1537
        return result                                                   1538
# ==== app/services/weight_service.py ====                            1539
# -*- coding: utf-8 -*-                                               1540
"""权重方案的存取与生效权重解析。                                       1541
系统有两套权重来源：                                                   1542
1. 系统内置的默认权重（app.utils.constants.DEFAULT_WEIGHTS），         1543
   是**权威取值来源**，案例复算与报告结论以此为准；                    1544
2. 通过层次分析法在本系统中实际算出的权重，保存在                      1545
   weight_schemes / weight_items 两张表中，可标记为「生效」。          1546
本模块负责把这两者合成为计算引擎直接可用的嵌套权重结构：              1547
以系统内置权重为底，用生效方案中已保存的条目覆盖。                    1548
没有任何生效方案时，即完全使用系统内置权重。                          1549
这样既保证默认行为与内置取值一致，又使 AHP 计算结果真正参与评价，     1550
```

```
而不是算完即弃。
"""
import copy
from app.models import db, WeightScheme, WeightItem
from app.utils.constants import DEFAULT_WEIGHTS
from app.services.indicator_service import TREE
#  方案名称：每个微地质单元类型固定一个可生效的方案
ACTIVE_NAME = "自定义权重方案（AHP 计算结果）"
def _code_maps():
    """由指标树导出 编码 → 名称 的映射，避免名称在两处各写一遍。"""
    l1, l2 = {}, {}
    for row in TREE:
        code, name, level, parent = row[0], row[1], row[2],
    row[3]
        if level == 1:
            l1[code] = name
        elif level == 2:
            l2[code] = (l1.get(parent, parent), name)
    return l1, l2
def effective_weights(unit_type):
    """返回该单元类型当前生效的权重。
    返回 (weights, source_label, scheme)
        weights       嵌套权重结构，可直接交给标准化与评价引擎
        source_label  权重来源说明，用于界面与报告标注
        scheme        生效的权重方案对象，无自定义方案时为 None
    """
    base = DEFAULT_WEIGHTS.get(unit_type)
    if base is None:
        return None, "未定义该微地质单元类型的权重", None
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ
    e,

    name=ACTIVE_NAME).first()
    if scheme is None:
        return copy.deepcopy(base), "系统默认权重", None
    merged = copy.deepcopy(base)
    l1_map, l2_map = _code_maps()
    applied = 0
    for item in WeightItem.query.filter_by(scheme_id=scheme.
    id).all():
        code, w = item.indicator_code, item.weight
        if w is None or w <= 0:
            continue
        if code in l1_map and l1_map[code] in merged["l1"]:
            merged["l1"][l1_map[code]] = w
            applied += 1
        elif code in l2_map:
            l1_name, l2_name = l2_map[code]
            if l1_name in merged["l2"] and l2_name in
    merged["l2"][l1_name]:
```

```
                merged["l2"][l1_name][l2_name] = w                  1601
                applied += 1                                        1602
    if applied == 0:                                                1603
        return copy.deepcopy(base), "系统默认权重", None              1604
    return merged, "自定义方案（%d 项覆盖系统内置权重）" % applied, scheme  1605
def save_weights(unit_type, code_weight_pairs, note=None):          1606
    """把一组「指标编码 → 权重」写入生效方案（已存在则更新对应条目）。          1607
    仅覆盖本次传入的条目，其余条目保留，便于分层次（一级 / 二级）              1608
    分次录入而无需一次填完。                                              1609
    """                                                             1610
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ        1611
    e,                                                              1612
                                                                    1613
    name=ACTIVE_NAME).first()                                       1614
    if scheme is None:                                              1615
        scheme = WeightScheme(unit_type=unit_type,                  1616
    name=ACTIVE_NAME,                                               1617
                                      level=1, is_default=True)     1618
        db.session.add(scheme)                                      1619
        db.session.flush()                                          1620
    if note:                                                        1621
        scheme.level = scheme.level or 1                            1622
    if not scheme.is_default:                                       1623
        scheme.is_default = True                                    1624
    written = 0                                                     1625
    for code, w in code_weight_pairs:                               1626
        if w is None or w <= 0:                                     1627
            continue                                                1628
        item = WeightItem.query.filter_by(scheme_id=scheme.i        1629
    d,                                                              1630
                                                                    1631
    indicator_code=code).first()                                    1632
        if item is None:                                            1633
            item = WeightItem(scheme_id=scheme.id,                  1634
    indicator_code=code)                                            1635
            db.session.add(item)                                    1636
        item.weight = float(w)                                      1637
        written += 1                                                1638
    db.session.commit()                                             1639
    return scheme, written                                          1640
def reset_weights(unit_type):                                       1641
    """删除该单元类型的自定义方案，恢复为系统默认权重。"""                      1642
    schemes = WeightScheme.query.filter_by(unit_type=unit_ty        1643
    pe,                                                             1644
                                                                    1645
    name=ACTIVE_NAME).all()                                         1646
    n = 0                                                           1647
    for s in schemes:                                               1648
        WeightItem.query.filter_by(scheme_id=s.id).delete(          1649
            synchronize_session=False)                              1650
```

```
            db.session.delete(s)                                                1651
            n += 1                                                              1652
        db.session.commit()                                                     1653
        return n                                                                1654
def scheme_items(unit_type):                                                    1655
        """返回该单元类型已保存的自定义权重条目 {指标编码： 权重}。"""                1656
        scheme = WeightScheme.query.filter_by(unit_type=unit_typ                1657
        e,                                                                      1658
                                                                                1659
        name=ACTIVE_NAME).first()                                               1660
        if scheme is None:                                                      1661
            return {}, None                                                     1662
        items = {i.indicator_code: i.weight                                     1663
                 for i in WeightItem.query.filter_by(scheme_id=s                1664
        cheme.id).all()}                                                        1665
        return items, scheme                                                    1666
def summary():                                                                  1667
        """各单元类型的权重来源概览，供权重首页展示。"""                            1668
        rows = []                                                               1669
        for unit_type in DEFAULT_WEIGHTS:                                       1670
            weights, label, scheme = effective_weights(unit_type                1671
        )                                                                       1672
            rows.append({                                                       1673
                "unit_type": unit_type,                                         1674
                "label": label,                                                 1675
                "custom": scheme is not None,                                   1676
            })                                                                  1677
        return rows                                                             1678
# ==== app/services/evaluation_service.py ====                                  1679
# -*- coding: utf-8 -*-                                                         1680
"""适宜性评价引擎。                                                             1681
按加权合成公式实现：                                                            1682
- 单元级：F' = Σ（二级指标权重 × 该二级指标下三级指标标准化值）                  1683
- 矿山级：F = Σ（Si × F'i），Si 为各微地质单元面积占比                          1684
- 等级判定按适宜性等级划分表。                                                  1685
"""                                                                             1686
from app.utils.constants import DEFAULT_WEIGHTS, grade_of                       1687
from app.services import standardization as std                                 1688
def score_unit(unit_type, data, weights=None,                                   1689
        settings=None):                                                         1690
        """计算单个微地质单元的评价因子 F'。                                    1691
        参数                                                                    1692
        ----                                                                    1693
        weights  : 该单元类型的权重；不传用系统默认权重                         1694
        settings : 图斑级标准化参数（基准值与理想取值）；不传全部按代码常量     1695
        返回                                                                    1696
        ----                                                                    1697
        dict：{ l1_scores: {一级指标: 得分}, unit_score: F', details:           1698
        [...] }                                                                 1699
        若某一级指标下存在"不适用"（标准化值为 None）的二级指标，                1700
```

```
    则按其余项权重归一化后计算，保证单元得分始终有定义。                                    1701
    """                                                                    1702
    if weights is None:                                                    1703
        weights = DEFAULT_WEIGHTS[unit_type]                               1704
    std_result = std.standardize_unit(unit_type, data,                     1705
weights, settings)                                                         1706
    l1_weights = weights["l1"]                                             1707
    l1_scores = {}                                                         1708
    details = []                                                           1709
    for l1_name, items in std_result.items():                              1710
        wsum = 0.0                                                         1711
        acc = 0.0                                                          1712
        for l2_name, (raw, std_value, w2) in items.items():                1713
            if std_value is None:                                          1714
                # 不适用项：不计入                                         1715
                details.append({                                           1716
                    "l1": l1_name, "l2": l2_name, "raw":                   1717
raw,                                                                       1718
                    "std_value": None, "weight": w2,                       1719
"contribution": None,                                                      1720
                })                                                         1721
                continue                                                   1722
            contribution = std_value * w2                                  1723
            acc += contribution                                            1724
            wsum += w2                                                     1725
            details.append({                                               1726
                "l1": l1_name, "l2": l2_name, "raw": raw,                  1727
                "std_value": std_value, "weight": w2,                      1728
                "contribution": contribution,                              1729
            })                                                             1730
        l1_scores[l1_name] = acc / wsum if wsum > 0 else 0.0               1731
    # 一级指标权重按实际参与计算的指标归一化。                                  1732
    # 内置权重本身已归一（和为 1），此处为恒等变换；                             1733
    # 若某层指标缺失或自定义权重不完整，可避免单元得分被整体压低。                 1734
    lw_sum = sum(l1_weights[k] for k in l1_scores if k in                  1735
l1_weights)                                                                1736
    if lw_sum > 0:                                                         1737
        unit_score = sum(l1_weights[k] * l1_scores[k]                      1738
                         for k in l1_scores if k in                        1739
l1_weights) / lw_sum                                                       1740
    else:                                                                  1741
        unit_score = 0.0                                                   1742
    return {"l1_scores": l1_scores, "unit_score":                          1743
unit_score, "details": details}                                            1744
def score_mine(units, weights_map=None, settings=None):                    1745
    """计算整个矿山的评价因子 F 与适宜性等级。                                  1746
    参数                                                                   1747
    ----                                                                   1748
    units : list[dict]，每个元素：                                          1749
        {unit_type, ratio, data}                                           1750
```

```
        ratio 为该单元面积占比（0~1），data 为实测数据 dict。
    weights_map : dict，{单元类型： 权重结构}。未提供的单元类型
        使用系统默认权重。权重可来自 AHP 计算并经
        weight_service 保存生效，从而真正参与评价。
    settings : 图斑级标准化参数（参照生态系统基准、指标理想取值），
        由 app.services.unit_config.for_patch() 生成；不传按代码常量。
    返回
    ----
    dict：{ final_score, grade, grade_info, unit_results }
    """
    weights_map = weights_map or {}
    total = 0.0
    unit_results = []
    for u in units:
        r = score_unit(u["unit_type"], u["data"],
                       weights_map.get(u["unit_type"]),
    settings)
        weighted = u["ratio"] * r["unit_score"]
        total += weighted
        unit_results.append({
            "unit_type": u["unit_type"],
            "ratio": u["ratio"],
            "unit_score": r["unit_score"],
            "weighted": weighted,
            "l1_scores": r["l1_scores"],
            "details": r["details"],
        })
    grade_info = grade_of(total)
    return {
        "final_score": total,
        "grade": grade_info["grade"],
        "grade_info": grade_info,
        "unit_results": unit_results,
    }
# ==== app/services/report_service.py ====
# -*- coding: utf-8 -*-
"""报告生成：生成 Word 评价报告。
格式（规范体例）：
- 题目：三号（16pt） 黑体 居中
- 一级标题：四号（14pt） 黑体 居中
- 正文/表格：小四号（12pt） 宋体
- 行距：单倍行距
- 全部黑色
"""
import io
import json
from docx import Document
from docx.shared import Pt, Cm, RGBColor
from docx.oxml.ns import qn
from docx.enum.text import WD_ALIGN_PARAGRAPH,
```

```
    WD_LINE_SPACING
from app.utils.timefmt import to_local, now_utc
from app.services import workflow as wf
BLACK = RGBColor(0, 0, 0)
def _set_run(run, text=None, size=12, bold=False,
    east_asia="宋体"):
    if text is not None:
        run.text = text
    run.font.name = "Times New Roman"
    run.font.size = Pt(size)
    run.font.bold = bold
    run.font.color.rgb = BLACK
    run._element.rPr.rFonts.set(qn("w:eastAsia"), east_asia)
    return run
def _set_line_spacing(p):
    """单倍行距。"""
    p.paragraph_format.line_spacing_rule =
    WD_LINE_SPACING.SINGLE
def _para(doc, text, size=12, bold=False, align=None,
    space_after=6, east_asia="宋体"):
    p = doc.add_paragraph()
    if align is not None:
        p.alignment = align
    p.paragraph_format.space_after = Pt(space_after)
    _set_line_spacing(p)
    _set_run(p.add_run(), text, size, bold, east_asia)
    return p
def _heading(doc, text, level=1):
    """level=0 题目（三号黑体居中）；level=1 一级标题（四号黑体居中）。"""
    p = doc.add_paragraph()
    p.alignment = WD_ALIGN_PARAGRAPH.CENTER
    p.paragraph_format.space_before = Pt(12)
    p.paragraph_format.space_after = Pt(8)
    _set_line_spacing(p)
    if level == 0:
        size = 16  # 三号
    else:
        size = 14  # 四号
    _set_run(p.add_run(), text, size=size, bold=True,
    east_asia="黑体")
    return p
def _table(doc, headers, rows, widths=None):
    t = doc.add_table(rows=1, cols=len(headers))
    t.style = "Table Grid"
    hdr = t.rows[0].cells
    for i, htext in enumerate(headers):
        cell_p = hdr[i].paragraphs[0]
        _set_line_spacing(cell_p)
        _set_run(cell_p.add_run(), htext, 12, bold=True)
    for row in rows:
```

```
        cells = t.add_row().cells
        for i, val in enumerate(row):
            cell_p = cells[i].paragraphs[0]
            _set_line_spacing(cell_p)
            _set_run(cell_p.add_run(), str(val), 12)
    return t

def build_report(patch, mine_result, record=None,
    weak=None, operator=None,
                 check=None):
    """生成评价报告 Word 文档，返回 BytesIO。
    check : 标准结论对照（app.services.standard_check.compare 的结果）。
            标准不参与计算，只在报告里回答「结论是否符合当地文件要求」。
    """
    doc = Document()
    for section in doc.sections:
        section.top_margin = Cm(2.5)
        section.bottom_margin = Cm(2.5)
        section.left_margin = Cm(2.5)
        section.right_margin = Cm(2.5)
    # 题目：三号黑体居中
    _heading(doc, "历史遗留废弃矿山自然恢复适宜性评价报告", 0)
    _para(doc, "软件：历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1．0",
          size=12, align=WD_ALIGN_PARAGRAPH.CENTER)
    _para(doc, "生成时间：%s" % to_local(now_utc(), "%Y-%m-%d
%H：%M：%S"),
          size=12, align=WD_ALIGN_PARAGRAPH.CENTER,
    space_after=12)
    # 计算人与参数来源
    if record is not None:
        from app.views.evaluate import param_source_summary
        try:
            params = json.loads(record.params_json) if
    record.params_json else {}
        except ValueError:
            params = {}
        meta = [
            ("计算人", operator or "—"),
            ("计算时间", to_local(record.created_at, "%Y-%m-%d
%H：%M：%S")),
            ("评价记录编号", record.id),
            ("计算参数", param_source_summary(params)),
        ]
        _table(doc, ["项目", "内容"], meta)
    # 图斑信息
    _heading(doc, "一、图斑基本信息", 1)
    info = [
        ("图斑名称", patch.name),
        ("所在行政区", patch.admin_region or "—"),
        ("图斑面积（公顷）", str(patch.area_ha) if patch.area_ha is
    not None else "—"),
```

```
        ("开采矿种", patch.mineral_type or "—"),
        ("开采方式", patch.mining_method or "—"),
    ]
    _table(doc, ["项目", "内容"], info)
    # 评价结论
    _heading(doc, "二、评价结论", 1)
    gi = mine_result["grade_info"]
    concl = [
        ("最终评价因子 F", "%.4f" % mine_result["final_score"]),
        ("适宜性等级", mine_result["grade"]),
        ("判定依据", gi["meaning"]),
        ("推荐措施", gi["measure"]),
    ]
    _table(doc, ["项目", "结果"], concl)
    # 各单元得分
    _heading(doc, "三、各微地质单元得分", 1)
    rows = []
    for u in mine_result["unit_results"]:
        rows.append((u["unit_type"], "%.2f%%" % (u["ratio"]
* 100),
                     "%.4f" % u["unit_score"], "%.4f" %
u["weighted"]))
    _table(doc, ["微地质单元", "面积占比", "单元评价因子 F'", "加权贡献"],
rows)
    # 与当地标准 / 政策文件的对照（不参与计算，只回答结论是否符合文件要求）
    if check and check.get("covered"):
        _heading(doc, "四、与地方标准 / 政策文件的对照", 1)
        _para(doc, "图斑所在行政区：%s（对照 %s 的标准与政策文件）。"
                   "对照只把文件里的判定项逐条映射到系统已有数据，"
                   "系统内没有对应数据的定性要求一律标为待人工判断。"
                  % (check.get("region") or "未填写",
check.get("province") or ""),
              size=12, space_after=6)
        for s in check.get("standards", []):
            _para(doc, "%s%s" % (s["name"],
                                 ("（%s）" % s["doc_number"])
if s.get("doc_number") else ""),
                  size=12, bold=True, space_after=4)
            _table(doc, ["文件判定项", "条件", "类型", "对照结果", "依据"],
                   [(r["item"], "%s %s" % (r["operator"],
r["threshold"] or ""),
                     r["condition_type"] or "—",
r["verdict"], r["evidence"])
                    for r in s["rules"]])
        t = check.get("totals", {})
        _para(doc, "合计：符合 %d 条、不符合 %d 条、待人工判断 %d 条。"
                  % (t.get("符合", 0), t.get("不符合", 0),
t.get("待人工判断", 0)),
              size=12)
    # 完整得分明细
```

```
    _heading(doc, "五、得分明细", 1)
    for u in mine_result["unit_results"]:
        _para(doc, "%s（占比 %.2f%%）" % (u["unit_type"],
u["ratio"] * 100),
              size=12, bold=True, space_after=4)
        detail_rows = []
        for d in u["details"]:
            raw = "—" if d["raw"] is None else d["raw"]
            std = "—" if d["std_value"] is None else "%.4f"
% d["std_value"]
            w = "%.4f" % d["weight"]
            c = "—" if d["contribution"] is None else
"%.4f" % d["contribution"]
            detail_rows.append((d["l1"], d["l2"], raw, std,
w, c))
        _table(doc, ["一级指标", "二级指标", "实测值", "标准化值", "权重",
"贡献"],
               detail_rows)
        l1_row = []
        for k, v in u["l1_scores"].items():
            l1_row.append("%s=%.4f" % (k, v))
        _para(doc, "一级指标得分：" + "，".join(l1_row), size=12,
space_after=10)
# 数据完整度
undetected = wf.undetected_metals(patch.id)
if undetected:
    _heading(doc, "六、数据完整度说明", 1)
    parts = ["%s：%s" % (t, "、".join(m)) for t, m in
undetected]
    _para(doc, "以下重金属项未检测，按方法约定以「未受污染」计入污染指标"
                "（并非实测达标）：%s。" % "；".join(parts),
size=12)
# 短板提示
if weak:
    _heading(doc, "七、短板提示", 1)
    _para(doc, "按「权重 ×（1 − 标准化值）」排序，提升下列指标对总分的拉动最大。",
          size=12, space_after=4)
    _table(doc, ["微地质单元", "指标", "实测值", "标准化值", "提升空间"],
           [(u, l2, "—" if raw is None else raw, "%.4f"
% std,
             "%.4f" % gain) for u, l2, raw, std, gain
in weak])
# 签字栏
_para(doc, "", size=12, space_after=18)
_para(doc, "评价人（签字）：_____", size=12)
_para(doc, "日期：_____", size=12)
buf = io.BytesIO()
doc.save(buf)
buf.seek(0)
return buf
```

```
# ==== app/services/indicator_service.py ====
# -*- coding: utf-8 -*-
"""指标定义维护。
把内置指标体系写入 indicators 表，使指标名称、单位、
理想值与取值依据可在界面上查看维护，而不再只存在于代码常量中。
代码常量（app.utils.constants）仍是计算时的**权威取值来源**；
本表用于展示、核对与记录取值依据。二者由 seed_indicators() 保持同步。
"""
from app.models import db, Indicator
from app.utils.constants import (
    UNIT_TYPES, UNIT_ENABLED_INDICATORS, IDEAL_SOIL_RATIO,
    IDEAL_THICKNESS,
    IDEAL_PH_RANGE, IDEAL_ORGANIC, IDEAL_COVERAGE,
    IDEAL_SHANNON,
    HEAVY_METAL_LIMIT, SLOPE_BREAK, ASPECT_SCORES,
)
ALL_UNITS = ",".join(UNIT_TYPES)
# (code, name, level, parent, unit, ideal_value,
#  ideal_type, enabled_units, desc, is_constraint)
TREE = [
    # ---- 一级指标 ----
    ("soil", "土壤指标", 1, None, None, None, None, "碎石边坡,平台",
     "评价指标体系的一级指标", False),
    ("veg", "植被指标", 1, None, None, None, None, ALL_UNITS,
     "评价指标体系的一级指标", False),
    ("terrain", "地形指标", 1, None, None, None, None,
     "岩质边坡,碎石边坡", "评价指标体系的一级指标", False),
    # ---- 二级指标：土壤 ----
    ("soil_ratio", "土石比", 2, "soil", "—",
     str(IDEAL_SOIL_RATIO), "benefit", "碎石边坡,平台",
     "取值规定：理想值 0.8，根据矿山修复实际情况确定", False),
    ("thickness", "土层厚度", 2, "soil", "cm",
     str(IDEAL_THICKNESS), "benefit", "碎石边坡,平台",
     "取值规定：理想值 20 cm", False),
    ("ph", "pH", 2, "soil", "—", "%g~%g" % IDEAL_PH_RANGE,
     "mid", "碎石边坡,平台",
     "取值规定：pH 用适中指标的标准化方法，理想区间 %g~%g" % IDEAL_PH_RANGE,
     False),
    ("organic_matter", "有机质", 2, "soil", "g/kg",
     str(IDEAL_ORGANIC), "benefit", "碎石边坡,平台",
     "《全国第二次土壤普查养分等级划分标准》二等（较丰富）最低值 30 g/kg", False),
    ("pollution", "污染指标", 2, "soil", "mg/kg", "见三级指标",
     "cost", "碎石边坡,平台",
     "同时作为限制性指标与评价指标；8 种重金属等权合成（各 0.125）", True),
    # ---- 二级指标：植被 ----
    ("coverage", "植被覆盖度", 2, "veg", "%",
     str(IDEAL_COVERAGE), "benefit", ALL_UNITS,
     "取值规定：参照生态系统取值 75%", False),
    ("shannon", "生物多样性", 2, "veg", "—", str(IDEAL_SHANNON),
     "benefit", ALL_UNITS,
```

```
        "Shannon-Wiener 多样性指数，参照生态系统取值 0.7", False),
    # ---- 二级指标：地形 ----
    ("slope", "坡度", 2, "terrain", "°", "%g° 分界" %
    SLOPE_BREAK, "cost", "岩质边坡,碎石边坡",
        "取值规定引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号：25° 为陡坡重要分界",
    False),
    ("aspect", "坡向", 2, "terrain", "—",
    "阳坡0.2/阴阳坡0.6/阴坡1.0", "discrete", "岩质边坡,碎石边坡",
        "取值规定：阴坡及半阴坡水分条件更优，故赋值高于阳坡", False),
]
# 污染指标下的三级指标（8 种重金属）
for _m, _limit in HEAVY_METAL_LIMIT.items():
    TREE.append((
        "hm_" + _m, _m, 3, "pollution", "mg/kg",
    str(_limit), "cost",
        "碎石边坡,平台",
        "《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值 %g mg/kg" %
    _limit,
        False,
    ))
def seed_indicators():
    """写入/更新指标定义（幂等）。返回写入条数。"""
    code_map = {
        "benefit": "越大越优", "cost": "越小越优",
        "mid": "适中型", "discrete": "离散赋值",
    }
    n = 0
    for (code, name, level, parent, unit, ideal, itype,
    units, desc, is_con) in TREE:
        ind = Indicator.query.filter_by(code=code).first()
        if ind is None:
            ind = Indicator(code=code)
            db.session.add(ind)
        ind.name = name
        ind.level = level
        ind.parent_code = parent
        ind.unit = unit
        ind.ideal_value = ideal
        ind.ideal_type = itype
        ind.enabled_units = units
        ind.description = desc
        ind.is_constraint = is_con
        ind.is_evaluative = (level in (2, 3))
        n += 1
    db.session.commit()
    return n
def ideal_type_label(code):
    return {"benefit": "越大越优", "cost": "越小越优",
            "mid": "适中型", "discrete": "离散赋值"}.get(code,
    code or "—")
```

```
# ==== app/services/import_service.py ====
# -*- coding: utf-8 -*-
"""指标数据批量导入。
支持 CSV 与 Excel（.xlsx/.xls）。一行对应一个微地质单元，
按「单元类型」匹配图斑内已有的微地质单元后写入指标数据。
表头约定（缺失列按不适用处理）：
    单元类型， 土石比， 土层厚度， pH， 有机质，
    Hg， As， Ni， Cu， Cd， Pb， Cr， Zn，
    植被覆盖度， Shannon， 坡度， 坡向
"""
import csv
import io
import json
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
    UNIT_TYPES
# 表头（顺序即导出模板顺序）
HEADERS = (["单元类型", "土石比", "土层厚度", "pH", "有机质"]
           + list(HEAVY_METALS)
           + ["植被覆盖度", "Shannon", "坡度", "坡向"])
FIELD_ALIASES = {
    "单元类型": "unit_type",
    "土石比": "soil_ratio",
    "土层厚度": "thickness",
    "pH": "ph", "PH": "ph", "ph": "ph",
    "有机质": "organic_matter",
    "植被覆盖度": "coverage",
    "Shannon": "shannon", "shannon": "shannon", "生物多样性":
    "shannon",
    "坡度": "slope",
    "坡向": "aspect",
}
for _m in HEAVY_METALS:
    FIELD_ALIASES[_m] = "hm_" + _m
def build_template_csv():
    """生成导入模板（含表头与示例行）。"""
    buf = io.StringIO()
    w = csv.writer(buf)
    w.writerow(HEADERS)
    # 示例：碎石边坡一行，说明每列含义
    row = {h: "" for h in HEADERS}
    row.update({"单元类型": "碎石边坡", "土石比": "0.34", "土层厚度": "10",
                "pH": "9.24", "有机质": "4.89", "Hg": "0.05",
    "As": "5.17",
                "Ni": "12.60", "Cu": "11.70", "Cd": "0.04",
    "Pb": "6.50",
                "植被覆盖度": "87.92", "Shannon": "0.60",
                "坡度": "28.75", "坡向": "阳坡"})
    w.writerow([row[h] for h in HEADERS])
    return buf.getvalue()
def _read_rows(filename, stream):
```

```
    """读取上传文件为「表头 -> 值」的字典列表。"""
    ext = filename.rsplit(".", 1)[-1].lower()
    if ext == "csv":
        text = stream.read().decode("utf-8-sig",
errors="replace")
        reader = csv.DictReader(io.StringIO(text))
        return [dict(r) for r in reader]
    if ext in ("xlsx", "xls"):
        from openpyxl import load_workbook
        wb = load_workbook(io.BytesIO(stream.read()),
data_only=True)
        ws = wb.active
        rows = list(ws.iter_rows(values_only=True))
        if not rows:
            return []
        header = [str(c).strip() if c is not None else ""
for c in rows[0]]
        out = []
        for r in rows[1:]:
            if all(c is None or str(c).strip() == "" for c
in r):
                continue
            out.append({header[i]: (r[i] if i < len(r) else
None)
                        for i in range(len(header)) if
header[i]})
        return out
    raise ValueError("不支持的文件类型：.%s" % ext)
def _num(value):
    if value is None:
        return None, None
    s = str(value).strip()
    if s == "":
        return None, None
    try:
        return float(s), None
    except ValueError:
        return None, "「%s」不是有效数值" % s
def parse_upload(filename, stream, unit_type_map):
    """解析上传文件。
    参数
    ----
    unit_type_map : dict, {单元类型: MicroUnit 对象}
    返回
    ----
    (parsed, errors)
      parsed : list[dict], 每项 {unit_type, fields{...},
    heavy_metals{}}
      errors : list[str], 逐行问题说明
    """
```

```
    rows = _read_rows(filename, stream)
    parsed, errors = [], []
    if not rows:
        return [], ["文件为空或没有数据行"]
    # 表头校验
    unknown = [h for h in (rows[0].keys() if rows else [])
    if h not in FIELD_ALIASES]
    if unknown:
        errors.append("存在无法识别的列，将被忽略：%s" %
"、".join(map(str, unknown)))
    for idx, raw in enumerate(rows, start=2):       # 第 1 行是表头
        rec = {}
        for k, v in raw.items():
            key = FIELD_ALIASES.get(str(k).strip())
            if key:
                rec[key] = v
        unit_type = str(rec.get("unit_type") or "").strip()
        if not unit_type:
            errors.append("第 %d 行：缺少「单元类型」" % idx)
            continue
        if unit_type not in unit_type_map:
            errors.append("第 %d 行：图斑内不存在单元类型「%s」（现有：%s）"
                          % (idx, unit_type,
"、".join(unit_type_map) or "无"))
            continue
        if unit_type not in UNIT_TYPES:
            errors.append("第 %d 行：单元类型「%s」不是系统支持的三种之一"
                          % (idx, unit_type))
            continue
        fields, heavy = {}, {}
        ok = True
        for key, val in rec.items():
            if key in ("unit_type",):
                continue
            if key.startswith("hm_"):
                n, err = _num(val)
                if err:
                    errors.append("第 %d 行：%s %s" % (idx,
key[3:], err))
                    ok = False
                elif n is not None:
                    heavy[key[3:]] = n
                continue
            if key == "aspect":
                s = str(val).strip() if val is not None
else ""
                if s == "":
                    fields["aspect"] = None
                elif s in ASPECT_TYPES:
                    fields["aspect"] = s
```

```
                    else:
                        errors.append("第 %d 行：坡向「%s」无效（应为 %s）"
                                      % (idx, s,
    "、".join(ASPECT_TYPES)))
                        ok = False
                    continue
                n, err = _num(val)
                if err:
                    errors.append("第 %d 行：%s %s" % (idx, key,
    err))
                    ok = False
                else:
                    fields[key] = n
            if ok:
                parsed.append({"unit_type": unit_type,
    "fields": fields,
                               "heavy_metals": heavy})
    return parsed, errors
# ==== app/services/workflow.py ====
# -*- coding: utf-8 -*-
"""评价流程前置条件（定量评价与报告输出的流程门禁）。
评价流程存在严格的先后关系：
    建立图斑 → 限制性条件判定（通过） → 划分微地质单元并计算面积占比
    → 逐单元录入指标数据 → 计算评价因子并判定等级 → 输出报告
流程约定：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
废弃矿山，按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡三种
地质微单元类型，并计算各种地形面积占比。」限制性条件判定（一票否决）
在前，微地质单元划分在后，顺序不可颠倒。
本模块把该顺序固化为可检查的状态。后续模块在入口处调用它，
前置步骤未完成时直接拦截并给出「缺什么、去哪补」的提示，
避免在条件不具备时产出看似合理的评价结论。
"""
from app.models import (db, MicroUnit, IndicatorData,
    ConstraintCheck,
                        EvaluationResult)
from app.services import unit_config
#  指标数据非空字段（用于判断某单元是否已录入数据）
DATA_FIELDS = ("soil_ratio", "thickness", "ph",
    "organic_matter",
               "coverage", "shannon", "slope", "aspect")
def _has_data(ind):
    if ind is None:
        return False
    if any(getattr(ind, f, None) is not None for f in
    DATA_FIELDS):
        return True
    return bool(ind.heavy_metals_dict())
def _data_of(ind):
    """把 IndicatorData 行转成标准化接口要的 dict。"""
    if ind is None:
```

```
        return {}
    data = {f: getattr(ind, f, None) for f in DATA_FIELDS}
    data["heavy_metals"] = ind.heavy_metals_dict()
    return data
def patch_progress(patch_id):
    """检查某图斑的评价流程完成情况。
    返回 dict：
        units          {done, count, ratio_sum, detail}
        constraint     {done, passed, detail}
        data           {done, ready, missing, detail}
        blocking       未满足的前置步骤列表（按流程顺序）
        ready          是否可进入定量评价
    """
    units = MicroUnit.query.filter_by(patch_id=patch_id).all()
    ratio_sum = sum(u.ratio or 0.0 for u in units)
    # ---- 步骤一：微地质单元 ----
    units_step = {
        "done": bool(units),
        "count": len(units),
        "ratio_sum": ratio_sum,
        "detail": ("已划分 %d 个微地质单元，面积占比合计 %.2f%%"
                   % (len(units), ratio_sum * 100))
                  if units else "尚未划分微地质单元",
    }
    # ---- 步骤二：限制性条件判定 ----
    cc = ConstraintCheck.query.filter_by(patch_id=patch_id)\
        .order_by(ConstraintCheck.id.desc()).first()
    if cc is None:
        constraint_step = {"done": False, "passed": False,
                           "detail": "尚未执行限制性条件判定"}
    elif cc.passed:
        constraint_step = {"done": True, "passed": True,
                           "detail": "限制性条件判定已通过"}
    else:
        constraint_step = {"done": True, "passed": False,
                           "detail": "限制性条件判定未通过（%s）" %
    (cc.reason or "")}
    # ---- 步骤三：指标数据 ----
    # 判断标准是「该单元类型启用的必填项是否齐全」，而不是「有没有填过任意一个字段」。
    # 早先只要 8 个字段里有一个非空就算齐，于是碎石边坡只填一个坡度也能跑出
    # F=0.69「基本适宜」；而 score_unit 又会把缺的项剔掉、把权重摊给其余指标，
    # 结论方向还偏乐观。污染指标按方法约定允许「未检测」，不算缺。
    settings = unit_config.for_patch(patch_id)
    missing = []                # 完全没录数据的单元类型
    incomplete = []             # 录了但不齐的单元类型 → [(单元类型, [缺的字段])]
    for u in units:
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi
    rst()
```

```
        data = _data_of(ind)
        gaps = settings.missing_fields(u.unit_type, data)
        if not _has_data(ind):
            missing.append(u.unit_type)
        elif gaps:
            incomplete.append((u.unit_type, gaps))
    if not units:
        data_step = {"done": False, "ready": False,
    "missing": [],
                     "detail": "尚无可录入数据的微地质单元"}
    elif missing:
        data_step = {"done": False, "ready": False,
    "missing": missing,
                     "detail": "以下微地质单元尚未录入指标数据：%s"
                               % "、".join(missing)}
    elif incomplete:
        data_step = {"done": False, "ready": False,
                     "missing": [t for t, _ in incomplete],
                     "detail": "以下微地质单元的必填指标未填齐：%s"
                               % "；".join("%s缺 %s" % (t,
    "、".join(g))
                                           for t, g in
    incomplete)}
    else:
        data_step = {"done": True, "ready": True,
    "missing": [],
                     "detail": "全部 %d 个微地质单元的指标数据已录入" %
    len(units)}
    #  占比合计必须落在 １００% 附近才算通过。只挡「偏小」是不够的：
    #  库里若出现合计 １.５ 的记录（手工改库、导入异常），F = Σ(Si×F'i)
    #  会被放大到 １.５ 倍，所以上下都要挡。
    ratio_ok = 0.999 <= ratio_sum <= 1.001 if units else
    False
    if units and not ratio_ok:
        units_step["detail"] += "（占比合计不等于 １００%，请检查面积录入）"
    #  ---- 汇总未满足的前置条件（按评价流程的顺序：限制性判定优先） ----
    #  reason 只写「缺什么」这一个事实，不写为什么重要的说明 ——
    #  缺少什么由 label 指明，怎么补由 fix_label 给出，页面不必再解释一遍。
    blocking = []
    if not constraint_step["done"]:
        blocking.append({
            "step": "constraint", "label": "限制性条件判定未执行",
            "reason": "五项限制性条件尚未逐项判定。",
            "fix_url": "/constraint/%d" % patch_id,
    "fix_label": "去执行判定",
        })
    elif not constraint_step["passed"]:
        blocking.append({
            "step": "constraint", "label": "限制性条件判定未通过",
            "reason": cc.reason or "",
```

```
            "fix_url": "/constraint/%d" % patch_id,
        "fix_label": "去复核判定结果",
        })
    if not units_step["done"]:
        blocking.append({
            "step": "units", "label": "未划分微地质单元",
            "reason": "该图斑下没有任何微地质单元。",
            "fix_url": "/unit/%d" % patch_id, "fix_label":
    "去划分",
        })
    elif not ratio_ok:
        blocking.append({
            "step": "units", "label": "面积占比合计不为 100%",
            "reason": "当前合计 %.2f%%。" % (ratio_sum * 100),
            "fix_url": "/unit/%d" % patch_id, "fix_label":
    "去核对面积",
        })
    if units_step["done"] and not data_step["ready"]:
        blocking.append({
            "step": "data", "label": "指标数据未录入完整",
            "reason": data_step["detail"],
            "fix_url": "/indicator/", "fix_label": "去录入",
        })
    return {
        "units": units_step,
        "constraint": constraint_step,
        "data": data_step,
        "blocking": blocking,
        "ready": not blocking,
    }
def undetected_metals(patch_id):
    """列出该图斑各单元「未检测」的重金属，供结果页与报告如实标注。
    方法约定允许未检测的重金属按未受污染计入，但报告里必须写明是哪些 ——
    否则读报告的人会以为这些项是实测达标。
    只列启用了土壤指标的单元：岩质边坡本来就不评价土壤，列 8 项未检测是噪音。
    """
    settings = unit_config.for_patch(patch_id)
    out = []
    for u in MicroUnit.query.filter_by(patch_id=patch_id).al
    l():
        if settings.group_enabled(u.unit_type, "土壤指标",
    "soil") is False:
            continue
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi
    rst()
        if ind is None:
            continue
        metals = settings.undetected_metals(_data_of(ind))
        if metals:
            out.append((u.unit_type, metals))
```

```
    return out
def latest_result(patch_id):
    """取该图斑最近一次的评价记录。"""
    return EvaluationResult.query.filter_by(patch_id=patch_id) \
        .order_by(EvaluationResult.id.desc()).first()
def report_progress(patch_id):
    """报告输出的前置条件：评价流程齐备，且已有评价记录。
    报告本质上是评价结论的呈现，因此必须先真实执行过定量评价，
    导出时读取已保存的评价结果，而不是在导出环节重新计算。
    否则评价完成后再修改指标数据，报告会与系统记录的评价结论不一致，
    同一份评价出现两个数字，结论无法追溯。
    返回 dict：{ready, blocking, result, flow}
        result 为最近一次评价记录（EvaluationResult），无记录时为空。
    """
    flow = patch_progress(patch_id)
    blocking = list(flow["blocking"])
    record = latest_result(patch_id)
    if not blocking and record is None:
        blocking.append({
            "step": "evaluate", "label": "尚未执行适宜性评价",
            "reason": "该图斑还没有评价记录。",
            "fix_url": "/evaluate/%d" % patch_id,
            "fix_label": "去执行评价",
        })
    return {
        "ready": not blocking,
        "blocking": blocking,
        "result": record,
        "flow": flow,
    }
# ==== app/services/dashboard.py ====
# -*- coding: utf-8 -*-
"""工作台数据：统计数字、评价流程步骤状态、侧边栏待办角标。
首页（工作台）需要把「现在有哪些数据」「流程走到哪一步」「哪一步还差活」
一次算清楚，供模板直接渲染。集中在此，避免在视图与模板里各算一遍。
"""
from flask import g
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    Standard)
from app.services import workflow as wf
#  评价流程的八个步骤（顺序即评价流程的作业顺序）
#     key          步骤标识
#     no           步骤编号（01~08）
#     title        名称
#     icon         图标名（见 templates/_macros.html 的 ICONS）
#     desc         一句话说明
#     href         入口地址
```

```
STEPS = [                                                                    2501
    dict(key="project", no="01", title="项目与矿山档案",                      2502
    icon="folder",                                                           2503
        href="/project/",                                                    2504
        desc="建立评价项目、录入矿山图斑、挂接遥感影像、设置参照生态系统"),    2505
    dict(key="standard", no="02", title="政策与标准知识库",                   2506
    icon="book",                                                             2507
        href="/standard/",                                                   2508
        desc="收录地方标准与政策文件，结构化判定规则，支持省际口径比对"),    2509
    dict(key="constraint", no="03", title="限制性指标判定",                   2510
    icon="shield",                                                           2511
        href="/constraint/",                                                 2512
        desc="政策、区位约束、土壤污染、地质灾害、耕地属性五项一票否决"),    2513
    dict(key="unit", no="04", title="微地质单元", icon="layers",              2514
        href="/unit/",                                                       2515
        desc="划分岩质边坡、碎石边坡、平台三类单元，自动计算面积占比"),      2516
    dict(key="ahp", no="05", title="指标体系与权重",                          2517
    icon="sliders",                                                          2518
        href="/ahp/",                                                        2519
        desc="指标定义维护，判断矩阵录入、权重计算与一致性检验、多专家汇总"),2520
    dict(key="indicator", no="06", title="指标数据采集",                      2521
    icon="database",                                                         2522
        href="/indicator/",                                                  2523
        desc="按微地质单元录入土壤、植被、地形指标，支持 CSV / Excel 批量导入"),2524
    dict(key="evaluate", no="07", title="适宜性定量评价",                     2525
    icon="target",                                                           2526
        href="/evaluate/",                                                   2527
        desc="指标标准化、分单元赋分、面积加权合成，判定适宜性等级"),        2528
    dict(key="report", no="08", title="报告与结果输出",                       2529
    icon="file",                                                             2530
        href="/report/",                                                     2531
        desc="结果图表与逐层得分明细，导出 PDF 与 Word 评价报告"),           2532
]                                                                            2533
def _has_data(ind):                                                          2534
    return wf._has_data(ind)                                                 2535
#  系统的九个功能模块（登录页介绍用，含不在评价流程内的账号与权限管理）      2536
MODULES = [                                                                  2537
    ("01", "账号与权限管理", "自助注册与登录注销、管理员与操作员两种职务、写操作自动留痕"),2538
    ("02", "项目与矿山档案", "建立项目、录入图斑、挂接遥感影像、设置参照生态系统"),2539
    ("03", "政策与标准知识库", "标准判定条件结构化为规则，支持省际口径比对"),2540
    ("04", "限制性指标判定", "政策、区位、污染、地质灾害、耕地五项一票否决"),2541
    ("05", "微地质单元管理", "划分岩质边坡、碎石边坡、平台，自动算面积占比"),2542
    ("06", "指标体系与权重", "指标定义维护、AHP 判断矩阵与一致性检验、多专家汇总"),2543
    ("07", "指标数据采集", "按单元类型录入土壤、植被、地形指标，支持批量导入"),2544
    ("08", "适宜性定量评价", "四类标准化、分单元赋分、面积加权、等级判定"),2545
    ("09", "报告与结果输出", "结果图表与逐层得分明细，导出 PDF 与 Word 报告"),2546
]                                                                            2547
def snapshot():                                                              2548
    """一次算出工作台需要的全部数据（同一请求内只算一次）。                  2549
    返回 dict：                                                              2550
```

```
        stats       {projects, patches, unit_types, standards}
        steps       八个步骤，附 available / pending / note
        badges      {key: 待办数量}，仅含有待办的项
        progress    {patches, units, data_units, ready,
    evaluated}
    """
    cached = getattr(g, "_dash_snapshot", None)
    if cached is None:
        cached = _compute()
        g._dash_snapshot = cached
    return cached
def _compute():
    projects = Project.query.count()
    patches = MinePatch.query.all()
    patch_ids = [p.id for p in patches]
    units = MicroUnit.query.filter(
        MicroUnit.patch_id.in_(patch_ids)).all() if
    patch_ids else []
    unit_by_patch = {}
    for u in units:
        unit_by_patch.setdefault(u.patch_id, []).append(u)
    unit_ids = [u.id for u in units]
    data_unit_ids = set()
    if unit_ids:
        for ind in IndicatorData.query.filter(
                IndicatorData.unit_id.in_(unit_ids)).all():
            if _has_data(ind):
                data_unit_ids.add(ind.unit_id)
    passed_constraint = set()
    if patch_ids:
        for c in ConstraintCheck.query.filter(
                ConstraintCheck.patch_id.in_(patch_ids)).all
    ():
            if c.passed:
                passed_constraint.add(c.patch_id)
    evaluated = set()
    if patch_ids:
        for r in EvaluationResult.query.filter(
                EvaluationResult.patch_id.in_(patch_ids)).al
    l():
            evaluated.add(r.patch_id)
    # ---- 各步骤的待办数量 ----
    pending = {
        "constraint": sum(1 for p in patches if p.id not in
    passed_constraint),
        "unit": sum(1 for p in patches if not
    unit_by_patch.get(p.id)),
        "indicator": 0,
        "evaluate": 0,
    }
```

```
    ready_patches = 0                                                    2601
    for p in patches:                                                    2602
        us = unit_by_patch.get(p.id, [])                                 2603
        if us and any(u.id not in data_unit_ids for u in                 2604
    us):                                                                 2605
            pending["indicator"] += 1                                    2606
        pr = wf.patch_progress(p.id)                                     2607
        if pr["ready"]:                                                  2608
            ready_patches += 1                                           2609
            if p.id not in evaluated:                                    2610
                pending["evaluate"] += 1                                 2611
    # ---- 步骤可用性：前置数据不具备时置灰 ----                          2612
    avail = {                                                            2613
        "project": True,                                                 2614
        "standard": True,                                                2615
        "constraint": bool(patches),                                     2616
        "unit": bool(patches),                                           2617
        "ahp": True,                                                     2618
        "indicator": bool(units),                                        2619
        "evaluate": ready_patches > 0,                                   2620
        "report": bool(evaluated),                                       2621
    }                                                                    2622
    notes = {                                                            2623
        "constraint": "需先建立矿山图斑",                                2624
        "unit": "需先建立矿山图斑",                                      2625
        "indicator": "需先划分微地质单元",                               2626
        "evaluate": "需前置步骤齐备的图斑",                              2627
        "report": "需先执行适宜性评价",                                  2628
    }                                                                    2629
    steps = []                                                           2630
    for s in STEPS:                                                      2631
        steps.append(dict(                                               2632
            s,                                                           2633
            available=avail[s["key"]],                                   2634
            pending=pending.get(s["key"], 0),                            2635
            note=None if avail[s["key"]] else                            2636
    notes.get(s["key"], ""),                                             2637
        ))                                                               2638
    unit_types = len({u.unit_type for u in units})                       2639
    stats = {                                                            2640
        "projects": projects,                                            2641
        "patches": len(patches),                                         2642
        "unit_types": unit_types,                                        2643
        "standards": Standard.query.count(),                             2644
    }                                                                    2645
    return {                                                             2646
        "stats": stats,                                                  2647
        "steps": steps,                                                  2648
        "badges": {k: v for k, v in pending.items() if v},               2649
        "progress": {                                                    2650
```

```
                "patches": len(patches),
                "units": len(units),
                "data_units": len(data_unit_ids),
                "ready": ready_patches,
                "evaluated": len(evaluated),
            },
        }
def nav_badges():
    """侧边栏待办角标：{入口地址前缀： 数量}，只含有待办的项。"""
    snap = snapshot()
    return {
        "/constraint/": snap["badges"].get("constraint", 0),
        "/unit/": snap["badges"].get("unit", 0),
        "/indicator/": snap["badges"].get("indicator", 0),
        "/evaluate/": snap["badges"].get("evaluate", 0),
    }
# ==== app/services/audit.py ====
# -*- coding: utf-8 -*-
"""操作审计。
在应用工厂中以 after_request 钩子统一埋点：凡是成功的写操作（POST/PUT/DELETE）
且已登录，即写入 operation_logs 表，无需在每个视图里手工调用。
动作名称由「端点 → 中文动作」映射表给出，未收录的端点回退为方法 + 路径。
"""
from flask import request
from flask_login import current_user
from app.models import db, OperationLog
# 端点 → 可读动作名
ACTION_NAMES = {
    "auth.login": "登录系统",
    "auth.register": "注册账号",
    "auth.logout": "退出登录",
    "auth.add_user": "新增账号",
    "auth.edit_user": "编辑账号",
    "auth.delete_user": "删除账号",
    "auth.reset_password": "重置用户密码",
    "auth.profile": "修改个人资料",
    "auth.upload_avatar": "更新头像",
    "auth.delete_avatar": "移除头像",
    "project.add": "新建项目",
    "project.edit": "编辑项目",
    "project.delete": "删除项目",
    "project.patch_add": "新建矿山图斑",
    "project.patch_edit": "编辑矿山图斑",
    "project.patch_delete": "删除矿山图斑",
    "project.upload_image": "上传遥感影像",
    "unit.add": "添加微地质单元",
    "unit.edit": "编辑微地质单元",
    "unit.delete": "删除微地质单元",
    "standard.add": "添加标准/政策文件",
    "standard.edit": "编辑标准/政策文件",
```

```
    "standard.delete": "删除标准/政策文件",
    "standard.rules": "维护标准规则",
    "standard.rule_delete": "删除标准规则",
    "constraint.check": "执行限制性条件判定",
    "ahp.save_expert": "录入专家判断矩阵",
    "ahp.delete_expert": "删除专家判断矩阵",
    "ahp.aggregate": "汇总多专家矩阵",
    "indicator.edit": "保存指标数据",
    "indicator.import_data": "批量导入指标数据",
    "indicator.update_meta": "维护指标定义",
    "evaluate.run": "执行适宜性评价",
}
# 无需记录的操作（查看类或纯导航）
SKIP_ENDPOINTS = {"static", "auth.index"}
def action_name(endpoint):
    # 个人设置页的两个表单共用一个端点，按提交的动作区分
    if endpoint == "auth.profile":
        return "修改密码" if request.form.get("action") ==
    "password" \
            else "修改个人资料"
    if endpoint in ACTION_NAMES:
        return ACTION_NAMES[endpoint]
    return "写操作 %s %s" % (request.method, request.path)
def extract_target_id():
    """从视图参数中取出被操作对象的主键，取不到则返回 None。"""
    args = request.view_args or {}
    for key in ("pid", "patch_id", "unit_id", "sid", "rid",
    "result_id", "user_id"):
        if key in args:
            try:
                return int(args[key])
            except (TypeError, ValueError):
                return None
    return None
def record(endpoint, status_code):
    """记录一次写操作。失败（4xx/5xx）也记录，便于追溯。"""
    if endpoint in SKIP_ENDPOINTS:
        return
    try:
        detail = "路径 %s" % request.path
        if status_code >= 400:
            detail += "；结果 HTTP %d" % status_code
        db.session.add(OperationLog(
            user_id=current_user.id if
    current_user.is_authenticated else None,
            action=action_name(endpoint),
            target_type=(endpoint or "").split(".")[0] or
    None,
            target_id=extract_target_id(),
            detail=detail,
```

```
            ))                                                          2751
            db.session.commit()                                         2752
        except Exception:                                               2753
            #  审计失败不应影响业务响应                                 2754
            db.session.rollback()                                       2755
#  ====  app/views/auth.py  ====                                        2756
#  -*-  coding:  utf-8  -*-                                            2757
"""账号与权限  ——  登录、注册、个人资料、用户维护、操作日志。            2758
页面组织：                                                              2759
   /login          登录页（含「点我注册」入口）                         2760
   /register       自助注册（默认「操作员」）                           2761
   /profile        个人设置，分类切换：info / security / users / logs   2762
                   后两个分类仅管理员可见，由本模块统一管控。           2763
"""                                                                     2764
from flask import (Blueprint, render_template, redirect,                2765
    url_for, request,                                                   2766
                       flash, abort, current_app)                       2767
from flask_login import login_user, logout_user,                        2768
    login_required, current_user                                        2769
from werkzeug.security import check_password_hash,                      2770
    generate_password_hash                                              2771
from app.forms import (LoginForm, EmptyForm, ProfileForm,               2772
    PasswordForm,                                                       2773
                           RegisterForm, email_ok)                      2774
from app.models import db, User, OperationLog                           2775
from app.utils import uploads as up                                     2776
from app.utils.constants import ROLE_LABELS, DEFAULT_ROLE,              2777
    GENDERS                                                             2778
bp = Blueprint("auth", __name__)                                        2779
#  「个人设置」的四个分类；后两个是管理员专属                           2780
PROFILE_TABS = ("info", "security", "users", "logs")                    2781
ADMIN_TABS = ("users", "logs")                                          2782
#  口令最短长度：自助注册与管理员建号保持一致，免得出现「a」这种口令   2783
MIN_PASSWORD_LEN = 6                                                    2784
def require_admin():                                                    2785
    """非管理员直接  403，避免误操作基础数据。"""                       2786
    if current_user.role != "admin":                                    2787
        abort(403)                                                      2788
def _intro_data():                                                      2789
    """登录页下方的系统介绍所用的静态内容。"""                          2790
    from app.services import dashboard as dash                          2791
    return {"modules": dash.MODULES}                                    2792
@bp.route("/")                                                          2793
def index():                                                            2794
    if current_user.is_authenticated:                                   2795
        from app.services import dashboard as dash                      2796
        return render_template("auth/home.html",                        2797
    dash=dash.snapshot())                                               2798
    return redirect(url_for("auth.login"))                              2799
@bp.route("/login", methods=["GET", "POST"])                            2800
```

```
def login():
    if current_user.is_authenticated:
        return redirect(url_for("auth.index"))
    form = LoginForm()
    if form.validate_on_submit():
        user = User.query.filter_by(username=form.username.data).first()
        if user and check_password_hash(user.password_hash, form.password.data):
            if not user.active:
                flash("该账号已停用，请联系管理员", "toast-error")
            else:
                login_user(user, remember=form.remember.data)
                # 用 toast 类别：渲染为页面中上方的浮层，淡入停留后淡出。
                # 登录成功只是告知性反馈，不需要用户停下来阅读。
                flash("登录成功", "toast")
                return redirect(url_for("auth.index"))
        else:
            flash("账户名称或密码错误", "toast-error")
    # 字段为空之类的校验失败，错误留在表单里逐项显示，不走浮层：
    # 那类消息要对着输入框看，一闪而过反而让人以为「点了没反应」。
    return render_template("auth/login.html", form=form, **_intro_data())

@bp.route("/register", methods=["GET", "POST"])
def register():
    """自助注册。新账号一律为「操作员」，权限只能由管理员事后调整。"""
    if current_user.is_authenticated:
        return redirect(url_for("auth.index"))
    form = RegisterForm()
    if form.validate_on_submit():
        username = form.username.data.strip()
        if User.query.filter_by(username=username).first():
            form.username.errors.append("该账户名称已被使用")
        else:
            user = User(username=username,
                        role=DEFAULT_ROLE,
                        password_hash=generate_password_hash(form.password.data),
                        display_name=(form.display_name.data or "").strip() or None,
                        gender=(form.gender.data or "").strip() or None,
                        email=(form.email.data or "").strip() or None,
                        phone=(form.phone.data or "").strip() or None)
            rel, err = _save_avatar(request.files.get("avatar"))
            if err:
```

```
                    flash(err, "error")
                else:
                    user.avatar = rel
                    db.session.add(user)
                    db.session.commit()
                    login_user(user)
                    flash("注册成功，欢迎使用", "toast")
                    return redirect(url_for("auth.index"))
    return render_template("auth/register.html", form=form)
@bp.route("/logout")
@login_required
def logout():
    logout_user()
    flash("已退出登录", "ok")
    return redirect(url_for("auth.login"))
# ---------------------------- 个人设置 ----------------------------
    ----
# 任何登录用户在这里维护自己；管理员额外能管理账号与查看日志。
# 「用户与操作日志」不再是独立菜单，而是本页的管理员分类。
@bp.route("/profile", methods=["GET", "POST"])
@login_required
def profile():
    is_admin = current_user.role == "admin"
    tab = request.values.get("tab") or "info"
    if tab not in PROFILE_TABS or (tab in ADMIN_TABS and
    not is_admin):
        tab = "info"
    # 表单放在同一页，用字段前缀区分，避免 submit / csrf_token 重名
    info_form = ProfileForm(prefix="info", obj=current_user)
    pwd_form = PasswordForm(prefix="pwd")
    if request.method == "POST":
        action = request.form.get("action")
        # 校验失败要重新渲染本页，此时必须回到提交表单所在的那个分类，
        # 否则错误提示会出现在另一个分类里，用户根本看不到。
        if action == "password":
            tab = "security"
        elif action == "info":
            tab = "info"
        if action == "info" and info_form.validate_on_submit
    ():
            current_user.display_name =
    (info_form.display_name.data or "").strip() or None
            current_user.gender = (info_form.gender.data or
    "").strip() or None
            current_user.email = (info_form.email.data or
    "").strip() or None
            current_user.phone = (info_form.phone.data or
    "").strip() or None
            db.session.commit()
            flash("个人资料已保存", "ok")
```

```
            return redirect(url_for("auth.profile",                2901
    tab="info"))                                                   2902
        if action == "password" and pwd_form.validate_on_sub       2903
    mit():                                                         2904
            if not check_password_hash(current_user.password       2905
    _hash,                                                         2906
                                                                   2907
    pwd_form.old_password.data):                                   2908
                pwd_form.old_password.errors.append("当前密码不正确 2909
    ")                                                             2910
            else:                                                  2911
                current_user.password_hash =                       2912
    generate_password_hash(                                        2913
                    pwd_form.new_password.data)                    2914
                db.session.commit()                                2915
                flash("密码已修改", "ok")                          2916
                return redirect(url_for("auth.profile",            2917
    tab="security"))                                               2918
    ctx = dict(info_form=info_form, pwd_form=pwd_form,             2919
    avatar_form=EmptyForm(),                                       2920
                tab=tab, role_labels=ROLE_LABELS,                  2921
    genders=GENDERS)                                               2922
    if tab == "users":                                             2923
        ctx.update(_users_context())                               2924
    elif tab == "logs":                                            2925
        ctx.update(_logs_context())                                2926
    return render_template("auth/profile.html", **ctx)             2927
def _users_context():                                              2928
    """管理员分类「账号」所需的数据。"""                           2929
    return {"users": User.query.order_by(User.id).all(),           2930
            "form": EmptyForm()}                                   2931
def _logs_context():                                               2932
    """管理员分类「操作日志」所需的数据。"""                       2933
    per_page = 50                                                  2934
    q = OperationLog.query.order_by(OperationLog.id.desc())        2935
    total = q.count()                                              2936
    pages = max(1, (total + per_page - 1) // per_page)             2937
    # 页码先夹到 [1, pages] 再算 offset：手工把 page 写成天文数字时，2938
    # (page-1)*per_page 会超出 SQLite 能绑定的整数范围，直接 500。  2939
    try:                                                           2940
        page = int(request.args.get("page", 1))                    2941
    except (TypeError, ValueError):                                2942
        page = 1                                                   2943
    page = max(1, min(page, pages))                                2944
    records = q.offset((page - 1) * per_page).limit(per_page       2945
    ).all()                                                        2946
    users_map = {u.id: u.username for u in User.query.all()}       2947
    return {"records": records, "page": page, "pages":             2948
    pages,                                                         2949
            "total": total, "users_map": users_map}                2950
```

```
# ---------------------------- 头像 ----------------------------
--
def _avatar_length_ok(file_storage):
    """按字节检查头像大小。超限时 werkzeug 读到内存里的内容也能量出来。"""
    limit = current_app.config.get("MAX_AVATAR_LENGTH", 4 * 1024 * 1024)
    stream = file_storage.stream
    pos = stream.tell()
    stream.seek(0, 2)
    size = stream.tell()
    stream.seek(pos)
    return size <= limit, size
def _save_avatar(file):
    """保存头像文件，返回 (相对路径, 错误信息)。没有文件时返回 (None, None)。"""
    if file is None or not file.filename:
        return None, None
    ok, size = _avatar_length_ok(file)
    if not ok:
        return None, ("头像不能超过 %.0f MB（当前 %.1f MB）"
                      % (current_app.config["MAX_AVATAR_LENGTH"] / 1024 / 1024,
                         size / 1024 / 1024))
    return up.save_upload(file, "avatars",
                          current_app.config["ALLOWED_AVATAR_EXT"],
                          current_app.config["UPLOAD_FOLDER"])
@bp.route("/profile/avatar", methods=["POST"])
@login_required
def upload_avatar():
    form = EmptyForm()
    back = url_for("auth.profile", tab=request.form.get("tab") or "info")
    if not form.validate_on_submit():
        flash("表单校验失败，请重试", "error")
        return redirect(back)
    rel, err = _save_avatar(request.files.get("avatar"))
    if err:
        flash(err, "error")
        return redirect(back)
    if rel is None:
        flash("请选择头像图片", "error")
        return redirect(back)
    # 换新头像时删掉旧文件，避免残留
    up.remove_upload(current_user.avatar,
    current_app.config["UPLOAD_FOLDER"])
    current_user.avatar = rel
    db.session.commit()
    flash("头像已更新", "ok")
    return redirect(back)
```

```
@bp.route("/profile/avatar/delete", methods=["POST"])
@login_required
def delete_avatar():
    form = EmptyForm()
    back = url_for("auth.profile", tab=request.form.get("tab
    ") or "info")
    if form.validate_on_submit():
        up.remove_upload(current_user.avatar,
    current_app.config["UPLOAD_FOLDER"])
        current_user.avatar = None
        db.session.commit()
        flash("头像已移除", "ok")
    return redirect(back)
# ---------------------------- 账号维护（仅管理员）
    ------------------------------
@bp.route("/users")
@login_required
def users():
    """旧地址保留，统一并入「个人设置 · 账号」。"""
    return redirect(url_for("auth.profile", tab="users"))
@bp.route("/users/add", methods=["GET", "POST"])
@login_required
def add_user():
    require_admin()
    if request.method == "POST":
        username = request.form.get("username", "").strip()
        password = request.form.get("password", "")
        role = request.form.get("role", DEFAULT_ROLE)
        if role not in ROLE_LABELS:
            role = DEFAULT_ROLE
        if not username or not password:
            flash("账户名称和密码不能为空", "error")
            return redirect(url_for("auth.add_user"))
        if len(username) > 64:
            flash("账户名称不能超过 64 个字符", "error")
            return redirect(url_for("auth.add_user"))
        if len(password) < MIN_PASSWORD_LEN:
            flash("密码至少 %d 位（与自助注册保持一致）" % MIN_PASSWORD_LEN,
                  "error")
            return redirect(url_for("auth.add_user"))
        if User.query.filter_by(username=username).first():
            flash("账户名称已存在", "error")
            return redirect(url_for("auth.add_user"))
        db.session.add(User(
            username=username, role=role,
            password_hash=generate_password_hash(password),
            display_name=request.form.get("display_name",
    "").strip() or None,
            gender=request.form.get("gender", "").strip()
    or None,
```

```
            email=request.form.get("email", "").strip() or
    None,
            phone=request.form.get("phone", "").strip() or
    None))
        db.session.commit()
        flash("账号已添加", "ok")
        return redirect(url_for("auth.profile",
    tab="users"))
    return render_template("auth/add_user.html",
    role_labels=ROLE_LABELS,
                           genders=GENDERS,
    default_role=DEFAULT_ROLE,
                           form=EmptyForm())
@bp.route("/users/<int:user_id>/edit", methods=["GET",
    "POST"])
@login_required
def edit_user(user_id):
    require_admin()
    user = db.get_or_404(User, user_id)
    if request.method == "POST":
        role = request.form.get("role", user.role)
        if role not in ROLE_LABELS:
            role = user.role
        active = request.form.get("active") == "1"
        # 保护：不能把最后一个管理员降级或停用，否则无人能管理系统
        if user.role == "admin" and (role != "admin" or not
    active):
            others = User.query.filter(User.role == "admin",
                                       User.id != user.id,

    User.active.is_(True)).count()
            if others == 0:
                flash("系统至少需保留一名启用状态的管理员", "error")
                return redirect(url_for("auth.edit_user",
    user_id=user.id))
        # 个人资料字段（用户自己也能在「个人设置」里改，管理员在此代改）
        email = request.form.get("email", "").strip()
        if not email_ok(email):
            flash("邮箱格式不正确", "error")
            return redirect(url_for("auth.edit_user",
    user_id=user.id))
        user.role = role
        user.active = active
        user.display_name = request.form.get("display_name",
    "").strip() or None
        user.gender = request.form.get("gender",
    "").strip() or None
        user.email = email or None
        user.phone = request.form.get("phone", "").strip()
    or None
```

```
        new_password = request.form.get("password",                    3101
    "").strip()                                                        3102
        if new_password:                                               3103
            if len(new_password) < MIN_PASSWORD_LEN:                   3104
                flash("密码至少 %d 位" % MIN_PASSWORD_LEN,              3105
    "error")                                                           3106
                return redirect(url_for("auth.edit_user",              3107
    user_id=user.id))                                                  3108
            user.password_hash = generate_password_hash(new_           3109
    password)                                                          3110
        db.session.commit()                                            3111
        flash("账号信息已保存", "ok")                                   3112
        return redirect(url_for("auth.profile",                        3113
    tab="users"))                                                      3114
    return render_template("auth/edit_user.html", user=user,           3115
                           role_labels=ROLE_LABELS,                    3116
    genders=GENDERS,                                                   3117
                           form=EmptyForm())                           3118
@bp.route("/users/<int:user_id>/delete", methods=["POST"])             3119
@login_required                                                        3120
def delete_user(user_id):                                              3121
    require_admin()                                                    3122
    user = db.get_or_404(User, user_id)                                3123
    if user.id == current_user.id:                                     3124
        flash("不能删除当前登录的账号", "error")                        3125
        return redirect(url_for("auth.profile",                        3126
    tab="users"))                                                      3127
    if user.role == "admin":                                           3128
        others = User.query.filter(User.role == "admin",               3129
                                   User.id != user.id,                 3130
                                   User.active.is_(True)).co           3131
    unt()                                                              3132
        if others == 0:                                                3133
            flash("系统至少需保留一名启用状态的管理员", "error")        3134
            return redirect(url_for("auth.profile",                    3135
    tab="users"))                                                      3136
    db.session.delete(user)                                            3137
    db.session.commit()                                                3138
    flash("账号「%s」已删除" % user.username, "ok")                     3139
    return redirect(url_for("auth.profile", tab="users"))              3140
# ---------------------------- 操作日志（仅管理员）                    3141
    ----------------------------                                       3142
@bp.route("/logs")                                                     3143
@login_required                                                        3144
def logs():                                                            3145
    """旧地址保留，统一并入「个人设置 · 操作日志」。"""                 3146
    return redirect(url_for("auth.profile", tab="logs"))               3147
# ==== app/views/project.py ====                                       3148
# -*- coding: utf-8 -*-                                                3149
"""项目与矿山档案 —— 项目、图斑、参照生态系统、遥感影像挂接。"""       3150
```

```
from flask import (Blueprint, render_template, request,
    redirect, url_for,
                       flash, current_app)
from flask_login import login_required, current_user
from app.forms import EmptyForm
from app.models import db, Project, MinePatch,
    ReferenceEcosystem
from app.services import cleanup
from app.utils.parsing import FieldError, parse_float
from app.utils.timefmt import utcnow
bp = Blueprint("project", __name__, url_prefix="/project")
@bp.route("/")
@login_required
def index():
    projects = Project.query.order_by(Project.id.desc()).all
    ()
    return render_template("project/index.html",
    projects=projects,
                           form=EmptyForm())
@bp.route("/add", methods=["GET", "POST"])
@login_required
def add():
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        p = Project(
            name=request.form.get("name", "").strip(),
            code=request.form.get("code", "").strip() or
    None,
            description=request.form.get("description",
    "").strip() or None,
            created_by=current_user.id,
        )
        if not p.name:
            flash("项目名称不能为空", "error")
        else:
            db.session.add(p)
            db.session.commit()
            flash("项目已创建", "ok")
            return redirect(url_for("project.index"))
    return render_template("project/edit.html", form=form,
    project=None)
@bp.route("/<int:pid>/edit", methods=["GET", "POST"])
@login_required
def edit(pid):
    p = db.get_or_404(Project, pid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        p.name = request.form.get("name", "").strip()
```

```
            p.code = request.form.get("code", "").strip() or
        None
            p.description = request.form.get("description",
        "").strip() or None
            p.status = request.form.get("status", "进行中")
            db.session.commit()
            flash("项目已保存", "ok")
            return redirect(url_for("project.index"))
        return render_template("project/edit.html", form=form,
        project=p)
@bp.route("/<int:pid>/delete", methods=["POST"])
@login_required
def delete(pid):
        p = db.get_or_404(Project, pid)
        # 项目下的图斑由 ORM 级联删除，但图斑下面的指标数据、限制性判定、
        # 评价结果、得分明细、参照生态系统不会跟着走，必须先逐个图斑清理，
        # 否则每删一个项目都在库里留下一批孤儿数据（见 app/services/cleanup.py）。
        cleanup.delete_project(p)
        flash("项目已删除", "ok")
        return redirect(url_for("project.index"))
# ---------------------------- 矿山图斑 ----------------------------
        ----
@bp.route("/<int:pid>/patches")
@login_required
def patches(pid):
        p = db.get_or_404(Project, pid)
        return render_template("project/patches.html",
        project=p,
                               form=EmptyForm())
@bp.route("/patch/<int:patch_id>/delete", methods=["POST"])
@login_required
def patch_delete(patch_id):
        """删除矿山图斑，及其微地质单元、指标数据与评价记录。"""
        patch = db.get_or_404(MinePatch, patch_id)
        pid = patch.project_id
        # 清理逻辑统一放在 app/services/cleanup.py：项目删除走的是同一个函数，
        # 免得两条删除路径各写一份、漏掉某张子表（曾经就漏过）。
        deleted, _counts = cleanup.delete_patch(patch.id)
        flash("图斑「%s」及其数据已删除" % (deleted.name if deleted else
        patch.name),
              "ok")
        return redirect(url_for("project.patches", pid=pid))
@bp.route("/<int:pid>/patch/add", methods=["GET", "POST"])
@login_required
def patch_add(pid):
        p = db.get_or_404(Project, pid)
        form = EmptyForm()
        if request.method == "POST" and form.validate_on_submit(
        ):
            def fv(name, cast=str):
```

```
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        try:
            area_ha = parse_float(request.form.get("area_ha"
    ), "图斑面积（公顷）",
                                  minimum=0)
            stop_years = parse_float(request.form.get("stop_
    years"), "停产年限",
                                     minimum=0)
        except FieldError as e:
            flash(str(e), "error")
            return redirect(url_for("project.patch_add",
    pid=p.id))
        patch = MinePatch(
            project_id=p.id,
            name=request.form.get("name", "").strip(),
            admin_region=fv("admin_region"),
            area_ha=area_ha,
            mineral_type=fv("mineral_type"),
            mining_method=fv("mining_method"),
            stop_years=stop_years,
            responsible_entity_missing=bool(request.form.get
    ("responsible_entity_missing")),
        )
        if not patch.name:
            flash("图斑名称不能为空", "error")
        else:
            db.session.add(patch)
            db.session.flush()
            #  参照生态系统默认取系统内置值
            db.session.add(ReferenceEcosystem(
                patch_id=patch.id, coverage_benchmark=75.0,
    shannon_benchmark=0.7))
            db.session.commit()
            flash("图斑已创建", "ok")
            return redirect(url_for("project.patches",
    pid=p.id))
    return render_template("project/patch_edit.html",
    form=form, project=p, patch=None)
@bp.route("/patch/<int:patch_id>/edit", methods=["GET",
    "POST"])
@login_required
def patch_edit(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        def fv(name, cast=str):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
```

```
        # 数值字段统一走 parse_float：填了非数字给中文提示，而不是 500                3301
        try:                                                                    3302
            area_ha = parse_float(request.form.get("area_ha"                    3303
), "图斑面积（公顷）",                                                           3304
                                  minimum=0)                                    3305
            stop_years = parse_float(request.form.get("stop_                    3306
years"), "停产年限",                                                             3307
                                     minimum=0)                                 3308
            cov = parse_float(request.form.get("coverage_ben                    3309
chmark"),                                                                       3310
                              "植被覆盖度基准（%）", minimum=0,                   3311
maximum=100)                                                                    3312
            sha = parse_float(request.form.get("shannon_benc                    3313
hmark"),                                                                        3314
                              "Shannon 基准", minimum=0)                        3315
        except FieldError as e:                                                 3316
            flash(str(e), "error")                                              3317
            return redirect(url_for("project.patch_edit",                       3318
patch_id=patch.id))                                                             3319
        patch.name = request.form.get("name", "").strip()                       3320
        patch.admin_region = fv("admin_region")                                 3321
        patch.area_ha = area_ha                                                 3322
        patch.mineral_type = fv("mineral_type")                                 3323
        patch.mining_method = fv("mining_method")                               3324
        patch.stop_years = stop_years                                           3325
        patch.responsible_entity_missing =                                      3326
bool(request.form.get("responsible_entity_missing"))                            3327
        # 参照生态系统                                                          3328
        ref = ReferenceEcosystem.query.filter_by(patch_id=pa                    3329
tch.id).first()                                                                 3330
        if ref is None:                                                         3331
            ref = ReferenceEcosystem(patch_id=patch.id)                         3332
            db.session.add(ref)                                                 3333
        # 基准值：用户填 0 是合法输入（不能写成 `or 75.0`，那样 0 会被悄悄换成默认值）  3334
        ref.coverage_benchmark = 75.0 if cov is None else                       3335
cov                                                                             3336
        ref.shannon_benchmark = 0.7 if sha is None else sha                     3337
        ref.dominant_species = fv("dominant_species")                           3338
        patch.image_note = fv("image_note")                                     3339
        db.session.commit()                                                     3340
        flash("图斑已保存", "ok")                                               3341
        return redirect(url_for("project.patches",                              3342
pid=patch.project_id))                                                          3343
    return render_template("project/patch_edit.html",                           3344
form=form,                                                                      3345
                           project=patch.project,                               3346
patch=patch,                                                                    3347
                           reference=ReferenceEcosystem.quer                    3348
y.filter_by(                                                                    3349
                               patch_id=patch.id).first())                      3350
```

```
# ---------------------------- 遥感影像挂接                                    3351
    ----------------------------                                              3352
@bp.route("/patch/<int:patch_id>/image", methods=["POST"])                    3353
@login_required                                                               3354
def upload_image(patch_id):                                                   3355
    """上传遥感底图 / 无人机正射影像并挂接到图斑。"""                                3356
    from app.utils import uploads as up                                       3357
    patch = db.get_or_404(MinePatch, patch_id)                                3358
    form = EmptyForm()                                                        3359
    if not form.validate_on_submit():                                         3360
        flash("表单校验失败，请重试", "error")                                    3361
        return redirect(url_for("project.patch_edit",                         3362
    patch_id=patch.id))                                                       3363
    rel, err = up.save_upload(request.files.get("image"),                     3364
    "patches",                                                                3365
                              current_app.config["ALLOWED_IM                  3366
    AGE_EXT"],                                                                3367
                              current_app.config["UPLOAD_FOL                  3368
    DER"])                                                                    3369
    if err:                                                                   3370
        flash(err, "error")                                                   3371
        return redirect(url_for("project.patch_edit",                         3372
    patch_id=patch.id))                                                       3373
    # 替换旧影像时清理原文件，避免残留                                              3374
    if patch.image_path:                                                      3375
        up.remove_upload(patch.image_path,                                    3376
    current_app.config["UPLOAD_FOLDER"])                                      3377
    patch.image_path = rel                                                    3378
    patch.image_uploaded_at = utcnow()                                        3379
    note = request.form.get("image_note", "").strip()                         3380
    if note:                                                                  3381
        patch.image_note = note                                               3382
    db.session.commit()                                                       3383
    flash("遥感影像已挂接", "ok")                                                3384
    return redirect(url_for("project.patch_edit",                             3385
    patch_id=patch.id))                                                       3386
@bp.route("/patch/<int:patch_id>/image/delete",                               3387
    methods=["POST"])                                                         3388
@login_required                                                               3389
def delete_image(patch_id):                                                   3390
    from app.utils import uploads as up                                       3391
    patch = db.get_or_404(MinePatch, patch_id)                                3392
    form = EmptyForm()                                                        3393
    if form.validate_on_submit():                                             3394
        up.remove_upload(patch.image_path,                                    3395
    current_app.config["UPLOAD_FOLDER"])                                      3396
        patch.image_path = None                                               3397
        patch.image_uploaded_at = None                                        3398
        db.session.commit()                                                   3399
        flash("遥感影像已移除", "ok")                                             3400
```

```
    return redirect(url_for("project.patch_edit",                3401
    patch_id=patch.id))                                          3402
# ==== app/views/unit.py ====                                    3403
# -*- coding: utf-8 -*-                                          3404
"""微地质单元管理 —— 岩质边坡 / 碎石边坡 / 平台的划分与面积占比。"""          3405
from flask import Blueprint, render_template, request,           3406
    redirect, url_for, flash                                     3407
from flask_login import login_required                           3408
from app.forms import EmptyForm                                  3409
from app.models import db, MinePatch, MicroUnit,                 3410
    IndicatorData                                                3411
from app.utils.constants import UNIT_TYPES                       3412
from app.utils.parsing import FieldError, parse_float            3413
bp = Blueprint("unit", __name__, url_prefix="/unit")             3414
def recalc_ratio(patch_id):                                      3415
    """按面积重算该图斑各单元面积占比。"""                                3416
    units = MicroUnit.query.filter_by(patch_id=patch_id).all     3417
    ()                                                           3418
    total = sum(u.area or 0.0 for u in units)                    3419
    if total > 0:                                                3420
        for u in units:                                          3421
            u.ratio = round((u.area or 0.0) / total, 6)          3422
    return total                                                 3423
@bp.route("/")                                                   3424
@login_required                                                  3425
def list_patches():                                              3426
    """微地质单元入口：列出各图斑及其单元划分情况。"""                        3427
    patches = MinePatch.query.order_by(MinePatch.id.desc()).     3428
    all()                                                        3429
    rows = []                                                    3430
    for p in patches:                                            3431
        units = MicroUnit.query.filter_by(patch_id=p.id).all     3432
    ()                                                           3433
        rows.append({                                            3434
            "patch": p,                                          3435
            "count": len(units),                                 3436
            "ratio_sum": sum(u.ratio or 0.0 for u in units),     3437
        })                                                       3438
    return render_template("unit/list.html", rows=rows)          3439
@bp.route("/<int:patch_id>")                                     3440
@login_required                                                  3441
def index(patch_id):                                             3442
    patch = db.get_or_404(MinePatch, patch_id)                   3443
    units = MicroUnit.query.filter_by(patch_id=patch_id).all     3444
    ()                                                           3445
    total = sum(u.area or 0.0 for u in units)                    3446
    ratio_sum = sum(u.ratio or 0.0 for u in units)               3447
    return render_template("unit/index.html", patch=patch,       3448
    units=units,                                                 3449
                           total_area=total,                     3450
```

```
    ratio_sum=ratio_sum,                                                    3451
                                unit_types=UNIT_TYPES,                      3452
    form=EmptyForm())                                                       3453
@bp.route("/<int:patch_id>/add", methods=["POST"])                          3454
@login_required                                                             3455
def add(patch_id):                                                          3456
    patch = db.get_or_404(MinePatch, patch_id)                              3457
    form = EmptyForm()                                                      3458
    if form.validate_on_submit():                                           3459
        unit_type = request.form.get("unit_type")                           3460
        if unit_type not in UNIT_TYPES:                                     3461
            flash("微地质单元类型不正确", "error")                          3462
            return redirect(url_for("unit.index",                           3463
    patch_id=patch_id))                                                     3464
        # 评价流程规定将图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。    3465
        # 允许重复会让面积占比与按单元类型组织的指标数据对不上号。            3466
        if MicroUnit.query.filter_by(patch_id=patch.id,                     3467
                                     unit_type=unit_type).fi                3468
    rst():                                                                  3469
            flash("该图斑已存在「%s」，同一类型只能划分一个" % unit_type,    3470
    "error")                                                                3471
            return redirect(url_for("unit.index",                           3472
    patch_id=patch_id))                                                     3473
        area = request.form.get("area", "").strip()                         3474
        try:                                                                3475
            area_value = parse_float(area, "面积（公顷）",                  3476
    minimum=0)                                                              3477
        except FieldError as e:                                             3478
            flash(str(e), "error")                                          3479
            return redirect(url_for("unit.index",                           3480
    patch_id=patch_id))                                                     3481
        u = MicroUnit(patch_id=patch.id,                                    3482
    unit_type=unit_type,                                                    3483
                      area=area_value,                                      3484
                      description=request.form.get("descript                3485
    ion", "").strip() or None)                                              3486
        db.session.add(u)                                                   3487
        db.session.flush()                                                  3488
        recalc_ratio(patch.id)                                              3489
        db.session.commit()                                                 3490
        flash("微地质单元已添加", "ok")                                     3491
    return redirect(url_for("unit.index",                                   3492
    patch_id=patch_id))                                                     3493
@bp.route("/edit/<int:unit_id>", methods=["GET", "POST"])                   3494
@login_required                                                             3495
def edit(unit_id):                                                          3496
    """单个微地质单元的编辑页。"""                                          3497
    u = db.get_or_404(MicroUnit, unit_id)                                   3498
    form = EmptyForm()                                                      3499
    if request.method == "POST" and form.validate_on_submit(                3500
```

```
    ):                                                                  3501
        new_type = request.form.get("unit_type",                        3502
    u.unit_type)                                                        3503
        if new_type not in UNIT_TYPES:                                  3504
            flash("微地质单元类型不正确", "error")                      3505
            return redirect(url_for("unit.edit",                        3506
    unit_id=u.id))                                                      3507
        # 改类型时同样不允许与本图斑已有的其他单元重复                  3508
        dup = MicroUnit.query.filter_by(patch_id=u.patch_id,            3509
                                                                        3510
    unit_type=new_type).first()                                         3511
        if dup is not None and dup.id != u.id:                          3512
            flash("该图斑已存在「%s」，同一类型只能划分一个" % new_type,  3513
    "error")                                                            3514
            return redirect(url_for("unit.edit",                        3515
    unit_id=u.id))                                                      3516
        u.unit_type = new_type                                          3517
        area = request.form.get("area", "").strip()                     3518
        try:                                                            3519
            u.area = parse_float(area, "面积（公顷）", minimum=0)       3520
        except FieldError as e:                                         3521
            flash(str(e), "error")                                      3522
            return redirect(url_for("unit.edit",                        3523
    unit_id=u.id))                                                      3524
        u.description = request.form.get("description",                 3525
    "").strip() or None                                                 3526
        recalc_ratio(u.patch_id)                                        3527
        db.session.commit()                                             3528
        flash("微地质单元已保存", "ok")                                 3529
        return redirect(url_for("unit.index",                           3530
    patch_id=u.patch_id))                                               3531
    return render_template("unit/edit.html", unit=u,                    3532
    patch=u.patch,                                                      3533
                                unit_types=UNIT_TYPES, form=form)       3534
@bp.route("/delete/<int:unit_id>", methods=["POST"])                    3535
@login_required                                                         3536
def delete(unit_id):                                                    3537
    u = db.get_or_404(MicroUnit, unit_id)                               3538
    pid = u.patch_id                                                    3539
    name = u.unit_type                                                  3540
    # 一并删除该单元的指标数据，避免残留                               3541
    IndicatorData.query.filter_by(unit_id=u.id).delete(synch            3542
    ronize_session=False)                                               3543
    db.session.delete(u)                                                3544
    db.session.flush()                                                  3545
    recalc_ratio(pid)                                                   3546
    db.session.commit()                                                 3547
    flash("微地质单元「%s」已删除" % name, "ok")                        3548
    return redirect(url_for("unit.index", patch_id=pid))                3549
# ==== app/views/standard.py ====                                       3550
```

```
# -*- coding: utf-8 -*-
"""政策与标准知识库 —— 标准 / 政策文件的结构化规则与省际口径比对。"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, Standard, StandardRule
from app.services import standard_check
bp = Blueprint("standard", __name__, url_prefix="/standard")
@bp.route("/")
@login_required
def index():
    standards = Standard.query.order_by(Standard.category,
    Standard.province).all()
    local = [s for s in standards if s.category == "地方标准"]
    policy = [s for s in standards if s.category != "地方标准"]
    return render_template("standard/index.html",
    local=local, policy=policy,
                           form=EmptyForm())
@bp.route("/add", methods=["GET", "POST"])
@login_required
def add():
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        s = Standard(
            province=request.form.get("province",
    "").strip(),
            name=request.form.get("name", "").strip(),
            doc_number=request.form.get("doc_number",
    "").strip() or None,
            publish_date=request.form.get("publish_date",
    "").strip() or None,
            implement_date=request.form.get("implement_date"
    , "").strip() or None,
            category=request.form.get("category", "地方标准"),
            summary=request.form.get("summary", "").strip()
    or None,
        )
        if not s.province or not s.name:
            flash("省份与名称不能为空", "error")
        else:
            db.session.add(s)
            db.session.commit()
            flash("标准已添加", "ok")
            return redirect(url_for("standard.index"))
    return render_template("standard/edit.html", form=form,
    standard=None)
@bp.route("/<int:sid>/edit", methods=["GET", "POST"])
@login_required
```

```
def edit(sid):                                                    3601
    s = db.get_or_404(Standard, sid)                              3602
    form = EmptyForm()                                            3603
    if request.method == "POST" and form.validate_on_submit(      3604
    ):                                                            3605
        s.province = request.form.get("province",                 3606
    "").strip()                                                   3607
        s.name = request.form.get("name", "").strip()             3608
        s.doc_number = request.form.get("doc_number",             3609
    "").strip() or None                                           3610
        s.publish_date = request.form.get("publish_date",         3611
    "").strip() or None                                           3612
        s.implement_date = request.form.get("implement_date"      3613
    , "").strip() or None                                         3614
        s.category = request.form.get("category", "地方标准")     3615
        s.summary = request.form.get("summary", "").strip()       3616
    or None                                                       3617
        db.session.commit()                                       3618
        flash("已保存", "ok")                                     3619
        return redirect(url_for("standard.rules", sid=s.id))      3620
    return render_template("standard/edit.html", form=form,       3621
    standard=s)                                                   3622
@bp.route("/<int:sid>/delete", methods=["POST"])                  3623
@login_required                                                   3624
def delete(sid):                                                  3625
    s = db.get_or_404(Standard, sid)                              3626
    StandardRule.query.filter_by(standard_id=sid).delete()        3627
    db.session.delete(s)                                          3628
    db.session.commit()                                           3629
    flash("已删除", "ok")                                         3630
    return redirect(url_for("standard.index"))                    3631
@bp.route("/<int:sid>/rules", methods=["GET", "POST"])            3632
@login_required                                                   3633
def rules(sid):                                                   3634
    s = db.get_or_404(Standard, sid)                              3635
    form = EmptyForm()                                            3636
    if request.method == "POST" and form.validate_on_submit(      3637
    ):                                                            3638
        r = StandardRule(                                         3639
            standard_id=s.id,                                     3640
            item=request.form.get("item", "").strip(),            3641
            operator=request.form.get("operator", ">="),          3642
            threshold=request.form.get("threshold",               3643
    "").strip() or None,                                          3644
            condition_type=request.form.get("condition_type"      3645
    , "必要条件"),                                               3646
            region=request.form.get("region", "").strip()         3647
    or None,                                                      3648
            note=request.form.get("note", "").strip() or          3649
    None,                                                         3650
```

```
            )                                                                    3651
            if not r.item:                                                       3652
                flash("判定项不能为空", "error")                                  3653
            else:                                                                3654
                db.session.add(r)                                                3655
                db.session.commit()                                              3656
                flash("规则已添加", "ok")                                         3657
            return redirect(url_for("standard.rules", sid=s.id))                 3658
        rs = StandardRule.query.filter_by(standard_id=s.id).all(                 3659
        )                                                                        3660
        return render_template("standard/rules.html",                            3661
        form=form, standard=s, rules=rs)                                         3662
@bp.route("/rule/<int:rid>/delete", methods=["POST"])                            3663
@login_required                                                                  3664
def rule_delete(rid):                                                            3665
    r = db.get_or_404(StandardRule, rid)                                         3666
    sid = r.standard_id                                                          3667
    db.session.delete(r)                                                         3668
    db.session.commit()                                                          3669
    flash("规则已删除", "ok")                                                     3670
    return redirect(url_for("standard.rules", sid=sid))                          3671
@bp.route("/compare")                                                            3672
@login_required                                                                  3673
def compare():                                                                   3674
    """省际口径比对：同一指标在不同省份的阈值横向对比。"""                                3675
    rules = StandardRule.query.join(Standard).all()                              3676
    groups = {}                                                                  3677
    for r in rules:                                                              3678
        groups.setdefault(r.item, []).append(r)                                  3679
    return render_template("standard/compare.html",                              3680
    groups=groups)                                                               3681
@bp.route("/check")                                                              3682
@login_required                                                                  3683
def check_index():                                                               3684
    """结论对照入口：选择一个图斑，看评价结论是否符合当地文件要求。"""                      3685
    from app.models import MinePatch                                             3686
    patches = MinePatch.query.order_by(MinePatch.id.desc()).                     3687
    all()                                                                        3688
    rows = []                                                                    3689
    for p in patches:                                                            3690
        found = standard_check.province_of(p.admin_region)                       3691
        rows.append({"patch": p, "province": found})                             3692
    return render_template("standard/check_index.html",                          3693
    rows=rows)                                                                   3694
@bp.route("/check/<int:patch_id>")                                               3695
@login_required                                                                  3696
def check(patch_id):                                                             3697
    """某个图斑的评价结论与当地标准 / 政策文件的逐条对照。"""                            3698
    result = standard_check.compare(patch_id)                                    3699
    if result is None:                                                           3700
```

```
            flash("未找到该矿山图斑", "error")                                3701
            return redirect(url_for("standard.check_index"))                3702
      return render_template("standard/check.html",                         3703
      check=result)                                                         3704
# ==== app/views/constraint.py ====                                         3705
# -*- coding: utf-8 -*-                                                     3706
"""限制性指标判定 —— 一票否决判定。                                          3707
按指标清单与评价流程，共 5 项：政策、区位约束、土壤污染、地质灾害、耕地属性。  3708
任一项不通过即判定为"人工辅助或生态重建"，不再进入定量评价。                3709
"""                                                                         3710
from flask import Blueprint, render_template, request,                      3711
     redirect, url_for, flash                                               3712
from flask_login import login_required                                      3713
from app.forms import EmptyForm                                             3714
from app.models import db, MinePatch, ConstraintCheck                       3715
from app.utils.constants import CONSTRAINT_ITEMS                            3716
bp = Blueprint("constraint", __name__,                                      3717
     url_prefix="/constraint")                                              3718
# 字段名 -> 中文名                                                          3719
FIELDS = [                                                                  3720
     ("policy_pass", "政策约束"),                                           3721
     ("location_pass", "区位约束"),                                         3722
     ("pollution_pass", "土壤污染"),                                        3723
     ("geo_pass", "地质灾害"),                                              3724
     ("farmland_pass", "耕地属性"),                                         3725
]                                                                           3726
@bp.route("/")                                                              3727
@login_required                                                             3728
def index():                                                                3729
     patches = MinePatch.query.order_by(MinePatch.id.desc()).               3730
     all()                                                                  3731
     latest = {}                                                            3732
     for p in patches:                                                      3733
          latest[p.id] = ConstraintCheck.query.filter_by(patch              3734
     _id=p.id) \                                                            3735
               .order_by(ConstraintCheck.id.desc()).first()                 3736
     return render_template("constraint/index.html",                        3737
     patches=patches, latest=latest)                                        3738
@bp.route("/<int:patch_id>", methods=["GET", "POST"])                       3739
@login_required                                                             3740
def check(patch_id):                                                        3741
     patch = db.get_or_404(MinePatch, patch_id)                             3742
     form = EmptyForm()                                                     3743
     record = ConstraintCheck.query.filter_by(patch_id=patch_               3744
     id) \                                                                  3745
          .order_by(ConstraintCheck.id.desc()).first()                      3746
     if request.method == "POST" and form.validate_on_submit(               3747
     ):                                                                     3748
          if record is None:                                                3749
               record = ConstraintCheck(patch_id=patch.id)                  3750
```

```
                db.session.add(record)                                    3751
            for field, _ in FIELDS:                                       3752
                setattr(record, field, request.form.get(field)            3753
    == "1")                                                               3754
            record.passed = all(getattr(record, f) for f, _ in            3755
    FIELDS)                                                               3756
            failed = [name for f, name in FIELDS if not                   3757
    getattr(record, f)]                                                   3758
            record.reason = ("全部通过" if record.passed                    3759
                             else "未通过：" + "、".join(failed))           3760
            db.session.commit()                                           3761
            flash("判定结果已保存", "ok")                                   3762
            return redirect(url_for("constraint.check",                   3763
    patch_id=patch.id))                                                   3764
    # 预处理各条件的当前取值，供模板直接使用（Jinja 无 getattr）                3765
    field_states = [(f, name, None if record is None else                 3766
    getattr(record, f))                                                   3767
                    for f, name in FIELDS]                                3768
    return render_template("constraint/check.html",                       3769
    patch=patch, form=form,                                               3770
                                fields=FIELDS,                            3771
    field_states=field_states,                                            3772
                                record=record)                            3773
# ==== app/views/ahp.py ====                                              3774
# -*- coding: utf-8 -*-                                                   3775
"""指标体系与 AHP 权重 —— 路由与页面逻辑。"""                                  3776
import json                                                               3777
import math                                                               3778
from flask import Blueprint, render_template, request,                    3779
    redirect, url_for, flash                                              3780
from app.forms import AHPSelectForm, EmptyForm                            3781
from app.models import db, Indicator, ExpertMatrix,                       3782
    WeightScheme                                                          3783
from app.utils.constants import DEFAULT_WEIGHTS, UNIT_TYPES               3784
from app.services import ahp_service as ahp                               3785
from app.services import indicator_service as ind_svc                     3786
from app.services import weight_service as ws                             3787
bp = Blueprint("ahp", __name__, url_prefix="/ahp")                        3788
# 一次计算的权重结果对应的指标编码：名称 → 编码                                 3789
_L1_CODE = {"土壤指标": "soil", "植被指标": "veg", "地形指标":                  3790
    "terrain"}                                                            3791
_L2_CODE = {                                                              3792
    ("土壤指标", "土石比"): "soil_ratio",                                     3793
    ("土壤指标", "土层厚度"): "thickness",                                    3794
    ("土壤指标", "pH"): "ph",                                             3795
    ("土壤指标", "有机质"): "organic_matter",                                 3796
    ("土壤指标", "污染指标"): "pollution",                                    3797
    ("植被指标", "植被覆盖度"): "coverage",                                   3798
    ("植被指标", "生物多样性"): "shannon",                                    3799
    ("地形指标", "坡度"): "slope",                                            3800
```

```
    ("地形指标", "坡向"): "aspect",
}
def to_code_pairs(unit_type, level, l1, names, weights):
    """把一次 AHP 计算的权重结果转成「指标编码 → 权重」，便于落库。"""
    pairs = []
    for name, w in zip(names, weights):
        if level == 1:
            code = _L1_CODE.get(name)
        else:
            code = _L2_CODE.get((l1, name))
        if code:
            pairs.append((code, w))
    return pairs
def normalize_l1(unit_type, level, l1):
    """把 l1 参数规整为该单元类型下确实存在的一级指标名。
    level=2 的页面（判断矩阵、多专家矩阵）都按 l1 取二级指标。
    直接带 ?level=2 访问而没给 l1 时，原先会 KeyError: None 直接 500；
    这里退回该单元类型的第一个一级指标，页面照常可用。
    """
    if level != 2:
        return None
    valid = list(DEFAULT_WEIGHTS.get(unit_type,
    {}).get("l2", {}))
    if not valid:
        return None
    return l1 if l1 in valid else valid[0]
def indicator_list(unit_type, level, l1=None):
    """返回 (指标名称列表, 默认权重列表)。
    level=1 时返回一级指标；level=2 时返回指定一级指标下的二级指标。
    l1 缺失或不属于该单元类型时，退回第一个一级指标，不抛异常。
    """
    weights = DEFAULT_WEIGHTS[unit_type]
    if level == 1:
        names = list(weights["l1"].keys())
        vec = [weights["l1"][n] for n in names]
        return names, vec
    l2_all = weights["l2"]
    if l1 not in l2_all:
        l1 = next(iter(l2_all))
    l2 = l2_all[l1]
    names = list(l2.keys())
    vec = [l2[n] for n in names]
    return names, vec
def l1_choices(unit_type):
    """某单元类型的一级指标名称列表（用于二级指标录入时选择父指标）。"""
    return list(DEFAULT_WEIGHTS[unit_type]["l1"].keys())
@bp.route("/", methods=["GET", "POST"])
def index():
    form = AHPSelectForm()
    # 先根据请求中的单元类型填充 l1 选项，再进行表单校验
```

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    unit = request.form.get("unit_type") or                          3851
    form.unit_type.choices[0][0]                                     3852
    form.l1.choices = [(n, n) for n in l1_choices(unit)]             3853
    if form.validate_on_submit():                                    3854
        unit_type = form.unit_type.data                              3855
        level = int(form.level.data)                                 3856
        if level == 1:                                               3857
            return redirect(url_for("ahp.matrix",                    3858
    unit_type=unit_type, level=1))                                   3859
        if not form.l1.data:                                         3860
            flash("选择「二级指标」时，请同时选择所属的一级指标", "error")      3861
            return render_template("ahp/index.html",                 3862
    form=form,                                                       3863
                                          weight_rows=ws.summary())) 3864
        return redirect(url_for("ahp.matrix",                        3865
    unit_type=unit_type,                                             3866
                                       level=2, l1=form.l1.data))    3867
    return render_template("ahp/index.html", form=form,              3868
                           weight_rows=ws.summary())                 3869
@bp.route("/matrix", methods=["GET", "POST"])                        3870
def matrix():                                                        3871
    unit_type = request.values.get("unit_type")                      3872
    # level 是手改地址栏就能改的参数，写成 abc 不该把页面打成 500          3873
    try:                                                             3874
        level = int(request.values.get("level", 1))                 3875
    except (TypeError, ValueError):                                  3876
        level = 1                                                    3877
    if level not in (1, 2):                                          3878
        level = 1                                                    3879
    l1 = normalize_l1(unit_type, level,                              3880
    request.values.get("l1"))                                        3881
    if unit_type not in DEFAULT_WEIGHTS:                             3882
        flash("无效的微地质单元类型", "error")                            3883
        return redirect(url_for("ahp.index"))                        3884
    names, vec = indicator_list(unit_type, level, l1)                3885
    n = len(names)                                                   3886
    form = EmptyForm()                                               3887
    # 预填默认值：系统默认权重对应的完全一致矩阵 a_ij = w_i / w_j          3888
    default = [[round(vec[i] / vec[j], 4) for j in                   3889
    range(n)] for i in range(n)]                                     3890
    def compute(upper):                                              3891
        A = ahp.matrix_from_upper(upper)                             3892
        r = ahp.weights_and_consistency(A)                           3893
        bad = ahp.find_most_inconsistent(A)                          3894
        return {                                                     3895
            "weights": [round(v, 4) for v in r["weights"]],          3896
            "lambda_max": round(r["lambda_max"], 4),                 3897
            "ci": r["ci"],                                           3898
            "cr": r["cr"],                                           3899
            "consistent": r["consistent"],                           3900
```

```
            "matrix": [[round(A[i, j], 4) for j in
range(n)] for i in range(n)],
            "most_inconsistent": bad,
        }
    result = None
    if request.method == "POST" and form.validate_on_submit(
):
        action = request.form.get("action", "calc")
        try:
            upper = [float(request.form.get("m_%d_%d" % (i,
j), "1"))
                     for i in range(n) for j in range(i +
1, n)]
            # 0 / 负数 / inf / nan 能过 float()，但会让 n 次方根算出 inf
或 nan，
            # 保存时直接撞上 NOT NULL 约束（weight 为 nan）→ 500
            if any((not math.isfinite(x)) or x <= 0 for x
in upper):
                raise ValueError("矩阵元素必须是正数")
            result = compute(upper)
        except (ValueError, ZeroDivisionError):
            flash("矩阵输入有误：请填入正数（对角线自动为 1，下三角自动取倒数）",
"error")
            return redirect(url_for("ahp.matrix",
unit_type=unit_type,
                                    level=level, l1=l1))
        if action == "save":
            if not result["consistent"]:
                flash("一致性比例 CR=%.4f ≥ 0.1，判断矩阵不满足一致性要求，"
                      "请调整后再保存" % result["cr"], "error")
            else:
                pairs = to_code_pairs(unit_type, level, l1,
names,
                                      result["weights"])
                scheme, cnt = ws.save_weights(unit_type,
pairs)
                flash("已保存并为「%s」生效 %d 项权重（CR=%.4f）。"
                      "此后该单元类型的评价将以此权重计算。"
                      % (unit_type, cnt, result["cr"]),
"ok")
                return redirect(url_for("ahp.matrix",
unit_type=unit_type,
                                        level=level, l1=l1))
    elif request.values.get("calc") == "1":
        # 按默认（系统默认）矩阵直接试算，便于一键查看权重与一致性
        result = compute([default[i][j] for i in range(n)
for j in range(i + 1, n)])
    _, source_label, custom_scheme = ws.effective_weights(un
it_type)
    return render_template(
```

```
            "ahp/matrix.html",                                                  3951
            form=form, unit_type=unit_type, level=level, l1=l1,                 3952
            names=names, vec=vec, default=default,                              3953
        result=result,                                                          3954
            source_label=source_label, custom_scheme=custom_sche                3955
        me,                                                                     3956
        )                                                                       3957
@bp.route("/weights/reset", methods=["POST"])                                   3958
def weights_reset():                                                            3959
    """删除自定义权重方案，恢复为该单元类型的系统默认权重。"""                                3960
    unit_type = request.form.get("unit_type")                                   3961
    form = EmptyForm()                                                          3962
    if form.validate_on_submit():                                               3963
        if unit_type not in DEFAULT_WEIGHTS:                                    3964
            flash("无效的微地质单元类型", "error")                                  3965
        else:                                                                   3966
            n = ws.reset_weights(unit_type)                                     3967
            if n:                                                               3968
                flash("已恢复「%s」为系统默认权重" % unit_type, "ok")                  3969
            else:                                                               3970
                flash("「%s」本就是系统默认权重，无需恢复" % unit_type,                 3971
    "ok")                                                                       3972
    return redirect(url_for("ahp.index"))                                       3973
# ---------------------------- 指标维护 ----------------------------           3974
    ----                                                                        3975
@bp.route("/indicators")                                                        3976
def indicators():                                                               3977
    """指标定义维护：查看指标树、单位、理想值与取值依据。"""                                  3978
    if Indicator.query.count() == 0:                                            3979
        ind_svc.seed_indicators()                                               3980
    all_ind = Indicator.query.order_by(Indicator.level,                         3981
    Indicator.id).all()                                                         3982
    by_parent = {}                                                              3983
    for i in all_ind:                                                           3984
        by_parent.setdefault(i.parent_code, []).append(i)                       3985
    return render_template("ahp/indicators.html",                               3986
    by_parent=by_parent,                                                        3987
                                   all_ind=all_ind,                             3988
    unit_types=UNIT_TYPES,                                                      3989
                                   type_label=ind_svc.ideal_type_lab            3990
    el,                                                                         3991
                                   form=EmptyForm())                            3992
@bp.route("/indicators/<code>/edit", methods=["GET",                            3993
    "POST"])                                                                    3994
def indicator_edit(code):                                                       3995
    ind = Indicator.query.filter_by(code=code).first_or_404(                    3996
    )                                                                           3997
    form = EmptyForm()                                                          3998
    if request.method == "POST" and form.validate_on_submit(                    3999
    ):                                                                          4000
```

```
        name = request.form.get("name", "").strip()                    4001
        if not name:                                                   4002
            flash("指标名称不能为空", "error")                          4003
            return redirect(url_for("ahp.indicator_edit",              4004
    code=code))                                                        4005
        ind.name = name                                                4006
        ind.unit = request.form.get("unit", "").strip() or             4007
    None                                                               4008
        ind.ideal_value = request.form.get("ideal_value",              4009
    "").strip() or None                                                4010
        ind.ideal_type = request.form.get("ideal_type") or             4011
    ind.ideal_type                                                     4012
        units = request.form.getlist("enabled_units")                  4013
        ind.enabled_units = ",".join(units) if units else              4014
    None                                                               4015
        ind.is_constraint = request.form.get("is_constraint"           4016
    ) == "1"                                                           4017
        ind.description = request.form.get("description",              4018
    "").strip() or None                                                4019
        db.session.commit()                                            4020
        flash("指标「%s」已保存" % ind.name, "ok")                      4021
        return redirect(url_for("ahp.indicators"))                     4022
    return render_template("ahp/indicator_edit.html",                  4023
    ind=ind,                                                           4024
                           unit_types=UNIT_TYPES,                      4025
                           type_label=ind_svc.ideal_type_lab           4026
    el, form=form)                                                     4027
@bp.route("/indicators/reset", methods=["POST"])                       4028
def indicators_reset():                                                4029
    """按系统默认值重置指标定义。"""                                    4030
    form = EmptyForm()                                                 4031
    if form.validate_on_submit():                                      4032
        n = ind_svc.seed_indicators()                                  4033
        flash("已按系统默认值重置 %d 项指标定义" % n, "ok")             4034
    return redirect(url_for("ahp.indicators"))                         4035
# ---------------------------- 多专家矩阵 ------------------------------4036
    -----                                                              4037
def expert_scope():                                                    4038
    """从请求中取（并规整）单元类型、层级与所属一级指标。"""           4039
    unit_type = request.values.get("unit_type") or                     4040
    UNIT_TYPES[0]                                                      4041
    if unit_type not in DEFAULT_WEIGHTS:                               4042
        unit_type = UNIT_TYPES[0]                                      4043
    try:                                                               4044
        level = int(request.values.get("level", 1))                   4045
    except (TypeError, ValueError):                                    4046
        level = 1                                                      4047
    if level not in (1, 2):                                            4048
        level = 1                                                      4049
    l1 = normalize_l1(unit_type, level,                                4050
```

```
    request.values.get("l1"))                                          4051
    return unit_type, level, l1                                        4052
def expert_query(unit_type, level, l1):                                4053
    q = ExpertMatrix.query.filter_by(unit_type=unit_type,              4054
    level=level)                                                       4055
    if level == 2:                                                     4056
        q = q.filter_by(l1_name=l1)                                    4057
    else:                                                              4058
        q = q.filter(ExpertMatrix.l1_name.is_(None))                   4059
    return q.order_by(ExpertMatrix.id)                                 4060
def default_scheme(unit_type, level, l1=None):                         4061
    """取（或建）该范围的权重方案，专家矩阵统一挂到它下面。"""                4062
    name = "%s · %s%s" % (unit_type,                                   4063
                          "一级指标" if level == 1 else "二级指标",        4064
                          "(%s)" % l1 if l1 else "")                   4065
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ           4066
    e,                                                                 4067
                                          level=level,                 4068
    name=name).first()                                                 4069
    if scheme is None:                                                 4070
        scheme = WeightScheme(unit_type=unit_type,                     4071
    level=level, name=name,                                            4072
                              is_default=False)                        4073
        db.session.add(scheme)                                         4074
        db.session.flush()                                             4075
    return scheme                                                      4076
@bp.route("/experts", methods=["GET", "POST"])                         4077
def experts():                                                         4078
    """多专家判断矩阵录入与汇总。"""                                        4079
    unit_type, level, l1 = expert_scope()                              4080
    if unit_type not in DEFAULT_WEIGHTS:                               4081
        flash("无效的微地质单元类型", "error")                             4082
        return redirect(url_for("ahp.index"))                          4083
    names, vec = indicator_list(unit_type, level, l1)                  4084
    n = len(names)                                                     4085
    form = EmptyForm()                                                 4086
    if request.method == "POST" and form.validate_on_submit(           4087
    ):                                                                 4088
        action = request.form.get("action")                            4089
        if action == "add":                                            4090
            expert_name = request.form.get("expert_name",              4091
    "").strip() or "未具名专家"                                          4092
            specialty = request.form.get("specialty",                  4093
    "").strip()                                                        4094
            try:                                                       4095
                upper = [float(request.form.get("m_%d_%d" %            4096
    (i, j), "1"))                                                      4097
                         for i in range(n) for j in range(i            4098
    + 1, n)]                                                           4099
            except ValueError:                                         4100
```

```
                flash("矩阵输入有误，请填入正数", "error")                    4101
                return redirect(url_for("ahp.experts",                      4102
unit_type=unit_type,                                                        4103
                                level=level, l1=l1))                        4104
            # 0 / 负数 / inf / nan 都能过 float()，但会让 n 次方根算出       4105
inf 或 nan,                                                                 4106
            # 于是库里存下 [[1,0],[Infinity,1]] 这种矩阵、CR 还显示         4107
0.0000。                                                                    4108
            if any((not math.isfinite(x)) or x <= 0 for x                   4109
in upper):                                                                  4110
                flash("判断矩阵只能填正数（大于 0 的有限数字）", "error")     4111
                return redirect(url_for("ahp.experts",                      4112
unit_type=unit_type,                                                        4113
                                level=level, l1=l1))                        4114
            A = ahp.matrix_from_upper(upper)                                4115
            r = ahp.weights_and_consistency(A)                              4116
            scheme = default_scheme(unit_type, level, l1)                   4117
            db.session.add(ExpertMatrix(                                    4118
                scheme_id=scheme.id,                                        4119
                unit_type=unit_type, l1_name=(l1 if level                   4120
== 2 else None),                                                            4121
                expert_name=expert_name,                                    4122
specialty=specialty or None, level=level,                                   4123
                matrix_json=json.dumps([list(map(float,                     4124
row)) for row in A]),                                                       4125
                cr=round(r["cr"], 6)))                                      4126
            db.session.commit()                                             4127
            flash("已录入专家「%s」%s的判断矩阵（CR=%.4f）" %                4128
(expert_name, ("（%s）" % specialty) if specialty else                       4129
"", r["cr"]), "ok")                                                         4130
        elif action == "save_agg":                                          4131
            # 把汇总权重落库并生效，使多专家结果真正参与评价计算           4132
            weights = _aggregate(unit_type, level, l1)                      4133
            if weights is None:                                             4134
                flash("暂无可汇总的专家矩阵", "error")                       4135
            elif not weights["consistent"]:                                 4136
                flash("汇总矩阵一致性比例 CR=%.4f ≥ 0.1，不满足要求，"       4137
                      "请先检查或调整专家矩阵" % weights["cr"],              4138
"error")                                                                    4139
            else:                                                           4140
                names_, _ = indicator_list(unit_type,                       4141
level, l1)                                                                  4142
                pairs = to_code_pairs(unit_type, level, l1,                 4143
names_,                                                                     4144
                                      weights["weights"])                   4145
                _, cnt = ws.save_weights(unit_type, pairs)                  4146
                flash("已保存并为「%s」生效 %d 项汇总权重（CR=%.4f）"        4147
                      % (unit_type, cnt, weights["cr"]),                    4148
"ok")                                                                       4149
        else:                                                               4150
```

```
                # action 缺失或值不认识时以前是静默跳回，用户只觉得「点了没反应」
                flash("未识别的操作，请从页面上的按钮重新提交", "error")
            return redirect(url_for("ahp.experts",
        unit_type=unit_type,
                                    level=level, l1=l1))
        rows = expert_query(unit_type, level, l1).all()
        default = [[round(vec[i] / vec[j], 4) for j in
        range(n)] for i in range(n)]
        return render_template("ahp/experts.html",
        unit_type=unit_type, level=level,
                                l1=l1, names=names, rows=rows,
        default=default,
                                n=n, form=form)
@bp.route("/experts/<int:mid>/delete", methods=["POST"])
def expert_delete(mid):
        row = db.get_or_404(ExpertMatrix, mid)
        unit_type, level, l1 = row.unit_type, row.level,
        row.l1_name
        form = EmptyForm()
        if form.validate_on_submit():
            db.session.delete(row)
            db.session.commit()
            flash("已删除专家矩阵", "ok")
        return redirect(url_for("ahp.experts",
        unit_type=unit_type,
                                level=level, l1=l1))
def _aggregate(unit_type, level, l1):
        """按几何平均汇总多专家矩阵并做一致性检验。无矩阵时返回 None。"""
        rows = expert_query(unit_type, level, l1).all()
        mats = []
        for r in rows:
            try:
                mats.append([list(map(float, line))
                             for line in json.loads(r.matrix_jso
        n)])
            except Exception:
                continue
        if not mats:
            return None
        agg = ahp.aggregate_matrices(mats)
        res = ahp.weights_and_consistency(agg)
        return {
            "count": len(mats),
            "weights": res["weights"],
            "lambda_max": res["lambda_max"],
            "ci": res["ci"], "cr": res["cr"],
            "consistent": res["consistent"],
            "most_inconsistent": ahp.find_most_inconsistent(agg)
        ,
            "expert_crs": [r.cr for r in rows if r.cr is not
```

```
    None],                                                          4201
    }                                                               4202
@bp.route("/experts/aggregate")                                     4203
def experts_aggregate():                                            4204
    """按几何平均汇总多专家矩阵，并做一致性检验。"""                    4205
    unit_type, level, l1 = expert_scope()                           4206
    if unit_type not in DEFAULT_WEIGHTS:                            4207
        flash("无效的微地质单元类型", "error")                        4208
        return redirect(url_for("ahp.index"))                       4209
    names, vec = indicator_list(unit_type, level, l1)               4210
    rows = expert_query(unit_type, level, l1).all()                 4211
    result = _aggregate(unit_type, level, l1)                       4212
    if result is not None:                                          4213
        result["weights"] = [round(v, 4) for v in                   4214
    result["weights"]]                                              4215
        result["lambda_max"] = round(result["lambda_max"],          4216
    4)                                                              4217
    return render_template("ahp/experts_aggregate.html",            4218
    unit_type=unit_type,                                            4219
                                level=level, l1=l1, names=names,    4220
    rows=rows,                                                      4221
                                result=result, form=EmptyForm())    4222
# ==== app/views/indicator.py ====                                  4223
# -*- coding: utf-8 -*-                                             4224
"""指标数据采集与管理 —— 路由与页面逻辑。                            4225
按微地质单元组织录入；系统按单元类型自动决定应录入哪些指标          4226
（岩质边坡无土壤指标、平台无地形指标）。                            4227
"""                                                                 4228
import io                                                           4229
import json                                                         4230
from flask import (Blueprint, render_template, request,             4231
    redirect, url_for,                                              4232
                      flash, send_file)                             4233
from app.forms import EmptyForm                                     4234
from app.models import db, MinePatch, MicroUnit,                    4235
    IndicatorData                                                   4236
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,         4237
    UNIT_ENABLED_INDICATORS                                         4238
from app.utils.parsing import FieldError, parse_float               4239
from app.services import import_service as imp                      4240
bp = Blueprint("indicator", __name__,                               4241
    url_prefix="/indicator")                                        4242
# 指标字段定义：字段名 -> (中文名, 单位, 是否可选)                   4243
SOIL_FIELDS = [                                                     4244
    ("soil_ratio", "土石比", ""),                                   4245
    ("thickness", "土层厚度", "cm"),                                4246
    ("ph", "pH", ""),                                               4247
    ("organic_matter", "有机质", "g/kg"),                           4248
]                                                                   4249
VEG_FIELDS = [                                                      4250
```

```
    ("coverage", "植被覆盖度", "%"),                                    4251
    ("shannon", "Shannon-Wiener 多样性指数", ""),                       4252
]                                                                       4253
TERRAIN_FIELDS = [                                                      4254
    ("slope", "坡度", "°"),                                             4255
    ("aspect", "坡向", ""),                                             4256
]                                                                       4257
def unit_field_config(unit_type):                                       4258
    """根据单元类型返回需录入的字段分组。"""                                4259
    enabled = UNIT_ENABLED_INDICATORS.get(unit_type, ())                4260
    groups = []                                                         4261
    if "土壤指标" in enabled:                                            4262
        groups.append(("土壤指标", SOIL_FIELDS, True))    # True        4263
    = 含重金属                                                          4264
    if "植被指标" in enabled:                                            4265
        groups.append(("植被指标", VEG_FIELDS, False))                  4266
    if "地形指标" in enabled:                                            4267
        groups.append(("地形指标", TERRAIN_FIELDS, False))              4268
    return groups                                                       4269
@bp.route("/")                                                          4270
def index():                                                            4271
    patches = MinePatch.query.order_by(MinePatch.id.desc()).            4272
    all()                                                               4273
    return render_template("indicator/index.html",                      4274
    patches=patches)                                                    4275
@bp.route("/<int:unit_id>", methods=["GET", "POST"])                    4276
def edit(unit_id):                                                      4277
    unit = db.get_or_404(MicroUnit, unit_id)                            4278
    ind = IndicatorData.query.filter_by(unit_id=unit.id).fir            4279
    st()                                                                4280
    form = EmptyForm()                                                  4281
    groups = unit_field_config(unit.unit_type)                          4282
    if request.method == "POST" and form.validate_on_submit(            4283
    ):                                                                  4284
        # 各字段的合理范围（覆盖度是百分数、pH 0–14、坡度 0–90，其余非负）。    4285
        # 以前这里直接 float(v)，用户填个「约 3.5」看到的就是英文 500 页。      4286
        LIMITS = {                                                      4287
            "soil_ratio": ("土石比", 0, None),                          4288
            "thickness": ("土层厚度（cm）", 0, None),                    4289
            "ph": ("pH", 0, 14),                                        4290
            "organic_matter": ("有机质（g/kg）", 0, None),               4291
            "coverage": ("植被覆盖度（%）", 0, 100),                     4292
            "shannon": ("Shannon 指数", 0, None),                       4293
            "slope": ("坡度（°）", 0, 90),                              4294
        }                                                               4295
        try:                                                            4296
            values = {}                                                 4297
            for name, (label, lo, hi) in LIMITS.items():                4298
                values[name] = parse_float(request.form.get(            4299
    name), label,                                                       4300
```

```
                                                    minimum=lo,
    maximum=hi)
            heavy = {}
            for m in HEAVY_METALS:
                v = parse_float(request.form.get("hm_" + m),
                                "重金属 %s (mg/kg)" % m,
    minimum=0)
                if v is not None:
                    heavy[m] = v
        except FieldError as e:
            flash(str(e), "error")
            return redirect(url_for("indicator.edit",
    unit_id=unit.id))
        if ind is None:
            ind = IndicatorData(unit_id=unit.id)
            db.session.add(ind)
        for name, value in values.items():
            setattr(ind, name, value)
        ind.heavy_metals = json.dumps(heavy,
    ensure_ascii=False) if heavy else None
        ind.aspect = request.form.get("aspect", "").strip()
    or None
        db.session.commit()
        flash("指标数据已保存", "ok")
        return redirect(url_for("indicator.index"))
    # 组装当前值供模板回显
    current = {}
    if ind:
        current = {
            "soil_ratio": ind.soil_ratio, "thickness":
    ind.thickness,
            "ph": ind.ph, "organic_matter":
    ind.organic_matter,
            "coverage": ind.coverage, "shannon":
    ind.shannon,
            "slope": ind.slope, "aspect": ind.aspect,
        }
        current["heavy"] = ind.heavy_metals_dict()
    return render_template("indicator/edit.html",
                           unit=unit, form=form,
    groups=groups,
                           heavy_metals=HEAVY_METALS,
                           aspect_types=ASPECT_TYPES,
    current=current)
# ------------------------------ 批量导入 ------------------------------
    ----
@bp.route("/import", methods=["GET", "POST"])
def import_data():
    """批量导入指标数据（CSV / Excel）。"""
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
```

```
    all()                                                          4351
    form = EmptyForm()                                             4352
    report = None                                                  4353
    if request.method == "POST" and form.validate_on_submit(       4354
    ):                                                             4355
        patch_id = request.form.get("patch_id", type=int)          4356
        patch = db.session.get(MinePatch, patch_id) if             4357
    patch_id else None                                             4358
        if patch is None:                                          4359
            flash("请选择矿山图斑", "error")                          4360
            return redirect(url_for("indicator.import_data")       4361
    )                                                              4362
        file = request.files.get("datafile")                       4363
        if file is None or not file.filename:                      4364
            flash("请选择要导入的文件", "error")                      4365
            return redirect(url_for("indicator.import_data")       4366
    )                                                              4367
        ext = file.filename.rsplit(".", 1)[-1].lower()             4368
        if ext not in ("csv", "xlsx", "xls"):                      4369
            flash("仅支持 CSV 或 Excel(.xlsx/.xls)文件", "error")     4370
            return redirect(url_for("indicator.import_data")       4371
    )                                                              4372
        units = MicroUnit.query.filter_by(patch_id=patch.id)       4373
    .all()                                                         4374
        unit_map = {}                                              4375
        for u in units:                                            4376
            unit_map.setdefault(u.unit_type, u)                    4377
        if not unit_map:                                           4378
            flash("该图斑尚未划分微地质单元，请先划分后再导入", "error")     4379
            return redirect(url_for("indicator.import_data")       4380
    )                                                              4381
        try:                                                       4382
            parsed, errors = imp.parse_upload(file.filename,       4383
    file.stream, unit_map)                                         4384
        except Exception as e:                                     4385
            flash("文件解析失败：%s" % e, "error")                    4386
            return redirect(url_for("indicator.import_data")       4387
    )                                                              4388
        saved = 0                                                  4389
        for rec in parsed:                                         4390
            unit = unit_map[rec["unit_type"]]                      4391
            ind = IndicatorData.query.filter_by(unit_id=unit       4392
    .id).first()                                                   4393
            if ind is None:                                        4394
                ind = IndicatorData(unit_id=unit.id)               4395
                db.session.add(ind)                                4396
            f = rec["fields"]                                      4397
            for key in ("soil_ratio", "thickness", "ph",           4398
    "organic_matter",                                              4399
                        "coverage", "shannon", "slope",            4400
```

```
    "aspect"):                                                          4401
                if key in f:                                            4402
                    setattr(ind, key, f[key])                           4403
            if rec["heavy_metals"]:                                     4404
                ind.heavy_metals = json.dumps(rec["heavy_met            4405
    als"],                                                              4406
                                                                        4407
    ensure_ascii=False)                                                 4408
            saved += 1                                                  4409
        db.session.commit()                                             4410
        report = {"patch": patch.name, "saved": saved,                  4411
    "errors": errors,                                                   4412
                  "total": len(parsed) + len(                           4413
                      [e for e in errors if e.startswith("第            4414
    ")])}                                                               4415
        if saved:                                                       4416
            flash("已导入 %d 个微地质单元的指标数据" % saved, "ok")          4417
        else:                                                           4418
            flash("没有可导入的数据行，请检查文件内容", "error")             4419
    return render_template("indicator/import.html",                     4420
    patches=patches,                                                    4421
                                form=form, report=report,               4422
    headers=imp.HEADERS)                                                4423
@bp.route("/import/template")                                           4424
def import_template():                                                  4425
    """下载导入模板（CSV，含一行示例）。"""                                4426
    csv_text = imp.build_template_csv()                                 4427
    data = csv_text.encode("utf-8-sig")       # 带 BOM，便于            4428
    Excel 正确识别中文                                                   4429
    return send_file(io.BytesIO(data), as_attachment=True,              4430
                     download_name="指标数据导入模板.csv",               4431
                     mimetype="text/csv")                               4432
# ==== app/views/evaluate.py ====                                       4433
# -*- coding: utf-8 -*-                                                 4434
"""适宜性定量评价引擎 —— 路由与页面逻辑。                                4435
从数据库加载某图斑的微地质单元与指标数据，调用评价引擎计算，              4436
展示各单元得分、最终评价因子 F、适宜性等级与推荐措施。                   4437
入口处按评价流程的顺序检查前置条件（见 app/services/workflow.py），     4438
前置步骤未完成时不予计算，避免在条件不具备时产出评价结论。               4439
"""                                                                     4440
import json                                                             4441
from flask import Blueprint, render_template, request,                  4442
    redirect, url_for, flash                                            4443
from flask_login import current_user                                    4444
from app.forms import EmptyForm                                         4445
from app.models import (db, MinePatch, MicroUnit,                       4446
    IndicatorData,                                                      4447
                        EvaluationResult, ScoreDetail)                  4448
from app.services import evaluation_service as ev                       4449
from app.services import workflow as wf                                 4450
```

```
from app.services import unit_config
from app.services import standard_check
bp = Blueprint("evaluate", __name__, url_prefix="/evaluate")
def load_patch_input(patch_id):
    """从数据库组装评价引擎输入。"""
    units = MicroUnit.query.filter_by(patch_id=patch_id).all()
    result = []
    for u in units:
        ind = IndicatorData.query.filter_by(unit_id=u.id).first()
        data = {
            "soil_ratio": ind.soil_ratio if ind else None,
            "thickness": ind.thickness if ind else None,
            "ph": ind.ph if ind else None,
            "organic_matter": ind.organic_matter if ind else None,
            "heavy_metals": ind.heavy_metals_dict() if ind else {},
            "coverage": ind.coverage if ind else None,
            "shannon": ind.shannon if ind else None,
            "slope": ind.slope if ind else None,
            "aspect": ind.aspect if ind else None,
        }
        result.append({
            "unit_id": u.id,
            "unit_type": u.unit_type,
            "ratio": u.ratio if u.ratio is not None else 0.0,
            "data": data,
        })
    return result
def _round(v, n=6):
    return None if v is None else round(v, n)
def load_weights(units):
    """解析各单元类型当前生效的权重与来源说明。
    权重可来自 AHP 计算并保存的生效方案；未配置时用系统默认值。
    来源一并记录，便于报告标注与事后复核。
    """
    from app.services import weight_service as ws
    weights, sources = {}, {}
    for u in units:
        ut = u["unit_type"]
        if ut not in weights:
            w, label, _ = ws.effective_weights(ut)
            weights[ut] = w
            sources[ut] = label
    return weights, sources
def pack_units(unit_results, sources=None):
    """把评价结果整理为可持久化的 JSON。
```

```
    完整保留各单元得分、一级指标得分与逐条二级指标明细，
    使结果页与报告能够复现「已记录的那一次评价」，
    而不是每次打开时重新推算。
    """
    sources = sources or {}
    out = []
    for u in unit_results:
        out.append({
            "unit_type": u["unit_type"],
            "ratio": _round(u["ratio"]),
            "unit_score": _round(u["unit_score"]),
            "weighted": _round(u["weighted"]),
            "weight_source": sources.get(u["unit_type"]),
            "l1_scores": {k: _round(v) for k, v in
u["l1_scores"].items()},
            "details": [{
                "l1": d["l1"], "l2": d["l2"],
                "raw": None if d["raw"] is None else
str(d["raw"]),
                "std_value": _round(d["std_value"]),
                "weight": _round(d["weight"]),
                "contribution": _round(d["contribution"]),
            } for d in u["details"]],
        })
    return out
def pack_params(settings, sources=None):
    """把本次计算用到的参数整理成可持久化的快照。
    参数（指标理想取值、图斑参照生态系统基准、权重方案）都能在界面上改，
    所以结果记录必须连「当时用的什么参数」一起留下，否则事后无法复现。
    """
    from app.utils.constants import (IDEAL_SOIL_RATIO,
    IDEAL_THICKNESS,
                                     IDEAL_ORGANIC,
    IDEAL_COVERAGE,
                                     IDEAL_SHANNON,
    HEAVY_METAL_LIMIT)
    defaults = {
        "soil_ratio": IDEAL_SOIL_RATIO, "thickness":
    IDEAL_THICKNESS,
        "organic_matter": IDEAL_ORGANIC, "coverage":
    IDEAL_COVERAGE,
        "shannon": IDEAL_SHANNON,
    }
    for metal, limit in HEAVY_METAL_LIMIT.items():
        defaults["hm_" + metal] = limit
    used = {}
    for code, fallback in sorted(defaults.items()):
        value = settings.params.get(code) if settings else
    None
        if value is None:
```

```
                used[code] = {"value": fallback, "default":
    fallback,
                              "source": "系统默认取值"}
            elif abs(float(value) - float(fallback)) < 1e-9:
                #  库里存着这一项，但取值与系统默认值一致 —— 不能标成「界面配置」，
                #  否则每份报告都会声称用了自定义参数，反而看不出真正的偏离。
                used[code] = {"value": value, "default":
    fallback,
                              "source": "系统默认取值"}
            else:
                used[code] = {"value": value, "default":
    fallback,
                              "source": "界面配置（系统默认值 %.4g）" %
    fallback}
    return {"ideals": used, "weight_sources": dict(sources
    or {})}
def save_evaluation(patch_id, units):
    """执行评价并把结果与逐层明细落库，返回评价记录。
    权重取各单元类型当前生效的方案（AHP 计算结果），未配置时用
    系统默认权重；来源一并记录。结果 JSON 保留各单元得分、
    一级指标得分与逐条二级指标明细，结果页与报告据此复现
    记录下来的那次评价，不再重新推算。
    """
    weights, sources = load_weights(units)
    #  标准化参数取自该图斑的参照生态系统与指标定义（界面上可改），
    #  没配过的项自动回落到代码常量，历史结果与验证算例不受影响。
    settings = unit_config.for_patch(patch_id)
    res = ev.score_mine(units, weights, settings)
    saved = EvaluationResult(
        patch_id=patch_id,
        final_score=round(res["final_score"], 4),
        grade=res["grade"],
        measures=res["grade_info"]["measure"],
        unit_scores=json.dumps(pack_units(res["unit_results"
    ], sources),
                               ensure_ascii=False),
        #  参数快照 + 计算人：参数现在可以在界面上改，不留快照就复现不了这次结果
        params_json=json.dumps(pack_params(settings,
    sources),
                               ensure_ascii=False),
        created_by=current_user.id if getattr(current_user,
    "is_authenticated", False)
        else None,
    )
    db.session.add(saved)
    db.session.flush()        #  先取得 result.id，再写明细
    unit_ids = {u["unit_type"]: u["unit_id"] for u in units}
    for u in res["unit_results"]:
        for d in u["details"]:
            db.session.add(ScoreDetail(
```

```
                    result_id=saved.id,
                    unit_id=unit_ids.get(u["unit_type"]),
                    l1_name=d["l1"], l2_name=d["l2"],
                    raw_value=None if d["raw"] is None else
    str(d["raw"]),
                    std_value=d["std_value"],
    weight=d["weight"],
                    contribution=d["contribution"]))
    db.session.commit()
    return saved
def weak_points(unit_scores, top=3):
    """从已保存的结果里挑出「提升空间最大」的指标。
    提升空间 = 权重 ×（1 − 标准化值）：标准化值越低、权重越大，抬它越划算。
    只读已保存的记录，不重新计算，保证与报告口径一致。
    """
    rows = []
    for u in unit_scores or []:
        for d in u.get("details") or []:
            std = d.get("std_value")
            weight = d.get("weight")
            if std is None or weight is None:
                continue                # 不适用项（未参与计算）不进短板
            rows.append({
                "unit_type": u.get("unit_type"),
                "l2": d.get("l2"),
                "raw": d.get("raw"),
                "std_value": std,
                "gain": weight * (1.0 - std),
            })
    rows.sort(key=lambda x: x["gain"], reverse=True)
    return [x for x in rows if x["gain"] > 1e-9][:top]
def user_label(user_id):
    """把计算人 id 换成「昵称（账户名称）」。"""
    from app.models import User
    if not user_id:
        return None
    u = db.session.get(User, user_id)
    if u is None:
        return "已删除的账号"
    return u.shown_name
def param_source_summary(params):
    """一句话说清参数从哪来：有没有偏离系统默认值。
    多数情况下全部等于系统默认值，报告应直接说明「未偏离」；
    只有真被改过时才点名，便于审阅者一眼看到差异。
    """
    ideals = (params or {}).get("ideals") or {}
    custom = sorted(k for k, v in ideals.items()
                    if v.get("source",
    "").startswith("界面配置"))
    if not custom:
```

```
            return "全部采用系统默认取值，未被改动"
        return "以下 %d 项被改为界面配置值：%s" % (len(custom),
        "、".join(custom))
def block_reasons(progress):
    """把未满足的前置条件转成可读的拦截原因。"""
    return ["%s：%s" % (b["label"], b["reason"]) for b in
    progress["blocking"]]
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    latest = {}
    progress = {}
    for p in patches:
        latest[p.id] = EvaluationResult.query.filter_by(patc
    h_id=p.id) \
            .order_by(EvaluationResult.id.desc()).first()
        progress[p.id] = wf.patch_progress(p.id)
    return render_template("evaluate/index.html",
    patches=patches,
                           latest=latest, progress=progress)
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
def run(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    progress = wf.patch_progress(patch_id)
    if request.method == "POST" and form.validate_on_submit(
    ):
        if not progress["ready"]:
            for reason in block_reasons(progress):
                flash("无法执行评价 —— " + reason, "error")
            return redirect(url_for("evaluate.run",
    patch_id=patch_id))
        units = load_patch_input(patch_id)
        saved = save_evaluation(patch.id, units)
        return redirect(url_for("evaluate.result",
    result_id=saved.id))
    # 展示将采用的权重来源，便于在执行前确认
    units_now = load_patch_input(patch_id)
    _, sources = load_weights(units_now) if units_now else
    ({}, {})
    return render_template("evaluate/run.html",
    patch=patch, form=form,
                           progress=progress,
    weight_sources=sources)
@bp.route("/result/<int:result_id>")
def result(result_id):
    r = db.get_or_404(EvaluationResult, result_id)
    patch = db.session.get(MinePatch, r.patch_id)
    unit_scores = json.loads(r.unit_scores) if
```

```
    r.unit_scores else []                                              4701
    params = json.loads(r.params_json) if r.params_json                4702
    else {}                                                            4703
    return render_template("evaluate/result.html", r=r,                4704
    patch=patch,                                                       4705
                                unit_scores=unit_scores,               4706
                                undetected=wf.undetected_metals(p      4707
    atch.id)                                                           4708
                                if patch else [],                      4709
                                weak=weak_points(unit_scores),         4710
                                params=params.get("ideals") or         4711
    {},                                                                4712
                                param_sources=param_source_summar      4713
    y(params),                                                         4714
                                check=standard_check.compare(patc      4715
    h.id) if patch else None,                                          4716
                                operator=user_label(r.created_by)      4717
    )                                                                  4718
# ==== app/views/report.py ====                                        4719
# -*- coding: utf-8 -*-                                              4720
"""报告与结果输出 —— 得分明细、图表、Word / PDF 导出。"""                4721
import io                                                              4722
import json                                                            4723
from datetime import datetime                                          4724
from flask import Blueprint, render_template, send_file,               4725
    flash, redirect, url_for                                           4726
from app.models import db, MinePatch, ReferenceEcosystem               4727
from app.services import report_service as rs                          4728
from app.services import standard_check                                4729
from app.services import workflow as wf                                4730
from app.utils import charts                                           4731
from app.utils import browser                                          4732
from app.utils.constants import grade_of                               4733
from app.utils.timefmt import now_utc                                  4734
bp = Blueprint("report", __name__, url_prefix="/report")               4735
def _prepare(patch_id):                                                4736
    """加载图斑并取出已记录的评价结果，供明细页与导出复用。                4737
    报告是评价结论的呈现，因此这里读取最近一次保存的评价记录，            4738
    而不是重新计算：评价完成后再改动指标数据，也不会让报告与              4739
    系统记录的评价结论对不上。没有任何评价记录时，返回流程检查            4740
    结果，由调用方拦截并提示先去执行评价。                               4741
    """                                                                4742
    patch = db.get_or_404(MinePatch, patch_id)                         4743
    prog = wf.report_progress(patch_id)                                4744
    if not prog["ready"]:                                              4745
        return patch, None, None, None, prog                           4746
    rec = prog["result"]                                               4747
    result = {                                                         4748
        "final_score": rec.final_score,                                4749
        "grade": rec.grade,                                            4750
```

```
            "grade_info": grade_of(rec.final_score),
            "unit_results": json.loads(rec.unit_scores) if
        rec.unit_scores else [],
        }
        reference = ReferenceEcosystem.query.filter_by(patch_id=
        patch_id).first()
        return patch, reference, result, rec, prog
def _deny(progress, patch_id, endpoint="report.detail"):
        """前置条件未满足时的统一处理：逐条提示并退回报告列表。"""
        for b in progress["blocking"]:
            flash("暂不能生成报告 —— %s：%s" % (b["label"],
        b["reason"]), "error")
        if progress["blocking"]:
            flash("请先完成上述步骤后再生成报告。", "error")
        return redirect(url_for("report.index"))
def _weak_points(result, top=3):
        """提升空间最大的指标：权重 ×（1 − 标准化值）。
        与结果页用同一套口径，便于向使用方解释「先改哪几项」。
        """
        rows = []
        for u in result.get("unit_results") or []:
            for d in u.get("details") or []:
                std, w = d.get("std_value"), d.get("weight")
                if std is None or w is None:
                    continue
                rows.append((u["unit_type"], d["l2"],
        d.get("raw"), std,
                             w * (1.0 - std)))
        rows.sort(key=lambda x: x[4], reverse=True)
        return [r for r in rows if r[4] > 1e-9][:top]
def _operator(record):
        """报告里的计算人：昵称（账户名称）。"""
        from app.models import User
        uid = getattr(record, "created_by", None)
        if not uid:
            return None
        u = db.session.get(User, uid)
        return u.shown_name if u else "已删除的账号"
def _charts(result):
        """生成两张图表：单元得分对比 + 一级指标平均得分分布。"""
        units = [(u["unit_type"], u["unit_score"],
                  "%.4f" % u["unit_score"]) for u in
        result["unit_results"]]
        chart_units = charts.hbars(units)
        # 各一级指标在各单元上的平均得分，反映其对评价结果的实际影响
        acc, cnt = {}, {}
        for u in result["unit_results"]:
            for k, v in (u.get("l1_scores") or {}).items():
                acc[k] = acc.get(k, 0.0) + v
                cnt[k] = cnt.get(k, 0) + 1
```

```
    items = [(k, acc[k] / cnt[k], "%.4f" % (acc[k] /
    cnt[k])) for k in acc]
    items.sort(key=lambda x: -x[1])
    chart_weight = charts.donut(items)
    return chart_units, chart_weight
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    progress = {p.id: wf.report_progress(p.id) for p in
    patches}
    return render_template("report/index.html",
    patches=patches,
                           progress=progress)
@bp.route("/<int:patch_id>")
def detail(patch_id):
    patch, reference, result, record, prog =
    _prepare(patch_id)
    if result is None:
        return _deny(prog, patch_id)
    chart_units, chart_weight = _charts(result)
    return render_template("report/detail.html",
    patch=patch, result=result,
                           record=record,
    reference=reference,
                           chart_units=chart_units,
    chart_weight=chart_weight,
                           check=standard_check.compare(patc
    h.id),
                           undetected=wf.undetected_metals(p
    atch.id))
@bp.route("/<int:patch_id>/word")
def word(patch_id):
    patch, reference, result, record, prog =
    _prepare(patch_id)
    if result is None:
        return _deny(prog, patch_id)
    buf = rs.build_report(patch, result, record=record,
                          weak=_weak_points(result),
                          operator=_operator(record),
                          check=standard_check.compare(patch
    .id))
    filename = "自然恢复适宜性评价报告_%s.docx" % patch.name
    return send_file(buf, as_attachment=True,
    download_name=filename,
                     mimetype="application/vnd.openxmlformat
    s-officedocument."
                              "wordprocessingml.document")
@bp.route("/<int:patch_id>/pdf")
def pdf(patch_id):
```

```
    """导出 PDF 报告：服务端渲染打印模板，再交浏览器无头打印。"""
    patch, reference, result, record, prog =
    _prepare(patch_id)
    if result is None:
        return _deny(prog, patch_id)
    chart_units, chart_weight = _charts(result)
    html = render_template(
        "report/print.html", patch=patch,
    reference=reference, result=result,
        chart_units=chart_units, chart_weight=chart_weight,
        check=standard_check.compare(patch.id),
        generated_at=now_utc())
    data, err = browser.html_to_pdf(html)
    if err:
        flash("PDF 导出失败：%s" % err, "error")
        return redirect(url_for("report.detail",
    patch_id=patch_id))
    return send_file(io.BytesIO(data), as_attachment=True,
                     download_name="自然恢复适宜性评价报告_%s.pdf" %
    patch.name,
                     mimetype="application/pdf")
# ==== app/views/search.py ====
# -*- coding: utf-8 -*-
"""全局搜索：在评价项目、矿山图斑、标准与政策文件中查找关键词。
顶栏的搜索框指向本模块。只做关键词包含匹配，不做分词，
但覆盖名称、编号、行政区、矿种、文号、摘要等常用检索字段。
"""
from sqlalchemy import or_
from flask import Blueprint, render_template, request
from app.models import db, Project, MinePatch, Standard
bp = Blueprint("search", __name__)
LIMIT = 50              # 每类结果上限，避免一次渲染过多
@bp.route("/search")
def index():
    q = (request.args.get("q") or "").strip()
    projects, patches, standards = [], [], []
    if q:
        like = "%" + q + "%"
        projects = Project.query.filter(or_(
            Project.name.like(like),
            Project.code.like(like),
            Project.description.like(like),
        )).order_by(Project.id.desc()).limit(LIMIT).all()
        patches = MinePatch.query.filter(or_(
            MinePatch.name.like(like),
            MinePatch.admin_region.like(like),
            MinePatch.mineral_type.like(like),
        )).order_by(MinePatch.id.desc()).limit(LIMIT).all()
        standards = Standard.query.filter(or_(
            Standard.name.like(like),
```

```
                Standard.doc_number.like(like),                    4901
                Standard.province.like(like),                      4902
                Standard.summary.like(like),                       4903
            )).order_by(Standard.id).limit(LIMIT).all()            4904
    total = len(projects) + len(patches) + len(standards)          4905
    return render_template("search.html", q=q,                     4906
    projects=projects,                                             4907
                                  patches=patches,                 4908
    standards=standards, total=total)                              4909
# ==== app/templates/base.html ====                                4910
{% import "_macros.html" as ui %}<!DOCTYPE html>                   4911
<html lang="zh-CN">                                                4912
<head>                                                             4913
  <meta charset="UTF-8">                                           4914
  <meta name="viewport" content="width=device-width,               4915
    initial-scale=1.0">                                            4916
  <link rel="icon" href="{{ url_for('static',                      4917
    filename='favicon.ico') }}">                                   4918
  <title>{% block title %}历史遗留废弃矿山自然恢复适宜性评价系统{% endblock     4919
    %}</title>                                                     4920
  <style>                                                          4921
    /* ---------- 设计变量：绿色主色 + 中性灰阶 ---------- */              4922
    :root {                                                        4923
      --accent: #2c6b5a;                  /* 主色：青绿，千里江山石绿偏青 */   4924
      --accent-hover: #1e5548;                                     4925
      --accent-soft: #e6f0ec;                                      4926
      --side-1: #1a4a4a;                  /* 侧边栏：深青墨 */          4927
      --side-2: #0f3338;                                           4928
      --danger: #a0392a;                  /* 朱砂红 */              4929
      --danger-soft: #fbeae6;                                      4930
      --ok: #2c6b5a;                                               4931
      --ok-soft: #e6f0ec;                                          4932
      --todo: #c08a3e;                    /* 赭石色待办 */            4933
      --ink: #2a2520;                     /* 深褐墨色 */              4934
      --ink-soft: #6b5d4f;                /* 褐灰 */                 4935
      --line: #e0d8c8;                    /* 绢本边线 */              4936
      --surface: #fdfaf2;                 /* 暖白绢本 */              4937
      --canvas: #f5efe0;                  /* 绢本米黄背景 */           4938
      --radius: 10px;                                              4939
      --shadow: 0 2px 8px rgba(42, 37, 32, .06);                   4940
      --header-h: 60px;                                            4941
      --nav-w: 226px;                                              4942
    }                                                              4943
    * { box-sizing: border-box; }                                  4944
    html, body { margin: 0; padding: 0; }                          4945
    body {                                                         4946
      font-family: "Microsoft YaHei", "PingFang SC", "Noto         4947
    Serif SC", "SimSun", serif;                                    4948
      font-size: 16.5px;                                           4949
      line-height: 1.6;                                            4950
```

```
    color: var(--ink);
    background: var(--canvas);
}
/* ---------- 跳到主要内容（键盘可达性） ---------- */
.skip-link {
    position: absolute;
    left: 12px;
    top: -60px;
    z-index: 30;
    padding: 8px 14px;
    background: var(--accent);
    color: #fff;
    border-radius: var(--radius);
    text-decoration: none;
    transition: top .15s ease-out;
}
.skip-link:focus { top: 12px; }
/* ---------- 整体骨架：左侧深绿导航 + 右侧内容 ----------
   固定为一屏高，页面滚动交给右侧内容区，
   这样左侧导航栏（含底部用户区）不会随长页面一起被拉到下面去。 */
.app-shell {
    display: flex;
    align-items: stretch;
    height: 100vh;
    height: 100dvh;
    overflow: hidden;
}
/* ---------- 侧边栏 ---------- */
.app-side {
    flex: 0 0 var(--nav-w);
    width: var(--nav-w);
    display: flex;
    flex-direction: column;
    background: linear-gradient(180deg, var(--side-1) 0%,
var(--side-2) 100%);
    color: #fff;
}
.app-brand {
    display: flex;
    align-items: center;
    gap: 11px;
    padding: 17px 16px 15px;
    border-bottom: 1px solid rgba(255, 255, 255, .10);
}
.app-brand__mark {
    flex: none;
    display: grid;
    place-items: center;
    width: 36px;
    height: 36px;
```

```
    border-radius: 10px;
    background: rgba(255, 255, 255, .15);
}
.app-brand__name {
    font-size: 16px;
    font-weight: 600;
    line-height: 1.35;
    letter-spacing: .2px;
}
.app-brand__sub {
    font-size: 11.5px;
    line-height: 1.35;
    color: rgba(255, 255, 255, .58);
}
.app-nav { flex: 1; padding: 6px 12px 14px; overflow-y:
auto; }
.app-nav__group {
   margin: 12px 0 5px;
    padding: 0 10px;
    font-size: 11.5px;
    letter-spacing: .05em;
    color: rgba(255, 255, 255, .48);
}
.app-nav ul { margin: 0; padding: 0; list-style: none; }
.app-nav a {
    display: flex;
    align-items: center;
    gap: 10px;
    padding: 9px 11px;
   margin-bottom: 2px;
    border-radius: 8px;
    color: rgba(255, 255, 255, .80);
    text-decoration: none;
    transition: background-color .16s ease, color .16s
ease;
}
.app-nav a:hover { background: rgba(255, 255, 255,
.10); color: #fff; }
.app-nav a:focus-visible {
    outline: 2px solid rgba(255, 255, 255, .60);
    outline-offset: 1px;
}
.app-nav a[aria-current="page"] {
    background: rgba(255, 255, 255, .17);
    color: #fff;
    font-weight: 600;
}
.app-nav .ico { flex: none; opacity: .92; }
.app-nav__badge {
   margin-left: auto;
```

```
    min-width: 20px;
    padding: 0 6px;
    font-size: 11px;
    font-weight: 600;
    line-height: 18px;
    text-align: center;
    border-radius: 9px;
    background: var(--todo);
    color: #fff;
    font-variant-numeric: tabular-nums;
}
.app-user {
    display: flex;
    align-items: center;
    gap: 8px;
    padding: 10px 14px;
    border-top: 1px solid rgba(255, 255, 255, .10);
}
.app-user__me {
    display: flex;
    flex: 1;
    min-width: 0;
    align-items: center;
    gap: 10px;
    padding: 4px 6px;
    border-radius: 8px;
    color: inherit;
    text-decoration: none;
    transition: background-color .16s ease;
}
.app-user__me:hover { background: rgba(255, 255, 255,
.10); }
.app-user__me:focus-visible {
    outline: 2px solid rgba(255, 255, 255, .60);
    outline-offset: 1px;
}
.app-user__avatar {
    flex: none;
    display: grid;
    place-items: center;
    width: 32px;
    height: 32px;
    border-radius: 50%;
    background: rgba(255, 255, 255, .16);
    font-size: 15.5px;
    font-weight: 600;
    object-fit: cover;
}
.app-user__name {
    display: block;
```

```
    font-size: 15.5px;
    line-height: 1.3;
    white-space: nowrap;
    overflow: hidden;
    text-overflow: ellipsis;
}
.app-user__state {
    display: flex;
    align-items: center;
    gap: 5px;
    font-size: 11.5px;
    line-height: 1.3;
    color: rgba(255, 255, 255, .60);
}
.app-user__dot {
    width: 6px;
    height: 6px;
    border-radius: 50%;
    background: #6ed49a;
}
.app-user__out {
    flex: none;
    padding: 6px;
    border-radius: 6px;
    color: rgba(255, 255, 255, .68);
    text-decoration: none;
}
.app-user__out:hover { color: #fff; background:
rgba(255, 255, 255, .12); }
/* ---------- 右侧内容区 ----------
   整个骨架固定一屏高，滚动只发生在内容区内部：
   左侧导航栏（含底部用户区）始终留在原位，右侧页面自己滚。 */
.app-content {
    flex: 1;
    min-width: 0;
    display: flex;
    flex-direction: column;
    overflow-y: auto;
    overflow-x: hidden;
    /* 滚动条槽位始终占位。滚动条是出现在这个滚动容器内部的，
       内容高的页面（工作台、标准库、操作日志）会因此把顶栏挤窄约 10px，
       内容刚好放得下的页面（限制性判定、项目档案…）则不会 ——
       结果就是「顶栏在不同页面不一样宽」。预留槽位后各页一致。 */
    scrollbar-gutter: stable;
    /* 细滚动条：默认半透明，悬停时加深，不抢视线 */
    scrollbar-width: thin;
    scrollbar-color: rgba(31, 107, 69, .28) transparent;
}
.app-content::-webkit-scrollbar { width: 12px; }
.app-content::-webkit-scrollbar-track { background:
```

```
transparent; }
.app-content::-webkit-scrollbar-thumb {
  background: rgba(31, 107, 69, .26);
  border: 3px solid transparent;
  background-clip: content-box;
  border-radius: 999px;
}
.app-content::-webkit-scrollbar-thumb:hover {
  background: rgba(31, 107, 69, .46);
  background-clip: content-box;
}
/* ---------- 顶栏：面包屑 + 搜索 ----------
   粘在内容区顶部，滚动时面包屑与搜索框始终可用 */
.app-header {
  position: sticky;
  top: 0;
  z-index: 20;
  display: flex;
  align-items: center;
  gap: 14px;
  height: var(--header-h);
  padding: 0 22px;
  background: var(--surface);
  border-bottom: 1px solid var(--line);
}
.crumb {
  display: flex;
  align-items: center;
  gap: 7px;
  font-size: 15.5px;
  color: var(--ink-soft);
  white-space: nowrap;
}
.crumb a { color: var(--ink-soft); text-decoration:
none; }
.crumb a:hover { color: var(--accent); }
.crumb strong { color: var(--ink); font-weight: 600; }
.crumb .ico { color: #c3c9d0; }
.app-header__spacer { flex: 1; }
.search-box {
  display: flex;
  align-items: center;
  gap: 8px;
  width: 268px;
  padding: 7px 12px;
  background: var(--canvas);
  border: 1px solid var(--line);
  border-radius: 9px;
  color: var(--ink-soft);
  transition: border-color .16s ease, background-color
```

```
.16s ease;                                                                    5201
}                                                                             5202
.search-box:focus-within { border-color: var(--accent);                       5203
background: #fff; }                                                           5204
.search-box input {                                                           5205
  flex: 1;                                                                    5206
  min-width: 0;                                                               5207
  border: none;                                                               5208
  outline: none;                                                              5209
  background: transparent;                                                    5210
  font-family: inherit;                                                       5211
  font-size: 15.5px;                                                          5212
  color: var(--ink);                                                          5213
}                                                                             5214
.search-box button {                                                          5215
  display: grid;                                                              5216
  place-items: center;                                                        5217
  padding: 0;                                                                 5218
  border: none;                                                               5219
  background: transparent;                                                    5220
  color: inherit;                                                             5221
  cursor: pointer;                                                            5222
}                                                                             5223
.icon-btn {                                                                   5224
  display: grid;                                                              5225
  place-items: center;                                                        5226
  width: 34px;                                                                5227
  height: 34px;                                                               5228
  border-radius: 9px;                                                         5229
  border: 1px solid transparent;                                              5230
  color: var(--ink-soft);                                                     5231
  text-decoration: none;                                                      5232
  transition: background-color .16s ease, color .16s                          5233
ease,                                                                         5234
              border-color .16s ease;                                         5235
}                                                                             5236
.icon-btn:hover {                                                             5237
  color: var(--accent);                                                       5238
  background: var(--accent-soft);                                             5239
  border-color: rgba(31, 107, 69, .18);                                       5240
}                                                                             5241
.app-main {                                                                   5242
  flex: 1;                                                                    5243
  min-width: 0;                                                               5244
  padding: 22px 26px 26px;                                                    5245
}                                                                             5246
.container { max-width: none; width: 100%; }                                  5247
/* ---------- 页脚 ---------- */                                              5248
.app-foot {                                                                   5249
  display: flex;                                                              5250
```

```
    align-items: center;
    justify-content: space-between;
    gap: 14px;
    padding: 0 26px 18px;
    font-size: 14.5px;
    color: var(--ink-soft);
}
/* ---------- 未登录时的裸页面（登录页） ---------- */
.login-shell {
    display: flex;
    flex-direction: column;
    min-height: 100dvh;
}
/* ---------- 卡片 ---------- */
.card {
    background: var(--surface);
    border: 1px solid var(--line);
    border-radius: var(--radius);
    box-shadow: var(--shadow);
    padding: 22px 24px;
    margin-bottom: 18px;
}
.card > h2 {
    margin: 0 0 14px;
    font-size: 18.5px;
    font-weight: 600;
    color: var(--accent);
    text-wrap: balance;
}
.card > h3 {
    margin: 20px 0 10px;
    font-size: 16.5px;
    font-weight: 600;
    color: var(--accent);
}
/* ---------- 表格 ---------- */
table {
    width: 100%;
    border-collapse: collapse;
    margin: 10px 0 16px;
    font-variant-numeric: tabular-nums;
}
caption {
    text-align: left;
    color: var(--ink-soft);
    font-size: 14.5px;
    padding-bottom: 6px;
}
th, td {
    border: 1px solid var(--line);
```

```
    padding: 8px 12px;                                                    5301
    text-align: center;                                                   5302
    vertical-align: middle;                                               5303
}                                                                         5304
th {                                                                      5305
    background: var(--accent-soft);                                       5306
    font-weight: 600;                                                     5307
    white-space: nowrap;                                                  5308
}                                                                         5309
tbody tr:hover { background: #fafbfc; }                                   5310
/* ---------- 表单控件 ---------- */                                      5311
label { display: inline-block; }                                          5312
.field { margin-bottom: 12px; }                                           5313
.field__label {                                                           5314
    display: block;                                                       5315
    margin-bottom: 4px;                                                   5316
    color: var(--ink-soft);                                               5317
    font-size: 15.5px;                                                    5318
}                                                                         5319
input[type="text"], input[type="number"],                                 5320
input[type="password"],                                                   5321
select, textarea {                                                        5322
    padding: 6px 9px;                                                     5323
    font-family: inherit;                                                 5324
    font-size: 16.5px;                                                    5325
    color: var(--ink);                                                    5326
    background: #fff;                                                     5327
    border: 1px solid #c6ccd4;                                            5328
    border-radius: 4px;                                                   5329
    max-width: 100%;                                                      5330
}                                                                         5331
input[type="number"] {                                                    5332
    width: 104px;                                                         5333
    text-align: center;                                                   5334
    font-variant-numeric: tabular-nums;                                   5335
}                                                                         5336
textarea { width: 100%; min-height: 64px; resize:                         5337
vertical; }                                                               5338
input:focus-visible, select:focus-visible,                                5339
textarea:focus-visible {                                                  5340
    outline: 2px solid var(--accent);                                     5341
    outline-offset: 1px;                                                  5342
    border-color: var(--accent);                                          5343
}                                                                         5344
.checkbox {                                                               5345
    display: inline-flex;                                                 5346
    align-items: center;                                                  5347
    gap: 7px;                                                             5348
    cursor: pointer;                                                      5349
    user-select: none;                                                    5350
```

```
}
.checkbox input { width: 16px; height: 16px; margin: 0; cursor: pointer; }
/* ---------- 按钮 ---------- 
   统一为圆角矩形：默认半透明底色，鼠标悬停时填充为实色。 */
.btn {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  gap: 6px;
  padding: 8px 18px;
  font-family: inherit;
  font-size: 16.5px;
  line-height: 1.4;
  color: #fff;
  background: rgba(31, 107, 69, .80);
  border: 1px solid rgba(31, 107, 69, .30);
  border-radius: 8px;
  cursor: pointer;
  text-decoration: none;
  transition: background-color .16s ease, border-color .16s ease,
              color .16s ease, box-shadow .16s ease, transform .16s ease;
}
.btn:hover {
  background: var(--accent);
  border-color: var(--accent);
  box-shadow: 0 3px 12px rgba(31, 107, 69, .22);
}
.btn:active { transform: translateY(1px); box-shadow: none; }
.btn:focus-visible { outline: 2px solid var(--accent); outline-offset: 2px; }
.btn--ghost {
  color: var(--accent);
  background: rgba(31, 107, 69, .06);
  border-color: rgba(31, 107, 69, .22);
}
.btn--ghost:hover {
  color: #fff;
  background: var(--accent);
  border-color: var(--accent);
}
.btn--danger {
  color: var(--danger);
  background: rgba(179, 38, 30, .07);
  border-color: rgba(179, 38, 30, .26);
}
.btn--danger:hover {
```

```
    color: #fff;
    background: var(--danger);
    border-color: var(--danger);
    box-shadow: 0 3px 12px rgba(179, 38, 30, .20);
}
.btn--sm { padding: 5px 13px; font-size: 15.5px;
border-radius: 7px; }
.btn:disabled { opacity: .45; cursor: not-allowed; }
.btn:disabled:hover {
    background: rgba(31, 107, 69, .80);
    border-color: rgba(31, 107, 69, .30);
    box-shadow: none;
}
.btn--ghost:disabled:hover { background: rgba(31, 107,
69, .06); color: var(--accent); }
/* ---------- 表格操作列的行内小按钮 ---------- */
    取代原来「文字 | 文字 | 文字」的写法：
    圆角矩形 + 半透明底色，悬停时变实色。 */
.row-actions {
    display: inline-flex;
    flex-wrap: nowrap;          /* 按钮不折行；宁可让旁边长名称换行 */
    align-items: center;
    gap: 6px;
    white-space: nowrap;
}
.row-actions form { display: inline-flex; margin: 0; }
.link-btn {
    display: inline-flex;
    align-items: center;
    padding: 4px 11px;
    font-family: inherit;
    font-size: 15.5px;
    line-height: 1.5;
    color: var(--accent);
    background: rgba(31, 107, 69, .07);
    border: 1px solid rgba(31, 107, 69, .20);
    border-radius: 7px;
    cursor: pointer;
    text-decoration: none;
    white-space: nowrap;
    transition: background-color .16s ease, border-color
.16s ease,
                color .16s ease, box-shadow .16s ease,
transform .16s ease;
}
.link-btn:hover {
    color: #fff;
    background: var(--accent);
    border-color: var(--accent);
    box-shadow: 0 2px 8px rgba(31, 107, 69, .20);
```

```
}
.link-btn:active { transform: translateY(1px);
box-shadow: none; }
.link-btn:focus-visible { outline: 2px solid
var(--accent); outline-offset: 2px; }
.link-btn--danger {
  color: var(--danger);
  background: rgba(179, 38, 30, .07);
  border-color: rgba(179, 38, 30, .24);
}
.link-btn--danger:hover {
  color: #fff;
  background: var(--danger);
  border-color: var(--danger);
  box-shadow: 0 2px 8px rgba(179, 38, 30, .22);
}
.actions { display: flex; flex-wrap: wrap; align-items:
center; gap: 8px; }
.actions form { display: inline; }
/* ---------- 提示条 ---------- */
.flash {
  padding: 10px 14px;
  margin-bottom: 14px;
  border-radius: var(--radius);
  border-left: 4px solid;
}
.flash.error {
  color: var(--danger);
  background: var(--danger-soft);
  border-left-color: var(--danger);
}
.flash.ok {
  color: var(--ok);
  background: var(--ok-soft);
  border-left-color: var(--ok);
}
/* ---------- 浮层提示（登录成功、各类操作完成、登录失败） ----------
   外层 toast-stack 固定在页面中上方，内层每条各自淡入 → 停留 → 淡出，
   多条提示会依次排开而不重叠。
   底色是浅绿半透明：既有一层底色把文字托住，又能透出后面的内容，
   不用高饱和色块，提示不该比页面内容更抢眼。
   pointer-events: none 让它不拦截鼠标，出现期间也能正常操作。 */
.toast-stack {
  position: fixed;
  top: 16px;
  left: 50%;
  z-index: 100;
  display: flex;
  flex-direction: column;
  align-items: center;
```

```
  gap: 6px;
  transform: translateX(-50%);
  pointer-events: none;
}
.toast-center {
  max-width: min(620px, 82vw);
  padding: 8px 22px;
  border-radius: 999px;
  /* 浅绿、半透明：不遮住背后的内容，也比纯透明更托得住文字 */
  background: rgba(223, 241, 231, .76);
  border: 1px solid rgba(31, 107, 69, .14);
  color: #2f4a3c;
  font-size: 18.5px;
  font-weight: 500;
  line-height: 1.65;
  letter-spacing: .03em;
  text-align: center;
  text-wrap: pretty;
  backdrop-filter: blur(6px) saturate(1.1);
  -webkit-backdrop-filter: blur(6px) saturate(1.1);
  box-shadow: 0 4px 16px rgba(23, 32, 27, .07);
  animation: toast-fade 2.6s ease-in-out forwards;
}
/* 失败类提示：同一套观感，换成浅红底，一眼能区分成败 */
.toast-center--error {
  background: rgba(252, 233, 230, .78);
  border-color: rgba(163, 43, 34, .16);
  color: #9c2b23;
}
@keyframes toast-fade {
  0%          { opacity: 0; transform: translateY(-6px); }
  14%, 74%    { opacity: 1; transform: none; }
  100%        { opacity: 0; transform: none; }
}
/* 系统设置里要求「减少动态效果」时，去掉位移，只做淡入淡出 */
@media (prefers-reduced-motion: reduce) {
  .toast-center { animation: toast-fade-plain 2.6s
linear forwards; }
  @keyframes toast-fade-plain {
    0%          { opacity: 0; }
    14%, 74%    { opacity: 1; }
    100%        { opacity: 0; }
  }
}
/* ---------- 文字工具类 ---------- */
.muted { color: var(--ink-soft); font-size: 14px; }
.mono { font-family: Consolas, "Courier New",
monospace; font-variant-numeric: tabular-nums; }
.num { font-variant-numeric: tabular-nums; }
.warn { color: var(--danger); font-weight: 600; }
```

```
.ok { color: var(--ok); font-weight: 600; }
.nowrap { white-space: nowrap; }
/* 表格操作列里常把删除按钮放在 <form> 中。form 默认是块级元素，
   会把「删除」挤到下一行，导致操作列参差不齐，这里让它留在行内。 */
.nowrap form { display: inline; margin: 0; }
.stack { display: flex; flex-direction: column; gap: 14px; }
/* 状态标签：不依赖颜色单独表意 */
.tag {
  display: inline-block;
  padding: 1px 9px;
  font-size: 14px;
  border-radius: 10px;
  border: 1px solid;
  white-space: nowrap;
}
.tag--ok { color: var(--ok); background: var(--ok-soft); border-color: #b8ddc6; }
.tag--warn { color: var(--danger); background: var(--danger-soft); border-color: #eec4c0; }
.tag--muted { color: var(--ink-soft); background: #f1f3f6; border-color: #dde1e6; }
/* ---------- 登录页 ---------- */
/* 仿 Apple 账户页：接近纯白的底，仅在顶部有一层极淡的绿色晕染，
   让彩色徽标成为整页唯一的视觉重心。排版见 auth/login.html。 */
.body--login {
  background:
    radial-gradient(1100px 640px at 50% 8%, #edf6f1 0%, rgba(237, 246, 241, 0) 72%),
    linear-gradient(180deg, #ffffff 0%, #fbfdfc 100%);
}
.body--login .app-main { padding: 0; }
.body--login .container { max-width: none; width: 100%; }
/* 仅供读屏软件的隐藏文本 */
.visually-hidden {
  position: absolute !important;
  width: 1px;
  height: 1px;
  margin: -1px;
  padding: 0;
  overflow: hidden;
  clip: rect(0 0 0 0);
  white-space: nowrap;
  border: 0;
}
/* ---------- 工作台：页首横幅 ---------- */
.hero {
  position: relative;
```

```
  overflow: hidden;                                                    5601
  min-height: 260px;                                                   5602
  padding: 26px 28px;                                                  5603
  margin-bottom: 16px;                                                 5604
  border-radius: 12px;                                                 5605
  /* 左侧压深保证文字可读，右侧透出远山插画；                                  5606
     背景按底部对齐，避免远山被居中裁掉 */                                    5607
  background:                                                          5608
    linear-gradient(100deg, rgba(15, 40, 45, .90) 0%,                  5609
                            rgba(20, 55, 58, .55) 45%,                 5610
                            rgba(20, 55, 58, .12) 100%),               5611
    url("{{ url_for('static', filename='hero_landscape.j               5612
pg') }}")                                                              5613
    center center / cover no-repeat;                                   5614
  color: #fff;                                                         5615
}                                                                      5616
.hero__body { max-width: 720px; }                                      5617
.hero__title {                                                         5618
  margin: 0 0 10px;                                                    5619
  font-size: 23.5px;                                                   5620
  font-weight: 700;                                                    5621
  line-height: 1.4;                                                    5622
  letter-spacing: .3px;                                                5623
}                                                                      5624
.hero__desc {                                                          5625
  margin: 0;                                                           5626
  max-width: 640px;                                                    5627
  font-size: 14px;                                                     5628
  line-height: 1.85;                                                   5629
  color: rgba(255, 255, 255, .86);                                     5630
}                                                                      5631
.hero__actions {                                                       5632
  display: flex;                                                       5633
  flex-wrap: wrap;                                                     5634
  gap: 10px;                                                           5635
  margin-top: 18px;                                                    5636
}                                                                      5637
.hero-btn {                                                            5638
  display: inline-flex;                                                5639
  align-items: center;                                                 5640
  gap: 6px;                                                            5641
  padding: 7px 15px;                                                   5642
  font-size: 15.5px;                                                   5643
  border-radius: 8px;                                                  5644
  text-decoration: none;                                               5645
  transition: background-color .16s ease, border-color                 5646
.16s ease,                                                             5647
              color .16s ease;                                         5648
}                                                                      5649
.hero-btn--solid {                                                     5650
```

```
    color: #14432f;
    background: rgba(255, 255, 255, .92);
    border: 1px solid rgba(255, 255, 255, .92);
}
.hero-btn--solid:hover { background: #fff; }
.hero-btn--ghost {
    color: #fff;
    background: rgba(255, 255, 255, .14);
    border: 1px solid rgba(255, 255, 255, .45);
}
.hero-btn--ghost:hover { background: rgba(255, 255,
255, .26); }
.hero-btn:focus-visible { outline: 2px solid #fff;
outline-offset: 2px; }
/* ---------- 工作台：数据概览 ---------- */
.stat-cards {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(190px,
1fr));
    gap: 12px;
    margin-bottom: 22px;
}
.stat-card {
    display: block;
    padding: 15px 17px;
    background: var(--surface);
    border: 1px solid var(--line);
    border-radius: var(--radius);
    box-shadow: var(--shadow);
    color: var(--ink);
    text-decoration: none;
    transition: border-color .16s ease, box-shadow .16s
ease;
}
.stat-card:hover {
    border-color: rgba(31, 107, 69, .45);
    box-shadow: 0 3px 12px rgba(19, 60, 43, .08);
}
.stat-card__label {
    display: block;
    font-size: 14px;
    color: var(--ink-soft);
}
.stat-card__value {
    display: block;
    margin: 3px 0 2px;
    font-size: 28.5px;
    font-weight: 700;
    line-height: 1.25;
    color: var(--accent);
```

```
    font-variant-numeric: tabular-nums;
}
.stat-card__unit {
    margin-left: 3px;
    font-size: 15.5px;
    font-weight: 500;
    color: var(--ink-soft);
}
.stat-card__note {
    display: block;
    font-size: 14.5px;
    color: var(--ink-soft);
}
/* ---------- 工作台：分区标题 ---------- */
.section { margin-bottom: 22px; }
.section__head {
    display: flex;
    align-items: baseline;
    flex-wrap: wrap;
    gap: 6px 14px;
    margin-bottom: 12px;
}
.section__title { margin: 0; font-size: 18.5px; }
.section__desc { margin: 0; font-size: 14px; color:
var(--ink-soft); }
/* ---------- 工作台：流程步骤卡 ---------- */
.step-grid {
    display: grid;
    grid-template-columns: repeat(auto-fill,
minmax(238px, 1fr));
    gap: 12px;
}
.step-card {
    position: relative;
    display: block;
    padding: 16px 16px 15px;
    background: var(--surface);
    border: 1px solid var(--line);
    border-radius: var(--radius);
    box-shadow: var(--shadow);
    color: var(--ink);
    text-decoration: none;
    transition: border-color .16s ease, box-shadow .16s
ease,
                transform .16s ease;
}
.step-card:hover {
    border-color: rgba(31, 107, 69, .55);
    box-shadow: 0 5px 16px rgba(19, 60, 43, .10);
    transform: translateY(-1px);
```

```
}
.step-card:focus-visible { outline: 2px solid var(--accent); outline-offset: 2px; }
.step-card__no {
  position: absolute;
  top: 13px;
  right: 15px;
  font-size: 19.5px;
  font-weight: 700;
  color: #d7e3db;
  font-variant-numeric: tabular-nums;
}
.step-card__icon {
  display: grid;
  place-items: center;
  width: 32px;
  height: 32px;
  margin-bottom: 10px;
  border-radius: 9px;
  background: var(--accent-soft);
  color: var(--accent);
}
.step-card__title {
  display: block;
  margin-bottom: 5px;
  font-size: 16.5px;
  font-weight: 600;
  color: var(--accent);
}
.step-card__desc {
  display: block;
  font-size: 17px;
  line-height: 1.65;
  color: var(--ink-soft);
  text-wrap: pretty;
}
.step-card__badge {
  display: inline-block;
  margin-top: 10px;
  padding: 1px 9px;
  font-size: 11.5px;
  border-radius: 9px;
  background: #fdf1e2;
  color: #9a5a12;
  border: 1px solid #f0d3ab;
}
/* 前置数据不具备时置灰，并说明缺什么 */
.step-card--locked {
  background: #fafbfa;
  border-color: #eceff0;
```

```
    box-shadow: none;                                                    5801
    cursor: default;                                                     5802
}                                                                        5803
.step-card--locked .step-card__no { color: #e6e9e8; }                    5804
.step-card--locked .step-card__icon {                                    5805
    background: #f1f3f2;                                                 5806
    color: #a9b2ae;                                                      5807
}                                                                        5808
.step-card--locked .step-card__title { color: #9aa3a0; }                 5809
.step-card--locked .step-card__desc { color: #b3bab7; }                  5810
.step-card__lock {                                                       5811
    display: inline-block;                                               5812
    margin-top: 10px;                                                    5813
    padding: 1px 9px;                                                    5814
    font-size: 11.5px;                                                   5815
    border-radius: 9px;                                                  5816
    background: #f1f3f2;                                                 5817
    color: #8b948f;                                                      5818
}                                                                        5819
/* ---------- 结果摘要 ---------- */                                     5820
.stat-row {                                                              5821
    display: flex;                                                       5822
    flex-wrap: wrap;                                                     5823
    gap: 28px;                                                           5824
    margin: 4px 0 6px;                                                   5825
}                                                                        5826
.stat__label { font-size: 14px; color: var(--ink-soft);                  5827
}                                                                        5828
.stat__value { font-size: 22.5px; font-weight: 700;                      5829
font-variant-numeric: tabular-nums; }                                    5830
/* ---------- 流程步骤条 ---------- */                                   5831
.steps { display: flex; flex-wrap: wrap; gap: 8px;                       5832
margin: 4px 0 18px; }                                                    5833
.step {                                                                  5834
    display: flex; align-items: center; gap: 7px;                        5835
    padding: 8px 12px; font-size: 15.5px;                                5836
    background: #fff; border: 1px solid var(--line);                     5837
    border-radius: var(--radius);                                        5838
}                                                                        5839
.step__no {                                                              5840
    display: inline-flex; align-items: center;                           5841
justify-content: center;                                                 5842
    width: 20px; height: 20px; border-radius: 50%;                       5843
    background: #eef1f5; color: var(--ink-soft);                         5844
    font-size: 14.5px; font-variant-numeric: tabular-nums;               5845
}                                                                        5846
.step__name { font-weight: 600; }                                        5847
.step__state { font-size: 14px; color: var(--ink-soft);                  5848
}                                                                        5849
.step--done { border-color: #b8ddc6; background:                         5850
```

```
var(--ok-soft); }
.step--done .step__no { background: var(--ok); color: #fff; }
.step--done .step__state { color: var(--ok); }
.step--fail { border-color: #eec4c0; background: var(--danger-soft); }
.step--fail .step__no { background: var(--danger); color: #fff; }
.step--fail .step__state { color: var(--danger); font-weight: 600; }
.step--next { border-color: var(--accent); }
.step--next .step__no { background: var(--accent); color: #fff; }
/* ---------- 提示块（中性底色，不用整块警示色压页面） ---------- */
.notice {
  padding: 14px 16px; margin: 0 0 16px;
  background: var(--surface);
  border: 1px solid var(--line);
  border-left: 3px solid var(--todo);
  border-radius: var(--radius);
}
.notice__title { margin: 0 0 8px; font-weight: 600; }
/* ---------- 前置条件表（与页面主表同款，只把待处理的行标成浅红） ---------- */
.prereq-table { margin: 0 0 18px; }
.prereq-table > caption { padding-bottom: 7px; }
.prereq-row > td { background: var(--danger-soft); }
.prereq-row:hover > td { background: #f6dcd7; }
.prereq-flag { color: var(--danger); font-weight: 600; }
.prereq-row .muted { color: #9a6b62; }
/* ---------- 图斑面板（按图斑分块展示，如指标数据采集） ---------- */
.patch-panel {
  border: 1px solid var(--line);
  border-radius: var(--radius);
  background: var(--surface);
  padding: 14px 16px;
  margin: 0 0 14px;
}
.patch-panel:last-child { margin-bottom: 0; }
.patch-panel__head {
  display: flex;
  align-items: baseline;
  flex-wrap: wrap;
  gap: 6px 14px;
  margin: 0 0 12px;
  padding-bottom: 9px;
  border-bottom: 1px solid var(--line);
}
.patch-panel__name {
  margin: 0;
  font-size: 17.5px;
```

```
        font-weight: 600;
      }
      .patch-panel__meta {
        font-size: 14px;
        color: var(--ink-soft);
      }
      .patch-panel__meta a { color: var(--accent); }
      .patch-panel__tail { margin-left: auto; }
      .patch-panel .notice { margin: 0; }
      .patch-panel table { margin: 0; }
      /* ---------- 窄屏适配 ---------- */
      @media (max-width: 960px) {
        /* 上下布局时取消「一屏高 + 内容区内部滚动」，
           否则导航会占掉半屏、内容被挤在剩下的小格子里滚动 */
        .app-shell {
          flex-direction: column;
          height: auto;
          min-height: 100vh;
          min-height: 100dvh;
          overflow: visible;
        }
        .app-content { overflow: visible; }
        .app-header { position: static; }
        .app-side { width: auto; flex: none; }
        .app-brand { padding: 12px 14px; }
        .app-nav { padding: 4px 10px 10px; }
        .app-nav ul { display: flex; flex-wrap: wrap; gap:
      4px; }
        .app-nav a { margin-bottom: 0; }
        .app-nav__group { margin: 8px 0 4px; }
        .app-user { padding: 10px 14px; }
        .app-header { padding: 0 14px; gap: 10px; }
        /* 搜索框保持与宽屏一致的宽度：原来这里缩到 180px，
           窗口一窄搜索框就变窄，同一台机器不同窗口大小看着像两个尺寸 */
        .search-box { width: 268px; }
        .app-main { padding: 16px 14px 20px; }
        .app-foot { padding: 0 14px 14px; }
      }
      @media (max-width: 640px) {
        .crumb { display: none; }
        .search-box { width: auto; flex: 1; }
      }
    </style>
</head>
<body class="{% if not current_user.is_authenticated
    %}body--login{% endif %}">
  <a class="skip-link" href="#main">跳到主要内容</a>
  {# 取一次即可（同一请求内重复调用返回相同内容）。
     类别为 toast / ok / toast-error 的渲染成中上方浮层，出现后自动淡出；
     其余仍是页面顶部的提示条 —— 
```

```
    需要对照着改表单的错误，留在页面上比一闪而过更有用。 #}
{% set flash_msgs = get_flashed_messages(with_categories=true) %}
<div class="toast-stack">
  {% for category, message in flash_msgs if category in ("toast", "ok", "toast-error") %}
  <div class="toast-center{% if category == 'toast-error' %} toast-center--error{% endif %}"
       role="status" aria-live="polite">{{ message }}</div>
  {% endfor %}
</div>
{% if current_user.is_authenticated %}
<div class="app-shell">
  <aside class="app-side">
    <div class="app-brand">
      <span class="app-brand__mark">{{ ui.icon("mountain", 21) }}</span>
      <span>
        <span class="app-brand__name">矿山自然恢复评价</span><br>
        <span class="app-brand__sub">适宜性评价系统 Ｖ1.0</span>
      </span>
    </div>
    <nav class="app-nav" aria-label="功能导航">
      <ul>
        <li>
          <a href="/"{% if home_active %}
  aria-current="page"{% endif %}>
            {{ ui.icon("home") }}<span>工作台</span>
          </a>
        </li>
      </ul>
      {% for g in nav_groups %}
      <p class="app-nav__group">{{ g.title }}</p>
      <ul>
        {% for item in g.links %}
        <li>
          <a href="{{ item.href }}"{% if item.active %}
  aria-current="page"{% endif %}>
            {{ ui.icon(item.icon) }}<span>{{ item.label
  }}</span>
            {% if item.badge %}<span
  class="app-nav__badge">{{ item.badge }}</span>{% endif
  %}
          </a>
        </li>
        {% endfor %}
      </ul>
      {% endfor %}
    </nav>
```

```
    <div class="app-user">                                                  6001
      <a class="app-user__me" href="/profile"                               6002
title="个人设置">                                                           6003
        {% if current_user.avatar %}                                        6004
        <img class="app-user__avatar"                                       6005
             src="{{ url_for('static',                                      6006
filename=current_user.avatar) }}" alt="">                                   6007
        {% else %}                                                          6008
        <span class="app-user__avatar"                                      6009
aria-hidden="true">{{ current_user.initial }}</span>                        6010
        {% endif %}                                                         6011
        <span style="min-width:0;">                                         6012
          <span class="app-user__name">{{                                  6013
current_user.shown_name }}</span>                                           6014
          <span class="app-user__state">                                    6015
            <span class="app-user__dot"                                     6016
aria-hidden="true"></span>                                                  6017
            {{ role_labels.get(current_user.role,                           6018
current_user.role) }}                                                       6019
          </span>                                                           6020
        </span>                                                             6021
      </a>                                                                  6022
      <a class="app-user__out" href="/logout" title="退出登录"              6023
         aria-label="退出登录">{{ ui.icon("logout", 17) }}</a>              6024
    </div>                                                                  6025
</aside>                                                                    6026
<div class="app-content">                                                   6027
    <header class="app-header">                                             6028
      <nav class="crumb" aria-label="当前位置">                             6029
        <a href="/">首页</a>                                                6030
        {{ ui.icon("chevron", 13) }}                                        6031
        <strong>{{ page_title }}</strong>                                   6032
      </nav>                                                                6033
      <span class="app-header__spacer"></span>                              6034
      <form class="search-box" method="get"                                 6035
action="/search" role="search">                                             6036
        {{ ui.icon("search", 16) }}                                         6037
        <label for="q" class="visually-hidden">搜索项目、图斑、标准</       6038
label>                                                                      6039
        <input id="q" type="search" name="q"                                6040
autocomplete="off"                                                          6041
               placeholder="搜索项目、图斑、标准" value="{{                 6042
search_q }}">                                                               6043
        <button type="submit" aria-label="搜索">{{                          6044
ui.icon("chevron", 15) }}</button>                                          6045
      </form>                                                               6046
      {% if current_user.is_admin %}                                        6047
      <a class="icon-btn" href="/profile?tab=logs"                          6048
title="操作日志"                                                            6049
         aria-label="操作日志">{{ ui.icon("bell", 18) }}</a>                6050
```

```
          {% endif %}
        </header>
    {% else %}
    <div class="login-shell">
    {% endif %}
      <main id="main" class="app-main">
        <div class="container">
          {% for category, message in flash_msgs if category
      not in ("toast", "ok", "toast-error") %}
          <p class="flash {{ category }}" role="status">{{
      message }}</p>
          {% endfor %}
          {% block content %}{% endblock %}
        </div>
      </main>
    {% if current_user.is_authenticated %}
        <footer class="app-foot">
          <span>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1．0</span>
          <span>数据截止 {{ today }}</span>
        </footer>
      </div>
    </div>
    {% else %}
    </div>
    {% endif %}
    {% block scripts %}{% endblock %}
</body>
</html>
# ==== app/templates/_macros.html ====
{# 共用宏：图标、流程状态面板 #}
{# ============================================================
   ==================
   图标：内联 SVG 线性图标（24×24 视图框，随文字颜色走）
   不引图标字体或外部图片，避免额外请求与资源依赖；
   stroke 用 currentColor，在深色侧边栏与浅色内容区都能自适应。
   ============================================================
   ================== #}
{% set ICONS = {
  "dot": '<circle cx="12" cy="12" r="3"/>',
  "home": '<path d="M3 9.6 12 3l9 6.6V20a2 2 0 0 1-2 2H5a2
    2 0 0 1-2-2z"/><path d="M9.5 22v-7h5v7"/>',
  "folder": '<path d="M3 7.5A2 2 0 0 1 5 5.5h3.6l2 2.2H19a2
    2 0 0 1 2 2v7.8a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2z"/>',
  "book": '<path d="M4 4.8A2.8 2.8 0 0 1 6.8
    2H20v17.5H6.8A2.8 2.8 0 0 0 4 22z"/><path d="M4
    19.5A2.8 2.8 0 0 1 6.8 16.7H20"/>',
  "shield": '<path d="M12 22s8-4 8-10V5.2L12 2 4 5.2V12c0 6
    8 10 8 10z"/><path d="m9 12 2.2 2.2L15.5 10"/>',
  "layers": '<path d="m12 2.8 9 4.7-9 4.7-9-4.7z"/><path
    d="m3 12.5 9 4.7 9-4.7"/><path d="m3 17 9 4.7 9-4.7"/>',
```

```
    "sliders": '<path d="M4 21v-6M4 11V3M12 21v-9M12 8V3M20
      21v-4M20 13V3"/><path d="M1.5 15h5M9.5 12h5M17.5
      17h5"/>',
    "database": '<ellipse cx="12" cy="5.5" rx="8"
      ry="3.2"/><path d="M4 5.5v13c0 1.8 3.6 3.2 8 3.2s8-1.4
      8-3.2v-13"/><path d="M4 12c0 1.8 3.6 3.2 8 3.2s8-1.4
      8-3.2"/>',
    "target": '<circle cx="12" cy="12" r="9"/><circle cx="12"
      cy="12" r="4.6"/><circle cx="12" cy="12" r="1"/>',
    "file": '<path d="M14 2.8H7a2 2 0 0 0-2 2v14.4a2 2 0 0 0
      2 2h10a2 2 0 0 0 2-2V7.8z"/><path d="M14
      2.8V7.8h5"/><path d="M8.8 13h6.4M8.8 17h6.4"/>',
    "users": '<circle cx="9.5" cy="8" r="3.6"/><path d="M3.5
      20.5v-1.4A3.6 3.6 0 0 1 7.1 15.5h4.8a3.6 3.6 0 0 1 3.6
      3.6v1.4"/><path d="M17 4.9a3.6 3.6 0 0 1 0 6.9"/><path
      d="M19 15.8a3.6 3.6 0 0 1 2.4 3.4v1.3"/>',
    "user": '<circle cx="12" cy="8" r="4"/><path d="M4.5
      21v-1.6A4.4 4.4 0 0 1 8.9 15h6.2a4.4 4.4 0 0 1 4.4
      4.4V21"/>',
    "mail": '<rect x="3" y="5" width="18" height="14"
      rx="2.5"/><path d="m3.8 7 8.2 5.8L20.2 7"/>',
    "phone": '<path d="M6.4 3h2.8l1.5 4.4-2 1.4a12.4 12.4 0 0
      0 6.5 6.5l1.4-2L21 14.8v2.8a2.4 2.4 0 0 1-2.6 2.4A16.6
      16.6 0 0 1 3.5 5.6 2.4 2.4 0 0 1 5.9 3z"/>',
    "lock": '<rect x="4" y="10.4" width="16" height="10.6"
      rx="2.6"/><path d="M8 10.4V7.6a4 4 0 0 1 8 0v2.8"/>',
    "briefcase": '<rect x="3" y="7.4" width="18"
      height="13.1" rx="2.6"/><path d="M8.4 7.4V6A2 2 0 0 1
      10.4 4h3.2a2 2 0 0 1 2 2v1.4"/><path d="M3 12.6h18"/>',
    "search": '<circle cx="11" cy="11" r="7"/><path d="m20.5
      20.5-4-4"/>',
    "bell": '<path d="M18 9.5a6 6 0 1 0-12 0c0 5-2.2 6.2-2.2
      6.2h16.4S18 14.5 18 9.5"/><path d="M13.8 19.5a2 2 0 0
      1-3.6 0"/>',
    "plus": '<path d="M12 5v14M5 12h14"/>',
    "upload": '<path d="M12 16V4"/><path d="m7.5 8.5 4.5-4.5
      4.5 4.5"/><path d="M4 16.5v2A2.5 2.5 0 0 0 6.5
      21h11a2.5 2.5 0 0 0 2.5-2.5v-2"/>',
    "logout": '<path d="M15 4h3.5A2.5 2.5 0 0 1 21 6.5v11A2.5
      2.5 0 0 1 18.5 20H15"/><path d="M10 16.5 5.5 12 10
      7.5"/><path d="M5.5 12H15"/>',
    "chevron": '<path d="m9.5 5.5 6.5 6.5-6.5 6.5"/>',
    "warn": '<path d="M12 3.5 21.5 20H2.5z"/><path d="M12
      10v4.2"/><path d="M12 17.4h.01"/>',
    "mountain": '<path d="m3 20 6.5-11L14 17l2.5-4L21
      20z"/><circle cx="17" cy="6" r="2.2"/>',
} %}
{% macro icon(name, size=18) %}
<svg class="ico" width="{{ size }}" height="{{ size }}"
     viewBox="0 0 24 24"
```

```
      fill="none" stroke="currentColor" stroke-width="1.7"
      stroke-linecap="round" stroke-linejoin="round"
      aria-hidden="true" focusable="false">{{
    ICONS.get(name, ICONS.dot)|safe }}</svg>
{% endmacro %}
{# 评价流程步骤条：按评价流程顺序显示各前置步骤的完成情况
   顺序为：限制性条件判定 → 划分微地质单元 → 录入指标数据 → 执行评价。
   流程约定：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
   废弃矿山，按照地形地貌划分……三种地质微单元类型」，故判定在前。 #}
{% macro workflow_steps(progress) %}
<div class="steps">
  <div class="step {% if progress.constraint.passed
    %}step--done{% elif progress.constraint.done
    %}step--fail{% else %}step--todo{% endif %}">
    <span class="step__no">1</span>
    <span class="step__name">限制性条件判定</span>
    <span class="step__state">
      {% if progress.constraint.passed %}已通过
      {% elif progress.constraint.done %}未通过
      {% else %}待完成{% endif %}
    </span>
  </div>
  <div class="step {% if progress.units.done %}step--done{%
    else %}step--todo{% endif %}">
    <span class="step__no">2</span>
    <span class="step__name">划分微地质单元</span>
    <span class="step__state">
      {% if progress.units.done %}已完成{% else %}待完成{% endif
    %}
    </span>
  </div>
  <div class="step {% if progress.data.ready %}step--done{%
    else %}step--todo{% endif %}">
    <span class="step__no">3</span>
    <span class="step__name">录入指标数据</span>
    <span class="step__state">
      {% if progress.data.ready %}已完成{% else %}待完成{% endif
    %}
    </span>
  </div>
  <div class="step step--next">
    <span class="step__no">4</span>
    <span class="step__name">执行适宜性评价</span>
    <span class="step__state">当前步骤</span>
  </div>
</div>
{% endmacro %}
{# 前置条件未满足时的提示：做成与页面主表同款的表格，只把待处理的那几行标成浅红，
   说明写在 caption 里，缺什么、去哪补都在行内。 #}
{% macro workflow_blocking(progress) %}
```

```
{% if progress.blocking %}
<table class="prereq-table">
  <caption>本步骤的前置条件尚未满足，按评价流程的顺序先补上：</caption>
  <thead>
    <tr>
      <th scope="col">前置步骤</th>
      <th scope="col">说明</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for b in progress.blocking %}
    <tr class="prereq-row">
      <td>{{ b.label }}</td>
      <td class="muted">{{ b.reason }}</td>
      <td><a class="btn btn--sm" href="{{ b.fix_url }}">{{
    b.fix_label }}</a></td>
    </tr>
    {% endfor %}
  </tbody>
</table>
{% endif %}
{% endmacro %}
{# 一行式状态摘要，用于列表页 #}
{% macro workflow_badge(progress) %}
{% if progress.ready %}
<span class="tag tag--ok">可执行评价</span>
{% elif progress.blocking %}
<span class="tag tag--warn">{{ progress.blocking[0].label
    }}</span>
<span class="muted">{{ progress.blocking[0].reason }}</span>
{% endif %}
{% endmacro %}
{# 层级引导：本系统是「项目 → 图斑」两级结构，评价针对的是图斑而非项目。
   只在流程里**第一个**按图斑操作的页面（限制性指标判定）列一次，后面各页
   不再重复；建好图斑才会出现在下面的表里，但离「能进入本页流程」还差别的步骤，
   所以措辞只说到「出现在这里」为止。 #}
{% macro patch_guide(projects_without_patch) %}
{% if projects_without_patch %}
<table class="prereq-table">
  <caption>以下项目下尚未建立矿山图斑，本页按图斑组织数据，建好图斑后才会出现在这里：</caption>
  <thead>
    <tr>
      <th scope="col">所属项目</th>
      <th scope="col">矿山图斑</th>
      <th scope="col">状态</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
```

```
      {% for p in projects_without_patch %}                                    6251
      <tr class="prereq-row">                                                  6252
        <td>项目「{{ p.name }}」</td>                                          6253
        <td class="muted">未建立</td>                                          6254
        <td><span class="prereq-flag">缺少矿山图斑</span></td>                 6255
        <td>                                                                   6256
          <a class="btn btn--sm" href="/project/{{ p.id                        6257
      }}/patch/add">                                                           6258
            {{ icon("plus", 14) }}去新建图斑                                   6259
          </a>                                                                 6260
        </td>                                                                  6261
      </tr>                                                                    6262
      {% endfor %}                                                             6263
    </tbody>                                                                   6264
</table>                                                                       6265
{% endif %}                                                                    6266
{% endmacro %}                                                                 6267
# ==== app/templates/_login_badge.svg ====                                     6268
<svg class="lp-badge__svg" xmlns="http://www.w3.org/2000/svg                   6269
      " viewBox="0 0 240 240" role="img"                                       6270
      aria-label="历史遗留废弃矿山自然恢复适宜性评价系统徽标">                 6271
    <title>系统徽标</title>                                                    6272
    <g class="lp-ring" style="--k: 2.7">                                       6273
      <circle cx="162.00" cy="120.00" r="2.60"                                 6274
      fill="#c4f5ab"/>                                                         6275
      <circle cx="157.84" cy="138.22" r="2.97"                                 6276
      fill="#b8f5ab"/>                                                         6277
      <circle cx="146.19" cy="152.84" r="3.06"                                 6278
      fill="#acf5ab"/>                                                         6279
      <circle cx="129.35" cy="160.95" r="2.80"                                 6280
      fill="#abf5b7"/>                                                         6281
      <circle cx="110.65" cy="160.95" r="2.40"                                 6282
      fill="#abf5c2"/>                                                         6283
      <circle cx="93.81" cy="152.84" r="2.14" fill="#abf5ce"/>                 6284
      <circle cx="82.16" cy="138.22" r="2.23" fill="#abf5da"/>                 6285
      <circle cx="78.00" cy="120.00" r="2.60" fill="#abf5e6"/>                 6286
      <circle cx="82.16" cy="101.78" r="2.97" fill="#abf5da"/>                 6287
      <circle cx="93.81" cy="87.16" r="3.06" fill="#abf5ce"/>                  6288
      <circle cx="110.65" cy="79.05" r="2.80" fill="#abf5c2"/>                 6289
      <circle cx="129.35" cy="79.05" r="2.40" fill="#abf5b7"/>                 6290
      <circle cx="146.19" cy="87.16" r="2.14" fill="#acf5ab"/>                 6291
      <circle cx="157.84" cy="101.78" r="2.23"                                 6292
      fill="#b8f5ab"/>                                                         6293
    </g>                                                                       6294
    <g class="lp-ring" style="--k: 2.15">                                      6295
      <circle cx="177.29" cy="129.07" r="3.92"                                 6296
      fill="#a4e28d"/>                                                         6297
      <circle cx="171.68" cy="146.33" r="4.01"                                 6298
      fill="#9be28d"/>                                                         6299
      <circle cx="161.01" cy="161.01" r="3.87"                                 6300
```

```
    fill="#91e28d"/>                                                    6301
    <circle cx="146.33" cy="171.68" r="3.55"                            6302
    fill="#8de292"/>                                                    6303
    <circle cx="129.07" cy="177.29" r="3.17"                            6304
    fill="#8de29c"/>                                                    6305
    <circle cx="110.93" cy="177.29" r="2.88"                            6306
    fill="#8de2a6"/>                                                    6307
    <circle cx="93.67" cy="171.68" r="2.79" fill="#8de2af"/>            6308
    <circle cx="78.99" cy="161.01" r="2.93" fill="#8de2b9"/>            6309
    <circle cx="68.32" cy="146.33" r="3.25" fill="#8de2c2"/>            6310
    <circle cx="62.71" cy="129.07" r="3.63" fill="#8de2cc"/>            6311
    <circle cx="62.71" cy="110.93" r="3.92" fill="#8de2cc"/>            6312
    <circle cx="68.32" cy="93.67" r="4.01" fill="#8de2c2"/>             6313
    <circle cx="78.99" cy="78.99" r="3.87" fill="#8de2b9"/>             6314
    <circle cx="93.67" cy="68.32" r="3.55" fill="#8de2af"/>             6315
    <circle cx="110.93" cy="62.71" r="3.17" fill="#8de2a6"/>            6316
    <circle cx="129.07" cy="62.71" r="2.88" fill="#8de29c"/>            6317
    <circle cx="146.33" cy="68.32" r="2.79" fill="#8de292"/>            6318
    <circle cx="161.01" cy="78.99" r="2.93" fill="#91e28d"/>            6319
    <circle cx="171.68" cy="93.67" r="3.25" fill="#9be28d"/>            6320
    <circle cx="177.29" cy="110.93" r="3.63"                            6321
    fill="#a4e28d"/>                                                    6322
  </g>                                                                  6323
  <g class="lp-ring" style="--k: 1.85">                                 6324
    <circle cx="194.00" cy="120.00" r="4.94"                            6325
    fill="#91cf72"/>                                                    6326
    <circle cx="191.85" cy="137.71" r="4.92"                            6327
    fill="#88cf72"/>                                                    6328
    <circle cx="185.52" cy="154.39" r="4.73"                            6329
    fill="#80cf72"/>                                                    6330
    <circle cx="175.39" cy="169.07" r="4.42"                            6331
    fill="#78cf72"/>                                                    6332
    <circle cx="162.04" cy="180.90" r="4.06"                            6333
    fill="#72cf73"/>                                                    6334
    <circle cx="146.24" cy="189.19" r="3.73"                            6335
    fill="#72cf7b"/>                                                    6336
    <circle cx="128.92" cy="193.46" r="3.51"                            6337
    fill="#72cf83"/>                                                    6338
    <circle cx="111.08" cy="193.46" r="3.45"                            6339
    fill="#72cf8b"/>                                                    6340
    <circle cx="93.76" cy="189.19" r="3.56" fill="#72cf93"/>            6341
    <circle cx="77.96" cy="180.90" r="3.81" fill="#72cf9c"/>            6342
    <circle cx="64.61" cy="169.07" r="4.16" fill="#72cfa4"/>            6343
    <circle cx="54.48" cy="154.39" r="4.51" fill="#72cfac"/>            6344
    <circle cx="48.15" cy="137.71" r="4.80" fill="#72cfb4"/>            6345
    <circle cx="46.00" cy="120.00" r="4.94" fill="#72cfbc"/>            6346
    <circle cx="48.15" cy="102.29" r="4.92" fill="#72cfb4"/>            6347
    <circle cx="54.48" cy="85.61" r="4.73" fill="#72cfac"/>             6348
    <circle cx="64.61" cy="70.93" r="4.42" fill="#72cfa4"/>             6349
    <circle cx="77.96" cy="59.10" r="4.06" fill="#72cf9c"/>             6350
```

```
    <circle cx="93.76" cy="50.81" r="3.73" fill="#72cf93"/> 6351
    <circle cx="111.08" cy="46.54" r="3.51" fill="#72cf8b"/> 6352
    <circle cx="128.92" cy="46.54" r="3.45" fill="#72cf83"/> 6353
    <circle cx="146.24" cy="50.81" r="3.56" fill="#72cf7b"/> 6354
    <circle cx="162.04" cy="59.10" r="3.81" fill="#72cf73"/> 6355
    <circle cx="175.39" cy="70.93" r="4.16" fill="#78cf72"/> 6356
    <circle cx="185.52" cy="85.61" r="4.51" fill="#80cf72"/> 6357
    <circle cx="191.85" cy="102.29" r="4.80"                 6358
    fill="#88cf72"/>                                        6359
  </g>                                                      6360
  <g class="lp-ring" style="--k: 1.62">                     6361
    <circle cx="209.57" cy="128.82" r="5.68"                6362
    fill="#76bb59"/>                                        6363
    <circle cx="206.12" cy="146.13" r="5.40"                6364
    fill="#6fbb59"/>                                        6365
    <circle cx="199.37" cy="162.43" r="5.06"                6366
    fill="#68bb59"/>                                        6367
    <circle cx="189.57" cy="177.10" r="4.71"                6368
    fill="#61bb59"/>                                        6369
    <circle cx="177.10" cy="189.57" r="4.41"                6370
    fill="#5abb59"/>                                        6371
    <circle cx="162.43" cy="199.37" r="4.19"                6372
    fill="#59bb5f"/>                                        6373
    <circle cx="146.13" cy="206.12" r="4.10"                6374
    fill="#59bb66"/>                                        6375
    <circle cx="128.82" cy="209.57" r="4.15"                6376
    fill="#59bb6d"/>                                        6377
    <circle cx="111.18" cy="209.57" r="4.32"                6378
    fill="#59bb73"/>                                        6379
    <circle cx="93.87" cy="206.12" r="4.60" fill="#59bb7a"/> 6380
    <circle cx="77.57" cy="199.37" r="4.94" fill="#59bb81"/> 6381
    <circle cx="62.90" cy="189.57" r="5.29" fill="#59bb88"/> 6382
    <circle cx="50.43" cy="177.10" r="5.59" fill="#59bb8f"/> 6383
    <circle cx="40.63" cy="162.43" r="5.81" fill="#59bb96"/> 6384
    <circle cx="33.88" cy="146.13" r="5.90" fill="#59bb9d"/> 6385
    <circle cx="30.43" cy="128.82" r="5.85" fill="#59bba4"/> 6386
    <circle cx="30.43" cy="111.18" r="5.68" fill="#59bba4"/> 6387
    <circle cx="33.88" cy="93.87" r="5.40" fill="#59bb9d"/>  6388
    <circle cx="40.63" cy="77.57" r="5.06" fill="#59bb96"/>  6389
    <circle cx="50.43" cy="62.90" r="4.71" fill="#59bb8f"/>  6390
    <circle cx="62.90" cy="50.43" r="4.41" fill="#59bb88"/>  6391
    <circle cx="77.57" cy="40.63" r="4.19" fill="#59bb81"/>  6392
    <circle cx="93.87" cy="33.88" r="4.10" fill="#59bb7a"/>  6393
    <circle cx="111.18" cy="30.43" r="4.15" fill="#59bb73"/> 6394
    <circle cx="128.82" cy="30.43" r="4.32" fill="#59bb6d"/> 6395
    <circle cx="146.13" cy="33.88" r="4.60" fill="#59bb66"/> 6396
    <circle cx="162.43" cy="40.63" r="4.94" fill="#59bb5f"/> 6397
    <circle cx="177.10" cy="50.43" r="5.29" fill="#5abb59"/> 6398
    <circle cx="189.57" cy="62.90" r="5.59" fill="#61bb59"/> 6399
    <circle cx="199.37" cy="77.57" r="5.81" fill="#68bb59"/> 6400
```

```
    <circle cx="206.12" cy="93.87" r="5.90" fill="#6fbb59" />
    <circle cx="209.57" cy="111.18" r="5.85"
    fill="#76bb59" />
  </g>
  <g class="lp-ring" style="--k: 1.46">
    <circle cx="226.00" cy="120.00" r="5.94"
    fill="#65a843" />
    <circle cx="224.55" cy="137.45" r="5.62"
    fill="#5fa843" />
    <circle cx="220.26" cy="154.42" r="5.29"
    fill="#59a843" />
    <circle cx="213.22" cy="170.45" r="5.00"
    fill="#53a843" />
    <circle cx="203.65" cy="185.11" r="4.77"
    fill="#4da843" />
    <circle cx="191.79" cy="197.99" r="4.63"
    fill="#47a843" />
    <circle cx="177.98" cy="208.74" r="4.59"
    fill="#43a846" />
    <circle cx="162.58" cy="217.07" r="4.67"
    fill="#43a84c" />
    <circle cx="146.02" cy="222.76" r="4.85"
    fill="#43a852" />
    <circle cx="128.75" cy="225.64" r="5.10"
    fill="#43a858" />
    <circle cx="111.25" cy="225.64" r="5.42"
    fill="#43a85e" />
    <circle cx="93.98" cy="222.76" r="5.75" fill="#43a864" />
    <circle cx="77.42" cy="217.07" r="6.06" fill="#43a86a" />
    <circle cx="62.02" cy="208.74" r="6.33" fill="#43a870" />
    <circle cx="48.21" cy="197.99" r="6.52" fill="#43a876" />
    <circle cx="36.35" cy="185.11" r="6.60" fill="#43a87c" />
    <circle cx="26.78" cy="170.45" r="6.58" fill="#43a882" />
    <circle cx="19.74" cy="154.42" r="6.45" fill="#43a888" />
    <circle cx="15.45" cy="137.45" r="6.23" fill="#43a88e" />
    <circle cx="14.00" cy="120.00" r="5.94" fill="#43a894" />
    <circle cx="15.45" cy="102.55" r="5.62" fill="#43a88e" />
    <circle cx="19.74" cy="85.58" r="5.29" fill="#43a888" />
    <circle cx="26.78" cy="69.55" r="5.00" fill="#43a882" />
    <circle cx="36.35" cy="54.89" r="4.77" fill="#43a87c" />
    <circle cx="48.21" cy="42.01" r="4.63" fill="#43a876" />
    <circle cx="62.02" cy="31.26" r="4.59" fill="#43a870" />
    <circle cx="77.42" cy="22.93" r="4.67" fill="#43a86a" />
    <circle cx="93.98" cy="17.24" r="4.85" fill="#43a864" />
    <circle cx="111.25" cy="14.36" r="5.10" fill="#43a85e" />
    <circle cx="128.75" cy="14.36" r="5.42" fill="#43a858" />
    <circle cx="146.02" cy="17.24" r="5.75" fill="#43a852" />
    <circle cx="162.58" cy="22.93" r="6.06" fill="#43a84c" />
    <circle cx="177.98" cy="31.26" r="6.33" fill="#43a846" />
    <circle cx="191.79" cy="42.01" r="6.52" fill="#47a843" />
```

```
      <circle cx="203.65" cy="54.89" r="6.60" fill="#4da843"/>
      <circle cx="213.22" cy="69.55" r="6.58" fill="#53a843"/>
      <circle cx="220.26" cy="85.58" r="6.45" fill="#59a843"/>
      <circle cx="224.55" cy="102.55" r="6.23"
      fill="#5fa843"/>
    </g>
    <g class="lp-badge__mark">
      <circle cx="120" cy="120" r="27" fill="#14432f"/>
      <g transform="translate(120 120) scale(1.5)
      translate(-12 -12)" fill="none" stroke="#ffffff"
      stroke-width="1.9" stroke-linecap="round"
      stroke-linejoin="round">
        <path d="m3 20 6.5-11L14 17l2.5-4L21 20z"/>
        <circle cx="17" cy="6" r="2.2"/>
      </g>
    </g>
  </svg>
  # ==== app/templates/error.html ====
  {% extends "base.html" %}
  {% block title %}{{ code }} · {{ message }}{% endblock %}
  {% block content %}
  <div class="card" style="max-width:560px;">
    <h2>{{ code }} · {{ message }}</h2>
    <p class="muted">
      请返回上一页重试；若确认该操作应当可用，请检查当前账号的角色权限。
    </p>
    <p class="actions" style="margin-top:16px;">
      <a class="btn" href="/">返回首页</a>
    </p>
  </div>
  {% endblock %}
  # ==== app/templates/auth/login.html ====
  {% extends "base.html" %}
  {% from "_macros.html" import icon %}
  {% block title %}登录 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
  {% block content %}
  <style>
    /* ================== 登录页（仿 Apple 账户页） ==================
       纯白底、居中、大量留白；色彩只用在徽标与按钮上。
       首屏放徽标、标题、一句话说明与登录按钮；系统介绍在下方，
       随滚动逐条弹入、往回滚时弹回。 */
    .lp {
      color: #1d1d1f;
      background:
        linear-gradient(180deg, rgba(245, 239, 224, .88) 0%,
                                rgba(245, 239, 224, .92)
      100%),
        url("../static/hero_qinglv.jpg")
        center top / cover no-repeat fixed;
    }
```

```
/* ---------- 首屏 ---------- */
.lp-hero {
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: center;
  text-align: center;
  min-height: 100dvh;
  padding: 56px 20px 72px;
  position: relative;
  background:
    linear-gradient(180deg, rgba(15, 40, 45, .72) 0%,
                            rgba(20, 55, 58, .50) 50%,
                            rgba(15, 40, 45, .76) 100%),
    url("../static/hero_qinglv.jpg")
    center center / cover no-repeat;
}
/* 徽标：滚动时各圈向四周散开并淡出。
   --p 为 0~1 的滚动进度，由页面脚本写入；
   每圈自带 --k（散开倍数，由内向外递减），使各圈逐渐拉开距离 */
.lp-badge {
  width: 212px;
  height: 212px;
  margin-bottom: 36px;
  --p: 0;
  animation: lp-in .7s ease both;
}
.lp-badge__svg {
  display: block;
  width: 100%;
  height: 100%;
  overflow: visible;                    /* 散开时圆点不被画布裁掉 */
  opacity: calc(1 - var(--p) * .94);
  filter: blur(calc(var(--p) * 2.6px));
}
.lp-ring {
  transform-box: fill-box;
  transform-origin: center;
  transform: scale(calc(1 + (var(--k) - 1) * var(--p)));
}
.lp-badge__mark {
  transform-box: fill-box;
  transform-origin: center;
  transform: scale(calc(1 + .18 * var(--p)));
}
.lp-title {
  margin: 0 0 18px;
  font-size: clamp(28px, 4.2vw, 42px);
  font-weight: 700;
  line-height: 1.3;
```

```
    letter-spacing: -.4px;
    color: #fff;
    text-shadow: 0 2px 12px rgba(0, 0, 0, .3);
    animation: lp-in .7s ease .06s both;
  }
  .lp-cta {
    padding: 13px 52px;
    font-family: inherit;
    font-size: 17px;
    font-weight: 500;
    color: #fff;
    background: var(--accent);
    border: none;
    border-radius: 980px;
    cursor: pointer;
    animation: lp-in .7s ease .18s both;
    transition: background-color .2s ease, box-shadow .2s
    ease,
                transform .2s ease;
  }
  .lp-cta:hover {
    background: var(--accent-hover);
    box-shadow: 0 10px 26px rgba(44, 107, 90, .4);
  }
  .lp-cta:active { transform: translateY(1px); box-shadow:
    none; }
  .lp-cta:focus-visible { outline: 3px solid rgba(31, 107,
    69, .40); outline-offset: 3px; }
  .lp-more {
    margin-top: 38px;
    font-size: 13px;
    color: rgba(255, 255, 255, .7);
    text-decoration: none;
    animation: lp-in .7s ease .24s both;
  }
  .lp-more:hover { color: #fff; }
  @keyframes lp-in {
    from { opacity: 0; transform: translateY(12px); }
    to   { opacity: 1; transform: none; }
  }
  /* ---------- 滚动进场 ----------
     进入视口时从屏外飞入，离开视口时飞回，可反复。
     交错延迟只加在「进入」的规则上，退出时不延迟，否则会显得拖沓。 */
  .lp-reveal {
    opacity: 0;
    transition: opacity .7s ease,
                transform 1.05s cubic-bezier(.18, 1.04, .3,
    1),
                border-color .62s ease, background-color
    .62s ease;
```

```
    transition-delay: 0s;                                                    6601
    will-change: opacity, transform;                                        6602
}                                                                            6603
/* 起始位置整体在屏外：横向按视口宽度取位移，纵向按视口高度取位移，                    6604
   再带一点缩小，落位时按贝塞尔曲线的过冲轻微回弹。 */                              6605
.lp-reveal[data-dir="left"]  { transform:                                    6606
  translate3d(-56vw, 5vh, 0) scale(.9); }                                    6607
.lp-reveal[data-dir="right"] { transform:                                    6608
  translate3d(56vw, 5vh, 0) scale(.9); }                                     6609
.lp-reveal[data-dir="up"]    { transform: translate3d(0,                     6610
  24vh, 0) scale(.94); }                                                     6611
.lp-reveal.is-in {                                                           6612
  opacity: 1;                                                                6613
  transform: none;                                                           6614
  transition-delay: calc(var(--i, 0) * 110ms);                               6615
}                                                                            6616
/* 标题颜色由淡到深，强化「浮现」的层次感 */                                       6617
.lp-reveal h2,                                                               6618
.lp-reveal h3,                                                               6619
.lp-reveal .lp-mod__name { color: #c2d2c8; transition:                       6620
  color .7s ease; }                                                          6621
.lp-reveal.is-in h3,                                                         6622
.lp-reveal.is-in .lp-mod__name { color: var(--accent); }                     6623
.lp-reveal.is-in h2 { color: #1d1d1f; }                                      6624
/* ---------- 系统介绍 ---------- */                                         6625
.lp-intro {                                                                  6626
  max-width: none;                                                           6627
  margin: 0;                                                                 6628
  padding: 56px 24px 72px;                                                   6629
  background: transparent;                                                   6630
  /* 卡片从屏外飞入，横向超出的部分裁掉，免得出现横向滚动条；                      6631
     用 clip 而不是 hidden，纵向滚动仍由页面本身负责 */                         6632
  overflow-x: clip;                                                          6633
}                                                                            6634
.lp-intro > * { max-width: 1040px; margin-left: auto;                        6635
  margin-right: auto; }                                                      6636
.lp-intro__title {                                                           6637
  margin: 0 auto 26px;                                                       6638
  font-size: clamp(22px, 2.6vw, 30px);                                       6639
  text-align: center;                                                        6640
  letter-spacing: -.3px;                                                     6641
}                                                                            6642
/* 三张特点卡与九个模块卡都按三等分排版，并整体居中 */                             6643
.lp-points {                                                                 6644
  display: grid;                                                             6645
  grid-template-columns: repeat(3, minmax(0, 1fr));                          6646
  gap: 16px;                                                                 6647
  margin-bottom: 46px;                                                       6648
}                                                                            6649
.lp-point {                                                                  6650
```

```
    padding: 30px 26px;                                                    6651
    background: rgba(253, 250, 242, .82);                                  6652
    backdrop-filter: blur(12px);                                           6653
    -webkit-backdrop-filter: blur(12px);                                   6654
    border: 1px solid rgba(224, 216, 200, .6);                             6655
    border-radius: 22px;                                                   6656
    box-shadow: 0 24px 60px rgba(42, 37, 32, .10), 0 8px                   6657
    20px rgba(42, 37, 32, .05);                                            6658
    transition: transform .25s ease, box-shadow .25s ease;                 6659
}                                                                          6660
.lp-point:hover {                                                          6661
    transform: translateY(-4px);                                           6662
    box-shadow: 0 32px 72px rgba(42, 37, 32, .14), 0 12px                  6663
    28px rgba(42, 37, 32, .07);                                            6664
}                                                                          6665
.lp-point__icon {                                                          6666
    display: grid;                                                         6667
    place-items: center;                                                   6668
    width: 40px;                                                           6669
    height: 40px;                                                          6670
    margin-bottom: 16px;                                                   6671
    border-radius: 12px;                                                   6672
    background: rgba(44, 107, 90, .1);                                     6673
    color: var(--accent);                                                  6674
}                                                                          6675
.lp-point h3 { margin: 0 0 8px; font-size: 16px; }                         6676
.lp-point p {                                                              6677
    margin: 0;                                                             6678
    font-size: 13.5px;                                                     6679
    line-height: 1.85;                                                     6680
    color: #6e6e73;                                                        6681
    text-wrap: pretty;                                                     6682
}                                                                          6683
.lp-sub-title {                                                            6684
    margin: 0 auto 20px;                                                   6685
    font-size: 17px;                                                       6686
    text-align: center;                                                    6687
}                                                                          6688
.lp-mods {                                                                 6689
    display: grid;                                                         6690
    grid-template-columns: repeat(3, minmax(0, 1fr));                      6691
    gap: 12px;                                                             6692
    margin-bottom: 46px;                                                   6693
    align-items: start;                                                    6694
}                                                                          6695
/* 模块卡：默认只显示编号与名称，鼠标悬停或键盘聚焦时才展开职责说明 */                    6696
.lp-mod {                                                                  6697
    position: relative;                                                    6698
    padding: 17px 18px 17px 48px;                                          6699
    background: rgba(253, 250, 242, .75);                                  6700
```

```
    backdrop-filter: blur(8px);                                    6701
    -webkit-backdrop-filter: blur(8px);                            6702
    border: 1px solid rgba(224, 216, 200, .6);                     6703
    border-radius: 14px;                                           6704
    outline: none;                                                 6705
    transition: border-color .3s ease, box-shadow .3s ease;        6706
  }                                                                6707
  .lp-mod:hover,                                                   6708
  .lp-mod:focus-within,                                            6709
  .lp-mod:focus {                                                  6710
    border-color: rgba(44, 107, 90, .5);                           6711
    box-shadow: 0 8px 22px rgba(42, 37, 32, .08);                  6712
  }                                                                6713
  .lp-mod__no {                                                    6714
    position: absolute;                                            6715
    left: 17px;                                                    6716
    top: 17px;                                                     6717
    font-size: 13px;                                               6718
    font-weight: 700;                                              6719
    color: #c8d6cd;                                                6720
    font-variant-numeric: tabular-nums;                            6721
  }                                                                6722
  .lp-mod__name {                                                  6723
    display: block;                                                6724
    font-size: 14px;                                               6725
    font-weight: 600;                                              6726
  }                                                                6727
  /* 用 grid-template-rows 0fr→1fr 展开，高度过渡自然 */             6728
  .lp-mod__body {                                                  6729
    display: grid;                                                 6730
    grid-template-rows: 0fr;                                       6731
    transition: grid-template-rows .34s ease, margin-top           6732
    .34s ease;                                                     6733
  }                                                                6734
  .lp-mod:hover .lp-mod__body,                                     6735
  .lp-mod:focus-within .lp-mod__body {                             6736
    grid-template-rows: 1fr;                                       6737
    margin-top: 7px;                                               6738
  }                                                                6739
  .lp-mod__desc {                                                  6740
    overflow: hidden;                                              6741
    margin: 0;                                                     6742
    font-size: 12.5px;                                             6743
    line-height: 1.7;                                              6744
    color: #6e6e73;                                                6745
    opacity: 0;                                                    6746
    transition: opacity .3s ease;                                  6747
    text-wrap: pretty;                                             6748
  }                                                                6749
  .lp-mod:hover .lp-mod__desc,                                     6750
```

```
.lp-mod:focus-within .lp-mod__desc { opacity: 1; }
/* 触摸设备没有悬停，直接把说明摊开 */
@media (hover: none) {
  .lp-mod__body { grid-template-rows: 1fr; margin-top:
  7px; }
  .lp-mod__desc { opacity: 1; }
}
.lp-foot {
  /* margin 简写会把 .lp-intro > * 里的左右 auto 一起清掉，块就贴到左边
     （文字自己居中，看着却整体偏左）—— 与标题那两处是同一个坑 */
  margin: 0 auto;
  padding-top: 26px;
  border-top: 1px solid #e6e6e8;
  text-align: center;
  font-size: 12.5px;
  color: #86868b;
}
/* ---------- 登录面板：点击后渐变浮现 ---------- */
.lp-auth {
  position: fixed;
  inset: 0;
  z-index: 60;
  display: grid;
  place-items: center;
  padding: 24px;
  opacity: 0;
  visibility: hidden;
  transition: opacity .42s ease, visibility 0s linear
  .42s;
}
.lp-auth.is-open {
  opacity: 1;
  visibility: visible;
  transition: opacity .42s ease, visibility 0s;
}
/* 遮罩用绿色径向渐变，形成「渐变渗出来」的观感 */
.lp-auth__scrim {
  position: absolute;
  inset: 0;
  background:
    radial-gradient(1000px 640px at 50% 0%,
                    rgba(31, 107, 69, .22), rgba(31, 107,
  69, 0) 70%),
    linear-gradient(180deg, rgba(255, 255, 255, .84),
                    rgba(246, 250, 247, .95));
  backdrop-filter: blur(16px) saturate(1.25);
  -webkit-backdrop-filter: blur(16px) saturate(1.25);
}
.lp-auth__panel {
  position: relative;
```

```
    width: 100%;                                                    6801
    max-width: 396px;                                               6802
    padding: 36px 36px 26px;                                        6803
    background: #fff;                                               6804
    border-radius: 20px;                                            6805
    box-shadow: 0 28px 80px rgba(14, 46, 32, .18);                  6806
    opacity: 0;                                                     6807
    transform: translateY(18px) scale(.975);                        6808
    transition: transform .55s cubic-bezier(.16, .84, .34,          6809
    1) .05s,                                                        6810
                opacity .55s ease .05s;                             6811
}                                                                   6812
.lp-auth.is-open .lp-auth__panel { opacity: 1; transform:           6813
    none; }                                                         6814
.lp-auth__close {                                                   6815
    position: absolute;                                             6816
    top: 14px;                                                      6817
    right: 14px;                                                    6818
    display: grid;                                                  6819
    place-items: center;                                            6820
    width: 30px;                                                    6821
    height: 30px;                                                   6822
    color: #86868b;                                                 6823
    background: transparent;                                        6824
    border: none;                                                   6825
    border-radius: 50%;                                             6826
    cursor: pointer;                                                6827
    transition: background-color .18s ease, color .18s ease;        6828
}                                                                   6829
.lp-auth__close:hover { background: #f5f5f7; color:                 6830
    #1d1d1f; }                                                      6831
.lp-auth__title {                                                   6832
    margin: 0 0 6px;                                                6833
    text-align: center;                                             6834
    font-size: 24px;                                                6835
    font-weight: 700;                                               6836
    letter-spacing: -.2px;                                          6837
}                                                                   6838
.lp-auth__sub {                                                     6839
    margin: 0 0 26px;                                               6840
    text-align: center;                                             6841
    font-size: 13px;                                                6842
    color: #86868b;                                                 6843
}                                                                   6844
/* 字段校验错误：贴着输入框显示，不做成会消失的浮层 ——                  6845
   这类消息要对着输入框看，一闪而过会让人以为「点了没反应」 */          6846
.lp-field__error {                                                  6847
    margin: 7px 0 0;                                                6848
    font-size: 12.5px;                                              6849
    color: #9c2b23;                                                 6850
```

```
}
.lp-field input[aria-invalid="true"] {
  background: #fdeeec;
  border-color: rgba(156, 43, 35, .35);
}
.lp-field { margin-bottom: 15px; }
.lp-field label {
  display: block;
  margin-bottom: 7px;
  font-size: 13px;
  color: #6e6e73;
}
.lp-field input {
  width: 100%;
  padding: 13px 16px;
  font-family: inherit;
  font-size: 16px;              /* 不小于 16px，避免手机端聚焦时自动放大 */
  color: #1d1d1f;
  background: #f5f5f7;
  border: 1px solid transparent;
  border-radius: 12px;
  outline: none;
  transition: background-color .18s ease, border-color .18s ease,
              box-shadow .18s ease;
}
.lp-field input::placeholder { color: #a1a1a6; }
.lp-field input:focus {
  background: #fff;
  border-color: rgba(31, 107, 69, .55);
  box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);
}
.lp-remember {
  display: flex;
  align-items: center;
  gap: 8px;
  margin: 4px 0 20px;
  font-size: 13px;
  color: #6e6e73;
  cursor: pointer;
}
.lp-remember input {
  width: 16px;
  height: 16px;
  margin: 0;
  accent-color: var(--accent);
  cursor: pointer;
}
.lp-submit {
  width: 100%;
```

```
    padding: 13px 20px;                                                    6901
    font-family: inherit;                                                  6902
    font-size: 16px;                                                       6903
    font-weight: 500;                                                      6904
    color: #fff;                                                           6905
    background: var(--accent);                                             6906
    border: none;                                                          6907
    border-radius: 980px;                                                  6908
    cursor: pointer;                                                       6909
    transition: background-color .2s ease, box-shadow .2s                  6910
    ease,                                                                  6911
                transform .2s ease;                                        6912
}                                                                          6913
.lp-submit:hover {                                                         6914
    background: var(--accent-hover);                                       6915
    box-shadow: 0 10px 24px rgba(31, 107, 69, .26);                        6916
}                                                                          6917
.lp-submit:active { transform: translateY(1px);                            6918
    box-shadow: none; }                                                    6919
.lp-submit:focus-visible { outline: 3px solid rgba(31,                     6920
    107, 69, .40); outline-offset: 3px; }                                  6921
.lp-auth__hint {                                                           6922
   margin: 18px 0 0;                                                       6923
    text-align: center;                                                    6924
    font-size: 12.5px;                                                     6925
    color: #a1a1a6;                                                        6926
}                                                                          6927
.lp-auth__reg {                                                            6928
    color: var(--accent);                                                  6929
    font-weight: 500;                                                      6930
    text-decoration: none;                                                 6931
    border-bottom: 1px solid rgba(31, 107, 69, .35);                       6932
    transition: border-color .18s ease;                                    6933
}                                                                          6934
.lp-auth__reg:hover { border-bottom-color: var(--accent);                  6935
    }                                                                      6936
/* 系统设置里要求减少动态效果时，去掉位移、缩放与滚动特效 */                                6937
@media (prefers-reduced-motion: reduce) {                                  6938
    .lp-badge, .lp-title, .lp-cta, .lp-more { animation:                   6939
    none; }                                                                6940
    .lp-badge__svg { opacity: 1; filter: none; }                           6941
    .lp-ring, .lp-badge__mark { transform: none; }                         6942
    .lp-reveal { opacity: 1; transform: none !important; }                 6943
    .lp-reveal h2, .lp-reveal h3, .lp-reveal .lp-mod__name                 6944
    { color: inherit; }                                                    6945
    .lp-reveal.is-in h2 { color: #1d1d1f; }                                6946
    .lp-reveal.is-in h3, .lp-reveal.is-in .lp-mod__name {                  6947
    color: var(--accent); }                                                6948
    .lp-auth, .lp-auth__panel { transition: opacity .2s                    6949
    linear; }                                                              6950
```

```
    .lp-auth__panel { transform: none; }
  }
  /* 窄屏退成两列、一列，避免卡片被挤变形 */
  @media (max-width: 900px) {
    .lp-points, .lp-mods { grid-template-columns: repeat(2, minmax(0, 1fr)); }
  }
  @media (max-width: 620px) {
    .lp-points, .lp-mods { grid-template-columns: minmax(0, 1fr); }
  }
  @media (max-width: 560px) {
    .lp-hero { padding: 36px 18px 48px; min-height: auto; }
    .lp-badge { width: 160px; height: 160px; margin-bottom: 26px; }
    .lp-auth__panel { padding: 30px 24px 22px; }
  }
</style>
<div class="lp">
  {# ---------------- 首屏：徽标 + 标题 + 说明 + 登录入口 ---------------- #}
  <section class="lp-hero">
    <div class="lp-badge" id="lp-badge" aria-hidden="true">
      {% include "_login_badge.svg" %}
    </div>
    <h1 class="lp-title">历史遗留废弃矿山<br>自然恢复适宜性评价系统</h1>
    <button type="button" class="lp-cta" id="lp-open"
            aria-expanded="false" aria-controls="lp-auth">登录</button>
    <a class="lp-more" href="#intro">
      向下滑动，了解系统 <span aria-hidden="true">↓</span>
    </a>
  </section>
  {# ---------------- 系统介绍：随滚动逐条弹入 ---------------- #}
  <section class="lp-intro" id="intro">
    <h2 class="lp-intro__title lp-reveal"
    data-dir="up">一套把评价流程固定下来的工具</h2>
    <div class="lp-points">
      {% set pdirs = ['left', 'up', 'right'] %}
      {% set points = [
        ('sliders', '方法可溯',
         '权重由层次分析法算出并做一致性检验，指标标准化、分单元赋分与面积加权合成的每一步都留有记录，可逐层回看。'),
        ('shield', '流程可控',
         '政策、区位约束、土壤污染、地质灾害、耕地属性五项限制性条件一票否决；未通过即终止评价，直接给出生态重建或人工辅助的结论。'),
        ('file', '结果可复算',
         '评价结果与逐层得分明细一并留存，报告取自已保存的评价记录，事后修改数据不会让报告与结论对不上。')
      ] %}
      {% for ic, title, text in points %}
```

```
      <div class="lp-point lp-reveal" data-dir="{{
    pdirs[loop.index0 % 3] }}"
          style="--i: {{ loop.index0 }}">
        <span class="lp-point__icon">{{ icon(ic, 20)
    }}</span>
        <h3>{{ title }}</h3>
        <p>{{ text }}</p>
      </div>
      {% endfor %}
    </div>
    <h3 class="lp-sub-title lp-reveal"
    data-dir="up">九个功能模块</h3>
    <div class="lp-mods">
      {% set mdirs = ['left', 'up', 'right'] %}
      {% for no, name, desc in modules %}
      <div class="lp-mod lp-reveal" data-dir="{{
    mdirs[loop.index0 % 3] }}"
          style="--i: {{ loop.index0 % 3 }}" tabindex="0">
        <span class="lp-mod__no">{{ no }}</span>
        <span class="lp-mod__name">{{ name }}</span>
        <div class="lp-mod__body"><p class="lp-mod__desc">{{
    desc }}</p></div>
      </div>
      {% endfor %}
    </div>
    <p class="lp-foot lp-reveal" data-dir="up">
      历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ１.０ ·
      运行于浏览器，数据保存在本机 ＳＱＬｉｔｅ 数据库
    </p>
  </section>
  {# ---------------- 登录面板：点击「登录」后渐变浮现 ---------------- #}
  <div class="lp-auth" id="lp-auth" aria-hidden="true">
    <div class="lp-auth__scrim" data-lp-close></div>
    <div class="lp-auth__panel" role="dialog"
    aria-modal="true"
        aria-labelledby="lp-auth-title">
      <button type="button" class="lp-auth__close"
    data-lp-close
          aria-label="关闭登录框">
        <svg width="16" height="16" viewBox="0 0 24 24"
    fill="none"
          stroke="currentColor" stroke-width="2"
    stroke-linecap="round"
          aria-hidden="true"><path d="M6 6l12 12M18 6L6
    18"/></svg>
      </button>
      <h2 class="lp-auth__title" id="lp-auth-title">登录</h2>
      <p class="lp-auth__sub">使用系统账号登录</p>
      <form method="post" novalidate>
        {{ form.hidden_tag() }}
```

```
          <div class="lp-field">                                          7051
            <label for="{{ form.username.id }}">账户名称</label>          7052
            {{ form.username(id=form.username.id,                         7053
    autocomplete="username",                                              7054
                             placeholder="请输入账户名称",                7055
                             **({'aria-invalid': 'true'} if               7056
    form.username.errors else {})) }}                                     7057
            {% for e in form.username.errors %}                           7058
            <p class="lp-field__error">{{ e }}</p>                        7059
            {% endfor %}                                                  7060
          </div>                                                          7061
          <div class="lp-field">                                          7062
            <label for="{{ form.password.id }}">密码</label>              7063
            {{ form.password(id=form.password.id,                         7064
    autocomplete="current-password",                                      7065
                             placeholder="请输入密码",                    7066
                             **({'aria-invalid': 'true'} if               7067
    form.password.errors else {})) }}                                     7068
            {% for e in form.password.errors %}                           7069
            <p class="lp-field__error">{{ e }}</p>                        7070
            {% endfor %}                                                  7071
          </div>                                                          7072
          <label class="lp-remember" for="{{ form.remember.id           7073
    }}">                                                                  7074
            {{ form.remember(id=form.remember.id) }}                      7075
            <span>记住我（下次自动登录，一个月内有效）</span>              7076
          </label>                                                        7077
          {{ form.submit(class_="lp-submit") }}                           7078
        </form>                                                           7079
        <p class="lp-auth__hint">                                         7080
          还没有账号？<a class="lp-auth__reg"                             7081
                        href="{{ url_for('auth.register')                 7082
    }}">点我注册</a>                                                      7083
        </p>                                                              7084
        <p class="lp-auth__hint" style="margin-top:10px;">                7085
          演示账号：admin 密码：admin123                                  7086
        </p>                                                              7087
      </div>                                                              7088
    </div>                                                                7089
</div>                                                                    7090
{% endblock %}                                                            7091
{% block scripts %}                                                       7092
<script>                                                                  7093
(function () {                                                            7094
  var reduce = window.matchMedia &&                                       7095
    window.matchMedia('(prefers-reduced-motion:                           7096
    reduce)').matches;                                                    7097
  /* ---------- 登录面板 ---------- */                                    7098
  var btn = document.getElementById('lp-open');                           7099
  var auth = document.getElementById('lp-auth');                          7100
```

```
var lastFocus = null;
/* 清掉上一次提交留下的字段错误（红框 + 输入框下方的提示）。
   登录失败时页面是直接渲染的，错误留在 DOM 里；如果只是把面板关掉再打开，
   那些红框还在，看起来就像「刚点开就报错」。 */
function clearFieldErrors(scope) {
  var root = scope || auth;
  Array.prototype.forEach.call(
    root.querySelectorAll('.lp-field__error'),
    function (el) { el.remove(); });
  Array.prototype.forEach.call(
    root.querySelectorAll('[aria-invalid="true"]'),
    function (el) { el.removeAttribute('aria-invalid'); });
}
function openPanel(keepErrors) {
  // keepErrors：由「登录失败自动展开」触发时保留错误提示；
  // 用户自己点「登录」重新打开时则清掉，给一个干净的开始。
  if (!keepErrors) clearFieldErrors();
  lastFocus = document.activeElement;
  auth.classList.add('is-open');
  auth.setAttribute('aria-hidden', 'false');
  btn.setAttribute('aria-expanded', 'true');
  document.body.style.overflow = 'hidden';
  // 记到地址栏，直接访问 /login#login 也能展开登录框
  if (location.hash !== '#login') {
    history.replaceState(null, '', location.pathname +
  '#login');
  }
  var first = auth.querySelector('input');
  if (first) window.setTimeout(function () {
  first.focus(); }, 300);
}
function closePanel() {
  auth.classList.remove('is-open');
  auth.setAttribute('aria-hidden', 'true');
  btn.setAttribute('aria-expanded', 'false');
  document.body.style.overflow = '';
  if (location.hash === '#login') {
    history.replaceState(null, '', location.pathname);
  }
  if (lastFocus && lastFocus.focus) lastFocus.focus();
}
if (btn && auth) {
  // 注意别把事件对象当成参数传进去（那样会被当成 keepErrors=true）
  btn.addEventListener('click', function () {
  openPanel(false); });
  Array.prototype.forEach.call(
    auth.querySelectorAll('[data-lp-close]'),
    function (el) { el.addEventListener('click',
  closePanel); }
```

```
  );
  document.addEventListener('keydown', function (e) {
    if (e.key === 'Escape' && auth.classList.contains('is-open')) closePanel();
  });
  // 登录失败回到本页、或直接带 #login 访问时，直接展开登录框。
  // 失败可能落在两处：字段校验错误在表单里，凭据错误在浮层里。
  // 这条路径保留错误提示（keepErrors），用户才知道刚才哪儿没填。
  if (location.hash === '#login'
      || document.querySelector('.lp-field__error')
      || document.querySelector('.toast-center')) {
    openPanel(true);
  }
  // 一开始输入就把这一项的红色状态撤掉：错误已经不被犯了，红框留着只会让人困惑
  Array.prototype.forEach.call(
    auth.querySelectorAll('input'),
    function (input) {
      input.addEventListener('input', function () {
        var field = input.closest('.lp-field');
        if (field) clearFieldErrors(field);
      });
    });
}
/* ---------- 内容进场 + 徽标散开：同一套滚动计算 ---------- */
  原先用 IntersectionObserver，实测在滚动到位置后回调并未生效，
  内容始终停在 opacity:0。改为直接按元素与视口的相对位置判断，
  与徽标用同一次滚动回调，行为可预期、也便于排查。
  判断「进没进视野」用的是**布局位置**，不是 getBoundingClientRect()：
  进场动画本身就是位移（左右两列横向 56vw、中列自下方 24vh 升起），
  用视觉位置判断时，中列的盒子一直比自己的位置低一截，
  于是它总要再往下滚一段才触发，看起来比左右两列慢半拍。
  transform 不影响 offsetTop，所以这里沿 offsetParent 把布局位置累加出来，
  只在加载时算一次；滚动过程中不再测量，也就没有读写交替带来的重排。 */
var items = document.querySelectorAll('.lp-reveal');
var badge = document.getElementById('lp-badge');
var ticking = false;
var slots = [];
function layoutTop(el) {
  var top = 0;
  var node = el;
  while (node) {
    top += node.offsetTop || 0;
    node = node.offsetParent;
  }
  return top;
}
if (!reduce) {
  for (var k = 0; k < items.length; k++) {
    slots.push({
      el: items[k],
```

```
          top: layoutTop(items[k]),
          height: items[k].offsetHeight
        });
      }
    }
    function applyScroll() {
      ticking = false;
      var vh = window.innerHeight || 800;
      var y = window.scrollY || window.pageYOffset || 0;
      /* 徽标：滚动前约三分之一屏内完成散开淡出 */
      if (badge) {
        var range = vh * 0.34;
        var p = range > 0 ? y / range : 0;
        badge.style.setProperty('--p', Math.min(1,
      Math.max(0, p)).toFixed(3));
      }
      if (reduce || !slots.length) return;
      for (var j = 0; j < slots.length; j++) {
        var s = slots[j];
        var viewTop = s.top - y;
        // 上沿进入视口下部、下沿尚未越出视口上部 → 视为在视野内
        var shown = viewTop < vh * 0.88 && viewTop + s.height
      > vh * 0.06;
        if (shown) s.el.classList.add('is-in');
        else s.el.classList.remove('is-in');
      }
    }
    function onScroll() {
      if (!ticking) {
        ticking = true;
        window.requestAnimationFrame(applyScroll);
      }
    }
    // 减少动态效果时不做滚动特效，内容直接显示
    if (reduce) {
      Array.prototype.forEach.call(items, function (el) {
        el.classList.add('is-in');
      });
      if (badge) badge.style.setProperty('--p', '0');
    } else {
      window.addEventListener('scroll', onScroll, { passive:
      true });
      window.addEventListener('resize', onScroll);
      applyScroll();
    }
  })();
  </script>
  {% endblock %}
  # ==== app/templates/auth/register.html ====
  {% extends "base.html" %}
```

```
{% block title %}注册账号 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  /* ================== 注册页 ================== */
     与登录页同一套观感：浅绿渐变底 + 白色卡片。
     必填只有账户名称与密码，其余收在「选填」分隔线以下，
     这样首次注册不会被一堆输入框挡住。 */
  .rg {
    min-height: 100dvh;
    display: grid;
    place-items: center;
    padding: 40px 20px 56px;
    background:
      radial-gradient(900px 520px at 50% 0%,
                      rgba(31, 107, 69, .14), rgba(31, 107,
    69, 0) 72%),
      #f6faf7;
  }
  .rg-card {
    width: 100%;
    max-width: 470px;
    padding: 38px 40px 30px;
    background: #fff;
    border-radius: 20px;
    box-shadow: 0 26px 70px rgba(14, 46, 32, .13);
    animation: rg-in .5s cubic-bezier(.16, .84, .34, 1)
    both;
  }
  @keyframes rg-in {
    from { opacity: 0; transform: translateY(16px)
    scale(.98); }
    to   { opacity: 1; transform: none; }
  }
  .rg-title {
    margin: 0 0 6px;
    text-align: center;
    font-size: 25px;
    font-weight: 700;
    letter-spacing: -.2px;
  }
  .rg-sub {
    margin: 0 0 28px;
    text-align: center;
    font-size: 13px;
    color: #86868b;
  }
  .rg-field { margin-bottom: 15px; }
  .rg-field > label {
    display: block;
    margin-bottom: 7px;
```

```
    font-size: 13px;                                                    7301
    color: #6e6e73;                                                     7302
  }                                                                     7303
  .rg-field > label .rg-opt {                                           7304
    margin-left: 5px;                                                   7305
    font-size: 12px;                                                    7306
    color: #a1a1a6;                                                     7307
  }                                                                     7308
  .rg-field input[type="text"],                                         7309
  .rg-field input[type="password"] {                                    7310
    width: 100%;                                                        7311
    padding: 12px 15px;                                                 7312
    font-family: inherit;                                               7313
    font-size: 16px;              /* 不小于 16px，避免手机端聚焦时自动放大 */    7314
    color: #1d1d1f;                                                     7315
    background: #f5f5f7;                                                7316
    border: 1px solid transparent;                                      7317
    border-radius: 12px;                                                7318
    outline: none;                                                      7319
    transition: background-color .18s ease, border-color                7320
    .18s ease,                                                          7321
                box-shadow .18s ease;                                   7322
  }                                                                     7323
  .rg-field input::placeholder { color: #a1a1a6; }                      7324
  .rg-field input:focus {                                               7325
    background: #fff;                                                   7326
    border-color: rgba(31, 107, 69, .55);                               7327
    box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);                       7328
  }                                                                     7329
  .rg-field input[aria-invalid="true"] {                                7330
    border-color: rgba(180, 60, 50, .55);                               7331
    background: #fff;                                                   7332
  }                                                                     7333
  .rg-field select {                                                    7334
    width: 100%;                                                        7335
    padding: 12px 15px;                                                 7336
    font-family: inherit;                                               7337
    font-size: 15px;                                                    7338
    color: #1d1d1f;                                                     7339
    background: #f5f5f7;                                                7340
    border: 1px solid transparent;                                      7341
    border-radius: 12px;                                                7342
    outline: none;                                                      7343
  }                                                                     7344
  .rg-field select:focus {                                              7345
    background: #fff;                                                   7346
    border-color: rgba(31, 107, 69, .55);                               7347
    box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);                       7348
  }                                                                     7349
  .rg-err {                                                             7350
```

```
    margin: 6px 0 0;
    font-size: 12.5px;
    color: var(--danger);
  }
  /* 「选填」分隔线 */
  .rg-split {
    display: flex;
    align-items: center;
    gap: 12px;
    margin: 24px 0 18px;
    font-size: 12.5px;
    color: #a1a1a6;
  }
  .rg-split::before,
  .rg-split::after {
    content: "";
    flex: 1;
    height: 1px;
    background: #ececee;
  }
  .rg-pair {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 0 14px;
  }
  @media (max-width: 480px) { .rg-pair {
    grid-template-columns: 1fr; } }
  /* 头像：圆形预览 + 选择文件 */
  .rg-avatar {
    display: flex;
    align-items: center;
    gap: 14px;
    margin-bottom: 15px;
  }
  .rg-avatar__face {
    display: grid;
    place-items: center;
    flex: none;
    width: 54px;
    height: 54px;
    border-radius: 50%;
    background: #eef2f0;
    color: #9aa8a1;
    font-size: 12px;
    overflow: hidden;
    background-size: cover;
    background-position: center;
  }
  .rg-avatar__pick { min-width: 0; }
  .rg-avatar__pick input[type="file"] {
```

```
      max-width: 100%;                                                    7401
      font-family: inherit;                                               7402
      font-size: 12.5px;                                                  7403
      color: #6e6e73;                                                     7404
    }                                                                     7405
    .rg-avatar__tip {                                                     7406
      margin: 4px 0 0;                                                    7407
      font-size: 12px;                                                    7408
      color: #a1a1a6;                                                     7409
    }                                                                     7410
    .rg-submit {                                                          7411
      width: 100%;                                                        7412
      margin-top: 24px;                                                   7413
      padding: 13px 20px;                                                 7414
      font-family: inherit;                                               7415
      font-size: 16px;                                                    7416
      font-weight: 500;                                                   7417
      color: #fff;                                                        7418
      background: var(--accent);                                          7419
      border: none;                                                       7420
      border-radius: 980px;                                               7421
      cursor: pointer;                                                    7422
      transition: background-color .2s ease, box-shadow .2s               7423
      ease,                                                               7424
                  transform .2s ease;                                     7425
    }                                                                     7426
    .rg-submit:hover {                                                    7427
      background: var(--accent-hover);                                    7428
      box-shadow: 0 10px 24px rgba(31, 107, 69, .26);                     7429
    }                                                                     7430
    .rg-submit:active { transform: translateY(1px);                       7431
      box-shadow: none; }                                                 7432
    .rg-submit:focus-visible { outline: 3px solid rgba(31,                7433
      107, 69, .40); outline-offset: 3px; }                               7434
    .rg-foot {                                                            7435
      margin: 20px 0 0;                                                   7436
      text-align: center;                                                 7437
      font-size: 13px;                                                    7438
      color: #86868b;                                                     7439
    }                                                                     7440
    .rg-foot a { color: var(--accent); text-decoration: none;             7441
      }                                                                   7442
    .rg-foot a:hover { text-decoration: underline; }                      7443
    @media (prefers-reduced-motion: reduce) {                             7444
      .rg-card { animation: none; }                                       7445
    }                                                                     7446
</style>                                                                  7447
<div class="rg">                                                          7448
    <div class="rg-card">                                                 7449
      <h1 class="rg-title">创建账号</h1>                                  7450
```

```
<p class="rg-sub">新账号为「操作员」，职务由管理员调整</p>
<form method="post" enctype="multipart/form-data"
novalidate>
  {{ form.hidden_tag() }}
  <div class="rg-field">
    <label for="{{ form.username.id }}">账户名称</label>
    {{ form.username(id=form.username.id,
autocomplete="username",
                     placeholder="2–64 个字符",
                     **({'aria-invalid': 'true'} if
form.username.errors else {})) }}
    {% for e in form.username.errors %}<p
class="rg-err">{{ e }}</p>{% endfor %}
  </div>
  <div class="rg-field">
    <label for="{{ form.password.id }}">密码</label>
    {{ form.password(id=form.password.id,
autocomplete="new-password",
                     placeholder="至少 6 位",
                     **({'aria-invalid': 'true'} if
form.password.errors else {})) }}
    {% for e in form.password.errors %}<p
class="rg-err">{{ e }}</p>{% endfor %}
  </div>
  <div class="rg-field">
    <label for="{{ form.confirm.id }}">确认密码</label>
    {{ form.confirm(id=form.confirm.id,
autocomplete="new-password",
                    placeholder="再次输入密码",
                    **({'aria-invalid': 'true'} if
form.confirm.errors else {})) }}
    {% for e in form.confirm.errors %}<p
class="rg-err">{{ e }}</p>{% endfor %}
  </div>
  <p class="rg-split">以下均为选填</p>
  <div class="rg-avatar">
    <span class="rg-avatar__face" id="rg-face"
aria-hidden="true">头像</span>
    <div class="rg-avatar__pick">
      <label class="visually-hidden"
for="rg-avatar">选择头像图片</label>
      <input id="rg-avatar" type="file" name="avatar"
             accept=".png,.jpg,.jpeg,.gif,.webp">
      <p class="rg-avatar__tip">PNG / JPG / GIF /
WebP，不超过 4 MB</p>
    </div>
  </div>
  <div class="rg-pair">
    <div class="rg-field">
      <label for="{{ form.display_name.id }}">昵称</label>
```

```
            {{ form.display_name(id=form.display_name.id,
                                 placeholder="留空则显示账户名称") }}
          </div>
          <div class="rg-field">
            <label for="{{ form.gender.id }}">性别</label>
            {{ form.gender(id=form.gender.id) }}
          </div>
        </div>
        <div class="rg-field">
          <label for="{{ form.email.id }}">邮箱</label>
          {{ form.email(id=form.email.id,
      autocomplete="email",
                        placeholder="用于联系，可留空",
                        **({'aria-invalid': 'true'} if
      form.email.errors else {})) }}
          {% for e in form.email.errors %}<p
      class="rg-err">{{ e }}</p>{% endfor %}
        </div>
        <div class="rg-field" style="margin-bottom:0;">
          <label for="{{ form.phone.id }}">手机号</label>
          {{ form.phone(id=form.phone.id, autocomplete="tel",
      placeholder="可留空") }}
        </div>
        {{ form.submit_reg(class_="rg-submit") }}
      </form>
      <p class="rg-foot">已有账号？<a href="{{
      url_for('auth.login') }}">返回登录</a></p>
    </div>
</div>
{% endblock %}
{% block scripts %}
<script>
// 选好头像后立刻在左侧圆框里预览，省得传上去才知道选错了
(function () {
  var input = document.getElementById('rg-avatar');
  var face = document.getElementById('rg-face');
  if (!input || !face || !window.FileReader) return;
  input.addEventListener('change', function () {
    var file = input.files && input.files[0];
    if (!file) return;
    var reader = new FileReader();
    reader.onload = function (e) {
      face.textContent = '';
      face.style.backgroundImage = 'url("' +
    e.target.result + '")';
    };
    reader.readAsDataURL(file);
  });
})();
</script>
```

```
{% endblock %}
# ==== app/templates/auth/home.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}工作台 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
{# ---------- 页首横幅 ---------- #}
<section class="hero">
  <div class="hero__body">
    <h1 class="hero__title">历史遗留废弃矿山自然恢复适宜性评价系统</h1>
    <div class="hero__actions">
      <a class="hero-btn hero-btn--ghost"
    href="/indicator/import">
        {{ icon("upload", 16) }}批量导入
      </a>
      <a class="hero-btn hero-btn--solid"
    href="/project/add">
        {{ icon("plus", 16) }}新建评价项目
      </a>
    </div>
  </div>
</section>
{# ---------- 数据概览 ---------- #}
<section class="stat-cards">
  <a class="stat-card" href="/project/">
    <span class="stat-card__label">评价项目</span>
    <span class="stat-card__value">{{ dash.stats.projects
    }}<span class="stat-card__unit">个</span></span>
    <span class="stat-card__note">
      {%- if projects_without_patch -%}
        其中 {{ projects_without_patch|length }} 个项目尚未建立图斑
      {%- elif dash.progress.patches -%}
        已建立 {{ dash.progress.patches }} 个矿山图斑
      {%- else -%}
        尚未建立矿山图斑
      {%- endif -%}
    </span>
  </a>
  <a class="stat-card" href="/project/">
    <span class="stat-card__label">矿山图斑</span>
    <span class="stat-card__value">{{ dash.stats.patches
    }}<span class="stat-card__unit">个</span></span>
    <span class="stat-card__note">
      {% if dash.progress.ready %}其中 {{ dash.progress.ready
    }} 个具备评价条件{% else %}暂无具备评价条件的图斑{% endif %}
    </span>
  </a>
  <a class="stat-card" href="/unit/">
    <span class="stat-card__label">微地质单元</span>
    <span class="stat-card__value">{{ dash.stats.unit_types
```

```
      }}<span class="stat-card__unit">类</span></span>                    7601
      <span class="stat-card__note">                                      7602
        {% if dash.progress.units %}共划分 {{                             7603
      dash.progress.units }} 个单元{% else %}尚未划分单元{% endif %}      7604
      </span>                                                             7605
   </a>                                                                   7606
   <a class="stat-card" href="/standard/">                                7607
      <span class="stat-card__label">标准与政策</span>                    7608
      <span class="stat-card__value">{{ dash.stats.standards              7609
      }}<span class="stat-card__unit">项</span></span>                    7610
      <span class="stat-card__note">地方标准与政策文件</span>             7611
   </a>                                                                   7612
</section>                                                               7613
{# ---------- 评价流程 ---------- #}                                     7614
<section class="section">                                                7615
   <div class="section__head">                                           7616
      <h2 class="section__title">评价流程</h2>                           7617
   </div>                                                                7618
   <div class="step-grid">                                               7619
      {% for s in dash.steps %}                                          7620
      {% if s.available %}                                               7621
      <a class="step-card" href="{{ s.href }}">                          7622
        <span class="step-card__no">{{ s.no }}</span>                    7623
        <span class="step-card__icon">{{ icon(s.icon, 18)                7624
      }}</span>                                                          7625
        <span class="step-card__title">{{ s.title }}</span>              7626
        <span class="step-card__desc">{{ s.desc }}</span>                7627
        {% if s.pending %}                                               7628
        <span class="step-card__badge">待处理 {{ s.pending               7629
      }}</span>                                                          7630
        {% endif %}                                                      7631
      </a>                                                               7632
      {% else %}                                                         7633
      <div class="step-card step-card--locked"                           7634
      aria-disabled="true">                                              7635
        <span class="step-card__no">{{ s.no }}</span>                    7636
        <span class="step-card__icon">{{ icon(s.icon, 18)                7637
      }}</span>                                                          7638
        <span class="step-card__title">{{ s.title }}</span>              7639
        <span class="step-card__desc">{{ s.desc }}</span>                7640
        <span class="step-card__lock">{{ s.note }}</span>                7641
      </div>                                                             7642
      {% endif %}                                                        7643
      {% endfor %}                                                       7644
   </div>                                                                7645
</section>                                                               7646
{# ---------- 系统管理 ---------- #}                                     7647
{# 个人设置、账号与操作日志的入口都在左侧导航栏，工作台不再重复列出 #}   7648
{% endblock %}                                                           7649
# ==== app/templates/auth/add_user.html ====                             7650
```

```
{% extends "base.html" %}
{% block title %}新增账号{% endblock %}
{% block content %}
<div class="card" style="max-width:560px;">
  <h2>新增账号</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="username">账户名称</label>
      <input id="username" type="text" name="username" required
             maxlength="64" autocomplete="off" style="width:280px;">
    </div>
    <div class="field">
      <label class="field__label" for="password">密码</label>
      <input id="password" type="password" name="password" required
             autocomplete="new-password" style="width:280px;">
    </div>
    <div class="field">
      <label class="field__label" for="role">职务</label>
      <select id="role" name="role">
        {% for code, label in role_labels.items() %}
        <option value="{{ code }}" {% if code == default_role %}selected{% endif %}>{{ label }}</option>
        {% endfor %}
      </select>
      <p class="muted" style="margin:6px 0 0;">
        管理员可管理系统、维护基础数据并查看操作日志；操作员可使用全部业务功能。
      </p>
    </div>
    <hr style="margin:22px 0;border:none;border-top:1px solid var(--line);">
    <h3 style="margin:0 0 12px;font-size:15px;">个人资料（选填）</h3>
    <div class="field">
      <label class="field__label" for="display_name">昵称</label>
      <input id="display_name" type="text" name="display_name" maxlength="64"
             placeholder="留空则显示账户名称">
    </div>
    <div class="field">
      <label class="field__label" for="gender">性别</label>
      <select id="gender" name="gender">
        <option value="">不填</option>
        {% for g in genders %}<option value="{{ g }}">{{ g
```

```
      }}</option>{% endfor %}                                          7701
        </select>                                                      7702
      </div>                                                           7703
      <div class="field">                                              7704
        <label class="field__label" for="email">邮箱</label>           7705
        <input id="email" type="text" name="email"                     7706
      maxlength="128">                                                 7707
      </div>                                                           7708
      <div class="field">                                              7709
        <label class="field__label" for="phone">手机号</label>         7710
        <input id="phone" type="text" name="phone"                     7711
      maxlength="32">                                                  7712
      </div>                                                           7713
      <p class="actions" style="margin-top:18px;">                     7714
        <button type="submit" class="btn">保存</button>                7715
        <a class="btn btn--ghost" href="{{                             7716
      url_for('auth.profile', tab='users') }}">返回</a>                7717
      </p>                                                             7718
    </form>                                                            7719
</div>                                                                 7720
{% endblock %}                                                         7721
# ==== app/templates/auth/edit_user.html ====                          7722
{% extends "base.html" %}                                              7723
{% block title %}编辑账号 · {{ user.username }}{% endblock %}          7724
{% block content %}                                                    7725
<div class="card" style="max-width:560px;">                            7726
  <h2>编辑账号 · {{ user.username }}</h2>                              7727
  <h3 style="margin:0 0 12px;font-size:15px;">个人资料</h3>            7728
  <p class="muted" style="margin:0 0 14px;font-size:12.5px;"           7729
    >                                                                  7730
    以下内容用户也可在「个人设置」中自行修改，此处为管理员代改。      7731
  </p>                                                                 7732
  <form method="post">                                                 7733
    {{ form.hidden_tag() }}                                            7734
    <div class="field">                                                7735
      <label class="field__label" for="display_name">昵称</lab         7736
    el>                                                                7737
      <input id="display_name" type="text"                             7738
    name="display_name" maxlength="64"                                 7739
             value="{{ user.display_name or '' }}"                     7740
             placeholder="留空则显示账户名称">                         7741
    </div>                                                             7742
    <div class="field">                                                7743
      <label class="field__label" for="gender">性别</label>            7744
      <select id="gender" name="gender">                               7745
        <option value="">不填</option>                                 7746
        {% for g in genders %}                                         7747
        <option value="{{ g }}" {% if user.gender == g                 7748
    %}selected{% endif %}>{{ g }}</option>                             7749
        {% endfor %}                                                   7750
```

```
    </select>
</div>
<div class="field">
    <label class="field__label" for="email">邮箱</label>
    <input id="email" type="text" name="email"
maxlength="128"
           value="{{ user.email or '' }}">
</div>
<div class="field">
    <label class="field__label" for="phone">手机号</label>
    <input id="phone" type="text" name="phone"
maxlength="32"
           value="{{ user.phone or '' }}">
</div>
<hr style="margin:22px 0;border:none;border-top:1px
solid var(--line);">
<h3 style="margin:0 0 12px;font-size:15px;">职务与权限</h3>
<p class="muted" style="margin:0 0
14px;font-size:12.5px;">
    职务只能由管理员设置：管理员可管理系统、维护基础数据并查看操作日志；
    操作员可使用全部业务功能。
</p>
<div class="field">
    <label class="field__label" for="role">职务</label>
    <select id="role" name="role">
      {% for code, label in role_labels.items() %}
      <option value="{{ code }}" {% if user.role == code
%}selected{% endif %}>{{ label }}</option>
      {% endfor %}
    </select>
</div>
<div class="field">
    <label class="checkbox" for="active">
      <input id="active" type="checkbox" name="active"
value="1"
             {% if user.active %}checked{% endif %}>
      <span>启用该账号</span>
    </label>
    <p class="muted" style="margin:6px 0 0;">
      停用后该账号无法登录，但其历史操作记录会保留。
      系统至少需保留一名启用状态的管理员。
    </p>
</div>
<div class="field">
    <label class="field__label" for="password">重置密码（留空则不修改
) </label>
    <input id="password" type="password" name="password"
           autocomplete="new-password"
style="width:280px;"
           placeholder="留空表示保持原密码">
```

```
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="{{
      url_for('auth.profile', tab='users') }}">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/auth/profile.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}个人设置 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  /* ================== 个人设置（仿 Apple 账户页） ================== */
      左栏是身份与分类，右栏是该分类下的功能卡。
      每张卡默认只显示当前值，点开才展开对应的修改表单 ——
      不点就不会误改，页面也不会一上来堆满输入框。 */
  .acct {
    display: grid;
    grid-template-columns: 244px 1fr;
    gap: 40px;
    align-items: start;
  }
  @media (max-width: 900px) {
    .acct { grid-template-columns: 1fr; gap: 24px; }
  }
  /* ---------- 左栏 ---------- */
  .acct-rail__avatar {
    display: grid;
    place-items: center;
    width: 88px;
    height: 88px;
    border-radius: 50%;
    background: #e9edeb;
    color: #97a09b;
    font-size: 32px;
    font-weight: 500;
    object-fit: cover;
  }
  .acct-rail__name {
    margin: 16px 0 2px;
    font-size: 16px;
    font-weight: 600;
  }
  .acct-rail__sub {
    margin: 0 0 24px;
    font-size: 13px;
    color: #86868b;
```

```
    word-break: break-all;
  }
  .acct-tabs {
    display: flex;
    flex-direction: column;
    gap: 2px;
    margin-bottom: 24px;
  }
  .acct-tabs a {
    padding: 7px 12px;
    border: 1px solid transparent;
    border-radius: 7px;
    font-size: 15px;
    color: #1d1d1f;
    text-decoration: none;
    transition: background-color .16s ease, border-color
    .16s ease;
  }
  .acct-tabs a:hover { background: #f5f5f7; }
  .acct-tabs a.is-active {
    font-weight: 600;
    border-color: #bcc9c1;
    background: #fff;
  }
  /* 更换头像：折叠在左栏，点开才出现文件选择 */
  .acct-avatar-edit > summary {
    display: inline-block;
    padding: 5px 12px;
    font-size: 13px;
    color: var(--accent);
    background: var(--accent-soft);
    border-radius: 7px;
    cursor: pointer;
    list-style: none;
  }
  .acct-avatar-edit > summary::-webkit-details-marker {
    display: none; }
  .acct-avatar-edit > summary:hover { background: #dcebe2; }
  .acct-avatar-edit__body { margin-top: 12px; }
  .acct-avatar-edit__body .btn { width: 100%;
    margin-bottom: 8px; }
  .acct-avatar-edit__file {
    margin: 0 0 10px;
    font-size: 12px;
    color: var(--ink-soft);
    word-break: break-all;
  }
  /* ---------- 右栏 ---------- */
  .acct-title {
    margin: 0 0 8px;
```

```
    font-size: 26px;
    font-weight: 700;
    letter-spacing: -.3px;
  }
  .acct-lead {
    margin: 0 0 26px;
    font-size: 14px;
    color: #6e6e73;
  }
  /* 固定两列，与 Apple 账户页的 2×2 一致；
     卡片也宽一些，邮箱之类的长值不至于被拆成两行 */
  .acct-grid {
    display: grid;
    grid-template-columns: repeat(2, minmax(0, 1fr));
    gap: 16px;
    align-items: start;
  }
  @media (max-width: 760px) {
    .acct-grid { grid-template-columns: 1fr; }
  }
  /* 功能卡：details/summary 实现，点开才展开表单 */
  .acct-card {
    background: #fff;
    border: 1px solid #e6e6e8;
    border-radius: 14px;
    transition: border-color .2s ease, box-shadow .2s ease;
  }
  .acct-card[open] {
    border-color: rgba(31, 107, 69, .45);
    box-shadow: 0 8px 22px rgba(19, 60, 43, .08);
  }
  .acct-card > summary {
    position: relative;
    display: block;
    padding: 18px 56px 18px 20px;
    border-radius: 14px;
    cursor: pointer;
    list-style: none;
  }
  .acct-card > summary::-webkit-details-marker { display:
    none; }
  .acct-card > summary:hover { background: #fafbfa; }
  .acct-card > summary:focus-visible {
    outline: 2px solid var(--accent);
    outline-offset: 2px;
  }
  .acct-card__title {
    display: block;
    font-size: 15px;
    font-weight: 600;
```

```
}
.acct-card__value {
  display: block;
  margin-top: 3px;
  font-size: 14px;
  color: #6e6e73;
  overflow-wrap: anywhere;
}
.acct-card__icon {
  position: absolute;
  top: 17px;
  right: 17px;
  display: grid;
  place-items: center;
  width: 27px;
  height: 27px;
  border-radius: 8px;
  background: var(--accent-soft);
  color: var(--accent);
}
.acct-card__body {
  padding: 4px 20px 20px;
  border-top: 1px solid #f0f1f2;
  margin-top: 2px;
}
.acct-card__body .field:first-child { margin-top: 16px; }
.acct-card__body .btn { width: 100%; }
/* 只读信息列表 */
.acct-kv {
  display: grid;
  grid-template-columns: 88px 1fr;
  gap: 7px 12px;
  margin: 16px 0 0;
  font-size: 13.5px;
}
.acct-kv dt { color: var(--ink-soft); }
.acct-kv dd { margin: 0; }
.field__error {
  margin: 5px 0 0;
  font-size: 12.5px;
  color: var(--danger);
}
/* 管理员分类（账号 / 操作日志）里的表格面板 */
.acct-panel {
  background: #fff;
  border: 1px solid #e6e6e8;
  border-radius: 14px;
  overflow: hidden;
}
.acct-panel__head {
```

```
      display: flex;                                                    8001
      align-items: center;                                              8002
      justify-content: space-between;                                   8003
      gap: 12px;                                                        8004
      padding: 13px 20px;                                               8005
      border-bottom: 1px solid #f0f1f2;                                 8006
    }                                                                   8007
    .acct-panel__count {                                                8008
      margin: 0;                                                        8009
      font-size: 13.5px;                                                8010
      color: #6e6e73;                                                   8011
    }                                                                   8012
    .acct-panel table { margin: 0; }                                    8013
    .acct-panel th,                                                     8014
    .acct-panel td { border: 0; border-bottom: 1px solid                8015
      #f0f1f2; }                                                        8016
    .acct-panel thead th { background: #f7f9f8; }                       8017
    .acct-panel tbody tr:last-child td { border-bottom: 0; }            8018
    .acct-panel tbody tr:hover { background: #fafbfa; }                 8019
</style>                                                                8020
<div class="acct">                                                      8021
    {# ---------------- 左栏：身份 + 分类 ---------------- #}           8022
    <aside class="acct-rail">                                           8023
      {% if current_user.avatar %}                                      8024
      <img class="acct-rail__avatar"                                    8025
           src="{{ url_for('static', filename=current_user.ava8026
      tar) }}"                                                          8027
           alt="{{ current_user.shown_name }} 的头像">                  8028
      {% else %}                                                        8029
      <span class="acct-rail__avatar" aria-hidden="true">{{             8030
      current_user.initial }}</span>                                    8031
      {% endif %}                                                       8032
      <p class="acct-rail__name">{{ current_user.shown_name             8033
      }}</p>                                                            8034
      <p class="acct-rail__sub">                                        8035
        {{ role_labels.get(current_user.role,                           8036
      current_user.role) }}                                             8037
        {% if current_user.email %} · {{ current_user.email             8038
      }}{% endif %}                                                     8039
      </p>                                                              8040
      <nav class="acct-tabs" aria-label="个人设置分类">                 8041
        <a href="{{ url_for('auth.profile', tab='info') }}"             8042
           {% if tab == 'info' %}class="is-active"                      8043
      aria-current="page"{% endif %}>个人信息</a>                       8044
        <a href="{{ url_for('auth.profile', tab='security')             8045
      }}"                                                               8046
           {% if tab == 'security' %}class="is-active"                  8047
      aria-current="page"{% endif %}>登录与安全性</a>                   8048
        {% if current_user.is_admin %}                                  8049
        <a href="{{ url_for('auth.profile', tab='users') }}"            8050
```

```
        {% if tab == 'users' %}class="is-active"                    8051
    aria-current="page"{% endif %}>账号</a>                         8052
      <a href="{{ url_for('auth.profile', tab='logs') }}"           8053
        {% if tab == 'logs' %}class="is-active"                     8054
    aria-current="page"{% endif %}>操作日志</a>                     8055
      {% endif %}                                                   8056
    </nav>                                                          8057
    <details class="acct-avatar-edit">                              8058
      <summary>更换头像</summary>                                   8059
      <div class="acct-avatar-edit__body">                          8060
        <form method="post" action="/profile/avatar"                8061
              enctype="multipart/form-data">                        8062
          {{ avatar_form.hidden_tag() }}                            8063
          <input type="hidden" name="tab" value="{{ tab }}">        8064
          <label class="visually-hidden"                            8065
    for="avatar-file">选择头像图片</label>                          8066
          <input id="avatar-file" type="file" name="avatar"         8067
                 accept=".png,.jpg,.jpeg,.gif,.webp">               8068
          <p class="acct-avatar-edit__file">PNG / JPG / GIF         8069
    / WebP，不超过 4 MB</p>                                         8070
          <button type="submit" class="btn                          8071
    btn--sm">上传头像</button>                                      8072
        </form>                                                     8073
        {% if current_user.avatar %}                                8074
        <form method="post" action="/profile/avatar/delete"         8075
              onsubmit="return confirm('确定移除当前头像吗？');">   8076
          {{ avatar_form.hidden_tag() }}                            8077
          <input type="hidden" name="tab" value="{{ tab }}">        8078
          <button type="submit" class="btn btn--sm                  8079
    btn--ghost">移除头像</button>                                   8080
        </form>                                                     8081
        {% endif %}                                                 8082
      </div>                                                        8083
    </details>                                                      8084
  </aside>                                                          8085
  {# ---------------- 右栏：功能卡 ---------------- #}              8086
  <div class="acct-main">                                           8087
    {% if tab == 'security' %}                                      8088
    <h1 class="acct-title">登录与安全性</h1>                        8089
    <p class="acct-lead">管理与登录账号、账号安全有关的设置。</p>  8090
    <div class="acct-grid">                                         8091
      <details class="acct-card"                                    8092
               {% if pwd_form.old_password.errors or                8093
    pwd_form.new_password.errors or pwd_form.confirm.errors         8094
    %}open{% endif %}>                                              8095
        <summary>                                                   8096
          <span class="acct-card__title">密码</span>                8097
          <span class="acct-card__value">修改登录密码</span>        8098
          <span class="acct-card__icon">{{ icon("lock", 15)         8099
    }}</span>                                                       8100
```

```
      </summary>                                                    8101
      <div class="acct-card__body">                                 8102
        <form method="post">                                        8103
          {{ pwd_form.hidden_tag() }}                               8104
          <input type="hidden" name="action"                        8105
value="password">                                                   8106
          <input type="hidden" name="tab"                           8107
value="security">                                                   8108
          <div class="field">                                       8109
            <label class="field__label" for="{{                     8110
pwd_form.old_password.id }}">当前密码</label>                        8111
            {{ pwd_form.old_password(id=pwd_form.old_passw          8112
ord.id,                                                             8113
                                                                    8114
autocomplete="current-password") }}                                 8115
            {% for e in pwd_form.old_password.errors %}             8116
            <p class="field__error">{{ e }}</p>                     8117
            {% endfor %}                                            8118
          </div>                                                    8119
          <div class="field">                                       8120
            <label class="field__label" for="{{                     8121
pwd_form.new_password.id }}">新密码</label>                          8122
            {{ pwd_form.new_password(id=pwd_form.new_passw          8123
ord.id,                                                             8124
                                                                    8125
autocomplete="new-password") }}                                     8126
            {% for e in pwd_form.new_password.errors %}             8127
            <p class="field__error">{{ e }}</p>                     8128
            {% endfor %}                                            8129
          </div>                                                    8130
          <div class="field">                                       8131
            <label class="field__label" for="{{                     8132
pwd_form.confirm.id }}">确认新密码</label>                           8133
            {{ pwd_form.confirm(id=pwd_form.confirm.id,             8134
                                autocomplete="new-password          8135
") }}                                                               8136
            {% for e in pwd_form.confirm.errors %}                  8137
            <p class="field__error">{{ e }}</p>                     8138
            {% endfor %}                                            8139
          </div>                                                    8140
          <p class="actions">{{ pwd_form.submit_pwd(class_          8141
="btn") }}</p>                                                      8142
        </form>                                                     8143
      </div>                                                        8144
    </details>                                                      8145
    <details class="acct-card">                                     8146
      <summary>                                                     8147
        <span class="acct-card__title">登录账号</span>               8148
        <span class="acct-card__value">{{                           8149
current_user.username }}</span>                                     8150
```

```
        <span class="acct-card__icon">{{ icon("user", 15) }}</span>
      </summary>
      <div class="acct-card__body">
        <dl class="acct-kv">
          <dt>登录账号</dt><dd class="mono">{{ current_user.username }}</dd>
          <dt>职务</dt>
          <dd>{{ role_labels.get(current_user.role, current_user.role) }}</dd>
          <dt>状态</dt>
          <dd>
            {% if current_user.active %}
            <span class="tag tag--ok">启用</span>
            {% else %}
            <span class="tag tag--warn">停用</span>
            {% endif %}
          </dd>
          <dt>创建时间</dt>
          <dd class="mono">{{ current_user.created_at | localdt }}</dd>
        </dl>
      </div>
    </details>
</div>
{% elif tab == 'users' %}
<h1 class="acct-title">账号</h1>
<p class="acct-lead">管理本系统的账号、职务与启用状态。新注册的账号默认为「操作员」。</p>
{% include "auth/_users.html" %}
{% elif tab == 'logs' %}
<h1 class="acct-title">操作日志</h1>
<p class="acct-lead">记录各账号在本系统中的关键操作。</p>
{% include "auth/_logs.html" %}
{% else %}
<h1 class="acct-title">个人信息</h1>
<p class="acct-lead">管理你的昵称、性别以及可以联系到你的邮箱与手机号。</p>
<form method="post">
  {{ info_form.hidden_tag() }}
  <input type="hidden" name="action" value="info">
  <input type="hidden" name="tab" value="info">
  <div class="acct-grid">
    <details class="acct-card" {% if info_form.display_name.errors %}open{% endif %}>
      <summary>
        <span class="acct-card__title">昵称</span>
        <span class="acct-card__value">
          {{ current_user.display_name or "未设置" }}
        </span>
        <span class="acct-card__icon">{{ icon("user",
```

```
15) }}</span>                                                    8201
        </summary>                                               8202
        <div class="acct-card__body">                            8203
          <div class="field">                                    8204
            <label class="field__label" for="{{                  8205
info_form.display_name.id }}">                                   8206
              昵称                                               8207
            </label>                                             8208
            {{ info_form.display_name(id=info_form.display       8209
_name.id,                                                        8210
                                                                 8211
placeholder="留空则显示账户名称") }}                               8212
            {% for e in info_form.display_name.errors %}         8213
            <p class="field__error">{{ e }}</p>                  8214
            {% endfor %}                                         8215
          </div>                                                 8216
          <p class="actions">{{ info_form.submit_info(clas       8217
s_="btn") }}</p>                                                 8218
        </div>                                                   8219
      </details>                                                 8220
      <details class="acct-card" {% if                           8221
info_form.gender.errors %}open{% endif %}>                       8222
        <summary>                                                8223
          <span class="acct-card__title">性别</span>              8224
          <span class="acct-card__value">{{                      8225
current_user.gender or "未设置" }}</span>                          8226
          <span class="acct-card__icon">{{ icon("user",          8227
15) }}</span>                                                    8228
        </summary>                                               8229
        <div class="acct-card__body">                            8230
          <div class="field">                                    8231
            <label class="field__label" for="{{                  8232
info_form.gender.id }}">性别</label>                               8233
            {{ info_form.gender(id=info_form.gender.id) }}       8234
            {% for e in info_form.gender.errors %}               8235
            <p class="field__error">{{ e }}</p>                  8236
            {% endfor %}                                         8237
          </div>                                                 8238
          <p class="actions">{{ info_form.submit_info(clas       8239
s_="btn") }}</p>                                                 8240
        </div>                                                   8241
      </details>                                                 8242
      <details class="acct-card" {% if                           8243
info_form.email.errors %}open{% endif %}>                        8244
        <summary>                                                8245
          <span class="acct-card__title">邮箱</span>              8246
          <span class="acct-card__value">{{                      8247
current_user.email or "未设置" }}</span>                           8248
          <span class="acct-card__icon">{{ icon("mail",          8249
15) }}</span>                                                    8250
```

```
            </summary>                                                    8251
            <div class="acct-card__body">                                 8252
              <div class="field">                                         8253
                <label class="field__label" for="{{                       8254
    info_form.email.id }}">邮箱</label>                                   8255
                {{ info_form.email(id=info_form.email.id) }}              8256
                {% for e in info_form.email.errors %}                     8257
                <p class="field__error">{{ e }}</p>                       8258
                {% endfor %}                                              8259
              </div>                                                      8260
              <p class="actions">{{ info_form.submit_info(clas            8261
    s_="btn") }}</p>                                                      8262
            </div>                                                        8263
          </details>                                                      8264
          <details class="acct-card" {% if                            8265
    info_form.phone.errors %}open{% endif %}>                             8266
            <summary>                                                     8267
              <span class="acct-card__title">手机号</span>                8268
              <span class="acct-card__value">{{                           8269
    current_user.phone or "未设置" }}</span>                              8270
              <span class="acct-card__icon">{{ icon("phone",              8271
    15) }}</span>                                                         8272
            </summary>                                                    8273
            <div class="acct-card__body">                                 8274
              <div class="field">                                         8275
                <label class="field__label" for="{{                       8276
    info_form.phone.id }}">手机号</label>                                 8277
                {{ info_form.phone(id=info_form.phone.id) }}              8278
                {% for e in info_form.phone.errors %}                     8279
                <p class="field__error">{{ e }}</p>                       8280
                {% endfor %}                                              8281
              </div>                                                      8282
              <p class="actions">{{ info_form.submit_info(clas            8283
    s_="btn") }}</p>                                                      8284
            </div>                                                        8285
          </details>                                                      8286
        </div>                                                            8287
    </form>                                                               8288
    {% endif %}                                                           8289
  </div>                                                                  8290
</div>                                                                    8291
{% endblock %}                                                            8292
{% block scripts %}                                                       8293
<script>                                                                  8294
// 同时只展开一张卡，避免点开多张后页面被撑得很长                                 8295
(function () {                                                            8296
  var cards = document.querySelectorAll('.acct-card');                    8297
  Array.prototype.forEach.call(cards, function (card) {                   8298
    card.addEventListener('toggle', function () {                         8299
      if (!card.open) return;                                             8300
```

```
        Array.prototype.forEach.call(cards, function (other) {
          if (other !== card && other.open) other.open =
      false;
        });
      });
    });
})();
</script>
{% endblock %}
# ==== app/templates/auth/_users.html ====
{# 管理员的「账号」分类。由 auth/profile.html 内联引入，不单独成页。 #}
<div class="acct-panel">
  <div class="acct-panel__head">
    <p class="acct-panel__count">共 {{ users|length }}
    个账号</p>
    <a class="btn btn--sm" href="{{ url_for('auth.add_user')
    }}">新增账号</a>
  </div>
  <table>
    <thead>
      <tr>
        <th scope="col">ID</th>
        <th scope="col">账户名称</th>
        <th scope="col">昵称</th>
        <th scope="col">职务</th>
        <th scope="col">状态</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for u in users %}
      <tr>
        <td class="num">{{ u.id }}</td>
        <td class="mono">
          {{ u.username }}
          {% if u.id == current_user.id %}<span
    class="muted">（当前登录）</span>{% endif %}
        </td>
        <td style="text-align:left;">
          {% if u.display_name %}{{ u.display_name }}
          {% else %}<span class="muted">—</span>{% endif %}
        </td>
        <td>
          <span class="tag {{ 'tag--ok' if u.role ==
    'admin' else 'tag--muted' }}">
            {{ role_labels.get(u.role, u.role) }}
          </span>
        </td>
        <td>
          {% if u.active %}<span class="tag
```

```
      tag--ok">启用</span>                                                    8351
              {% else %}<span class="tag tag--muted">停用</span>{%           8352
      endif %}                                                                8353
          </td>                                                               8354
          <td class="nowrap">                                                 8355
            <span class="row-actions">                                        8356
              <a class="link-btn" href="{{                                    8357
      url_for('auth.edit_user', user_id=u.id) }}">编辑</a>                    8358
              <form method="post"                                             8359
                    action="{{ url_for('auth.delete_user',                    8360
      user_id=u.id) }}"                                                       8361
                    onsubmit="return confirm('确定删除账号「{{                8362
      u.username }}」吗？\n该操作无法恢复。');">                              8363
                {{ form.hidden_tag() }}                                       8364
                <button type="submit" class="link-btn                         8365
      link-btn--danger"                                                       8366
                        {% if u.id == current_user.id                         8367
      %}disabled                                                              8368
                        title="不能删除当前登录的账号"{% endif                  8369
      %}>删除</button>                                                        8370
              </form>                                                         8371
            </span>                                                           8372
          </td>                                                               8373
        </tr>                                                                 8374
        {% endfor %}                                                          8375
      </tbody>                                                                8376
    </table>                                                                  8377
</div>                                                                        8378
#  ==== app/templates/auth/_logs.html ====                                    8379
{# 管理员的「操作日志」分类。由 auth/profile.html 内联引入，不单独成页。 #}  8380
<div class="acct-panel">                                                      8381
    <div class="acct-panel__head">                                            8382
      <p class="acct-panel__count">                                           8383
        共 {{ total }} 条记录，当前第 {{ page }} / {{ pages }} 页             8384
      </p>                                                                    8385
      {% if pages > 1 %}                                                      8386
      <nav class="row-actions" aria-label="日志分页">                         8387
        {% if page > 1 %}                                                     8388
        <a class="link-btn"                                                   8389
           href="{{ url_for('auth.profile', tab='logs',                       8390
      page=page - 1) }}">上一页</a>                                           8391
        {% endif %}                                                           8392
        {% if page < pages %}                                                 8393
        <a class="link-btn"                                                   8394
           href="{{ url_for('auth.profile', tab='logs',                       8395
      page=page + 1) }}">下一页</a>                                           8396
        {% endif %}                                                           8397
      </nav>                                                                  8398
      {% endif %}                                                             8399
    </div>                                                                    8400
```

```
  {% if not records %}
  <p class="muted" style="padding:18px 20px;">暂无操作记录。</p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">时间</th>
        <th scope="col">操作人</th>
        <th scope="col">操作</th>
        <th scope="col">对象</th>
        <th scope="col">说明</th>
      </tr>
    </thead>
    <tbody>
      {% for r in records %}
      <tr>
        <td class="mono nowrap">{{ r.created_at |
    localdt("%Y-%m-%d %H:%M:%S") }}</td>
        <td>{{ users_map.get(r.user_id, "—") }}</td>
        <td style="text-align:left;">{{ r.action }}</td>
        <td>
          {% if r.target_type %}{{ r.target_type }}{% if
    r.target_id %}#{{ r.target_id }}{% endif %}
          {% else %}—{% endif %}
        </td>
        <td style="text-align:left;">{{ r.detail or "—"
    }}</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endif %}
</div>
# ==== app/templates/project/index.html ====
{% extends "base.html" %}
{% block title %}项目与矿山档案{% endblock %}
{% block content %}
<div class="card">
  <h2>项目与矿山档案</h2>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm" href="/project/add">新建项目</a>
  </p>
  {% if not projects %}
  <p class="muted">
    暂无项目。点击上方「新建项目」开始建立评价项目，
    或运行 <span class="mono">python seed_demo.py</span>
    载入示例案例。
  </p>
  {% else %}
  <table>
```

```
      <caption class="muted">共 {{ projects|length }}
      个项目</caption>
      <thead>
        <tr>
          <th scope="col">项目名称</th>
          <th scope="col">编号</th>
          <th scope="col">状态</th>
          <th scope="col">图斑数</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for p in projects %}
        <tr>
          <td style="text-align:left;">{{ p.name }}</td>
          <td class="mono">{{ p.code or "—" }}</td>
          <td><span class="tag tag--muted">{{ p.status
      }}</span></td>
          <td class="num">
            {% if p.patches.count() %}
              {{ p.patches.count() }}
            {% else %}
              <span class="tag tag--warn">0</span>
            {% endif %}
          </td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/project/{{ p.id
      }}/patches">{% if p.patches.count() %}矿山图斑{% else %}+
      新建图斑{% endif %}</a>
              <a class="link-btn" href="/project/{{ p.id
      }}/edit">编辑</a>
              <form method="post" action="/project/{{ p.id
      }}/delete"
                    onsubmit="return confirm('确定删除项目「{{
      p.name }}」吗？\n该项目下的图斑、微地质单元与评价数据将一并删除，且无法恢复。');">
                {{ form.hidden_tag() }}
                <button type="submit" class="link-btn
      link-btn--danger">删除</button>
              </form>
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% endif %}
</div>
{% endblock %}
# ==== app/templates/project/edit.html ====
```

```
{% extends "base.html" %}
{% block title %}{{ "编辑项目" if project else "新建项目" }}{% endblock %}
{% block content %}
<div class="card" style="max-width:640px;">
  <h2>{{ "编辑项目" if project else "新建项目" }}</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="name">项目名称</label>
      <input id="name" type="text" name="name"
    style="width:100%;" required
             value="{{ project.name if project else '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="code">项目编号</label>
      <input id="code" type="text" name="code"
    style="width:240px;"
             value="{{ project.code or '' if project else
    '' }}">
    </div>
    {% if project %}
    <div class="field">
      <label class="field__label" for="status">状态</label>
      <select id="status" name="status">
        {% for s in ["进行中", "已完成", "已归档"] %}
        <option value="{{ s }}" {% if project.status == s
    %}selected{% endif %}>{{ s }}</option>
        {% endfor %}
      </select>
    </div>
    {% endif %}
    <div class="field">
      <label class="field__label" for="description">项目说明</la
    bel>
      <textarea id="description" name="description">{{
    project.description or '' if project else ''
    }}</textarea>
    </div>
    <p class="actions" style="margin-top:18px;">
      <button type="submit" class="btn">保存</button>
      <a class="btn btn--ghost" href="/project/">返回</a>
    </p>
  </form>
</div>
{% endblock %}
# ==== app/templates/project/patches.html ====
{% extends "base.html" %}
{% block title %}矿山图斑 · {{ project.name }}{% endblock %}
{% block content %}
```

```
<div class="card">
  <h2>矿山图斑 · {{ project.name }}</h2>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm" href="/project/{{ project.id
    }}/patch/add">新建图斑</a>
    <a class="btn btn--sm btn--ghost"
    href="/project/">返回项目列表</a>
  </p>
  {% if not project.patches.count() %}
  <p class="muted">该项目下暂无图斑。点击上方「新建图斑」建立第一个矿山图斑。</p>
  {% else %}
  <table>
    <caption class="muted">共 {{ project.patches.count() }}
    个图斑</caption>
    <thead>
      <tr>
        <th scope="col">图斑名称</th>
        <th scope="col">行政区</th>
        <th scope="col">面积（公顷）</th>
        <th scope="col">开采矿种</th>
        <th scope="col">微地质单元</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in project.patches %}
      <tr>
        <td style="text-align:left;">{{ p.name }}</td>
        <td>{{ p.admin_region or "—" }}</td>
        <td class="num">{{ p.area_ha if p.area_ha is not
    none else "—" }}</td>
        <td>{{ p.mineral_type or "—" }}</td>
        <td class="num">{{ p.units.count() }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/unit/{{ p.id
    }}">微地质单元</a>
            <a class="link-btn" href="/constraint/{{ p.id
    }}">限制性判定</a>
            <a class="link-btn" href="/project/patch/{{
    p.id }}/edit">编辑</a>
            <form method="post" action="/project/patch/{{
    p.id }}/delete"
                  onsubmit="return confirm('确定删除图斑「{{
    p.name }}」吗？\n该图斑的微地质单元、指标数据与评价结果将一并删除，且无法恢复。');">
              {{ form.hidden_tag() }}
              <button type="submit" class="link-btn
    link-btn--danger">删除</button>
            </form>
          </span>
```

```
          </td>                                                    8601
        </tr>                                                      8602
        {% endfor %}                                               8603
      </tbody>                                                     8604
    </table>                                                       8605
    {% endif %}                                                    8606
</div>                                                             8607
{% endblock %}                                                     8608
# ==== app/templates/project/patch_edit.html ====                  8609
{% extends "base.html" %}                                          8610
{% block title %}{{ "编辑图斑" if patch else "新建图斑" }}{%       8611
    endblock %}                                                    8612
{% block content %}                                                8613
<div class="card" style="max-width:720px;">                        8614
  <h2>{{ "编辑图斑" if patch else "新建图斑" }} · {{ project.name  8615
    }}</h2>                                                        8616
  <form method="post">                                             8617
    {{ form.hidden_tag() }}                                        8618
    <div class="field">                                            8619
      <label class="field__label" for="patch-name">图斑名称</lab8620
    el>                                                            8621
      <input id="patch-name" type="text" name="name"               8622
    style="width:100%;" required                                   8623
           value="{{ patch.name if patch else '' }}">              8624
    </div>                                                         8625
    <div class="field">                                            8626
      <label class="field__label" for="admin-region">所在行政区</ 8627
    label>                                                         8628
      <input id="admin-region" type="text"                         8629
    name="admin_region" style="width:100%;"                        8630
           value="{{ patch.admin_region or '' if patch             8631
    else '' }}">                                                   8632
    </div>                                                         8633
    <div class="field">                                            8634
      <label class="field__label" for="area-ha">图斑面积（公顷）</la8635
    bel>                                                           8636
      <input id="area-ha" type="number" step="any"                 8637
    name="area_ha"                                                 8638
           value="{{ patch.area_ha if patch and                    8639
    patch.area_ha is not none else '' }}">                         8640
    </div>                                                         8641
    <div class="field">                                            8642
      <label class="field__label" for="mineral-type">开采矿种</l8643
    abel>                                                          8644
      <input id="mineral-type" type="text"                         8645
    name="mineral_type" style="width:240px;"                       8646
           value="{{ patch.mineral_type or '' if patch             8647
    else '' }}">                                                   8648
    </div>                                                         8649
    <div class="field">                                            8650
```

```
    <label class="field__label" for="mining-method">开采方式</
label>
    <input id="mining-method" type="text"
name="mining_method" style="width:240px;"
           value="{{ patch.mining_method or '' if patch
else '' }}">
</div>
<div class="field">
    <label class="field__label" for="stop-years">停产年限</lab
el>
    <input id="stop-years" type="number" step="any"
name="stop_years"
           value="{{ patch.stop_years if patch and
patch.stop_years is not none else '' }}">
</div>
<div class="field">
    <label class="checkbox" for="responsible-missing">
      <input id="responsible-missing" type="checkbox"
name="responsible_entity_missing" value="1"
             {% if not patch or patch.responsible_entity_m
issing %}checked{% endif %}>
      <span>责任主体缺失</span>
    </label>
</div>
<h3>参照生态系统（指标标准化基准）</h3>
<p class="muted">
    系统默认取值：植被覆盖度 ７５%、Shannon-Wiener 多样性指数 ０.７。
    该基准与图斑绑定管理，可按不同参照生态系统分别设定。
</p>
<div class="field">
    <label class="field__label" for="coverage-benchmark">植
被覆盖度基准（%）</label>
    <input id="coverage-benchmark" type="number"
step="any" name="coverage_benchmark"
           value="{{ reference.coverage_benchmark if
reference else 75 }}">
</div>
<div class="field">
    <label class="field__label" for="shannon-benchmark">Sh
annon-Wiener 多样性指数基准</label>
    <input id="shannon-benchmark" type="number"
step="any" name="shannon_benchmark"
           value="{{ reference.shannon_benchmark if
reference else 0.7 }}">
</div>
<div class="field">
    <label class="field__label" for="dominant-species">建群种
</label>
    <input id="dominant-species" type="text"
name="dominant_species" style="width:100%;"
```

```
                value="{{ reference.dominant_species or '' if reference else '' }}">
    </div>
    <p class="actions" style="margin-top:18px;">
      <button type="submit" class="btn">保存</button>
      <a class="btn btn--ghost" href="/project/{{ project.id }}/patches">返回</a>
    </p>
  </form>
</div>
{% if patch %}
<div class="card" style="max-width:720px;">
  <h2>遥感影像挂接</h2>
  <p class="muted">
    挂接遥感底图或无人机正射影像，用于辅助微地质单元划分与植被覆盖度判读。
    支持 PNG、JPG、GIF、WebP、BMP、TIFF，单个文件不超过 32 MB。
  </p>
  {% if patch.image_path %}
  <figure style="margin:0 0 16px;">
    <img src="{{ url_for('static', filename=patch.image_path) }}"
         alt="图斑「{{ patch.name }}」已挂接的遥感影像"
         style="max-width:100%;border:1px solid var(--line);border-radius:6px;">
    <figcaption class="muted" style="margin-top:6px;">
      {{ patch.image_note or "遥感影像" }}
      {% if patch.image_uploaded_at %}
      · 上传于 {{ patch.image_uploaded_at | localdt }}
      {% endif %}
    </figcaption>
  </figure>
  <form method="post" action="/project/patch/{{ patch.id }}/image/delete"
        onsubmit="return confirm('确定移除该遥感影像吗？');">
    {{ form.hidden_tag() }}
    <button type="submit" class="btn btn--sm btn--danger">移除影像</button>
  </form>
  <hr style="margin:16px 0;border:none;border-top:1px solid var(--line);">
  {% endif %}
  <form method="post" action="/project/patch/{{ patch.id }}/image"
        enctype="multipart/form-data">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="image-file">
        {{ "替换影像文件" if patch.image_path else "选择影像文件" }}
      </label>
      <input id="image-file" type="file" name="image"
```

```
                accept=".png,.jpg,.jpeg,.gif,.webp,.bmp,.tif,.t
    iff" required>
    </div>
    <div class="field">
      <label class="field__label" for="image-note">影像说明（如来源、
    时相）</label>
      <input id="image-note" type="text" name="image_note"
    style="width:100%;"
             placeholder="如：2025 年 6 月无人机正射影像"
             value="{{ patch.image_note or '' }}">
    </div>
    <p class="actions">
      <button type="submit" class="btn">{{ "替换影像" if
    patch.image_path else "上传并挂接" }}</button>
    </p>
  </form>
</div>
{% endif %}
{% endblock %}
# ==== app/templates/unit/list.html ====
{% extends "base.html" %}
{% block title %}微地质单元{% endblock %}
{% block content %}
<div class="card">
  <h2>微地质单元</h2>
  {% if not rows %}
  <p class="muted">
    暂无矿山图斑。请先到「项目与矿山档案」建立图斑。
  </p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">所属项目</th>
        <th scope="col">矿山图斑</th>
        <th scope="col">已划分单元</th>
        <th scope="col">占比合计</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rows %}
      <tr>
        <td>{{ r.patch.project.name if r.patch.project else
    "—" }}</td>
        <td style="text-align:left;">{{ r.patch.name }}</td>
        <td class="num">{{ r.count }} 个</td>
        <td class="num">
          {% if r.count %}
            {{ "%.2f%%"|format(r.ratio_sum * 100) }}
```

```
                {% if r.ratio_sum < 0.999 or r.ratio_sum >
    1.001 %}
                <span class="warn">（不等于 100%）</span>
                {% endif %}
              {% else %}—{% endif %}
            </td>
            <td class="nowrap">
              <span class="row-actions">
                <a class="link-btn" href="/unit/{{ r.patch.id
    }}">划分与维护</a>
                <a class="link-btn" href="/indicator/">录入指标数据</a
    >
              </span>
            </td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
      {% endif %}
</div>
{% endblock %}
# ==== app/templates/unit/index.html ====
{% extends "base.html" %}
{% block title %}微地质单元 · {{ patch.name }}{% endblock %}
{% block content %}
<div class="card">
  <h2>微地质单元 · {{ patch.name }}</h2>
  {% if units %}
  <table>
    <caption class="muted">
      共 {{ units|length }} 个单元；
      面积合计 <span class="num">{{ "%.4f"|format(total_area)
    }}</span>；
      占比合计 <span class="num">{{ "%.2f%%"|format(ratio_sum *
    100) }}</span>
      {% if ratio_sum and (ratio_sum < 0.999 or ratio_sum >
    1.001) %}
      <span class="warn">（占比合计不等于 100%，请检查面积录入）</span>
      {% endif %}
    </caption>
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积</th>
        <th scope="col">面积占比</th>
        <th scope="col">说明</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
```

```
      {% for u in units %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">{{ u.area if u.area is not none
  else "—" }}</td>
        <td class="num">
          {% if u.ratio is not none %}{{
  "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}
        </td>
        <td style="text-align:left;">{{ u.description or
  "—" }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/unit/edit/{{ u.id
  }}">编辑</a>
            <form method="post" action="/unit/delete/{{
  u.id }}"
                  onsubmit="return confirm('确定删除微地质单元「{{
  u.unit_type }}」吗？\n该单元已录入的指标数据将一并删除，且无法恢复。');">
              {{ form.hidden_tag() }}
              <button type="submit" class="link-btn
  link-btn--danger">删除</button>
            </form>
          </span>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">尚未划分微地质单元。请在下方添加。</p>
  {% endif %}
  <h3>添加微地质单元</h3>
  <form method="post" action="/unit/{{ patch.id }}/add">
    {{ form.hidden_tag() }}
    <div class="actions">
      <label class="field__label" for="add-unit-type"
    style="margin:0;">类型</label>
      <select id="add-unit-type" name="unit_type">
        {% for t in unit_types %}<option value="{{ t }}">{{
    t }}</option>{% endfor %}
      </select>
      <label class="field__label" for="add-unit-area"
    style="margin:0;">面积</label>
      <input id="add-unit-area" type="number" step="any"
    name="area">
      <label class="field__label" for="add-unit-desc"
    style="margin:0;">说明</label>
      <input id="add-unit-desc" type="text"
    name="description" style="width:220px;">
```

```
        <button type="submit" class="btn btn--sm">添加</button>
      </div>
    </form>
    <p class="actions" style="margin-top:20px;">
      <a class="btn btn--sm btn--ghost" href="/project/{{
      patch.project_id }}/patches">返回图斑列表</a>
      <a class="btn btn--sm btn--ghost"
      href="/indicator/">去录入指标数据</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/unit/edit.html ====
{% extends "base.html" %}
{% block title %}编辑微地质单元{% endblock %}
{% block content %}
<div class="card">
    <h2>编辑微地质单元 · {{ patch.name }}</h2>
    <p class="muted">修改面积后，系统会自动重算本图斑各单元的面积占比。</p>
    <form method="post">
      {{ form.hidden_tag() }}
      <div class="field">
        <label class="field__label" for="unit-type">微地质单元类型</l
      abel>
        <select id="unit-type" name="unit_type">
          {% for t in unit_types %}
          <option value="{{ t }}" {% if unit.unit_type == t
      %}selected{% endif %}>{{ t }}</option>
          {% endfor %}
        </select>
      </div>
      <div class="field">
        <label class="field__label" for="unit-area">面积</label>
        <input id="unit-area" type="number" step="any"
      name="area"
               value="{{ unit.area if unit.area is not none
      else '' }}">
      </div>
      <div class="field">
        <label class="field__label" for="unit-desc">说明</label>
        <input id="unit-desc" type="text" name="description"
      style="width:420px;"
               value="{{ unit.description or '' }}">
      </div>
      <p class="actions" style="margin-top:16px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/unit/{{
      unit.patch_id }}">返回</a>
      </p>
    </form>
</div>
```

```
{% endblock %}
# ==== app/templates/standard/index.html ====
{% extends "base.html" %}
{% block title %}政策与标准知识库{% endblock %}
{% block content %}
<div class="card">
  <h2>政策与标准知识库</h2>  <p class="actions"
    style="margin-bottom:14px;">
    <a class="btn btn--sm" href="/standard/add">添加标准</a>
    <a class="btn btn--sm btn--ghost"
    href="/standard/compare">省际口径比对</a>
    <a class="btn btn--sm btn--ghost"
    href="/standard/check">结论对照</a>
  </p>
  <h3>地方标准</h3>
  {% if local %}
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">名称</th>
        <th scope="col">编号</th>
        <th scope="col">规则数</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for s in local %}
      <tr>
        <td>{{ s.province }}</td>
        <td style="text-align:left;">{{ s.name }}</td>
        <td class="mono">{{ s.doc_number or "—" }}</td>
        <td class="num">{{ s.rules|length }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/standard/{{ s.id
    }}/rules">规则</a>
            <a class="link-btn" href="/standard/{{ s.id
    }}/edit">编辑</a>
            <form method="post" action="/standard/{{ s.id
    }}/delete"
                  onsubmit="return confirm('确定删除「{{
    s.province }} {{ s.name }}」吗？\n该标准下的 {{ s.rules|length
    }} 条规则将一并删除，且无法恢复。');">
              {{ form.hidden_tag() }}
              <button type="submit" class="link-btn
    link-btn--danger">删除</button>
            </form>
          </span>
        </td>
```

```
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">暂无地方标准。运行 <span class="mono">python
    seed_demo.py</span> 可载入内置的 12 项标准与政策文件。</p>
  {% endif %}
  <h3>政策文件</h3>
  {% if policy %}
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">名称</th>
        <th scope="col">文号</th>
        <th scope="col">规则数</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for s in policy %}
      <tr>
        <td>{{ s.province }}</td>
        <td style="text-align:left;">{{ s.name }}</td>
        <td class="mono">{{ s.doc_number or "—" }}</td>
        <td class="num">{{ s.rules|length }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/standard/{{ s.id
    }}/rules">规则</a>
            <a class="link-btn" href="/standard/{{ s.id
    }}/edit">编辑</a>
            <form method="post" action="/standard/{{ s.id
    }}/delete"
                  onsubmit="return confirm('确定删除「{{
    s.province }} {{ s.name }}」吗？\n该文件下的 {{ s.rules|length
    }} 条规则将一并删除，且无法恢复。');">
              {{ form.hidden_tag() }}
              <button type="submit" class="link-btn
    link-btn--danger">删除</button>
            </form>
          </span>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">暂无政策文件。</p>
```

```
  {% endif %}
</div>
{% endblock %}
# ==== app/templates/standard/edit.html ====
{% extends "base.html" %}
{% block title %}{{ "编辑标准" if standard else "添加标准" }}{%
    endblock %}
{% block content %}
<div class="card" style="max-width:800px;">
  <h2>{{ "编辑标准" if standard else "添加标准" }}</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="province">省份</label>
      <input id="province" type="text" name="province"
    style="width:160px;" required
             value="{{ standard.province if standard else
    '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="std-name">名称</label>
      <input id="std-name" type="text" name="name"
    style="width:100%;" required
             value="{{ standard.name if standard else ''
    }}">
    </div>
    <div class="field">
      <label class="field__label" for="doc-number">标准编号 /
    文号</label>
      <input id="doc-number" type="text" name="doc_number"
    style="width:320px;"
             value="{{ standard.doc_number or '' if
    standard else '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="category">类别</label>
      <select id="category" name="category">
        {% for c in ["地方标准", "政策文件"] %}
        <option value="{{ c }}" {% if standard and
    standard.category == c %}selected{% endif %}>{{ c
    }}</option>
        {% endfor %}
      </select>
    </div>
    <div class="field">
      <label class="field__label" for="publish-date">发布时间</l
    abel>
      <input id="publish-date" type="text"
    name="publish_date" style="width:200px;"
             placeholder="如 2023-03-30"
```

```
                value="{{ standard.publish_date or '' if
    standard else '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="implement-date">实施时间<
    /label>
      <input id="implement-date" type="text"
    name="implement_date" style="width:200px;"
                placeholder="如 2023-04-30（未明确可留空）"
                value="{{ standard.implement_date or '' if
    standard else '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="summary">特点 /
    判定标准</label>
      <textarea id="summary" name="summary"
    style="min-height:96px;">{{ standard.summary or '' if
    standard else '' }}</textarea>
    </div>
    <p class="actions" style="margin-top:18px;">
      <button type="submit" class="btn">保存</button>
      {% if standard %}
      <form method="post" action="/standard/{{ standard.id
    }}/delete" style="display:inline;"
            onsubmit="return confirm('确定删除「{{
    standard.province }} {{ standard.name
    }}」吗？\n该标准下的规则将一并删除，且无法恢复。');">
        {{ form.hidden_tag() }}
        <button type="submit" class="btn
    btn--danger">删除该标准</button>
      </form>
      {% endif %}
      <a class="btn btn--ghost" href="/standard/">返回</a>
    </p>
  </form>
</div>
{% endblock %}
# ==== app/templates/standard/rules.html ====
{% extends "base.html" %}
{% block title %}结构化规则 · {{ standard.province }}{% endblock
    %}
{% block content %}
<div class="card">
  <h2>结构化规则 · {{ standard.province }} {{ standard.name
    }}</h2>
  {% if rules %}
  <table>
    <caption class="muted">共 {{ rules|length }}
    条规则</caption>
    <thead>
```

```
      <tr>
        <th scope="col">判定项</th>
        <th scope="col">条件</th>
        <th scope="col">条件类型</th>
        <th scope="col">适用分区</th>
        <th scope="col">备注</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rules %}
      <tr>
        <td style="text-align:left;">{{ r.item }}</td>
        <td class="mono nowrap">{{ r.operator }} {{
    r.threshold or "" }}</td>
        <td>{{ r.condition_type }}</td>
        <td>{{ r.region or "—" }}</td>
        <td style="text-align:left;">{{ r.note or "—"
    }}</td>
        <td>
          <form method="post" action="/standard/rule/{{
    r.id }}/delete"
                onsubmit="return confirm('确定删除规则「{{ r.item
    }} {{ r.operator }} {{ r.threshold or '' }}」吗?');">
            {{ form.hidden_tag() }}
            <button type="submit" class="btn btn--sm
    btn--danger">删除</button>
          </form>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">该标准暂无规则。请在下方添加第一条规则。</p>
  {% endif %}
  <h3>添加规则</h3>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="actions">
      <div>
        <label class="field__label" for="r-item">判定项</label>
        <input id="r-item" type="text" name="item"
    placeholder="如：植被覆盖度" required>
      </div>
      <div>
        <label class="field__label" for="r-op">比较符</label>
        <select id="r-op" name="operator">
          {% for o in [">=", ">", "<=", "<", "=="]
    %}<option value="{{ o }}">{{ o }}</option>{% endfor %}
```

```
          </select>
        </div>
        <div>
          <label class="field__label" for="r-thr">阈值</label>
          <input id="r-thr" type="text" name="threshold"
    placeholder="如：50" style="width:110px;">
        </div>
        <div>
          <label class="field__label" for="r-cond">条件类型</label
    >
          <select id="r-cond" name="condition_type">
            <option value="必要条件">必要条件</option>
            <option value="充分条件">充分条件</option>
          </select>
        </div>
        <div>
          <label class="field__label" for="r-region">适用分区</lab
    el>
          <input id="r-region" type="text" name="region"
    placeholder="如宁夏分区" style="width:130px;">
        </div>
        <div>
          <label class="field__label" for="r-note">备注</label>
          <input id="r-note" type="text" name="note"
    style="width:220px;">
        </div>
        <div>
          <button type="submit" class="btn">添加规则</button>
        </div>
      </div>
    </form>
    <p class="actions" style="margin-top:20px;">
      <a class="btn btn--sm btn--ghost"
      href="/standard/">返回标准列表</a>
      <a class="btn btn--sm btn--ghost"
      href="/standard/compare">省际口径比对</a>
    </p>
</div>
{% endblock %}
#  ==== app/templates/standard/compare.html ====
{% extends "base.html" %}
{% block title %}省际口径比对{% endblock %}
{% block content %}
<div class="card">
    <h2>省际口径比对</h2>
    {% if not groups %}
    <p class="muted">
      暂无规则数据。请先在「标准知识库」中为标准录入规则，
      或运行 <span class="mono">python seed_demo.py</span>
      载入内置标准库数据。
```

```
  </p>
  {% else %}
  {% for item, rs in groups.items() %}
  <h3>{{ item }}</h3>
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">标准 / 文件</th>
        <th scope="col">条件</th>
        <th scope="col">条件类型</th>
        <th scope="col">备注</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rs %}
      <tr>
        <td>{{ r.standard.province }}</td>
        <td style="text-align:left;">{{ r.standard.name
    }}</td>
        <td class="mono nowrap">{{ r.operator }} {{
    r.threshold or "" }}</td>
        <td>{{ r.condition_type }}</td>
        <td style="text-align:left;">{{ r.note or "—"
    }}</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endfor %}
  {% endif %}
  <p class="actions" style="margin-top:16px;">
    <a class="btn btn--ghost" href="/standard/">返回标准列表</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/constraint/index.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}限制性指标判定{% endblock %}
{% block content %}
<div class="card">
  <h2>限制性指标判定</h2>
  {{ patch_guide(projects_without_patch) }}
  {% if not patches %}
  <p class="muted">暂无矿山图斑。请先在「项目与矿山档案」中建立图斑。</p>
  {% else %}
  <table>
    <thead>
      <tr>
```

```
            <th scope="col">所属项目</th>                                    9301
            <th scope="col">矿山图斑</th>                                    9302
            <th scope="col">判定结果</th>                                    9303
            <th scope="col">操作</th>                                        9304
          </tr>                                                              9305
        </thead>                                                             9306
        <tbody>                                                              9307
          {% for p in patches %}                                             9308
          {% set r = latest.get(p.id) %}                                     9309
          <tr>                                                               9310
            <td>{{ p.project.name if p.project else "—" }}</td>              9311
            <td style="text-align:left;">{{ p.name }}</td>                   9312
            <td style="text-align:left;">                                    9313
              {% if r %}                                                     9314
                {% if r.passed %}                                            9315
                <span class="tag tag--ok">通过</span>                        9316
                {% else %}                                                   9317
                <span class="tag tag--warn">不通过</span>                    9318
                {% endif %}                                                  9319
                <span class="muted">{{ r.reason }}</span>                    9320
              {% else %}                                                     9321
              <span class="tag tag--muted">未判定</span>                     9322
              {% endif %}                                                    9323
            </td>                                                            9324
            <td class="nowrap">                                              9325
              <span class="row-actions">                                     9326
                <a class="link-btn" href="/constraint/{{ p.id                9327
        }}">执行判定</a>                                                     9328
              </span>                                                        9329
            </td>                                                            9330
          </tr>                                                              9331
          {% endfor %}                                                       9332
        </tbody>                                                             9333
      </table>                                                               9334
      {% endif %}                                                            9335
    </div>                                                                   9336
    {% endblock %}                                                           9337
    # ==== app/templates/constraint/check.html ====                          9338
    {% extends "base.html" %}                                                9339
    {% block title %}限制性条件判定 · {{ patch.name }}{% endblock %}         9340
    {% block content %}                                                      9341
    <div class="card" style="max-width:760px;">                              9342
      <h2>限制性条件判定 · {{ patch.name }}</h2>                             9343
      <form method="post">                                                   9344
        {{ form.hidden_tag() }}                                              9345
        {% for field, name, val in field_states %}                           9346
        <fieldset style="border:1px solid                                    9347
        var(--line);border-radius:6px;padding:10px                           9348
        14px;margin:0 0 10px;">                                              9349
          <legend style="padding:0 6px;font-weight:600;">{{                  9350
```

```
      name }}</legend>                                                    9351
        <div class="actions">                                             9352
          <label class="checkbox">                                        9353
            <input type="radio" name="{{ field }}" value="1"              9354
                   {% if val is none or val %}checked{% endif             9355
      %}>                                                                 9356
            <span>通过</span>                                             9357
          </label>                                                        9358
          <label class="checkbox">                                        9359
            <input type="radio" name="{{ field }}" value="0"              9360
                   {% if val is not none and not val                      9361
      %}checked{% endif %}>                                               9362
            <span>不通过</span>                                           9363
          </label>                                                        9364
        </div>                                                            9365
      </fieldset>                                                         9366
      {% endfor %}                                                        9367
      <p class="actions" style="margin-top:16px;">                        9368
        <button type="submit" class="btn">保存判定</button>               9369
        <a class="btn btn--ghost" href="/constraint/">返回</a>            9370
      </p>                                                                9371
    </form>                                                               9372
    {% if record %}                                                       9373
    <div style="margin-top:18px;padding:14px;border-radius:6px9374
      ;border-left:4px solid;                                             9375
                 {% if record.passed %}background:var(--ok-soft9376
      );border-left-color:var(--ok);                                      9377
                 {% else %}background:var(--danger-soft);border9378
      -left-color:var(--danger);{% endif %}">                             9379
      <p style="margin:0 0 4px;font-weight:600;">                         9380
        判定结论：                                                        9381
        {% if record.passed %}                                            9382
        <span class="ok">通过，可进入定量评价</span>                      9383
        {% else %}                                                        9384
        <span class="warn">不通过 —— 建议生态重建或人工辅助</span>        9385
        {% endif %}                                                       9386
      </p>                                                                9387
      <p class="muted" style="margin:0;">{{ record.reason                 9388
      }}</p>                                                              9389
    </div>                                                                9390
    {% endif %}                                                           9391
</div>                                                                    9392
{% endblock %}                                                            9393
# ==== app/templates/ahp/index.html ====                                  9394
{% extends "base.html" %}                                                 9395
{% block title %}AHP 权重计算{% endblock %}                               9396
{% block content %}                                                       9397
<div class="card" style="max-width:720px;">                               9398
  <h2>AHP 权重计算 · 选择指标层级</h2>                                    9399
  <form method="post">                                                    9400
```

```
      {{ form.hidden_tag() }}                                                  9401
      <div class="field">                                                      9402
        <label class="field__label" for="{{ form.unit_type.id                  9403
      }}">微地质单元类型</label>                                               9404
        {{ form.unit_type(id=form.unit_type.id) }}                             9405
      </div>                                                                   9406
      <div class="field">                                                      9407
        <label class="field__label" for="{{ form.level.id                      9408
      }}">指标层级</label>                                                     9409
        {{ form.level(id=form.level.id) }}                                     9410
      </div>                                                                   9411
      <div class="field">                                                      9412
        <label class="field__label" for="{{ form.l1.id }}">                    9413
          一级指标（层级选「二级指标」时需选择）                               9414
        </label>                                                               9415
        {{ form.l1(id=form.l1.id) }}                                           9416
      </div>                                                                   9417
      <p class="actions" style="margin-top:16px;">                             9418
        {{ form.submit(class_="btn") }}                                        9419
        <a class="btn btn--ghost" href="/ahp/indicators">指标定义维            9420
      护</a>                                                                   9421
        <a class="btn btn--ghost" href="/ahp/experts">多专家矩阵</a            9422
      >                                                                        9423
      </p>                                                                     9424
    </form>                                                                    9425
</div>                                                                         9426
<div class="card">                                                             9427
    <h2>各微地质单元当前生效权重</h2>                                          9428
    <table>                                                                    9429
      <thead>                                                                  9430
        <tr>                                                                   9431
          <th scope="col">微地质单元</th>                                      9432
          <th scope="col">当前权重来源</th>                                    9433
          <th scope="col">操作</th>                                            9434
        </tr>                                                                  9435
      </thead>                                                                 9436
      <tbody>                                                                  9437
        {% for r in weight_rows %}                                             9438
        <tr>                                                                   9439
          <td style="text-align:left;">{{ r.unit_type }}</td>                  9440
          <td style="text-align:left;">                                        9441
            {% if r.custom %}                                                  9442
            <span class="tag tag--warn">{{ r.label }}</span>                   9443
            {% else %}                                                         9444
            <span class="tag tag--muted">{{ r.label }}</span>                  9445
            {% endif %}                                                        9446
          </td>                                                                9447
          <td class="nowrap">                                                  9448
            <span class="row-actions">                                         9449
              <a class="link-btn" href="/ahp/matrix?unit_type=                 9450
```

```
      {{ r.unit_type }}&level=1">一级指标权重</a>                                    9451
                <a class="link-btn" href="/ahp/matrix?unit_type=                    9452
      {{ r.unit_type }}&level=1&calc=1">查看一致性</a>                               9453
                {% if r.custom %}                                                   9454
                <form method="post" action="/ahp/weights/reset"                     9455
                      onsubmit="return confirm('确定把「{{                          9456
      r.unit_type }}」恢复为系统默认权重吗？');">                                    9457
                  <input type="hidden" name="csrf_token"                            9458
      value="{{ csrf_token() }}">                                                   9459
                  <input type="hidden" name="unit_type"                             9460
      value="{{ r.unit_type }}">                                                    9461
                  <button type="submit" class="link-btn                             9462
      link-btn--danger">恢复默认</button>                                           9463
                </form>                                                             9464
                {% endif %}                                                         9465
              </span>                                                               9466
            </td>                                                                   9467
          </tr>                                                                     9468
          {% endfor %}                                                              9469
        </tbody>                                                                    9470
      </table>                                                                      9471
</div>                                                                              9472
{% endblock %}                                                                      9473
# ==== app/templates/ahp/matrix.html ====                                           9474
{% extends "base.html" %}                                                           9475
{% block title %}判断矩阵录入{% endblock %}                                          9476
{% block content %}                                                                 9477
<div class="card">                                                                  9478
  <h2>                                                                              9479
    判断矩阵录入                                                                    9480
    · {{ unit_type }}                                                               9481
    · {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{%                   9482
    endif %}                                                                        9483
  </h2>                                                                             9484
  <p class="muted">                                                                 9485
    采用 1–9 标度：1 同等重要、3 稍微重要、5 明显重要、7 强烈重要、9 极端重要，    9486
    2、4、6、8 为相邻判断的中间值。只需录入上三角，对角线为 1，下三角自动取倒数。  9487
    矩阵已按系统默认权重预填。                                                      9488
  </p>                                                                              9489
  <form method="post">                                                              9490
    {{ form.hidden_tag() }}                                                         9491
    <table style="width:auto;">                                                     9492
      <thead>                                                                       9493
        <tr>                                                                        9494
          <th></th>                                                                 9495
          {% for name in names %}<th scope="col">{{ name                            9496
    }}</th>{% endfor %}                                                             9497
        </tr>                                                                       9498
      </thead>                                                                      9499
      <tbody>                                                                       9500
```

```
          {% for i in range(names|length) %}                                  9501
          <tr>                                                                9502
            <th scope="row">{{ names[i] }}</th>                               9503
            {% for j in range(names|length) %}                                9504
              {% if i == j %}                                                 9505
              <td class="muted num">1</td>                                    9506
              {% elif i < j %}                                                9507
              <td>                                                            9508
                <label class="muted" style="position:absolute;                9509
    left:-9999px;"                                                            9510
                       for="m-{{ i }}-{{ j }}">{{ names[i] }}                 9511
    相对 {{ names[j] }} 的重要性</label>                                        9512
                <input id="m-{{ i }}-{{ j }}" type="number"                   9513
    step="0.01" min="0.01"                                                    9514
                       name="m_{{ i }}_{{ j }}" value="{{                     9515
    default[i][j] }}">                                                        9516
              </td>                                                           9517
              {% else %}                                                      9518
              <td class="muted num">{{ default[i][j] }}</td>                  9519
              {% endif %}                                                     9520
            {% endfor %}                                                      9521
          </tr>                                                               9522
          {% endfor %}                                                        9523
        </tbody>                                                              9524
    </table>                                                                  9525
    <p class="actions">                                                       9526
      <button type="submit" class="btn" name="action"                         9527
    value="calc">计算权重</button>                                              9528
      {% if result and result.consistent %}                                   9529
      <button type="submit" class="btn" name="action"                         9530
    value="save">保存并生效</button>                                            9531
      {% endif %}                                                             9532
      <a class="btn btn--ghost" href="/ahp/">重新选择</a>                       9533
    </p>                                                                      9534
</form>                                                                       9535
<p class="muted" style="margin-top:10px;">                                    9536
    当前「{{ unit_type }}」生效权重：<strong>{{ source_label                  9537
    }}</strong>。                                                             9538
    点击「保存并生效」后，本单元类型的适宜性评价将采用此处算出的权重；                  9539
    如需回到系统默认值，可在 <a href="/ahp/">权重首页</a> 恢复默认。                 9540
</p>                                                                          9541
{% if result %}                                                               9542
<hr style="margin:22px 0;border:none;border-top:1px solid                     9543
    var(--line);">                                                            9544
<h3>计算结果</h3>                                                             9545
<table style="width:auto;">                                                   9546
    <thead>                                                                   9547
      <tr>                                                                    9548
        <th scope="col">指标</th>                                             9549
        {% for name in names %}<th scope="col">{{ name                        9550
```

```
    }}</th>{% endfor %}                                                    9551
      </tr>                                                                9552
    </thead>                                                               9553
    <tbody>                                                                9554
      <tr>                                                                 9555
        <th scope="row">权重</th>                                          9556
        {% for w in result.weights %}<td class="mono">{{ w                 9557
    }}</td>{% endfor %}                                                    9558
      </tr>                                                                9559
    </tbody>                                                               9560
  </table>                                                                 9561
  <div class="stat-row">                                                   9562
    <div>                                                                  9563
      <div class="stat__label">最大特征值 λmax</div>                       9564
      <div class="stat__value mono">{{ result.lambda_max                   9565
    }}</div>                                                               9566
    </div>                                                                 9567
    <div>                                                                  9568
      <div class="stat__label">一致性指标 CI</div>                         9569
      <div class="stat__value mono">{{                                     9570
    "%.4f"|format(result.ci) }}</div>                                      9571
    </div>                                                                 9572
    <div>                                                                  9573
      <div class="stat__label">一致性比例 CR</div>                         9574
      <div class="stat__value mono">{{                                     9575
    "%.4f"|format(result.cr) }}</div>                                      9576
    </div>                                                                 9577
    <div>                                                                  9578
      <div class="stat__label">一致性检验（要求 CR &lt; 0.1）</div>        9579
      <div class="stat__value">                                            9580
        {% if result.consistent %}<span class="ok">通过</span>             9581
        {% else %}<span class="warn">不通过</span>{% endif %}              9582
      </div>                                                               9583
    </div>                                                                 9584
  </div>                                                                   9585
  {% if not result.consistent and result.most_inconsistent                 9586
    %}                                                                     9587
  <p style="margin-top:14px;padding:12px;border-radius:6px;b               9588
    ackground:var(--danger-soft);                                          9589
              border-left:4px solid var(--danger);">                       9590
    <strong class="warn">建议优先调整：</strong>                           9591
    第 {{ result.most_inconsistent.i + 1 }} 行、第 {{                      9592
    result.most_inconsistent.j + 1 }} 列的比较                             9593
    （当前 <span class="mono num">{{ result.most_inconsistent.v            9594
    alue|round(3) }}</span>，                                              9595
    理想约 <span class="mono num">{{ result.most_inconsistent.e            9596
    xpected|round(3) }}</span>）                                           9597
  </p>                                                                     9598
  {% endif %}                                                              9599
  {% endif %}                                                              9600
```

```
</div>
{% endblock %}
# ==== app/templates/ahp/indicators.html ====
{% extends "base.html" %}
{% block title %}指标定义维护{% endblock %}
{% block content %}
<div class="card">
  <h2>指标体系维护</h2>
  <p class="actions" style="margin-bottom:14px;">
    <a class="btn btn--sm btn--ghost"
    href="/ahp/">返回权重计算</a>
    <a class="btn btn--sm btn--ghost"
    href="/ahp/experts">多专家矩阵</a>
    <form method="post" action="/ahp/indicators/reset"
    style="display:inline;"
          onsubmit="return confirm('将按系统默认值重置全部指标定义，自定义修改会丢失
    。确定继续？');">
      {{ form.hidden_tag() }}
      <button type="submit" class="btn btn--sm
    btn--danger">按系统默认值重置</button>
    </form>
  </p>
  {% for level, title in [(1, "一级指标"), (2, "二级指标"), (3,
    "三级指标")] %}
  {% set level_items = all_ind|selectattr("level",
    "equalto", level)|list %}
  {% if level_items %}
  <h3>{{ title }}（{{ level_items|length }} 项）</h3>
  <table>
    <thead>
      <tr>
        <th scope="col">指标</th>
        <th scope="col">上级指标</th>
        <th scope="col">单位</th>
        <th scope="col">标准化类型</th>
        <th scope="col">理想取值</th>
        <th scope="col">启用单元</th>
        <th scope="col">限制性</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for i in level_items %}
      <tr>
        <td style="text-align:left;">
          <span class="mono muted">{{ i.code
    }}</span><br>{{ i.name }}
        </td>
        <td>{{ (all_ind|selectattr("code", "equalto",
    i.parent_code)|list|first).name if i.parent_code else
```

```
      "—" }}</td>                                                    9651
          <td>{{ i.unit or "—" }}</td>                               9652
          <td>{{ type_label(i.ideal_type) }}</td>                    9653
          <td class="mono">{{ i.ideal_value or "—" }}</td>           9654
          <td style="text-align:left;">{{ i.enabled_units or         9655
      "—" }}</td>                                                    9656
          <td>                                                       9657
            {% if i.is_constraint %}<span class="tag                 9658
      tag--warn">是</span>                                           9659
            {% else %}<span class="tag tag--muted">否</span>{%       9660
      endif %}                                                       9661
          </td>                                                      9662
          <td class="nowrap">                                        9663
            <span class="row-actions">                               9664
              <a class="link-btn" href="/ahp/indicators/{{           9665
      i.code }}/edit">编辑</a>                                       9666
            </span>                                                  9667
          </td>                                                      9668
        </tr>                                                        9669
        {% endfor %}                                                 9670
      </tbody>                                                       9671
    </table>                                                         9672
    {% endif %}                                                      9673
    {% endfor %}                                                     9674
</div>                                                               9675
{% endblock %}                                                       9676
# ==== app/templates/ahp/indicator_edit.html ====                    9677
{% extends "base.html" %}                                            9678
{% block title %}编辑指标 · {{ ind.name }}{% endblock %}             9679
{% block content %}                                                  9680
<div class="card" style="max-width:680px;">                          9681
  <h2>编辑指标 · {{ ind.name }}</h2>                                 9682
  <p class="muted">                                                  9683
    指标编码 <span class="mono">{{ ind.code }}</span>、层级 {{       9684
    ind.level }} 级，                                                9685
    由系统指标体系确定，不可修改。                                   9686
  </p>                                                               9687
  <form method="post">                                               9688
    {{ form.hidden_tag() }}                                          9689
    <div class="field">                                              9690
      <label class="field__label" for="name">指标名称</label>        9691
      <input id="name" type="text" name="name"                       9692
    style="width:100%;" required                                     9693
             value="{{ ind.name }}">                                 9694
    </div>                                                           9695
    <div class="field">                                              9696
      <label class="field__label" for="unit">计量单位</label>        9697
      <input id="unit" type="text" name="unit"                       9698
    style="width:180px;"                                             9699
             value="{{ ind.unit or '' }}">                           9700
```

```
</div>
<div class="field">
  <label class="field__label" for="ideal-type">标准化类型</label>
  <select id="ideal-type" name="ideal_type">
    {% for code, label in [("benefit","越大越优"),("cost","越小越优"),("mid","适中型"),("discrete","离散赋值")] %}
    <option value="{{ code }}" {% if ind.ideal_type == code %}selected{% endif %}>{{ label }}</option>
    {% endfor %}
  </select>
</div>
<div class="field">
  <label class="field__label" for="ideal-value">理想取值</label>
  <input id="ideal-value" type="text" name="ideal_value" style="width:100%;"
         value="{{ ind.ideal_value or '' }}">
  <p class="muted" style="margin:4px 0 0;">
    如土石比 0.8、土层厚度 20、pH 5~9、植被覆盖度 75。
  </p>
</div>
<fieldset style="border:1px solid var(--line);border-radius:6px;padding:10px 14px;margin:0 0 12px;">
  <legend style="padding:0 6px;font-weight:600;">启用该指标的微地质单元</legend>
  {% for t in unit_types %}
  <label class="checkbox" style="margin-right:18px;">
    <input type="checkbox" name="enabled_units" value="{{ t }}"
           {% if t in ind.units_list() %}checked{% endif %}>
    <span>{{ t }}</span>
  </label>
  {% endfor %}
  <p class="muted" style="margin:8px 0 0;">
    系统设定：岩质边坡不启用土壤指标，平台不启用地形指标。
  </p>
</fieldset>
<div class="field">
  <label class="checkbox" for="is-constraint">
    <input id="is-constraint" type="checkbox" name="is_constraint" value="1"
           {% if ind.is_constraint %}checked{% endif %}>
    <span>同时作为限制性指标（参与一票否决判定）</span>
  </label>
</div>
<div class="field">
  <label class="field__label" for="description">取值依据说明</
```

```
      label>                                                                    9751
        <textarea id="description" name="description"                           9752
      style="min-height:88px;">{{ ind.description or ''                         9753
      }}</textarea>                                                             9754
      </div>                                                                    9755
      <p class="actions" style="margin-top:16px;">                              9756
        <button type="submit" class="btn">保存</button>                         9757
        <a class="btn btn--ghost" href="/ahp/indicators">返回指标列9758
      表</a>                                                                    9759
      </p>                                                                      9760
    </form>                                                                     9761
</div>                                                                          9762
{% endblock %}                                                                  9763
# ==== app/templates/ahp/experts.html ====                                      9764
{% extends "base.html" %}                                                       9765
{% block title %}多专家判断矩阵{% endblock %}                                    9766
{% block content %}                                                             9767
<div class="card">                                                              9768
  <h2>多专家判断矩阵 · 录入与汇总</h2>                                          9769
  <form method="get" class="actions" style="margin-bottom:14 9770
    px;">                                                                       9771
      <div>                                                                     9772
        <label class="field__label" for="unit-type">微地质单元类型</l9773
      abel>                                                                     9774
        <select id="unit-type" name="unit_type"                                 9775
      onchange="this.form.submit()">                                            9776
          {% for t in ["岩质边坡", "碎石边坡", "平台"] %}                       9777
          <option value="{{ t }}" {% if unit_type == t                          9778
      %}selected{% endif %}>{{ t }}</option>                                    9779
          {% endfor %}                                                          9780
        </select>                                                               9781
      </div>                                                                    9782
      <div>                                                                     9783
        <label class="field__label" for="level">指标层级</label>               9784
        <select id="level" name="level"                                         9785
      onchange="this.form.submit()">                                            9786
          <option value="1" {% if level == 1 %}selected{%                       9787
      endif %}>一级指标</option>                                                9788
          <option value="2" {% if level == 2 %}selected{%                       9789
      endif %}>二级指标</option>                                                9790
        </select>                                                               9791
      </div>                                                                    9792
  </form>                                                                       9793
  <p class="muted">                                                             9794
    当前范围：<strong>{{ unit_type }}</strong> ·                                9795
    {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif          9796
    %}。                                                                        9797
    若选二级指标，请先到                                                        9798
    <a href="/ahp/">权重计算</a> 选择一级指标后进入本页。                       9799
  </p>                                                                          9800
```

```
{% if rows %}
<table>
  <caption class="muted">已录入 {{ rows|length }} 位专家的矩阵</caption>
  <thead>
    <tr>
      <th scope="col">专家</th>
      <th scope="col">专业</th>
      <th scope="col">矩阵规模</th>
      <th scope="col">该矩阵 CR</th>
      <th scope="col">一致性</th>
      <th scope="col">录入时间</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for r in rows %}
    <tr>
      <td>{{ r.expert_name }}</td>
      <td>{{ r.specialty or "—" }}</td>
      <td class="num">{{ n }} × {{ n }}</td>
      <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is not none else "—" }}</td>
      <td>
        {% if r.cr is not none and r.cr < 0.1 %}<span class="tag tag--ok">通过</span>{% else %}<span class="tag tag--warn">不通过</span>{% endif %}
      </td>
      <td class="mono nowrap">{{ r.created_at | localdt }}</td>
      <td>
        <form method="post" action="/ahp/experts/{{ r.id }}/delete"
              onsubmit="return confirm('确定删除专家「{{ r.expert_name }}」的矩阵吗？');">
          {{ form.hidden_tag() }}
          <button type="submit" class="btn btn--sm btn--danger">删除</button>
        </form>
      </td>
    </tr>
    {% endfor %}
  </tbody>
</table>
<p class="actions" style="margin-bottom:10px;">
  <a class="btn" href="/ahp/experts/aggregate?unit_type={{ unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}">
    汇总计算（几何平均）
```

```
    </a>                                                                    9851
</p>                                                                        9852
{% else %}                                                                  9853
<p class="muted">尚未录入任何专家矩阵。请在下方录入第一位专家的判断矩阵。</p>          9854
{% endif %}                                                                 9855
<h3>录入一位专家的判断矩阵</h3>                                               9856
<form method="post">                                                        9857
    {{ form.hidden_tag() }}                                                 9858
    <input type="hidden" name="action" value="add">                         9859
    <input type="hidden" name="unit_type" value="{{                         9860
    unit_type }}">                                                          9861
    <input type="hidden" name="level" value="{{ level }}">                  9862
    {% if l1 %}<input type="hidden" name="l1" value="{{ l1                  9863
    }}">{% endif %}                                                         9864
    <div style="display:flex; gap:16px;">                                   9865
      <div class="field">                                                   9866
        <label class="field__label" for="expert-name">专家姓名</          9867
    label>                                                                  9868
        <input id="expert-name" type="text"                                 9869
    name="expert_name"                                                      9870
               style="width:180px;" placeholder="如：张工">                 9871
      </div>                                                                9872
      <div class="field">                                                   9873
        <label class="field__label" for="expert-specialty">专            9874
    家专业（选填）</label>                                                    9875
        <input id="expert-specialty" type="text"                            9876
    name="specialty"                                                        9877
               style="width:220px;" placeholder="如：地质/生态/土壤"        9878
    >                                                                       9879
      </div>                                                                9880
    </div>                                                                  9881
    <table style="width:auto;">                                             9882
      <thead>                                                               9883
        <tr>                                                                9884
          <th></th>                                                         9885
          {% for name in names %}<th scope="col">{{ name                    9886
    }}</th>{% endfor %}                                                     9887
        </tr>                                                               9888
      </thead>                                                              9889
      <tbody>                                                               9890
        {% for i in range(n) %}                                             9891
        <tr>                                                                9892
          <th scope="row">{{ names[i] }}</th>                               9893
          {% for j in range(n) %}                                           9894
            {% if i == j %}                                                 9895
            <td class="muted num">1</td>                                    9896
            {% elif i < j %}                                                9897
            <td>                                                            9898
              <label class="muted" style="position:absolute;                9899
    left:-9999px;"                                                          9900
```

```
                            for="e-{{ i }}-{{ j }}">{{ names[i] }}
      相对 {{ names[j] }} 的重要性</label>
                  <input id="e-{{ i }}-{{ j }}" type="number"
      step="0.01" min="0.01"
                            name="m_{{ i }}_{{ j }}" value="{{
      default[i][j] }}">
                </td>
                {% else %}
                <td class="muted num">{{ default[i][j] }}</td>
                {% endif %}
              {% endfor %}
            </tr>
            {% endfor %}
          </tbody>
      </table>
      <p class="actions" style="margin-top:12px;">
        <button type="submit" class="btn">录入该专家矩阵</button>
        <a class="btn btn--ghost" href="/ahp/indicators">指标维护</
      a>
        <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/ahp/experts_aggregate.html ====
{% extends "base.html" %}
{% block title %}多专家矩阵汇总{% endblock %}
{% block content %}
<div class="card">
    <h2>多专家矩阵汇总 · {{ unit_type }} · {% if level == 1 %}一级指标{%
      else %}二级指标（{{ l1 }}）{% endif %}</h2>
    {% if not result %}
    <p class="muted">
      该范围内尚未录入专家矩阵。
      <a href="/ahp/experts?unit_type={{ unit_type
      }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif
      %}">去录入</a>
    </p>
    {% else %}
    <div class="stat-row">
      <div>
        <div class="stat__label">参与汇总的专家</div>
        <div class="stat__value">{{ result.count }} 位</div>
      </div>
      <div>
        <div class="stat__label">λmax</div>
        <div class="stat__value mono">{{ result.lambda_max
      }}</div>
      </div>
      <div>
```

```
      <div class="stat__label">CI</div>
      <div class="stat__value mono">{{
  "%.4f"|format(result.ci) }}</div>
  </div>
  <div>
      <div class="stat__label">CR（要求 &lt; 0.1）</div>
      <div class="stat__value mono">{{
  "%.4f"|format(result.cr) }}</div>
  </div>
  <div>
      <div class="stat__label">一致性检验</div>
      <div class="stat__value">
        {% if result.consistent %}<span class="ok">通过</span>
        {% else %}<span class="warn">不通过</span>{% endif %}
      </div>
  </div>
</div>
<h3>汇总后权重</h3>
<table style="width:auto;">
  <thead>
    <tr>
      <th scope="col">指标</th>
      {% for name in names %}<th scope="col">{{ name
  }}</th>{% endfor %}
    </tr>
  </thead>
  <tbody>
    <tr>
      <th scope="row">权重</th>
      {% for w in result.weights %}<td class="mono">{{ w
  }}</td>{% endfor %}
    </tr>
  </tbody>
</table>
<h3>各专家矩阵的一致性</h3>
<table style="width:auto;">
  <thead>
    <tr><th scope="col">专家</th><th scope="col">该矩阵
  CR</th><th scope="col">判定</th></tr>
  </thead>
  <tbody>
    {% for r in rows %}
    <tr>
      <td>{{ r.expert_name }}</td>
      <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is
  not none else "—" }}</td>
      <td>
        {% if r.cr is not none and r.cr < 0.1 %}<span
  class="ok">通过</span>
        {% else %}<span class="warn">不通过</span>{% endif %}
```

```
            </td>                                                    10001
          </tr>                                                      10002
          {% endfor %}                                               10003
        </tbody>                                                     10004
      </table>                                                       10005
      {% if not result.consistent and result.most_inconsistent       10006
        %}                                                           10007
      <p style="padding:12px;border-radius:6px;background:var(--     10008
        danger-soft);                                                10009
                 border-left:4px solid var(--danger);">              10010
        <strong class="warn">汇总矩阵一致性不通过，建议优先调整：</strong>  10011
        第 {{ result.most_inconsistent.i + 1 }} 行、第 {{            10012
        result.most_inconsistent.j + 1 }} 列的比较                   10013
        （当前 <span class="mono">{{ result.most_inconsistent.value  10014
        |round(3) }}</span>，                                        10015
        理想约 <span class="mono">{{ result.most_inconsistent.exped  10016
        ted|round(3) }}</span>）。                                   10017
        该偏差通常来自个别专家打分，可先核对一致性较差的专家矩阵。          10018
      </p>                                                           10019
      {% endif %}                                                    10020
      {% endif %}                                                    10021
      <p class="actions" style="margin-top:16px;">                   10022
        <form method="post"                                          10023
              action="/ahp/experts?unit_type={{ unit_type            10024
        }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}"      10025
              style="display:inline;"                                10026
              onsubmit="return confirm('保存后，「{{ unit_type       10027
        }}」的适宜性评价将改用这组汇总权重计算。确定继续吗？');">          10028
          {{ form.hidden_tag() }}                                    10029
          <input type="hidden" name="action" value="save_agg">       10030
          <button type="submit" class="btn" {% if not                10031
        result.consistent %}disabled{% endif %}>                     10032
            保存为生效权重                                             10033
          </button>                                                  10034
        </form>                                                      10035
        <a class="btn btn--ghost" href="/ahp/experts?unit_type={     10036
        { unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1         10037
        }}{% endif %}">返回录入</a>                                  10038
        <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>          10039
      </p>                                                           10040
</div>                                                               10041
{% endblock %}                                                       10042
# ==== app/templates/indicator/index.html ====                       10043
{% extends "base.html" %}                                            10044
{% block title %}指标数据采集{% endblock %}                              10045
{% block content %}                                                  10046
<div class="card">                                                   10047
  <h2>指标数据采集与管理</h2>                                            10048
  <p class="actions" style="margin-bottom:12px;">                    10049
    <a class="btn btn--sm" href="/indicator/import">批量导入（CSV      10050
```

```
    /  Excel）</a>
</p>
{% if not patches %}
<p class="muted">
    暂无矿山图斑。请先在「项目与矿山档案」中建立图斑，
    或运行 <span class="mono">python seed_demo.py</span>
    载入示例案例。
</p>
{% else %}
{% for p in patches %}
<div class="patch-panel">
    <div class="patch-panel__head">
        <h3 class="patch-panel__name">{{ p.name }}</h3>
        <span class="patch-panel__meta">
            所属项目：{{ p.project.name if p.project else "—" }}
        </span>
        <span class="patch-panel__meta">
            微地质单元：{{ p.units.count() }} 个
        </span>
        {% if p.units.count() %}
        <span class="patch-panel__meta patch-panel__tail">
            <a class="link-btn" href="/indicator/import">批量导入本图斑
    数据</a>
        </span>
        {% endif %}
    </div>
    {% if not p.units.count() %}
    <div class="notice">
        <p class="notice__title">该图斑尚未划分微地质单元，暂不能录入指标数据。</p>
        <p style="margin:0;">
            指标数据是按微地质单元（岩质边坡 / 碎石边坡 / 平台）分别组织的，
            请先完成划分，再回到本页录入。
        </p>
        <p style="margin:10px 0 0;">
            <a class="btn btn--sm" href="/unit/{{ p.id
    }}">去划分微地质单元</a>
        </p>
    </div>
    {% else %}
    <table>
        <thead>
            <tr>
                <th scope="col">微地质单元</th>
                <th scope="col">面积占比</th>
                <th scope="col">数据状态</th>
                <th scope="col">操作</th>
            </tr>
        </thead>
        <tbody>
            {% for u in p.units %}
```

```
            <tr>                                                                10101
              <td>{{ u.unit_type }}</td>                                        10102
              <td class="num">                                                  10103
                {% if u.ratio is not none %}{{                                  10104
      "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}                   10105
              </td>                                                             10106
              <td>                                                              10107
                {% if u.indicator_data %}                                       10108
                <span class="tag tag--ok">已录入</span>                          10109
                {% else %}                                                      10110
                <span class="tag tag--muted">未录入</span>                       10111
                {% endif %}                                                     10112
              </td>                                                             10113
              <td class="nowrap">                                               10114
                <span class="row-actions">                                      10115
                  <a class="link-btn" href="/indicator/{{ u.id                  10116
      }}">录入 / 编辑</a>                                                       10117
                </span>                                                         10118
              </td>                                                             10119
            </tr>                                                               10120
            {% endfor %}                                                        10121
          </tbody>                                                              10122
        </table>                                                                10123
        {% endif %}                                                             10124
    </div>                                                                      10125
    {% endfor %}                                                                10126
    {% endif %}                                                                 10127
</div>                                                                          10128
{% endblock %}                                                                  10129
# ==== app/templates/indicator/edit.html ====                                   10130
{% extends "base.html" %}                                                       10131
{% block title %}指标数据录入 · {{ unit.unit_type }}{% endblock %}              10132
{% block content %}                                                             10133
<div class="card" style="max-width:820px;">                                     10134
    <h2>指标数据录入 · {{ unit.unit_type }}</h2>                                10135
    <p class="muted">                                                           10136
      除标注「可选」的项外，本页字段均为必填：填齐后评价才能执行，缺项会在评价页逐条列出。   10137
      <br>标注「未检测」的重金属可以留空，按方法约定以「未受污染」计入污染指标，          10138
      评价结果与导出报告里会如实写明是哪几项未检测 —— 它不等于实测达标。              10139
      岩质边坡不设土壤指标、平台不设地形指标，系统已按单元类型隐藏不相关字段。          10140
    </p>                                                                        10141
    <form method="post">                                                        10142
      {{ form.hidden_tag() }}                                                   10143
      {% for group_name, fields, has_hm in groups %}                            10144
      <h3>{{ group_name }}</h3>                                                 10145
      {% for fname, cname, unit_ in fields %}                                   10146
      <div class="field">                                                       10147
        <label class="field__label" for="f-{{ fname }}">                        10148
          {{ cname }}{% if unit_ %} ({{ unit_ }}) {% endif %}                   10149
        </label>                                                                10150
```

```
    {% if fname == "aspect" %}
    <select id="f-{{ fname }}" name="aspect">
      <option value="">— 不适用 —</option>
      {% for a in aspect_types %}
      <option value="{{ a }}" {% if current.get("aspect")
== a %}selected{% endif %}>{{ a }}</option>
      {% endfor %}
    </select>
    {% else %}
    <input id="f-{{ fname }}" type="number" step="any"
name="{{ fname }}"
           value="{{ current.get(fname) if
current.get(fname) is not none else '' }}">
    {% endif %}
</div>
{% endfor %}
{% endfor %}
{% for group_name, fields, has_hm in groups %}
    {% if has_hm %}
    <h3>重金属含量（mg/kg）<span class="muted">（可选，未检测可留空）</span>
</h3>
    <p class="muted">未检测的金属可留空：按方法约定以「未受污染」计入污染指标；
      评价结果页与导出报告中会写明未检测的是哪几项，避免被误读成实测达标。</p>
    <table style="width:auto;">
      <thead>
        <tr>
          {% for m in heavy_metals %}<th scope="col">{{ m
}}</th>{% endfor %}
        </tr>
      </thead>
      <tbody>
        <tr>
          {% for m in heavy_metals %}
          <td>
            <label class="muted" style="position:absolute;
left:-9999px;"
                   for="hm-{{ m }}">{{ m }} 含量</label>
            <input id="hm-{{ m }}" type="number"
step="any" name="hm_{{ m }}"
                   value="{{ current.get('heavy',
{}).get(m) if current.get('heavy', {}).get(m) is not
none else '' }}">
          </td>
          {% endfor %}
        </tr>
      </tbody>
    </table>
    {% endif %}
{% endfor %}
<p class="actions" style="margin-top:18px;">
```

```
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/indicator/">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/indicator/import.html ====
{% extends "base.html" %}
{% block title %}批量导入指标数据{% endblock %}
{% block content %}
<div class="card" style="max-width:860px;">
  <h2>批量导入指标数据</h2>
  <p class="muted">
    支持 CSV 与 Excel（.xlsx/.xls）。文件每行对应一个微地质单元，
    系统按「单元类型」匹配图斑内已划分的单元后写入数据。
    先下载模板按格式填写，可减少出错。
  </p>
  <p class="actions" style="margin-bottom:14px;">
    <a class="btn btn--sm btn--ghost"
    href="/indicator/import/template">下载导入模板（CSV）</a>
    <a class="btn btn--sm btn--ghost"
    href="/indicator/">返回指标数据</a>
  </p>
  <form method="post" enctype="multipart/form-data">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="patch-id">目标矿山图斑</label
      >
      <select id="patch-id" name="patch_id" required>
        <option value="">— 请选择 —</option>
        {% for p in patches %}
        <option value="{{ p.id }}">{{ p.name }}（{{
    p.units.count() }} 个单元）</option>
        {% endfor %}
      </select>
    </div>
    <div class="field">
      <label class="field__label" for="datafile">数据文件</label
      >
      <input id="datafile" type="file" name="datafile"
             accept=".csv,.xlsx,.xls" required>
    </div>
    <p class="actions" style="margin-top:16px;">
      <button type="submit" class="btn">开始导入</button>
    </p>
  </form>
  <h3>模板列说明</h3>
  <p class="muted">
    共 {{ headers|length }} 列。留空表示该指标「不适用」。
    坡向须填「阳坡 / 阴阳坡 / 阴坡」之一。未检测的重金属留空即可
```

```
      （按方法约定视为未受污染）。
    </p>
    <p class="mono muted" style="line-height:1.9;">{{
      headers|join(" , ") }}</p>
    <h3>导入规则</h3>
    <ul class="muted" style="padding-left:20px;line-height:1.9
      ;">
      <li>「单元类型」必须与图斑内已划分的单元一致，否则该行会被跳过并在结果中提示。</li>
      <li>同一单元类型若在文件中出现多行，以最后一行为准。</li>
      <li>已有数据的单元会被覆盖更新；没有数据的单元将新建记录。</li>
      <li>数值列请填纯数字，不要带单位（如填 <span class="mono">10</span> 而非
      <span class="mono">10cm</span>）。</li>
    </ul>
</div>
{% if report %}
<div class="card" style="max-width:860px;">
    <h2>导入结果</h2>
    <div class="stat-row">
      <div>
        <div class="stat__label">目标图斑</div>
        <div style="font-size:15px;">{{ report.patch }}</div>
      </div>
      <div>
        <div class="stat__label">成功导入</div>
        <div class="stat__value ok">{{ report.saved }}
      个单元</div>
      </div>
      <div>
        <div class="stat__label">问题行</div>
        <div class="stat__value {% if report.errors %}warn{%
      endif %}">{{ report.errors|length }}</div>
      </div>
    </div>
    {% if report.errors %}
    <h3>未导入的行</h3>
    <ul class="muted" style="padding-left:20px;line-height:1.9
      ;">
      {% for e in report.errors %}<li>{{ e }}</li>{% endfor %}
    </ul>
    {% else %}
    <p class="ok" style="margin-top:12px;">全部数据行均已成功导入。</p>
    {% endif %}
    <p class="actions" style="margin-top:16px;">
      <a class="btn btn--ghost" href="/indicator/">查看指标数据</a>
    </p>
</div>
{% endif %}
{% endblock %}
# ==== app/templates/evaluate/index.html ====
{% extends "base.html" %}
```

```
{% block title %}适宜性定量评价{% endblock %}
{% block content %}
<div class="card">
  <h2>适宜性定量评价</h2>
  {% if not patches %}
  <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">所属项目</th>
        <th scope="col">矿山图斑</th>
        <th scope="col">前置条件</th>
        <th scope="col">最近评价结果</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in patches %}
      {% set pr = progress.get(p.id) %}
      <tr>
        <td>{{ p.project.name if p.project else "—" }}</td>
        <td style="text-align:left;">{{ p.name }}</td>
        <td style="text-align:left;">
          {% if pr and pr.ready %}
          <span class="tag tag--ok">已具备</span>
          {% elif pr %}
          <span class="tag tag--warn">缺：{{
    pr.blocking[0].label }}</span>
          <span class="muted">{{ pr.blocking[0].reason
    }}</span>
          {% endif %}
        </td>
        <td>
          {% if latest.get(p.id) %}
          F = <span class="mono num">{{
    latest[p.id].final_score }}</span>,
          <span class="tag tag--ok">{{ latest[p.id].grade
    }}</span>
          {% else %}
          <span class="tag tag--muted">未评价</span>
          {% endif %}
        </td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/evaluate/{{ p.id
    }}">执行评价</a>
            {% if latest.get(p.id) %}
            <a class="link-btn" href="/evaluate/result/{{
    latest[p.id].id }}">查看结果</a>
```

```
                {% endif %}                                                    10351
              </span>                                                          10352
            </td>                                                              10353
          </tr>                                                                10354
          {% endfor %}                                                         10355
        </tbody>                                                               10356
      </table>                                                                 10357
      {% endif %}                                                              10358
</div>                                                                         10359
{% endblock %}                                                                 10360
# ==== app/templates/evaluate/run.html ====                                    10361
{% extends "base.html" %}                                                      10362
{% import "_macros.html" as ui %}                                              10363
{% block title %}执行评价 · {{ patch.name }}{% endblock %}                     10364
{% block content %}                                                            10365
<div class="card" style="max-width:880px;">                                    10366
  <h2>执行适宜性评价 · {{ patch.name }}</h2>                                   10367
  {{ ui.workflow_steps(progress) }}                                            10368
  {{ ui.workflow_blocking(progress) }}                                         10369
  {% if progress.ready %}                                                      10370
  <form method="post">                                                         10371
    {{ form.hidden_tag() }}                                                    10372
    <p class="actions">                                                        10373
      <button type="submit" class="btn">开始评价</button>                      10374
      <a class="btn btn--ghost" href="/evaluate/">返回</a>                     10375
    </p>                                                                       10376
  </form>                                                                      10377
  {% else %}                                                                   10378
  <p class="actions">                                                          10379
    <button type="button" class="btn" disabled>开始评价</button>               10380
    <a class="btn btn--ghost" href="/evaluate/">返回</a>                       10381
  </p>                                                                         10382
  {% endif %}                                                                  10383
</div>                                                                         10384
<div class="card" style="max-width:880px;">                                    10385
  <h2>当前流程状态</h2>                                                       10386
  <table>                                                                      10387
    <thead>                                                                    10388
      <tr><th scope="col">步骤</th><th scope="col">状态</th><th               10389
    scope="col">说明</th></tr>                                                 10390
    </thead>                                                                   10391
    <tbody>                                                                    10392
      <tr>                                                                     10393
        <td>限制性条件判定</td>                                               10394
        <td>                                                                   10395
          {% if progress.constraint.passed %}<span                             10396
    class="tag tag--ok">已通过</span>                                          10397
          {% elif progress.constraint.done %}<span                             10398
    class="tag tag--warn">未通过</span>                                        10399
          {% else %}<span class="tag tag--muted">待完成</span>{%               10400
```

```
      % endif %}                                                    10401
            </td>                                                  10402
            <td style="text-align:left;">{{                       10403
      progress.constraint.detail }}</td>                           10404
         </tr>                                                     10405
         <tr>                                                      10406
            <td>划分微地质单元</td>                                 10407
            <td>                                                   10408
               {% if progress.units.done %}<span class="tag        10409
      tag--ok">已完成</span>                                       10410
               {% else %}<span class="tag tag--muted">待完成</span>{ 10411
      % endif %}                                                   10412
            </td>                                                  10413
            <td style="text-align:left;">{{                       10414
      progress.units.detail }}</td>                                10415
         </tr>                                                     10416
         <tr>                                                      10417
            <td>录入指标数据</td>                                   10418
            <td>                                                   10419
               {% if progress.data.ready %}<span class="tag        10420
      tag--ok">已完成</span>                                       10421
               {% else %}<span class="tag tag--muted">待完成</span>{ 10422
      % endif %}                                                   10423
            </td>                                                  10424
            <td style="text-align:left;">{{                       10425
      progress.data.detail }}</td>                                 10426
         </tr>                                                     10427
      </tbody>                                                     10428
   </table>                                                        10429
</div>                                                            10430
{% if weight_sources %}                                           10431
<div class="card" style="max-width:880px;">                       10432
   <h2>本次将采用的指标权重</h2>  <table style="max-width:720px;">  10433
      <thead>                                                      10434
         <tr><th scope="col">微地质单元</th><th                    10435
      scope="col">权重来源</th></tr>                               10436
      </thead>                                                     10437
      <tbody>                                                      10438
         {% for ut, label in weight_sources.items() %}             10439
         <tr>                                                      10440
            <td style="text-align:left;">{{ ut }}</td>             10441
            <td style="text-align:left;">                          10442
               {% if "自定义" in label %}                          10443
               <span class="tag tag--warn">{{ label }}</span>      10444
               {% else %}                                          10445
               <span class="tag tag--muted">{{ label }}</span>     10446
               {% endif %}                                         10447
            </td>                                                  10448
         </tr>                                                     10449
         {% endfor %}                                              10450
```

```
      </tbody>                                                                  10451
    </table>                                                                    10452
</div>                                                                          10453
{% endif %}                                                                     10454
{% endblock %}                                                                  10455
# ==== app/templates/evaluate/result.html ====                                  10456
{% extends "base.html" %}                                                       10457
{% block title %}评价结果{% endblock %}                                          10458
{% block content %}                                                             10459
<div class="card">                                                              10460
  <h2>适宜性评价结果 · {{ patch.name if patch else "" }}</h2>                    10461
  <div class="stat-row">                                                        10462
    <div>                                                                       10463
      <div class="stat__label">最终评价因子 F</div>                             10464
      <div class="stat__value mono">{{ r.final_score                            10465
    }}</div>                                                                    10466
    </div>                                                                      10467
    <div>                                                                       10468
      <div class="stat__label">适宜性等级</div>                                 10469
      <div class="stat__value ok">{{ r.grade }}</div>                           10470
    </div>                                                                      10471
    <div style="flex:1;min-width:240px;">                                       10472
      <div class="stat__label">推荐措施</div>                                   10473
      <div style="font-size:15px;">{{ r.measures }}</div>                       10474
    </div>                                                                      10475
  </div>                                                                        10476
  {% if unit_scores %}                                                          10477
  <h3>各微地质单元得分</h3>                                                     10478
  <table style="max-width:860px;">                                              10479
    <thead>                                                                     10480
      <tr>                                                                      10481
        <th scope="col">微地质单元</th>                                         10482
        <th scope="col">面积占比</th>                                           10483
        <th scope="col">单元评价因子 F'</th>                                    10484
        <th scope="col">加权贡献</th>                                           10485
        <th scope="col">权重来源</th>                                           10486
      </tr>                                                                     10487
    </thead>                                                                    10488
    <tbody>                                                                     10489
      {% for u in unit_scores %}                                                10490
      <tr>                                                                      10491
        <td>{{ u.unit_type }}</td>                                              10492
        <td class="num">{{ "%.2f%%"|format(u.ratio * 100)                       10493
    }}</td>                                                                     10494
        <td class="mono">{{ u.unit_score }}</td>                                10495
        <td class="mono">{{ "%.4f"|format(u.ratio *                             10496
    u.unit_score) }}</td>                                                       10497
        <td class="muted">{{ u.weight_source or "系统默认权重"                  10498
    }}</td>                                                                     10499
      </tr>                                                                     10500
```

```
      {% endfor %}                                                    10501
    </tbody>                                                          10502
  </table>                                                            10503
  {% endif %}                                                         10504
  {% if weak %}                                                       10505
  <h3>短板提示</h3>                                                   10506
  <p class="muted" style="margin-top:-4px;">                          10507
    按「权重 ×（1 − 标准化值）」排序，提升这些指标对总分的拉动最大：  10508
  </p>                                                                10509
  <table style="max-width:760px;">                                    10510
    <thead>                                                           10511
      <tr>                                                            10512
        <th scope="col">微地质单元</th>                               10513
        <th scope="col">指标</th>                                     10514
        <th scope="col">实测值</th>                                   10515
        <th scope="col">标准化值</th>                                 10516
        <th scope="col">提升空间</th>                                 10517
      </tr>                                                           10518
    </thead>                                                          10519
    <tbody>                                                           10520
      {% for w in weak %}                                             10521
      <tr>                                                            10522
        <td>{{ w.unit_type }}</td>                                    10523
        <td>{{ w.l2 }}</td>                                           10524
        <td class="mono">{{ w.raw if w.raw is not none else 10525
    "—" }}</td>                                                       10526
        <td class="mono">{{ "%.4f"|format(w.std_value)                10527
    }}</td>                                                           10528
        <td class="mono">{{ "%.4f"|format(w.gain) }}</td>             10529
      </tr>                                                           10530
      {% endfor %}                                                    10531
    </tbody>                                                          10532
  </table>                                                            10533
  {% endif %}                                                         10534
  {% if check and check.covered %}                                    10535
  {% include "standard/_check_block.html" %}                          10536
  {% endif %}                                                         10537
  <p class="muted" style="margin-top:14px;">                          10538
    计算人：{{ operator or "—" }} ·                                   10539
    计算时间：{{ r.created_at | localdt("%Y-%m-%d %H:%M:%S") }} ·     10540
    参数来源：{{ param_sources }}                                     10541
  </p>                                                                10542
  {% if params %}                                                     10543
  <details style="margin-top:8px;">                                   10544
    <summary class="muted">本次计算使用的参数快照</summary>           10545
    <table style="max-width:520px;">                                  10546
      <thead>                                                         10547
        <tr><th scope="col">参数</th><th                              10548
    scope="col">取值</th><th scope="col">来源</th></tr>               10549
      </thead>                                                        10550
```

```
        <tbody>
          {% for code, item in params.items() %}
          <tr>
            <td class="mono">{{ code }}</td>
            <td class="mono">{{ item.value }}</td>
            <td class="muted">{{ item.source }}</td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
    </details>
    {% endif %}
    <p class="actions" style="margin-top:18px;">
      <a class="btn btn--ghost" href="/report/{{ r.patch_id
      }}">查看得分明细</a>
      <a class="btn btn--ghost" href="/evaluate/">返回评价列表</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/report/index.html ====
{% extends "base.html" %}
{% block title %}报告输出{% endblock %}
{% block content %}
<div class="card">
    <h2>报告与结果输出</h2>
    {% if not patches %}
    <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>
    {% else %}
    <table>
      <thead>
        <tr>
          <th scope="col">所属项目</th>
          <th scope="col">矿山图斑</th>
          <th scope="col">可生成报告</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for p in patches %}
        {% set pr = progress.get(p.id) %}
        <tr>
          <td>{{ p.project.name if p.project else "—" }}</td>
          <td style="text-align:left;">{{ p.name }}</td>
          <td style="text-align:left;">
            {% if pr and pr.ready %}
            <span class="tag tag--ok">可以</span>
            {% elif pr %}
            <span class="tag tag--warn">暂不可</span>
            <span class="muted">{{ pr.blocking[0].reason
      }}</span>
```

```
            {% endif %}                                                    10601
          </td>                                                            10602
          <td class="nowrap">                                              10603
            {% if pr and pr.ready %}                                       10604
            <span class="row-actions">                                     10605
              <a class="link-btn" href="/report/{{ p.id                    10606
    }}">查看得分明细</a>                                                  10607
              <a class="link-btn" href="/report/{{ p.id                    10608
    }}/pdf">导出 PDF</a>                                                  10609
              <a class="link-btn" href="/report/{{ p.id                    10610
    }}/word">导出 Word</a>                                                10611
            </span>                                                        10612
            {% else %}                                                     10613
            <span class="muted">—</span>                                   10614
            {% endif %}                                                    10615
          </td>                                                            10616
        </tr>                                                              10617
        {% endfor %}                                                       10618
      </tbody>                                                             10619
    </table>                                                               10620
    {% endif %}                                                            10621
</div>                                                                     10622
{% endblock %}                                                             10623
# ==== app/templates/report/detail.html ====                               10624
{% extends "base.html" %}                                                  10625
{% block title %}得分明细 · {{ patch.name }}{% endblock %}                 10626
{% block content %}                                                        10627
<div class="card">                                                         10628
  <h2>得分明细 · {{ patch.name }}</h2>                                     10629
  <div class="stat-row">                                                   10630
    <div>                                                                  10631
      <div class="stat__label">最终评价因子 F</div>                        10632
      <div class="stat__value mono">{{                                     10633
    "%.4f"|format(result.final_score) }}</div>                             10634
    </div>                                                                 10635
    <div>                                                                  10636
      <div class="stat__label">适宜性等级</div>                            10637
      <div class="stat__value ok">{{ result.grade }}</div>                 10638
    </div>                                                                 10639
    <div style="flex:1;min-width:240px;">                                  10640
      <div class="stat__label">推荐措施</div>                              10641
      <div style="font-size:15px;">{{                                      10642
    result.grade_info.measure }}</div>                                     10643
    </div>                                                                 10644
  </div>                                                                   10645
  <p class="muted" style="margin-top:12px;">                               10646
    本页与导出报告均取自                                                   10647
    <strong>{% if record %}{{ record.created_at |                          10648
    localdt("%Y-%m-%d %H:%M:%S") }}{% endif %}                             10649
    记录的那次评价结果</strong>（评价记录编号 {{ record.id if record else  10650
```

```
      "—"  }}),                                                    10651
      不会因事后修改指标数据而变化。                                10652
    </p>                                                           10653
    {% if undetected %}                                            10654
    <p class="muted" style="margin-top:8px;">                      10655
      数据完整度说明：以下重金属项<strong>未检测</strong>，按方法约定以「未受污染」计入污染指标，  10656
      并非实测达标 ——                                             10657
      {% for unit_type, metals in undetected %}                    10658
        {{ unit_type }}：{{ metals|join("、") }}{% if not          10659
      loop.last %}；{% endif %}                                    10660
      {% endfor %}。                                               10661
    </p>                                                           10662
    {% endif %}                                                    10663
    {% if check and check.covered %}                               10664
    {% include "standard/_check_block.html" %}                     10665
    {% endif %}                                                    10666
    <p class="actions" style="margin-top:16px;">                   10667
      <a class="btn" href="/report/{{ patch.id }}/pdf">导出 PDF    10668
      报告</a>                                                     10669
      <a class="btn btn--ghost" href="/report/{{ patch.id          10670
      }}/word">导出 Word 报告</a>                                  10671
      <a class="btn btn--ghost" href="/report/">返回</a>           10672
    </p>                                                           10673
</div>                                                             10674
<div class="card">                                                 10675
    <h2>结果图表</h2>                                             10676
    <h3>各微地质单元得分对比</h3>                                 10677
    <div>{{ chart_units|safe }}</div>                              10678
    <h3>各一级指标平均得分</h3>                                   10679
    <p class="muted">反映各一级指标对评价结果的实际影响，便于识别关键控制项。</p>  10680
    <div>{{ chart_weight|safe }}</div>                             10681
</div>                                                             10682
{% for u in result.unit_results %}                                 10683
<div class="card">                                                 10684
    <h2>{{ u.unit_type }} · 占比 {{ "%.2f%%"|format(u.ratio *      10685
      100) }} · 单元得分 {{ "%.4f"|format(u.unit_score) }}</h2>    10686
    {% if u.weight_source %}                                       10687
    <p class="muted" style="margin-top:-6px;">权重来源：{{         10688
      u.weight_source }}</p>                                       10689
    {% endif %}                                                    10690
    <table>                                                        10691
      <thead>                                                      10692
        <tr>                                                       10693
          <th scope="col">一级指标</th>                            10694
          <th scope="col">二级指标</th>                            10695
          <th scope="col">实测值</th>                              10696
          <th scope="col">标准化值</th>                            10697
          <th scope="col">权重</th>                                10698
          <th scope="col">贡献</th>                                10699
        </tr>                                                      10700
```

```
      </thead>                                                                  10701
      <tbody>                                                                   10702
        {% for d in u.details %}                                                10703
        <tr>                                                                    10704
          <td>{{ d.l1 }}</td>                                                   10705
          <td style="text-align:left;">{{ d.l2 }}</td>                          10706
          <td class="mono" style="text-align:left;">{{ "—" if 10707
      d.raw is none else d.raw }}</td>                                          10708
          <td class="mono">{{ "—" if d.std_value is none else 10709
      "%.4f"|format(d.std_value) }}</td>                                        10710
          <td class="mono">{{ "%.4f"|format(d.weight) }}</td> 10711
          <td class="mono">{{ "—" if d.contribution is none 10712
      else "%.4f"|format(d.contribution) }}</td>                                10713
        </tr>                                                                   10714
        {% endfor %}                                                            10715
      </tbody>                                                                  10716
    </table>                                                                    10717
    <p class="muted" style="margin:0;">                                         10718
      一级指标得分：                                                            10719
      {% for k, v in u.l1_scores.items() %}{{ k }} = <span      10720
      class="num">{{ "%.4f"|format(v) }}</span>{% if not       10721
      loop.last %};{% endif %}{% endfor %}                                      10722
    </p>                                                                        10723
</div>                                                                          10724
{% endfor %}                                                                    10725
{% endblock %}                                                                  10726
# ==== app/templates/report/print.html ====                                     10727
<!DOCTYPE html>                                                                 10728
<html lang="zh-CN">                                                             10729
<head>                                                                          10730
<meta charset="UTF-8">                                                          10731
<title>自然恢复适宜性评价报告 · {{ patch.name }}</title>                        10732
<style>                                                                         10733
  @page { size: A4; margin: 25mm 25mm; }                                        10734
  * { box-sizing: border-box; }                                                 10735
  body {                                                                        10736
    margin: 0; color: #000; background: #fff;                                   10737
    font-family: "SimSun", "宋体", serif;                                       10738
    font-size: 12pt; line-height: 1.0;                                          10739
  }                                                                             10740
  h1 {                                                                          10741
    font-family: "SimHei", "黑体", sans-serif;                                  10742
    font-size: 16pt; text-align: center;                                        10743
    font-weight: bold; color: #000;                                             10744
    margin: 0 0 6px;                                                            10745
  }                                                                             10746
  h2 {                                                                          10747
    font-family: "SimHei", "黑体", sans-serif;                                  10748
    font-size: 14pt; text-align: center;                                        10749
    font-weight: bold; color: #000;                                             10750
```

```
    margin: 16px 0 8px;                                                    10751
    border-bottom: none; padding-bottom: 0;                                10752
  }                                                                        10753
  h3 {                                                                     10754
    font-family: "SimSun", "宋体", serif;                                  10755
    font-size: 12pt; color: #000; font-weight: bold;                       10756
    margin: 12px 0 6px;                                                    10757
  }                                                                        10758
  .sub { text-align: center; color: #000; font-size: 12pt;                 10759
    margin-bottom: 14px; }                                                 10760
  table { width: 100%; border-collapse: collapse; margin:                  10761
    8px 0 14px;                                                            10762
          font-size: 12pt; font-variant-numeric:                           10763
    tabular-nums; }                                                        10764
  th, td { border: 1px solid #000; padding: 3px 6px;                       10765
    text-align: center; }                                                  10766
  th { background: #fff; font-weight: bold; }                              10767
  .left { text-align: left; }                                              10768
  .mono { font-family: "SimSun", "宋体", serif; }                          10769
  .kv { width: auto; }                                                     10770
  .kv th { width: 30%; text-align: left; }                                 10771
  .kv td { text-align: left; }                                             10772
  ul { margin: 6px 0 12px; padding-left: 20px; font-size:                  10773
    12pt; }                                                                10774
  .note { color: #000; font-size: 12pt; }                                  10775
  .chart { margin: 10px 0 16px; }                                          10776
  .sig { margin-top: 28px; font-size: 12pt; }                              10777
  .avoid-break { break-inside: avoid; }                                    10778
  svg { max-width: 100%; }                                                 10779
</style>                                                                   10780
</head>                                                                    10781
<body>                                                                     10782
  <h1>历史遗留废弃矿山自然恢复适宜性评价报告</h1>                                  10783
  <p class="sub">                                                          10784
    {{ patch.name }}                                                       10785
    {% if patch.admin_region %} {{ patch.admin_region }}{%                 10786
    endif %}                                                               10787
    <br>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ１.０                                10788
    生成时间 {{ generated_at | localdt }}                                  10789
  </p>                                                                     10790
  <h2>一、图斑基本信息</h2>                                                   10791
  <table class="kv">                                                       10792
    <tr><th>图斑名称</th><td>{{ patch.name }}</td></tr>                      10793
    <tr><th>所在行政区</th><td>{{ patch.admin_region or "—"                 10794
    }}</td></tr>                                                           10795
    <tr><th>图斑面积（公顷）</th><td>{{ patch.area_ha if                    10796
    patch.area_ha is not none else "—" }}</td></tr>                        10797
    <tr><th>开采矿种</th><td>{{ patch.mineral_type or "—"                  10798
    }}</td></tr>                                                           10799
    <tr><th>开采方式</th><td>{{ patch.mining_method or "—"                 10800
```

```
    }}</td></tr>
    <tr><th>责任主体缺失</th><td>{{ "是" if patch.responsible_entity_missing else "否" }}</td></tr>
    {% if reference %}
    <tr><th>参照生态系统</th>
        <td>植被覆盖度基准 {{ reference.coverage_benchmark }}%，
            多样性指数基准 {{ reference.shannon_benchmark
    }}</td></tr>
    {% endif %}
</table>
<h2>二、评价结论</h2>
<table class="kv">
    <tr><th>最终评价因子 F</th><td class="mono"><strong>{{ "%.4f"|format(result.final_score) }}</strong></td></tr>
    <tr><th>适宜性等级</th><td><strong>{{ result.grade }}</strong></td></tr>
    <tr><th>判定依据</th><td>{{ result.grade_info.meaning }}</td></tr>
    <tr><th>推荐措施</th><td>{{ result.grade_info.measure }}</td></tr>
</table>
{% if check and check.covered %}
<h2>三、与地方标准 / 政策文件的对照</h2>
<p class="note">
  图斑所在行政区：{{ check.region or "未填写" }}（对照 {{ check.province }} 的标准与政策文件）。
  对照只把文件里的判定项逐条映射到系统已有数据，系统内没有对应数据的定性要求一律标为待人工判断。
  合计：符合 {{ check.totals["符合"] }} 条、不符合 {{ check.totals["不符合"] }} 条、待人工判断 {{ check.totals["待人工判断"] }} 条。
</p>
{% for s in check.standards %}
<h3>{{ s.name }}{% if s.doc_number %}（{{ s.doc_number }}）{% endif %}</h3>
<table>
  <thead>
    <tr><th>文件判定项</th><th>条件</th><th>类型</th><th>对照结果</th><th>依据</th></tr>
  </thead>
  <tbody>
    {% for r in s.rules %}
    <tr>
      <td class="left">{{ r.item }}</td>
      <td class="mono">{{ r.operator }} {{ r.threshold or "" }}</td>
      <td>{{ r.condition_type or "—" }}</td>
      <td>{{ r.verdict }}</td>
      <td class="left">{{ r.evidence }}</td>
    </tr>
    {% endfor %}
```

```
    </tbody>
</table>
{% endfor %}
{% endif %}
<h2>四、各微地质单元得分</h2>
<table>
    <thead>
        <tr><th>微地质单元</th><th>面积占比</th><th>单元评价因子
    F'</th><th>加权贡献</th></tr>
    </thead>
    <tbody>
        {% for u in result.unit_results %}
        <tr>
            <td>{{ u.unit_type }}</td>
            <td class="mono">{{ "%.2f%%"|format(u.ratio * 100)
    }}</td>
            <td class="mono">{{ "%.4f"|format(u.unit_score)
    }}</td>
            <td class="mono">{{ "%.4f"|format(u.weighted)
    }}</td>
        </tr>
        {% endfor %}
    </tbody>
</table>
<div class="chart">{{ chart_units|safe }}</div>
<h2>五、一级指标权重分布</h2>
<div class="chart">{{ chart_weight|safe }}</div>
<h2>六、得分明细</h2>
{% for u in result.unit_results %}
<h3>{{ u.unit_type }}（占比 {{ "%.2f%%"|format(u.ratio *
    100) }}，单元得分 {{ "%.4f"|format(u.unit_score) }}）</h3>
<table>
    <thead>
        <tr>
            <th>一级指标</th><th>二级指标</th><th>实测值</th>
            <th>标准化值</th><th>权重</th><th>贡献</th>
        </tr>
    </thead>
    <tbody>
        {% for d in u.details %}
        <tr>
            <td>{{ d.l1 }}</td>
            <td class="left">{{ d.l2 }}</td>
            <td class="mono left">{{ "—" if d.raw is none else
    d.raw }}</td>
            <td class="mono">{{ "—" if d.std_value is none else
    "%.4f"|format(d.std_value) }}</td>
            <td class="mono">{{ "%.4f"|format(d.weight) }}</td>
            <td class="mono">{{ "—" if d.contribution is none
    else "%.4f"|format(d.contribution) }}</td>
```

```
        </tr>
        {% endfor %}
      </tbody>
    </table>
    <p class="note">
      一级指标得分：
      {% for k, v in u.l1_scores.items() %}{{ k }} = {{
      "%.4f"|format(v) }}{% if not loop.last %};{% endif %}{%
      endfor %}
    </p>
    {% endfor %}
    <h2>七、评价依据说明</h2>
    <ol class="note">
      <li>指标标准化按标准化公式执行，按指标性质分为越大越优、越小越优、
          适中型与离散赋值四类。</li>
      <li>单元评价因子按加权合成公式由二级指标权重加权求得；
          整矿评价因子由各微地质单元按面积占比加权合成。</li>
      <li>适宜性等级按等级划分表判定：F&lt;0.3 不适宜、0.3≤F≤0.7 基本适宜、F≥0.7
      适宜。</li>
      <li>指标权重取自系统内置权重，按岩质边坡、碎石边坡、平台三类微地质单元分别确定。</li>
    </ol>
    <p class="sig">评价人（签字）：_____</p>
    <p class="sig">日期：_____</p>
</body>
</html>
# ==== app/templates/search.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}搜索 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<div class="card">
    <h2>搜索结果</h2>
    {% if not q %}
    <p class="muted">请输入关键词。可检索项目名称与编号、图斑名称与行政区 / 矿种、
      标准名称与文号。</p>
    {% else %}
    <p class="muted">
      关键词「<strong>{{ q }}</strong>」共找到
      <span class="num">{{ total }}</span> 条结果。
    </p>
    {% if not total %}
    <p class="muted">没有匹配的内容。换个词试试，或检查是否有错别字。</p>
    {% endif %}
    {% endif %}
</div>
{% if projects %}
<div class="card">
    <h2>评价项目（{{ projects|length }}）</h2>
    <table>
      <thead>
```

```
        <tr>                                                                    10951
          <th scope="col">项目名称</th>                                          10952
          <th scope="col">编号</th>                                              10953
          <th scope="col">状态</th>                                              10954
          <th scope="col">图斑数</th>                                            10955
          <th scope="col">操作</th>                                              10956
        </tr>                                                                   10957
      </thead>                                                                  10958
      <tbody>                                                                   10959
        {% for p in projects %}                                                 10960
        <tr>                                                                    10961
          <td style="text-align:left;">{{ p.name }}</td>                        10962
          <td class="mono">{{ p.code or "—" }}</td>                             10963
          <td><span class="tag tag--muted">{{ p.status                          10964
      }}</span></td>                                                            10965
          <td class="num">{{ p.patches.count() }}</td>                          10966
          <td class="nowrap">                                                   10967
            <span class="row-actions">                                          10968
              <a class="link-btn" href="/project/{{ p.id                        10969
      }}/patches">矿山图斑</a>                                                  10970
              <a class="link-btn" href="/project/{{ p.id                        10971
      }}/edit">编辑</a>                                                         10972
            </span>                                                             10973
          </td>                                                                 10974
        </tr>                                                                   10975
        {% endfor %}                                                            10976
      </tbody>                                                                  10977
    </table>                                                                    10978
</div>                                                                          10979
{% endif %}                                                                     10980
{% if patches %}                                                                10981
<div class="card">                                                              10982
  <h2>矿山图斑（{{ patches|length }}）</h2>                                     10983
  <table>                                                                       10984
    <thead>                                                                     10985
      <tr>                                                                      10986
        <th scope="col">所属项目</th>                                           10987
        <th scope="col">图斑名称</th>                                           10988
        <th scope="col">行政区</th>                                             10989
        <th scope="col">矿种</th>                                               10990
        <th scope="col">操作</th>                                               10991
      </tr>                                                                     10992
    </thead>                                                                    10993
    <tbody>                                                                     10994
      {% for m in patches %}                                                    10995
      <tr>                                                                      10996
        <td>{{ m.project.name if m.project else "—" }}</td>                     10997
        <td style="text-align:left;">{{ m.name }}</td>                          10998
        <td>{{ m.admin_region or "—" }}</td>                                    10999
        <td>{{ m.mineral_type or "—" }}</td>                                    11000
```

```
            <td class="nowrap">                                          11001
              <span class="row-actions">                                 11002
                <a class="link-btn" href="/constraint/{{ m.id            11003
      }}">限制性判定</a>                                                  11004
                <a class="link-btn" href="/unit/{{ m.id                  11005
      }}">微地质单元</a>                                                  11006
                <a class="link-btn" href="/evaluate/{{ m.id              11007
      }}">执行评价</a>                                                    11008
              </span>                                                    11009
            </td>                                                        11010
          </tr>                                                          11011
          {% endfor %}                                                   11012
        </tbody>                                                         11013
    </table>                                                             11014
</div>                                                                   11015
{% endif %}                                                              11016
{% if standards %}                                                       11017
<div class="card">                                                       11018
    <h2>标准与政策文件（{{ standards|length }}）</h2>                      11019
    <table>                                                              11020
      <thead>                                                            11021
        <tr>                                                             11022
          <th scope="col">类别</th>                                      11023
          <th scope="col">省份</th>                                      11024
          <th scope="col">名称</th>                                      11025
          <th scope="col">文号</th>                                      11026
          <th scope="col">操作</th>                                      11027
        </tr>                                                            11028
      </thead>                                                           11029
      <tbody>                                                            11030
        {% for s in standards %}                                         11031
        <tr>                                                             11032
          <td>{{ s.category }}</td>                                      11033
          <td>{{ s.province }}</td>                                      11034
          <td style="text-align:left;">{{ s.name }}</td>                 11035
          <td class="mono">{{ s.doc_number or "—" }}</td>                11036
          <td class="nowrap">                                            11037
            <span class="row-actions">                                   11038
              <a class="link-btn" href="/standard/{{ s.id                11039
      }}/rules">规则</a>                                                  11040
            </span>                                                      11041
          </td>                                                          11042
        </tr>                                                            11043
        {% endfor %}                                                     11044
      </tbody>                                                           11045
    </table>                                                             11046
</div>                                                                   11047
{% endif %}                                                              11048
{% endblock %}                                                           11049
# ==== seed_demo.py ====                                                 11050
```

```
# -*- coding: utf-8 -*-
"""内置玉田县示例案例：把示例数据写入数据库。
用于功能验证与操作演示。数据逐字取自内置示例数据。
运行：python seed_demo.py                    幂等，只补写缺失的演示数据，
                                                 不会删除用户自建的项目与图斑
      python seed_demo.py --reset                连同标准库与指标定义一并重建
"""
import json
import sys
sys.stdout.reconfigure(encoding="utf-8")
from app import create_app
from app.models import (db, User, Project, MinePatch,
    ReferenceEcosystem,
                        MicroUnit, IndicatorData, Standard,
    StandardRule,
                        ConstraintCheck, EvaluationResult,
    ScoreDetail)
app = create_app()
# 内置演示案例的项目编号，用于区分「演示数据」与「用户自建数据」
DEMO_CODE = "DEMO-001"
# 内置标准库：12 项地方标准与政策文件
STANDARDS = [
    dict(category="地方标准", province="辽宁",
    name="矿山废弃地植被自然恢复技术规程",
        doc_number="DB21/T 3702—2023",
    publish_date="2023-03-30",
        implement_date="2023-04-30",
        summary="首个正式发布的矿山自然恢复地方标准；提出必要条件和充分条件",
        rules=[("植被覆盖度", ">=", "50", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None),
               ("与参照系差距", "<", "20", "充分条件", "%")]),
    dict(category="地方标准", province="辽宁",
    name="废弃矿山自然恢复评估规程",
        doc_number="DB21/T 4421—2026",
    publish_date="2026-05-07",
        implement_date="2026-06-07",
        summary="按水热条件分成湿润区、半湿润区、半干旱区，分区确定植被覆盖度要求；"
                "评估考虑地质灾害、水土污染、立地条件、土地损毁程度、植被发育情况、"
                "植被群落六个方面",
        # 植被覆盖度按水热条件分区给阈值，分区写进判定项名称，便于对照时看清适用分区
        rules=[("植被覆盖度（湿润区）", ">=", "70", "必要条件", "湿润区"),
               ("植被覆盖度（半湿润区）", ">=", "60", "必要条件", "半湿润区"),
               ("植被覆盖度（半干旱区）", ">=", "50", "必要条件", "半干旱区"),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None)]),
    dict(category="地方标准", province="湖北",
    name="废弃露天矿山生态环境自然恢复标准",
        doc_number="DB42/T 2421—2025",
    publish_date="2025-08-20",
```

```
        implement_date="2025-10-20",                                    11101
        summary="结合矿山生态需求进行自然恢复适宜性分级判断",                    11102
        rules=[("自然恢复适宜性分级", "==", "是", "必要条件", None)]),    11103
dict(category="地方标准", province="内蒙古",                             11104
name="历史遗留废弃矿山自然恢复判定规范",                                   11105
        doc_number="DB15/T 4280—2026",                                  11106
publish_date="2026-01-30",                                              11107
        implement_date="2026-03-01",                                    11108
        summary="调查内容包含参照生态系统调查",                           11109
        rules=[("地质灾害发育程度", "<=", "弱发育", "必要条件", None),     11110
               ("地形地貌与周边协调", "==", "是", "必要条件", None),       11111
               ("有利于植被恢复的表层土壤", "==", "是", "必要条件", None),  11112
               ("已形成稳定植被覆盖", "==", "是", "必要条件", None)]),    11113
dict(category="地方标准", province="江西",                               11114
name="废弃矿山自然恢复技术规范",                                          11115
        doc_number="DB36/T 2221—2026",                                  11116
publish_date="2026-02-02",                                              11117
        implement_date="2026-08-01",                                    11118
        summary="对参照生态系统进行了要求；提出自然恢复适宜性指数",           11119
        rules=[("参照生态系统调查", "==", "是", "必要条件", None),         11120
               ("自然恢复适宜性指数", ">=", "阈值", "充分条件", None)]),  11121
dict(category="地方标准", province="宁夏", name="矿山自然恢复认定规程",11122
        doc_number=None, publish_date="2026-06-18",                     11123
implement_date=None,                                                    11124
        summary="分区差异化：三个一级生态修复分区及 22 个二级分区",        11125
        rules=[("无地质灾害风险", "==", "是", "必要条件", None),         11126
               ("植被覆盖度趋势", "==", "稳定或正增长", "必要条件", None),  11127
               ("与参照系差距", "<", "20", "充分条件", "%")]),           11128
dict(category="政策文件", province="安徽",                               11129
        name="安徽省自然资源厅关于规范废弃露天矿山采取自然恢复方式进行治理的通知",  11130
        doc_number="皖自然资修函[2020]34号",                             11131
publish_date="2020-03-03", implement_date=None,                         11132
        summary="自然恢复条件：责任主体为政府、无地质灾害、无水土污染、不在三线三边、水土条件良好",11133
        rules=[("责任主体", "==", "政府", "必要条件", None),             11134
               ("无地质灾害", "==", "是", "必要条件", None),             11135
               ("无水土污染", "==", "是", "必要条件", None)]),           11136
dict(category="政策文件", province="山东",                               11137
        name="山东省自然资源厅关于进一步规范历史遗留废弃露天开采矿山采取自然恢复方式进行治理的通知",11138
        doc_number="鲁自然资字[2022]104号",                              11139
publish_date="2022-08-02", implement_date=None,                         11140
        summary="条件与安徽省相同，另增加自然恢复认定标准",               11141
        rules=[("无地质灾害", "==", "是", "必要条件", None),             11142
               ("无水土污染", "==", "是", "必要条件", None)]),           11143
dict(category="政策文件", province="湖南",                               11144
        name="关于印发《湖南省历史遗留矿山自然恢复指导意见（试行）》的通知",  11145
        doc_number="湘自资办发[2023]58号",                               11146
publish_date="2023-07-20", implement_date=None,                         11147
        summary="自然恢复认定标准：无地质灾害或无威胁对象、无水土污染、设备拆除、符合规划、植被覆盖度11148
>75%、连续3年正向增长",                                                  11149
        rules=[("植被覆盖度", ">", "75", "必要条件", "%"),               11150
```

```
                    ("植被覆盖率趋势", "==", "连续3年正向增长", "必要条件", None),
                    ("无地质灾害", "==", "是", "必要条件", None),
                    ("符合国土空间规划", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="新疆",
         name="自治区历史遗留废弃矿山自然恢复方式销号图斑工作指南（试行）",
         doc_number="新自然资源办函〔2024〕219号",
    publish_date="2024-12-20", implement_date=None,
         summary="荒漠区沙地/裸土地/裸岩石砾地满足无地质灾害、地形协调、无水土污染即可判定",
         rules=[("植被覆盖度", ">=", "周边30%", "必要条件", None),
                    ("无地质灾害", "==", "是", "必要条件", None),
                    ("地形平缓地貌稳定", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="广西",
         name="广西壮族自治区历史遗留矿山自然恢复技术指南",
         doc_number="桂自然资办〔2024〕250号",
    publish_date="2024-12-30", implement_date=None,
         summary="除高陡边坡和裸露岩体外，水土水热条件良好、植被自然恢复趋势明显",
         rules=[("无地质灾害", "==", "是", "必要条件", None),
                    ("植被自然恢复趋势", "==", "明显", "必要条件", None),
                    ("生产设备拆除完毕", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="西藏",
         name="西藏自治区历史遗留废弃矿山生态自然恢复认定指导意见（试行）",
         doc_number="藏自然资发〔2025〕19号",
    publish_date="2025-09-24", implement_date=None,
         summary="含修复成本高、生态修复效益不显著等适用条件",
         rules=[("责任主体", "==", "政府", "必要条件", None),
                    ("地形地貌完整与周边协调", "==", "是", "必要条件", None),
                    ("无地质灾害", "==", "是", "必要条件", None)]),
]
CASE = {
    "name": "玉田县典型历史遗留废弃矿山案例",
    "patch_name": "玉田县孤树镇白云岩采石场",
    "area_ha": 5.6,
    "mineral_type": "白云岩",
    # 三类微地质单元的面积占比与各项实测值，与最新版评价方法表 5 一致：
    # 平台 26.3%、碎石边坡 69.5%、岩质边坡 4.2%；
    # 单元面积按占比 × 图斑面积（5.6 公顷）换算，保证面积与占比互相对得上。
    "units": [
        {
            "unit_type": "岩质边坡", "ratio": 0.042, "area":
    0.2352,
            "data": {"coverage": 68.88, "shannon": 0.51,
                     "slope": 63.33, "aspect": "阴阳坡"},
        },
        {
            "unit_type": "碎石边坡", "ratio": 0.695, "area":
    3.892,
            "data": {"soil_ratio": 0.34, "thickness": 10,
    "ph": 9.24,
                     "organic_matter": 4.89,
                     "heavy_metals": {"Hg": 0.05, "As":
```

```
        5.17, "Ni": 12.60,                                                  11201
                                                "Cu": 11.70, "Cd":          11202
        0.04, "Pb": 6.50},                                                  11203
                            "coverage": 87.92, "shannon": 0.60,             11204
                            "slope": 28.75, "aspect": "阳坡"},              11205
                },                                                          11206
                {                                                           11207
                    "unit_type": "平台", "ratio": 0.263, "area":            11208
        1.4728,                                                             11209
                    "data": {"soil_ratio": 0.56, "thickness": 30,           11210
        "ph": 8.72,                                                         11211
                            "organic_matter": 10.92,                        11212
                            "heavy_metals": {"Hg": 0.02, "As":              11213
        3.13, "Ni": 12.33,                                                  11214
                                                "Cu": 11.14, "Cd":          11215
        0.09, "Pb": 9.98},                                                  11216
                            "coverage": 58.42, "shannon": 0.72},            11217
                },                                                          11218
        ],                                                                  11219
}                                                                           11220
def clean_case(reset=False):                                                11221
        """清理数据。                                                       11222
        默认只清理内置演示案例（code = DEMO-001）自身的数据，**不触碰       11223
        用户自己建立的项目与图斑**。加 --reset 时才额外清空标准库与指标定义，  11224
        以便把演示环境恢复到初始状态。                                      11225
        注意：不要在此处执行 Project.query.delete()，否则用户新建的项目     11226
        会在每次启动时被删除。                                              11227
        """                                                                 11228
        from app.models import EvaluationResult, ScoreDetail,               11229
        ConstraintCheck                                                     11230
        demo_projects = Project.query.filter_by(code=DEMO_CODE).            11231
        all()                                                               11232
        demo_ids = [p.id for p in demo_projects]                            11233
        patch_ids = []                                                      11234
        if demo_ids:                                                        11235
            patch_ids = [p.id for p in MinePatch.query.filter(              11236
                MinePatch.project_id.in_(demo_ids)).all()]                  11237
        if patch_ids:                                                       11238
            unit_ids = [u.id for u in MicroUnit.query.filter(               11239
                MicroUnit.patch_id.in_(patch_ids)).all()]                   11240
            if unit_ids:                                                    11241
                IndicatorData.query.filter(                                 11242
                    IndicatorData.unit_id.in_(unit_ids)).delete(            11243
                    synchronize_session=False)                              11244
            result_ids = [r.id for r in EvaluationResult.query.f            11245
        ilter(                                                              11246
                EvaluationResult.patch_id.in_(patch_ids)).all()]            11247
            if result_ids:                                                  11248
                ScoreDetail.query.filter(                                   11249
                    ScoreDetail.result_id.in_(result_ids)).delet            11250
```

```
    e(                                                                  11251
            synchronize_session=False)                                  11252
        EvaluationResult.query.filter(                                  11253
            EvaluationResult.patch_id.in_(patch_ids)).delete        11254
    (                                                                   11255
            synchronize_session=False)                                  11256
        ConstraintCheck.query.filter(                                   11257
            ConstraintCheck.patch_id.in_(patch_ids)).delete(        11258
            synchronize_session=False)                                  11259
        ReferenceEcosystem.query.filter(                                11260
            ReferenceEcosystem.patch_id.in_(patch_ids)).dele        11261
    te(                                                                 11262
            synchronize_session=False)                                  11263
        MicroUnit.query.filter(MicroUnit.patch_id.in_(patch_        11264
    ids)).delete(                                                       11265
            synchronize_session=False)                                  11266
        MinePatch.query.filter(MinePatch.id.in_(patch_ids))         11267
    delete(                                                             11268
            synchronize_session=False)                                  11269
    if demo_ids:                                                        11270
        Project.query.filter(Project.id.in_(demo_ids)).delet        11271
    e(                                                                  11272
            synchronize_session=False)                                  11273
    if reset:                                                           11274
        StandardRule.query.delete()                                     11275
        Standard.query.delete()                                         11276
    #  兜底：清理指向已不存在评价记录的孤儿得分明细，                  11277
    #  避免长期运行后 score_details 表堆积无效数据。                   11278
    _purge_orphan_details()                                             11279
    db.session.commit()                                                 11280
    return len(demo_ids)                                                11281
def demo_patch():                                                       11282
    """取内置演示案例的图斑。                                          11283
    库中可能存在用户自建项目或验证算例（编号 VERIFY-），               11284
    故不能简单用 MinePatch.query.first()，否则取到的对象不确定。       11285
    没有演示案例时才退回第一个图斑。                                   11286
    """                                                                 11287
    proj = Project.query.filter_by(code=DEMO_CODE).first()          11288
    if proj is not None:                                                11289
        m = MinePatch.query.filter_by(project_id=proj.id).fi        11290
    rst()                                                               11291
        if m is not None:                                               11292
            return m                                                    11293
    return MinePatch.query.first()                                      11294
def _purge_orphan_details():                                            11295
    """清理指向已不存在评价记录的孤儿得分明细。                        11296
    长期运行后 score_details 表可能堆积无效明细，启动时顺带清一次。    11297
    """                                                                 11298
    live_ids = [r.id for r in EvaluationResult.query.all()]         11299
    if live_ids:                                                        11300
```

```
            ScoreDetail.query.filter(~ScoreDetail.result_id.in_(l
        live_ids)).delete(
                synchronize_session=False)
        else:
            ScoreDetail.query.delete()
        db.session.commit()
    def seed(reset=False):
        with app.app_context():
            db.create_all()        #  全新环境下先建表，保证本脚本可独立运行
            _purge_orphan_details()
            admin = User.query.filter_by(username="admin").first
        ()
            if not admin:
                from werkzeug.security import
        generate_password_hash
                admin = User(username="admin",
                             password_hash=generate_password_has
        h("admin123"),
                             role="admin")
                db.session.add(admin)
                db.session.commit()
            #  幂等：演示案例已存在就不重复写入，更不清空用户自己的项目
            existing = Project.query.filter_by(code=DEMO_CODE).f
        irst()
            if existing and not reset:
                print("演示案例已存在，跳过写入（用户自建项目不受影响）")
                _report()
                _seed_reference_data(reset=False)
                return
            clean_case(reset=reset)
            project = Project(name=CASE["name"], code=DEMO_CODE,
                              description="内置示例案例",
        created_by=admin.id)
            db.session.add(project)
            db.session.flush()
            patch = MinePatch(project_id=project.id,
        name=CASE["patch_name"],
                              admin_region="河北省唐山市玉田县",
                              area_ha=CASE["area_ha"],
                              mineral_type=CASE["mineral_type"],
                              mining_method="露天开采",
        stop_years=None,
                              responsible_entity_missing=True)
            db.session.add(patch)
            db.session.flush()
            ref = ReferenceEcosystem(patch_id=patch.id,
                                     coverage_benchmark=75.0,
        shannon_benchmark=0.7)
            db.session.add(ref)
            for u in CASE["units"]:
```

```
            unit = MicroUnit(patch_id=patch.id,
    unit_type=u["unit_type"],
                             ratio=u["ratio"],
    area=u.get("area"))
            db.session.add(unit)
            db.session.flush()
            d = u["data"]
            ind = IndicatorData(
                unit_id=unit.id,
                soil_ratio=d.get("soil_ratio"),
                thickness=d.get("thickness"),
                ph=d.get("ph"),
                organic_matter=d.get("organic_matter"),
                heavy_metals=json.dumps(d["heavy_metals"],
    ensure_ascii=False)
                               if d.get("heavy_metals") else
    None,
                coverage=d.get("coverage"),
                shannon=d.get("shannon"),
                slope=d.get("slope"),
                aspect=d.get("aspect"),
            )
            db.session.add(ind)
        #  限制性条件判定记录：示例案例属「无地质灾害隐患点」、
        #  具备自然恢复条件，故五项全部通过。缺此记录会使后续评价被流程门禁拦住。
        db.session.add(ConstraintCheck(
            patch_id=patch.id,
            policy_pass=True, location_pass=True,
    pollution_pass=True,
            geo_pass=True, farmland_pass=True,
            passed=True, reason="全部通过"))
        db.session.commit()
        print("已写入案例：", CASE["name"])
        print("   图斑：", CASE["patch_name"])
        print("   微地质单元：平台 26.3% / 碎石边坡 69.5% / 岩质边坡 4.2%")
        print("   管理员账号：admin / admin123")
        _seed_reference_data(reset=reset)
        _report()
def _seed_reference_data(reset=False):
    """写入标准库与指标定义。
    参考资料只在库为空（或显式 --reset）时写入，避免覆盖用户自行维护
    的标准条目与指标定义。
    """
    if reset or Standard.query.count() == 0:
        seed_standards()
    from app.services.indicator_service import
    seed_indicators
    n = seed_indicators()
    print("指标定义：%d 项" % n)
def _report():
```

```
    total_p = Project.query.count()                                    11401
    total_patch = MinePatch.query.count()                              11402
    print()                                                            11403
    print("当前库内：项目 %d 个、图斑 %d 个" % (total_p, total_patch))  11404
    if total_p > total_patch:                                          11405
        print("  提示：项目是归档单位，评价针对的是图斑。")             11406
        print("        新项目需先建立图斑，才会出现在后续各模块中。")   11407
def seed_standards():                                                  11408
    """写入内置标准库的 12 项标准与政策文件及其结构化规则。"""          11409
    StandardRule.query.delete()                                        11410
    Standard.query.delete()                                            11411
    db.session.commit()                                                11412
    for s in STANDARDS:                                                11413
        std = Standard(category=s["category"],                         11414
    province=s["province"],                                            11415
                       name=s["name"],                                 11416
    doc_number=s["doc_number"],                                        11417
                       publish_date=s["publish_date"],                 11418
                       implement_date=s["implement_date"],             11419
                       summary=s["summary"])                           11420
        db.session.add(std)                                            11421
        db.session.flush()                                             11422
        for item, op, thr, ctype, note in s["rules"]:                  11423
            db.session.add(StandardRule(standard_id=std.id,            11424
    item=item,                                                         11425
                                        operator=op,                   11426
    threshold=thr,                                                     11427
                                                                       11428
    condition_type=ctype, note=note))                                  11429
    db.session.commit()                                                11430
    print("已写入标准/政策文件：%d 项，规则 %d 条"                     11431
          % (len(STANDARDS), sum(len(s["rules"]) for s in              11432
    STANDARDS)))                                                       11433
if __name__ == "__main__":                                             11434
    # --reset：把演示环境恢复到初始状态（清空标准库与指标定义后重建）。  11435
    # 无论是否 --reset，都不会删除用户自建的项目与图斑。                 11436
    seed(reset="--reset" in sys.argv[1:])                              11437
# ==== 测试/test_ahp.py ====                                           11438
# -*- coding: utf-8 -*-                                                11439
"""地基自检：验证 AHP 引擎与应用可启动。"""                            11440
import sys                                                             11441
sys.stdout.reconfigure(encoding="utf-8")                               11442
import numpy as np                                                     11443
import os                                                              11444
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，            11445
# 使 `from app import ...` 与 app/app.db 的定位不受影响。              11446
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil           11447
    e__)))                                                             11448
if ROOT not in sys.path:                                               11449
    sys.path.insert(0, ROOT)                                           11450
```

```
from app.services import ahp_service as ahp
LINE = "=" * 66
# ------------------------------------------------------------------
# 1） 完全一致矩阵 → 权重应精确还原，CR=0
# ------------------------------------------------------------------
print(LINE)
print("测试 1　完全一致判断矩阵的权重还原与一致性")
print(LINE)
# 由权重构造完全一致矩阵：a_ij = w_i / w_j
w_true = np.array([0.44, 0.31, 0.25])
A = np.outer(w_true, 1.0 / w_true)
r = ahp.weights_and_consistency(A)
print("　判断矩阵 A：")
for row in A:
    print("    " + "  ".join(f"{v:8.4f}" for v in row))
print()
print("　计算权重 :", "  ".join(f"{v:.4f}" for v in
    r["weights"]))
print("　真实权重 :", "  ".join(f"{v:.4f}" for v in w_true))
print("　λmax = %.6f　CI = %.3e　CR = %.3e　一致: %s"
      % (r["lambda_max"], r["ci"], r["cr"],
    r["consistent"]))
assert np.allclose(r["weights"], w_true, atol=1e-6),
    "权重未精确还原"
assert r["consistent"], "完全一致矩阵应通过检验"
# ------------------------------------------------------------------
# 2） 上三角输入 → 完整矩阵 → 一致性定位
# ------------------------------------------------------------------
print()
print(LINE)
print("测试 2　上三角录入与最不一致项定位")
print(LINE)
# 一个不太一致的 3 阶矩阵：上三角 [2, 5, 3]（a12=2, a13=5, a23=3）
M = ahp.matrix_from_upper([2, 5, 3])
r2 = ahp.weights_and_consistency(M)
bad = ahp.find_most_inconsistent(M)
print("　由 [a12=2, a13=5, a23=3] 构造矩阵：")
for row in M:
    print("    " + "  ".join(f"{v:8.4f}" for v in row))
print("　权重:", "  ".join(f"{v:.4f}" for v in r2["weights"]),
      "　CR = %.3f　一致: %s" % (r2["cr"], r2["consistent"]))
print("　最不一致比较项: 第(%d,%d) 项，当前 %.3f，理想 %.3f"
      % (bad["i"] + 1, bad["j"] + 1, bad["value"],
    bad["expected"]))
# ------------------------------------------------------------------
```

```
# 3） 多专家聚合（几何平均）
# ------------------------------------------------------------
    ----------
print()
print(LINE)
print("测试 3  多专家判断矩阵聚合（几何平均）")
print(LINE)
m1 = ahp.matrix_from_upper([2, 4, 3])
m2 = ahp.matrix_from_upper([3, 5, 2])
agg = ahp.aggregate_matrices([m1, m2])
print("  专家1 上三角 [2,4,3]，专家2 上三角 [3,5,2]")
print("  聚合后上三角： a12=%.4f a13=%.4f a23=%.4f"
      % (agg[0, 1], agg[0, 2], agg[1, 2]))
assert abs(agg[0, 1] - np.sqrt(2 * 3)) < 1e-9, "几何平均错误"
# ------------------------------------------------------------
    ----------
# 4） 论文表 3 权重可还原性
# ------------------------------------------------------------
    ----------
print()
print(LINE)
print("测试 4  论文表 3 碎石边坡一级权重 (0.44, 0.31, 0.25)")
print(LINE)
j = ahp.judge_from_weights([0.44, 0.31, 0.25])
print("  λmax = %.6f  CR = %.3e  一致： %s" %
    (j["lambda_max"], j["cr"], j["consistent"]))
assert j["consistent"], "论文权重应能作为完全一致矩阵主特征向量"
# ------------------------------------------------------------
    ----------
# 5） 应用可启动 + 建表
# ------------------------------------------------------------
    ----------
print()
print(LINE)
print("测试 5  Flask 应用启动与建表")
print(LINE)
from app import create_app
from app.models import db
app = create_app()
with app.app_context():
    db.create_all()
    tables = sorted(db.metadata.tables.keys())
print("  已创建 %d 张表：" % len(tables))
for t in tables:
    print("    -", t)
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== 测试/test_evaluate.py ====
```

```
# -*- coding: utf-8 -*-
"""评价引擎自检：用论文玉田县案例跑通完整计算链。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app.services import evaluation_service as ev
from app.services import standardization as std
LINE = "=" * 70
# ------------------------------------------------------------------
# 论文第 5 节案例数据（表 5：占比与各单元实测值逐字抄录）
# ------------------------------------------------------------------
# 三类单元面积占比：平台 26.3%、碎石边坡 69.5%、岩质边坡 4.2%
# 注：岩质边坡无土壤指标、平台无地形指标，对应"不适用"。
case = [
    {
        "unit_type": "岩质边坡",
        "ratio": 0.042,
        "data": {
            "coverage": 68.88, "shannon": 0.51,
            "slope": 63.33, "aspect": "阴阳坡",
        },
    },
    {
        "unit_type": "碎石边坡",
        "ratio": 0.695,
        "data": {
            "soil_ratio": 0.34, "thickness": 10, "ph":
    9.24, "organic_matter": 4.89,
            "heavy_metals": {"Hg": 0.05, "As": 5.17, "Ni":
    12.60,
                             "Cu": 11.70, "Cd": 0.04, "Pb":
    6.50},
            "coverage": 87.92, "shannon": 0.60,
            "slope": 28.75, "aspect": "阳坡",
        },
    },
    {
        "unit_type": "平台",
        "ratio": 0.263,
        "data": {
            "soil_ratio": 0.56, "thickness": 30, "ph":
    8.72, "organic_matter": 10.92,
```

```
                "heavy_metals": {"Hg": 0.02, "As": 3.13, "Ni": 11601
    12.33,                                                          11602
                                    "Cu": 11.14, "Cd": 0.09, "Pb":11603
    9.98},                                                          11604
                "coverage": 58.42, "shannon": 0.72,                 11605
            },                                                      11606
    },                                                              11607
]                                                                   11608
# ------------------------------------------------------------------11609
    ----------------                                                11610
# 逐单元输出标准化与得分                                            11611
# ------------------------------------------------------------------11612
    ----------------                                                11613
print(LINE)                                                         11614
print("玉田县案例 · 评价引擎完整计算")                              11615
print(LINE)                                                         11616
for u in case:                                                      11617
    r = ev.score_unit(u["unit_type"], u["data"])                    11618
    print()                                                         11619
    print("  【%s】  (面积占比 %.2f%%)" % (u["unit_type"],          11620
    u["ratio"] * 100))                                              11621
    print("  " + "-" * 60)                                         11622
    for l1, score in r["l1_scores"].items():                        11623
        print("    %s 得分 = %.4f" % (l1, score))                  11624
    print("    单元评价因子 F' = %.4f" % r["unit_score"])          11625
print()                                                             11626
print(LINE)                                                         11627
print("面积加权合成（式(9)）")                                      11628
print(LINE)                                                         11629
res = ev.score_mine(case)                                           11630
total = res["final_score"]                                          11631
print()                                                             11632
for ur in res["unit_results"]:                                      11633
    print("  %s  %.2f%% × %.4f = %.4f"                             11634
          % (ur["unit_type"], ur["ratio"] * 100,                    11635
    ur["unit_score"], ur["weighted"]))                              11636
print("  " + "-" * 60)                                             11637
print("  最终评价因子 F = %.4f" % total)                           11638
print()                                                             11639
print("  适宜性等级：%s" % res["grade"])                           11640
print("  判定依据  ：%s" % res["grade_info"]["meaning"])           11641
print("  推荐措施  ：%s" % res["grade_info"]["measure"])           11642
print()                                                             11643
print(LINE)                                                         11644
print("结论")                                                       11645
print(LINE)                                                         11646
print("  - 评价引擎按论文式(8)(9) 与表 3 权重完整跑通")            11647
print("  - 坡度标准化采用论文 25° 分界的线性折减默认值（见 standardization.py）")11648
print("  - 最终 F ≈ %.2f，判定为「%s」，与论文「最终适宜自然恢复」的结论一致"11649
      % (total, res["grade"]))                                      11650
```

```
print("  - 论文表 5 载 0.77、正文载 0.76，而按表 5 单元得分×占比加权为 0.7614；")
print("      系统差异集中在坡度、坡向的标准化口径（standardization.std_slope
    可配置）")
print(LINE)
#  ==== 测试/test_web_ahp.py ====
#  -*- coding: utf-8 -*-
"""AHP 网页界面自检：走通选择→录入→计算 全流程。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file
    __)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
#  测试配置：关闭 CSRF 以简化测试客户端（真实运行仍开启 CSRF）
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///:memory:"
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
#  建立表并创建测试用户，登录后再访问业务页面
with app.app_context():
    from app.models import db, User
    from werkzeug.security import generate_password_hash
    db.create_all()
    if not User.query.filter_by(username="admin").first():
        db.session.add(User(username="admin",
                            password_hash=generate_password
    hash("admin123"),
                            role="admin"))
        db.session.commit()
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  首页与 AHP 入口")
print(LINE)
r = client.get("/")
print("  首页状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
r = client.get("/ahp/")
print("  AHP 入口状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
assert "微地质单元类型" in r.get_data(as_text=True)
```

```
print()
print(LINE)
print("测试 2  选择碎石边坡一级指标 → 跳转矩阵页")
print(LINE)
r = client.post("/ahp/", data={
    "unit_type": "碎石边坡", "level": "1", "submit": "生成判断矩阵",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转后页面含「判断矩阵录入」:", "判断矩阵录入" in body)
print("  含三个一级指标（土壤/植被/地形）:", all(n in body for n in
    ("土壤指标", "植被指标", "地形指标")))
assert "判断矩阵录入" in body
assert all(n in body for n in ("土壤指标", "植被指标", "地形指标"))
print()
print(LINE)
print("测试 3  提交论文表 3 一致矩阵 → 还原权重")
print(LINE)
# 碎石边坡一级权重 (0.44, 0.31, 0.25) 对应的完全一致矩阵上三角：
# a12 = 0.44/0.31 = 1.4194, a13 = 0.44/0.25 = 1.76, a23 =
    0.31/0.25 = 1.24
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  结果页含「计算结果」:", "计算结果" in body)
print("  含权重 0.44:", "0.44" in body, " 含 0.31:", "0.31" in
    body, " 含 0.25:", "0.25" in body)
print("  含一致性检验通过:", "通过" in body)
assert "计算结果" in body
assert "0.44" in body and "0.31" in body and "0.25" in body
print()
print(LINE)
print("测试 4  不一致矩阵应提示不通过并定位")
print(LINE)
# 明显不一致：a12=9, a13=1, a23=9
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "9", "m_0_2": "1", "m_1_2": "9",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  含「不通过」:", "不通过" in body)
print("  含「建议优先调整」:", "建议优先调整" in body)
assert "不通过" in body
assert "建议优先调整" in body
print()
print(LINE)
print("网页界面全部测试通过 √")
print(LINE)
# ==== 测试/test_web_modules.py ====
# -*- coding: utf-8 -*-
"""模块自检：登录、项目档案、微地质单元、标准库、限制性判定。"""
```

```
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于  测试/  子目录，先把仓库根目录加入模块搜索路径，
#  使  `from app import ...`  与  app/app.db  的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
print(LINE)
print("测试  1   未登录应被重定向到登录页")
print(LINE)
r = client.get("/project/")
print("   /project/  状态码:", r.status_code, "  重定向到:",
    r.headers.get("Location"))
assert r.status_code == 302 and "login" in
    r.headers.get("Location", "")
print()
print(LINE)
print("测试  2   登录")
print(LINE)
r = client.post("/login", data={"username": "admin",
    "password": "admin123"},
                follow_redirects=True)
body = r.get_data(as_text=True)
#  工作台：页首横幅  +  评价流程分区  +  数据概览
print("   登录后含「评价流程」:", "评价流程" in body)
print("   含账户 admin:", "admin" in body)
print("   含页首横幅:", 'class="hero"' in body)
print("   含数据概览:", 'class="stat-card"' in body)
assert "评价流程" in body and 'class="hero"' in body
print()
print(LINE)
print("测试  3   项目与矿山档案")
print(LINE)
r = client.get("/project/")
body = r.get_data(as_text=True)
print("   项目列表状态码:", r.status_code)
print("   含案例项目:", "玉田县典型历史遗留废弃矿山案例" in body)
```

```
assert "玉田县典型历史遗留废弃矿山案例" in body
# 新建项目
client.post("/project/add", data={"name": "测试项目A", "code": "T-001"})
r = client.get("/project/")
print("  新建项目后含「测试项目A」:", "测试项目A" in r.get_data(as_text=True))
assert "测试项目A" in r.get_data(as_text=True)
# 清理测试数据，避免污染演示库
with app.app_context():
    from app.models import db, Project
    for p in Project.query.filter_by(name="测试项目A").all():
        db.session.delete(p)
    db.session.commit()
print("  已清理测试项目")
print()
print(LINE)
print("测试 4  微地质单元（面积占比自动计算）")
print(LINE)
from app.models import MinePatch
with app.app_context():
    patch = MinePatch.query.filter_by(name="玉田县孤树镇白云岩采石场").first()
    pid = patch.id
r = client.get("/unit/%d" % pid)
body = r.get_data(as_text=True)
print("  单元列表含三类单元:", all(t in body for t in ("岩质边坡", "碎石边坡", "平台")))
print("  含占比合计:", "占比合计" in body)
assert "岩质边坡" in body and "占比合计" in body
print()
print(LINE)
print("测试 5  标准知识库与省际口径比对")
print(LINE)
r = client.get("/standard/")
body = r.get_data(as_text=True)
print("  含辽宁/湖南/新疆标准:", all(p in body for p in ("辽宁", "湖南", "新疆")))
assert "辽宁" in body
r = client.get("/standard/compare")
body = r.get_data(as_text=True)
print("  口径比对页含「植被覆盖度」分组:", "植被覆盖度" in body)
print("  含多省份并列:", "湖南" in body and "新疆" in body)
assert "植被覆盖度" in body
print()
print(LINE)
print("测试 6  限制性指标判定（一票否决）")
print(LINE)
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
```

```
print("   含五项条件：", all(x in body for x in ("政策约束", "区位约束",
    "土壤污染", "地质灾害", "耕地属性")))
assert all(x in body for x in ("政策约束", "区位约束", "土壤污染",
    "地质灾害", "耕地属性"))
#  全部通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("   全通过后显示「通过，可进入定量评价」：", "可进入定量评价" in body)
assert "可进入定量评价" in body
#  一项不通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "0",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("   一项不通过后显示「不通过」：", "不通过" in body)
print("   含失败项「区位约束」：", "区位约束" in body)
assert "不通过" in body
#  恢复为全部通过，避免把演示数据留在「不通过」状态影响后续步骤
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
print("   已恢复为全部通过：", "可进入定量评价" in r.get_data(as_text=True))
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
#  ==== 测试/test_web_evaluate.py ====
#  -*- coding: utf-8 -*-
"""端到端自检：录数据 → 执行评价 → 查看结果。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
#  与真实配置共用同一个 app.db，但关闭 CSRF 便于测试客户端
class TestConfig:
```

```
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  指标数据列表")
print(LINE)
r = client.get("/indicator/")
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含案例图斑：", "玉田县孤树镇白云岩采石场" in body)
print("  含三个单元：", all(n in body for n in ("岩质边坡", "碎石边坡",
    "平台")))
assert r.status_code == 200
assert "岩质边坡" in body and "碎石边坡" in body and "平台" in body
print()
print(LINE)
print("测试 2  指标数据录入页（按单元类型显示字段）")
print(LINE)
from app.models import MicroUnit, MinePatch
with app.app_context():
    unit_rock = MicroUnit.query.filter_by(unit_type="岩质边坡").
    first()
    unit_gravel = MicroUnit.query.filter_by(unit_type="碎石边坡"
    ).first()
r = client.get("/indicator/%d" % unit_rock.id)
b_rock = r.get_data(as_text=True)
print("  岩质边坡页：无「土石比」字段：", "土石比" not in b_rock, "（应 True）")
print("  岩质边坡页：含「植被覆盖度」：", "植被覆盖度" in b_rock)
assert "土石比" not in b_rock and "植被覆盖度" in b_rock
r = client.get("/indicator/%d" % unit_gravel.id)
b_gravel = r.get_data(as_text=True)
print("  碎石边坡页：含「土石比」字段：", "土石比" in b_gravel)
print("  碎石边坡页：含 8 种重金属：", all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn")))
assert "土石比" in b_gravel and all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn"))
print()
print(LINE)
print("测试 3  执行评价")
print(LINE)
with app.app_context():
    import seed_demo
```

```
    patch = seed_demo.demo_patch()
r = client.post("/evaluate/%d" % patch.id, data={},
    follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转至结果页：", "适宜性评价结果" in body)
print("  含最终评价因子 F：", "最终评价因子 F" in body)
print("  含等级「适宜」：", "适宜" in body)
print("  含推荐措施「自然恢复」：", "自然恢复" in body)
assert "最终评价因子 F" in body and "自然恢复" in body
print()
print(LINE)
print("测试 4  结果持久化")
print(LINE)
from app.models import EvaluationResult
with app.app_context():
    res = EvaluationResult.query.order_by(EvaluationResult.i
    d.desc()).first()
    print("  最终得分 F =", res.final_score)
    print("  等级 =", res.grade)
    assert res.grade == "适宜"
    assert 0.7 <= res.final_score < 1.0
print()
print(LINE)
print("端到端全部测试通过 √")
print(LINE)
# ==== 测试/test_web_report.py ====
# -*- coding: utf-8 -*-
"""报告输出自检：得分明细页 + Word 导出。"""
import os
import sys
import io
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
```

```
client.post("/login", data={"username": "admin",
    "password": "admin123"})
from app.models import MinePatch, EvaluationResult
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
    pid = patch.id
#  报告取自已保存的评价记录，因此先按网页路径真实执行一次评价
from app.views.evaluate import load_patch_input,
    save_evaluation
with app.app_context():
    units = load_patch_input(pid)
    rec = save_evaluation(pid, units)
    rid = rec.id
    print("前置：已执行评价并保存记录 id=%d，F=%.4f（%s）"
          % (rid, rec.final_score, rec.grade))
    from app.models import ScoreDetail
    n_detail = ScoreDetail.query.filter_by(result_id=rid).co
    unt()
    print("        逐层得分明细已落库：%d 条" % n_detail)
print()
print(LINE)
print("测试 1  报告列表")
print(LINE)
r = client.get("/report/")
print("  状态码：", r.status_code)
body = r.get_data(as_text=True)
print("  含「所属项目」列：", "所属项目" in body)
assert r.status_code == 200
print()
print(LINE)
print("测试 2  得分明细页")
print(LINE)
r = client.get("/report/%d" % pid)
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含「得分明细」：", "得分明细" in body)
print("  含「最终评价因子 F」：", "最终评价因子 F" in body)
print("  含二级指标明细（植被覆盖度 / 坡度 / 坡向）：", all(n in body for n in
    ("植被覆盖度", "坡度", "坡向")))
print("  含权重 / 标准化值列：", "标准化值" in body and "贡献" in body)
assert r.status_code == 200
assert "植被覆盖度" in body and "标准化值" in body
print()
print(LINE)
print("测试 3  Word 报告导出")
print(LINE)
r = client.get("/report/%d/word" % pid)
print("  状态码：", r.status_code)
print("  Content-Type:", r.headers.get("Content-Type",
```

```
        "")[:60])                                                    12051
data = r.data                                                        12052
print("   文件大小:", len(data), "字节")                              12053
print("   是否为有效 docx（PK 头）:", data[:2] == b"PK")              12054
assert r.status_code == 200 and data[:2] == b"PK"                    12055
# 用 python-docx 打开验证内容                                        12056
import docx                                                          12057
d = docx.Document(io.BytesIO(data))                                  12058
text = "\n".join(p.text for p in d.paragraphs)                       12059
print("   含报告标题:", "自然恢复适宜性评价报告" in text)             12060
print("   含评价结论章节:", "评价结论" in text)                       12061
print("   含得分明细章节:", "得分明细" in text)                       12062
print("   段落数:", len(d.paragraphs), " 表格数:", len(d.tables))     12063
assert "自然恢复适宜性评价报告" in text and len(d.tables) >= 3        12064
print()                                                              12065
print(LINE)                                                          12066
print("报告输出全部测试通过 √")                                     12067
print(LINE)                                                          12068
# ==== 测试/test_ui_fixes.py ====                                    12069
# -*- coding: utf-8 -*-                                              12070
"""界面修复自检：「记住我」与各模块删除功能。"""                     12071
import io                                                            12072
import os                                                            12073
import re                                                            12074
import sys                                                           12075
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，          12076
# 使 `from app import ...` 与 app/app.db 的定位不受影响。            12077
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil12078
    e__)))                                                           12079
if ROOT not in sys.path:                                             12080
    sys.path.insert(0, ROOT)                                         12081
from app import create_app                                           12082
sys.stdout.reconfigure(encoding="utf-8")                             12083
LINE = "=" * 70                                                      12084
class TestConfig:                                                    12085
    SECRET_KEY = "test"                                              12086
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(           12087
        ROOT, "app", "app.db")                                       12088
    SQLALCHEMY_TRACK_MODIFICATIONS = False                           12089
    WTF_CSRF_ENABLED = False                                         12090
    TESTING = True                                                   12091
    REMEMBER_COOKIE_DURATION = __import__("datetime").timede12092
    lta(days=30)                                                     12093
app = create_app(TestConfig)                                         12094
client = app.test_client()                                           12095
# ------------------------------------------------------------------12096
    ----------------                                                 12097
print(LINE)                                                          12098
print("一、「记住我」")                                             12099
print(LINE)                                                          12100
```

```
html = client.get("/login").get_data(as_text=True)
# 1) 复选框必须有可点击关联的 label（这是原先失效的根因）
#    只判断「复选框与文字是否落在同一个 <label> 内」，不绑定具体类名 ——
#    否则换个样式类就会误报，而真正的回归点其实与类名无关。
labels = re.findall(r"<label[^>]*>(.*?)</label>", html,
    re.S)
wrapped = any(('name="remember"' in b and "记住我" in b) for b
    in labels)
print("  %s 「记住我」文字与复选框包在同一 label 内" % ("[符合]" if wrapped
    else "[不符合]"))
assert wrapped, "复选框未与文字关联，点击文字不会勾选"
has_for = 'for="remember"' in html
print("  %s label 带 for 属性指向复选框" % ("[符合]" if has_for else
    "[不符合]"))
# 2) 勾选后应下发带过期时间的 remember_token
c = app.test_client()
r = c.post("/login", data={"username": "admin", "password":
    "admin123",
                           "remember": "y"},
    follow_redirects=False)
cookies = r.headers.getlist("Set-Cookie")
rt = [h for h in cookies if h.startswith("remember_token=")]
print("  %s 勾选后下发 remember_token" % ("[符合]" if rt else
    "[不符合]"))
assert rt, "未下发 remember_token"
exp = re.search(r"Expires=([^;]+)", rt[0])
print("  %s 该 cookie 带过期时间：%s"
      % ("[符合]" if exp else "[不符合]", exp.group(1) if exp
    else "无（会话 cookie）"))
assert exp, "remember_token 无过期时间，关浏览器即失效"
# 3) 不勾选时不应下发 remember_token
c2 = app.test_client()
r2 = c2.post("/login", data={"username": "admin",
    "password": "admin123"},
             follow_redirects=False)
rt2 = [h for h in r2.headers.getlist("Set-Cookie") if
    h.startswith("remember_token=")]
print("  %s 未勾选时不下发 remember_token" % ("[符合]" if not rt2
    else "[不符合]"))
assert not rt2
# 4) 携带 remember_token 的新会话应自动登录
c3 = app.test_client()
c3.set_cookie("remember_token",
              rt[0].split(";")[0].split("=", 1)[1],
    domain="localhost")
r3 = c3.get("/", follow_redirects=False)
print("  %s 仅凭 remember_token 访问首页可自动登录（状态码 %d）"
      % ("[符合]" if r3.status_code == 200 else "[不符合]",
    r3.status_code))
assert r3.status_code == 200
```

```
# ------------------------------------------------------------------
    ----------------
print()
print(LINE)
print("二、删除功能")
print(LINE)
client.post("/login", data={"username": "admin",
    "password": "admin123"})
# 项目：列表页须有删除按钮
html = client.get("/project/").get_data(as_text=True)
has_del = "/delete" in html and "btn--danger" in html
print("  %s 项目列表含删除按钮" % ("[符合]" if has_del else "[不符合]"))
assert has_del, "项目列表缺少删除按钮"
print("  %s 删除前有二次确认" % ("[符合]" if "confirm(" in html else
    "[不符合]"))
assert "confirm(" in html
# 图斑列表：地址用的是项目主键，不是图斑主键
with app.app_context():
    from app.models import MinePatch, Project
    import seed_demo
    patch = seed_demo.demo_patch()
    patch_id = patch.id
    project_id = patch.project_id
html = client.get("/project/%d/patches" %
    project_id).get_data(as_text=True)
print("  %s 图斑列表含删除按钮" % ("[符合]" if "/project/patch/" in
    html and "btn--danger" in html else "[不符合]"))
assert "/project/patch/" in html and "btn--danger" in html
# 微地质单元
html = client.get("/unit/%d" % patch_id).get_data(as_text=Tr
    ue)
print("  %s 微单元列表含删除按钮" % ("[符合]" if "/unit/delete/" in
    html else "[不符合]"))
assert "/unit/delete/" in html
print("  %s 微单元编辑为独立页面（表单不再嵌在表格行内）"
      % ("[符合]" if "/unit/edit/" in html else "[不符合]"))
# 标准
html = client.get("/standard/").get_data(as_text=True)
print("  %s 标准列表含删除按钮" % ("[符合]" if "btn--danger" in html
    else "[不符合]"))
assert "btn--danger" in html
# 实际执行一次删除：新建项目 → 删除 → 确认消失
client.post("/project/add", data={"name": "待删除测试项目",
    "code": "DEL-1"})
with app.app_context():
    from app.models import db, Project
    tmp = Project.query.filter_by(name="待删除测试项目").first()
    tmp_id = tmp.id
html = client.get("/project/").get_data(as_text=True)
print("  %s 新建项目成功" % ("[符合]" if "待删除测试项目" in html else
```

```
        "［不符合］"））
r = client.post("/project/%d/delete" % tmp_id,
        follow_redirects=True)
html = r.get_data(as_text=True)
gone = "待删除测试项目" not in html
print("  %s 删除项目后列表中不再出现" % ("［符合］" if gone else "［不符合］"))
assert gone
print("  %s 有删除成功提示" % ("［符合］" if "已删除" in html else
        "［不符合］"))
print()
print(LINE)
print("六、顶栏宽度在各页面保持一致")
print(LINE)
#  滚动条出现在 .app-content 内部：内容高的页面会因此把顶栏挤窄约 10px，
#  于是「顶栏在不同页面不一样宽」。scrollbar-gutter: stable 预留槽位后各页一致。
base_src = io.open(os.path.join(ROOT, "app", "templates",
        "base.html"),
                           encoding="utf-8").read()
has_gutter = "scrollbar-gutter: stable" in base_src
print("  %s 内容区预留滚动条槽位（顶栏宽度不再随页面变化）"
        % ("［符合］" if has_gutter else "［不符合］"))
assert has_gutter, "缺少 scrollbar-gutter:
        stable，顶栏宽度会因滚动条而不一致"
#  搜索框宽度不随窗口大小变：原来 960px 断点把它缩到 180px，同一台机器换窗口
#  大小看着就像两个尺寸。统一保持 268px（640px 以下才改为铺满，属手机版式）。
no_shrink = "width: 180px" not in base_src
print("  %s 搜索框宽度在窄屏断点下不再缩小"
        % ("［符合］" if no_shrink else "［不符合］"))
assert no_shrink, "搜索框在 960px 断点被缩成 180px，宽度会随窗口变化"
print()
print(LINE)
print("界面修复自检全部通过 √")
print(LINE)
#  ==== 测试/test_new_features.py ====
#  -*- coding: utf-8 -*-
"""补齐模块自检：操作日志与用户维护、遥感影像、指标与多专家矩阵、
批量导入、图表与 PDF 导出。
对应计划书功能清单与系统实现的差异核对结论。
"""
import io
import json
import os
import re
import sys
sys.stdout.reconfigure(encoding="utf-8")
import fitz
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__
        e__)))
```

```
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, User, Project, MinePatch,
    MicroUnit, IndicatorData,
                            Indicator, ExpertMatrix,
    OperationLog)
from app.utils.schema import ensure_schema
from app.utils import browser, charts
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    UPLOAD_FOLDER = os.path.join(BASE, "app", "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
app = create_app(TestConfig)
with app.app_context():
    db.create_all()
    ensure_schema(db)
    ExpertMatrix.query.delete()
    db.session.commit()
    # 固定取内置演示案例的图斑，避免受库中用户自建项目 / 验证算例的影响
    import seed_demo
    patch_id = seed_demo.demo_patch().id
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
def ok(flag, label, extra=""):
    print("  %s %-44s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    assert flag, label
# ------------------------------------------------------------------
    ------------------
print(LINE)
print("操作日志与用户维护")
print(LINE)
before = None
with app.app_context():
    before = OperationLog.query.count()
c.post("/project/add", data={"name": "日志核对项目", "code":
    "LOG-T"})
with app.app_context():
```

```
    after = OperationLog.query.count()
    last = OperationLog.query.order_by(OperationLog.id.desc(
    )).first()
ok(after == before + 1, "写操作被写入操作日志", "%d -> %d" % (before,
    after))
ok(last.action == "新建项目", "日志记录了可读动作名", last.action)
ok("project" in (last.target_type or ""), "日志记录了对象类型",
    last.target_type)
r = c.get("/profile?tab=logs")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "操作日志" in body, "操作日志分类可访问")
ok("新建项目" in body, "日志页显示操作记录")
# 账号编辑 / 删除（职务只有管理员与操作员两种）
with app.app_context():
    u = User(username="tmp_user", role="operator",
             password_hash=User.query.first().password_hash)
    db.session.add(u); db.session.commit()
    uid = u.id
r = c.get("/users/%d/edit" % uid)
ok(r.status_code == 200, "账号编辑页可访问")
# 保护校验要趁系统里只有一名管理员时做：
# 先降级唯一的 admin 应被拒绝，再把临时账号提为管理员看修改是否生效
with app.app_context():
    admin_id = User.query.filter_by(role="admin").first().id
c.post("/users/%d/edit" % admin_id,
       data={"role": "operator", "active": "1"})
with app.app_context():
    ok(db.session.get(User, admin_id).role == "admin",
    "拒绝把最后一个管理员降级")
c.post("/users/%d/edit" % uid, data={"role": "admin",
    "active": "1"})
with app.app_context():
    ok(db.session.get(User, uid).role == "admin",
    "职务已修改并保存")
c.post("/users/%d/delete" % uid, follow_redirects=True)
with app.app_context():
    ok(db.session.get(User, uid) is None, "账号删除生效")
# 清理日志核对用的项目
with app.app_context():
    for p in Project.query.filter_by(name="日志核对项目").all():
        db.session.delete(p)
    db.session.commit()
# ----------------------------------------------------------------
    ------------------
print()
print(LINE)
print("遥感影像挂接")
print(LINE)
doc = fitz.open()
pg = doc.new_page(width=320, height=200)
```

```
pg.insert_text((20, 60), "RS image", fontname="china-s",
    fontsize=16)
png = doc[0].get_pixmap(dpi=100).tobytes("png")
doc.close()
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(png), "影像.png"),
                 "image_note": "2025年6月无人机正射影像"},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("已挂接" in body and "<img" in body, "影像上传并显示")
with app.app_context():
    rel = db.session.get(MinePatch, patch_id).image_path
ok(bool(rel), "数据库记录影像路径", rel)
abs_path = os.path.join(TestConfig.UPLOAD_FOLDER,
    rel.replace("uploads/", "", 1))
ok(os.path.isfile(abs_path), "影像文件落盘")
ok(c.get("/static/" + rel).status_code == 200, "影像可经 URL
    访问")
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(b"x"), "bad.txt")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("不支持" in r.get_data(as_text=True), "拒绝非图片格式")
c.post("/project/patch/%d/image/delete" % patch_id,
    follow_redirects=True)
with app.app_context():
    ok(db.session.get(MinePatch, patch_id).image_path is
    None, "影像移除后字段清空")
ok(not os.path.isfile(abs_path), "影像文件已删除")
# ----------------------------------------------------------------
    ------------------
print()
print(LINE)
print("指标定义维护与多专家矩阵")
print(LINE)
r = c.get("/ahp/indicators")
body = r.get_data(as_text=True)
with app.app_context():
    n_ind = Indicator.query.count()
ok(r.status_code == 200 and n_ind > 0, "指标定义已入库并可查看", "%d
    项" % n_ind)
ok(all(k in body for k in ("一级指标", "二级指标", "三级指标")),
   "指标树按层级展示")
ok("岩质边坡" in body and "平台" in body, "显示各指标启用的微地质单元")
c.post("/ahp/indicators/coverage/edit",
       data={"name": "植被覆盖度", "unit": "%", "ideal_type":
    "benefit",
             "ideal_value": "75", "enabled_units": ["岩质边坡"],
             "description": "自检修改"})
```

```
with app.app_context():
    ok(Indicator.query.filter_by(code="coverage").first().de
    scription == "自检修改",
        "指标定义可编辑")
c.post("/ahp/indicators/reset")
with app.app_context():
    ok("自检修改" not in (Indicator.query.filter_by(code="covera
    ge").first().description or ""),
        "可按系统默认值重置")
base = {"m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24"}
for name in ("张工", "李工", "王工"):
    d = dict(base); d.update({"action": "add", "unit_type":
    "碎石边坡",
                              "level": "1", "expert_name":
    name})
    c.post("/ahp/experts", data=d)
with app.app_context():
    ok(ExpertMatrix.query.filter_by(unit_type="碎石边坡",
    level=1).count() == 3,
        "录入 3 位专家的判断矩阵")
r = c.get("/ahp/experts/aggregate?unit_type=碎石边坡&level=1")
body = r.get_data(as_text=True)
ok("汇总后权重" in body and "各专家矩阵的一致性" in body,
   "多专家矩阵几何平均汇总可用")
ws = re.search(r"<th scope=\"row\">权重</th>\s*(.*?)</tr>",
    body, re.S)
vals = re.findall(r">([0-9.]+)</td>", ws.group(1)) if ws
    else []
ok(len(vals) == 3 and abs(sum(map(float, vals)) - 1.0) <
    0.01,
   "汇总权重归一", "、".join(vals))
ok(abs(float(vals[0]) - 0.44) < 0.02, "汇总权重接近论文表 3 取值",
    vals[0] if vals else "")
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("批量导入")
print(LINE)
r = c.get("/indicator/import/template")
tpl = r.data.decode("utf-8-sig")
ok(r.status_code == 200 and "单元类型" in tpl, "导入模板可下载")
ok("土石比" in tpl and "植被覆盖度" in tpl, "模板包含指标列")
csv_text = (
    "单元类型,土石比,土层厚度,pH,有机质,Hg,As,Ni,Cu,Cd,Pb,植被覆盖度,Shannon,坡度
    ,坡向\n"
    "碎石边坡,0.34,10,9.24,4.89,0.05,5.17,12.60,11.70,0.04,6.50,
    87.92,0.60,28.75,阳坡\n"
    "平台,0.56,30,8.72,10.92,0.02,3.13,12.33,11.14,0.09,9.98,5
    8.42,0.72,,,\n"
```

```
        "不存在单元，1，1，7，1，，，，，，，，，\n"                                12451
)                                                                                12452
r = c.post("/indicator/import",                                                  12453
                data={"patch_id": str(patch_id),                                 12454
                      "datafile": (io.BytesIO(csv_text.encode("ut                12455
        f-8")), "d.csv")},                                                       12456
                content_type="multipart/form-data",                              12457
        follow_redirects=True)                                                   12458
body = r.get_data(as_text=True)                                                  12459
ok("导入结果" in body, "导入结果页展示")                                          12460
ok("不存在单元" in body, "错误行被报告")                                          12461
with app.app_context():                                                          12462
        u = MicroUnit.query.filter_by(patch_id=patch_id,                         12463
        unit_type="碎石边坡").first()                                             12464
        ind = IndicatorData.query.filter_by(unit_id=u.id).first(                 12465
        )                                                                        12466
        ok(abs(ind.soil_ratio - 0.34) < 1e-6 and                                 12467
        abs(ind.coverage - 87.92) < 1e-6,                                        12468
            "导入数据已写入数据库")                                               12469
r = c.post("/indicator/import",                                                  12470
                data={"patch_id": str(patch_id),                                 12471
                      "datafile": (io.BytesIO(b"x"), "d.pdf")},                  12472
                content_type="multipart/form-data",                              12473
        follow_redirects=True)                                                   12474
ok("仅支持" in r.get_data(as_text=True), "拒绝不支持的导入格式")                  12475
# ------------------------------------------------------------------------------12476
        ------------------                                                       12477
print()                                                                          12478
print(LINE)                                                                      12479
print("图表与 PDF 导出")                                                         12480
print(LINE)                                                                      12481
svg = charts.hbars([("岩质边坡", 0.54, "0.5401"), ("平台", 0.84,                 12482
        "0.8376")])                                                              12483
ok("<svg" in svg and "0.8376" in svg, "条形图生成且带数值标签")                  12484
svg2 = charts.donut([("植被指标", 0.6, "0.6000"), ("土壤指标", 0.4,               12485
        "0.4000")])                                                              12486
ok("<svg" in svg2 and "占比" in svg2, "环形图生成且带图例")                       12487
# 报告是评价结论的呈现，需先真实执行一次评价才能打开                              12488
with app.app_context():                                                          12489
        from app.views.evaluate import load_patch_input,                         12490
        save_evaluation                                                          12491
        save_evaluation(patch_id, load_patch_input(patch_id))                    12492
r = c.get("/report/%d" % patch_id)                                               12493
body = r.get_data(as_text=True)                                                  12494
ok(body.count("<svg") >= 2, "得分明细页含图表", "%d 个 SVG" %                     12495
        body.count("<svg"))                                                      12496
exe = browser.browser_path()                                                     12497
if exe is None:                                                                  12498
        print("  [跳过] 未找到浏览器，PDF 导出无法验证")                         12499
else:                                                                            12500
```

```
    r = c.get("/report/%d/pdf" % patch_id)
    data = r.data
    ok(data[:4] == b"%PDF", "PDF 报告导出成功", "%d 字节" %
    len(data))
    d = fitz.open(stream=data, filetype="pdf")
    txt = "".join(d[i].get_text() for i in
    range(d.page_count))
    ok("自然恢复适宜性评价报告" in txt, "PDF 含报告标题")
    ok("评价结论" in txt and "得分明细" in txt, "PDF 含结论与明细章节")
    ok(d.page_count >= 2, "PDF 页数合理", "%d 页" % d.page_count)
    d.close()
print()
print(LINE)
print("补齐模块自检全部通过 √")
print(LINE)
#  ==== 测试/test_workflow.py ====
#  -*- coding: utf-8 -*-
"""流程门禁自检：前置步骤未完成时，后续模块必须拦得住。
论文 4.1 节确定的评价流程存在先后关系，缺任何一环都不应产出评价结论。
本测试逐个构造「缺前置」的场景，验证系统不会放行。
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__
    )))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult)
from app.services import workflow as wf
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
```

```
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
def purge_fixtures():
    """开跑前先清掉上一次可能残留的自检项目。
    自检项目按 code 前缀命名（WF-）；如果上一次跑到一半被打断（报错、被中断），
    清理函数没执行到，残留的项目就会出现在工作台与项目列表里 ——
    生成软著截图时这些测试数据会被拍进去。
    """
    with app.app_context():
        leftovers = Project.query.filter(Project.code.like('
    WF-%')).all()
        for p in leftovers:
            for m in p.patches.all():
                db.session.delete(m)
            db.session.delete(p)
        db.session.commit()
        return len(leftovers)
def new_patch(tag):
    with app.app_context():
        p = Project(name="门禁-" + tag, code="WF-" + tag)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="门禁图斑-" +
    tag)
        db.session.add(patch); db.session.flush()
        pid = patch.id
        db.session.commit()
    return pid
def add_units(pid):
    with app.app_context():
        db.session.add(MicroUnit(patch_id=pid,
    unit_type="碎石边坡",
                                 area=1.0, ratio=1.0))
        db.session.commit()
def add_constraint(pid, passed=True):
    with app.app_context():
        db.session.add(ConstraintCheck(
            patch_id=pid, policy_pass=True,
    location_pass=passed,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=passed, reason="全部通过" if passed else
    "未通过：区位约束"))
        db.session.commit()
def add_data(pid):
    with app.app_context():
        u = MicroUnit.query.filter_by(patch_id=pid).first()
        db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.34, thickness=10,
```

```
                                                ph=9.24,                    12601
        organic_matter=4.89, coverage=87.92,                                12602
                                                shannon=0.60,               12603
        slope=28.75, aspect="阳坡"))                                        12604
            db.session.commit()                                             12605
def cleanup(pid, tag):                                                      12606
    with app.app_context():                                                 12607
        from app.models import ScoreDetail                                  12608
        # 先删得分明细，再删评价记录，避免留下孤儿明细                      12609
        rids = [r.id for r in EvaluationResult.query.filter_                12610
    by(patch_id=pid)]                                                       12611
        if rids:                                                            12612
            ScoreDetail.query.filter(ScoreDetail.result_id.i                12613
    n_(rids)).delete(                                                       12614
                synchronize_session=False)                                  12615
        EvaluationResult.query.filter_by(patch_id=pid).delet                12616
    e()                                                                     12617
        ConstraintCheck.query.filter_by(patch_id=pid).delete                12618
    ()                                                                      12619
        uids = [u.id for u in MicroUnit.query.filter_by(pat                 12620
    h_id=pid)]                                                              12621
        if uids:                                                            12622
            IndicatorData.query.filter(IndicatorData.unit_id                12623
    .in_(uids)).delete(                                                     12624
                synchronize_session=False)                                  12625
        MicroUnit.query.filter_by(patch_id=pid).delete()                    12626
        m = db.session.get(MinePatch, pid)                                  12627
        if m:                                                               12628
            db.session.delete(m)                                            12629
        pr = Project.query.filter_by(name="门禁-" +                         12630
    tag).first()                                                            12631
        if pr:                                                              12632
            db.session.delete(pr)                                           12633
        db.session.commit()                                                 12634
def results_of(pid):                                                        12635
    with app.app_context():                                                 12636
        return EvaluationResult.query.filter_by(patch_id=pid                12637
    ).count()                                                               12638
def post_eval(pid):                                                         12639
    return c.post("/evaluate/%d" % pid, data={},                            12640
    follow_redirects=False)                                                 12641
def get_report(pid):                                                        12642
    return c.get("/report/%d" % pid, follow_redirects=False)                12643
# ------------------------------------------------------------------------12644
    ----------------                                                        12645
print(LINE)                                                                 12646
print("一、前置条件缺失时，评价必须被拦住")                                 12647
print(LINE)                                                                 12648
print("  （开跑前清掉 %d 个上一次残留的自检项目）" % purge_fixtures())       12649
print()                                                                     12650
```

```
print("场景 1：连微地质单元都没划分")
pid = new_patch("1")
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未划分单元 → 不产生评价结果",
   "HTTP %d，结果数 %d" % (r.status_code, n))
with app.app_context():
    pr = wf.patch_progress(pid)
steps = [b["step"] for b in pr["blocking"]]
ok(not pr["ready"] and "units" in steps,
   "拦原因包含「划分微地质单元」", "、".join(steps))
ok(steps == ["constraint", "units"],
   "拦截顺序符合论文 4.1 节：限制性判定在前，划分单元在后",
   "、".join(steps))
cleanup(pid, "1")
print()
print("场景 2：有单元，但未做限制性条件判定")
pid = new_patch("2"); add_units(pid); add_data(pid)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未做限制性判定 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "constraint" for b in pr["blocking"]),
   "拦原因指向「执行限制性条件判定」")
cleanup(pid, "2")
print()
print("场景 3：限制性条件判定【未通过】")
pid = new_patch("3"); add_units(pid); add_data(pid);
    add_constraint(pid, passed=False)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "判定未通过 → 不进入定量评价",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(not pr["constraint"]["passed"], "流程状态显示判定未通过")
blocked = [b for b in pr["blocking"] if b["step"] ==
    "constraint"]
# 提示只写「缺什么」：label 说明未通过，reason 直接引用判定记录里没过的项
ok(bool(blocked) and "未通过" in blocked[0]["label"],
   "拦提示指明限制性条件判定未通过", blocked[0]["label"] if blocked else
    "")
ok(bool(blocked) and "区位约束" in (blocked[0]["reason"] or ""),
   "并把具体没通过的条件带出来", blocked[0]["reason"] if blocked else "")
cleanup(pid, "3")
print()
print("场景 4：判定通过，但指标数据全空")
pid = new_patch("4"); add_units(pid); add_constraint(pid,
```

```
        passed=True)                                                    12701
r = post_eval(pid)                                                      12702
n = results_of(pid)                                                     12703
ok(r.status_code == 302 and n == 0, "数据未录入 → 不产生评价结果",          12704
    "结果数 %d" % n)                                                    12705
with app.app_context():                                                 12706
    pr = wf.patch_progress(pid)                                         12707
ok(any(b["step"] == "data" for b in pr["blocking"]),                    12708
    "拦原因指向「录入指标数据」")                                        12709
ok(pr["data"]["missing"] == ["碎石边坡"],                               12710
    "指出具体缺哪个单元的数据", "、".join(pr["data"]["missing"]))        12711
cleanup(pid, "4")                                                       12712
print()                                                                 12713
print("场景 5：单元面积占比合计不为 100%")                              12714
pid = new_patch("5"); add_constraint(pid, passed=True)                  12715
with app.app_context():                                                 12716
    u = MicroUnit(patch_id=pid, unit_type="碎石边坡", area=1.0,         12717
    ratio=0.5)                                                          12718
    db.session.add(u); db.session.commit()                              12719
    db.session.add(IndicatorData(unit_id=u.id,                          12720
    soil_ratio=0.3, thickness=10,                                       12721
                                 ph=8.0,                                12722
    organic_matter=5.0, coverage=80.0,                                  12723
                                 shannon=0.6, slope=20.0,               12724
    aspect="阳坡"))                                                     12725
    db.session.commit()                                                 12726
r = post_eval(pid)                                                      12727
ok(results_of(pid) == 0, "占比不足 100% → 不产生评价结果")              12728
cleanup(pid, "5")                                                       12729
print()                                                                 12730
print(LINE)                                                             12731
print("二、前置条件齐备时才放行")                                       12732
print(LINE)                                                             12733
pid = new_patch("OK"); add_units(pid); add_constraint(pid,              12734
    passed=True); add_data(pid)                                         12735
with app.app_context():                                                 12736
    pr = wf.patch_progress(pid)                                         12737
ok(pr["ready"], "单元 + 判定通过 + 数据齐备 → 判定为可执行")            12738
r = post_eval(pid)                                                      12739
n = results_of(pid)                                                     12740
ok(n == 1, "此时执行评价成功产生结果", "结果数 %d" % n)                  12741
print("  重定向到：", r.headers.get("Location"))                        12742
print()                                                                 12743
print(LINE)                                                             12744
print("三、报告不得先于评价产生")                                       12745
print(LINE)                                                             12746
pid2 = new_patch("R")    # 什么都不做                                   12747
r = get_report(pid2)                                                    12748
ok(r.status_code == 302 and "/report/" ==                               12749
    (r.headers.get("Location") or ""),                                  12750
```

```
    "前置缺失时 /report/<id> 被退回列表", "HTTP %d" % r.status_code)
for suffix in ("/pdf", "/word"):
    rr = c.get("/report/%d%s" % (pid2, suffix),
    follow_redirects=False)
    ok(rr.status_code == 302, "导出 %s 同样被拦住" % suffix, "HTTP
    %d" % rr.status_code)
cleanup(pid2, "R")
cleanup(pid, "OK")
print()
print(LINE)
print("四、前置齐备但从未执行评价时，报告仍不得产生")
print(LINE)
#  报告是评价结论的呈现：即使前置步骤都做完了，只要没真正执行过定量评价，
#  也不应凭空推算出结论。
pid3 = new_patch("N")
add_units(pid3); add_constraint(pid3, passed=True);
    add_data(pid3)
with app.app_context():
    pr = wf.report_progress(pid3)
ok(not pr["ready"], "无评价记录 → 报告流程判定为未就绪")
ok(bool(pr["blocking"]) and pr["blocking"][0]["step"] ==
    "evaluate",
    "拦原因指向「执行适宜性评价」")
r = get_report(pid3)
ok(r.status_code == 302 and results_of(pid3) == 0,
    "直接访问 /report/<id> 被退回，且不产生任何评价结果",
    "HTTP %d" % r.status_code)
#  真正执行评价之后，报告才放行，且取值与评价记录一致
r = post_eval(pid3)
ok(results_of(pid3) == 1, "先执行评价 → 产生评价记录")
with app.app_context():
    pr = wf.report_progress(pid3)
    rec = pr["result"]
ok(pr["ready"] and rec is not None, "有评价记录后，报告流程就绪")
rr = get_report(pid3)
body = rr.get_data(as_text=True)
ok(rr.status_code == 200, "此时可打开得分明细页", "HTTP %d" %
    rr.status_code)
ok(("%.4f" % rec.final_score) in body,
    "报告数值取自评价记录，而非重新计算",
    "F=%.4f" % rec.final_score)
cleanup(pid3, "N")
print()
print(LINE)
print("五、内置演示数据的清理不得波及用户自建数据")
print(LINE)
#  回归保护：seed_demo.clean_case() 曾执行 Project.query.delete()，
#  导致用户新建的项目在每次启动时被连带删除。
import seed_demo as sd
with app.app_context():
```

```
    keep_p = Project(name="用户自建项目-守护", code="KEEP-001")
    db.session.add(keep_p); db.session.flush()
    keep_patch = MinePatch(project_id=keep_p.id,
    name="用户自建图斑-守护")
    db.session.add(keep_patch); db.session.flush()
    keep_pid = keep_p.id
    keep_patch_id = keep_patch.id
    db.session.commit()
with app.app_context():
    sd.clean_case(reset=False)
with app.app_context():
    still_p = db.session.get(Project, keep_pid)
    still_patch = db.session.get(MinePatch, keep_patch_id)
ok(still_p is not None, "清理演示数据后，用户自建项目仍存在")
ok(still_patch is not None, "清理演示数据后，用户自建图斑仍存在")
#  演示数据清掉后应能重新写入，保证演示环境可恢复
sd.seed()
with app.app_context():
    demo_back = Project.query.filter_by(code=sd.DEMO_CODE).f
    irst()
ok(demo_back is not None, "演示案例可重新写入（幂等恢复）")
with app.app_context():
    m = db.session.get(MinePatch, keep_patch_id)
    if m:
        db.session.delete(m)
    p = db.session.get(Project, keep_pid)
    if p:
        db.session.delete(p)
    db.session.commit()
print()
print(LINE)
print("六、同一图斑不得重复划分同类型微地质单元")
print(LINE)
#  论文 4.1 节把图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
#  允许重复会让面积占比、按类型组织的指标数据与批量导入对不上号。
pid4 = new_patch("D")
c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "1.0"},
       follow_redirects=True)
with app.app_context():
    n1 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n1 == 1, "首次添加「平台」成功", "单元数 %d" % n1)
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "2.0"},
           follow_redirects=True)
with app.app_context():
    n2 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n2 == 1, "重复添加同类型被拒绝", "单元数 %d" % n2)
ok("同一类型只能划分一个" in r.get_data(as_text=True), "给出重复提示")
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "火山口",
```

```
        "area": "1.0"},                                                  12851
                  follow_redirects=True)                                 12852
with app.app_context():                                                  12853
        n3 = MicroUnit.query.filter_by(patch_id=pid4).count()            12854
ok(n3 == 1, "非法单元类型被拒绝", "单元数 %d" % n3)                          12855
c.post("/unit/%d/add" % pid4, data={"unit_type": "碎石边坡",              12856
        "area": "3.0"},                                                  12857
            follow_redirects=True)                                       12858
with app.app_context():                                                  12859
        uid = MicroUnit.query.filter_by(patch_id=pid4,                   12860
                                        unit_type="碎石边坡").first(     12861
        ).id                                                             12862
r = c.post("/unit/edit/%d" % uid,                                        12863
                  data={"unit_type": "平台", "area": "3.0"},             12864
        follow_redirects=True)                                           12865
with app.app_context():                                                  12866
        types = sorted(u.unit_type for u in                              12867
                       MicroUnit.query.filter_by(patch_id=pid4)          12868
        all())                                                           12869
ok(types == ["平台", "碎石边坡"], "改类型时同样拦截重复", "、".join(types))    12870
cleanup(pid4, "D")                                                       12871
print()                                                                  12872
print(LINE)                                                              12873
if _fails:                                                               12874
        print("存在未通过项：")                                             12875
        for f in _fails:                                                 12876
            print("  -", f)                                             12877
        sys.exit(1)                                                      12878
print("流程门禁自检全部通过 √")                                             12879
print(LINE)                                                              12880
#  ==== 测试/test_weights.py ====                                        12881
#  -*- coding: utf-8 -*-                                                 12882
"""权重生效链路自检：AHP 算出的权重必须真正参与评价计算。                          12883
此前 weight_schemes / weight_items 两张表从未被写入，评价引擎也只用              12884
代码里的论文表 3 默认权重，相当于权重模块算完即弃。本测试守住这条链路：           12885
    默认 → 论文表 3                                                      12886
    保存 AHP 权重 → 评价改用新权重，结果随之变化                          12887
    恢复默认 → 结果回到论文取值                                          12888
"""                                                                      12889
import os                                                                12890
import sys                                                               12891
sys.stdout.reconfigure(encoding="utf-8")                                 12892
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，                  12893
#  使 `from app import ...` 与 app/app.db 的定位不受影响。                12894
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil             12895
    e__)))                                                               12896
if ROOT not in sys.path:                                                 12897
    sys.path.insert(0, ROOT)                                             12898
from app import create_app                                               12899
from app.models import (db, Project, MinePatch, MicroUnit,               12900
```

```
    IndicatorData,                                                  12901
                        ConstraintCheck, EvaluationResult,          12902
    WeightScheme,                                                   12903
                        WeightItem)                                 12904
from app.services import workflow as wf                            12905
from app.services import weight_service as ws                      12906
LINE = "=" * 72                                                    12907
BASE = ROOT                                                        12908
class TestConfig:                                                  12909
    SECRET_KEY = "test"                                            12910
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +                       12911
    os.path.join(BASE, "app", "app.db")                            12912
    SQLALCHEMY_TRACK_MODIFICATIONS = False                         12913
    WTF_CSRF_ENABLED = False                                       12914
    TESTING = True                                                 12915
app = create_app(TestConfig)                                       12916
c = app.test_client()                                              12917
c.post("/login", data={"username": "admin", "password":            12918
    "admin123"})                                                   12919
_fails = []                                                        12920
def ok(flag, label, extra=""):                                     12921
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",      12922
    label, extra))                                                 12923
    if not flag:                                                   12924
        _fails.append(label)                                       12925
TAG = "W"                                                          12926
UNIT_TYPE = "碎石边坡"                                             12927
def build_case():                                                  12928
    """建一个只含单一微地质单元的图斑，便于观察权重变化对得分的影响。"""  12929
    with app.app_context():                                        12930
        p = Project(name="权重测试-%s" % TAG, code="WEIGHT-%s"      12931
    % TAG)                                                         12932
        db.session.add(p); db.session.flush()                      12933
        patch = MinePatch(project_id=p.id, name="权重测试图斑-%s"   12934
    % TAG)                                                         12935
        db.session.add(patch); db.session.flush()                  12936
        db.session.add(MicroUnit(patch_id=patch.id,                12937
    unit_type=UNIT_TYPE,                                           12938
                                 area=1.0, ratio=1.0))             12939
        db.session.flush()                                         12940
        u = MicroUnit.query.filter_by(patch_id=patch.id).fi        12941
    st()                                                           12942
        db.session.add(IndicatorData(                              12943
            unit_id=u.id, soil_ratio=0.34, thickness=10,           12944
    ph=9.24,                                                       12945
            organic_matter=4.89, coverage=87.92,                   12946
    shannon=0.60,                                                  12947
            slope=28.75, aspect="阳坡"))                           12948
        db.session.add(ConstraintCheck(                            12949
            patch_id=patch.id, policy_pass=True,                   12950
```

```
    location_pass=True,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=True, reason="全部通过"))
        db.session.commit()
        return patch.id
def purge_fixtures():
    """开跑前清掉上一次可能残留的自检项目（code 前缀 WEIGHT-）。
    上一次跑到一半被打断时清理函数执行不到，残留项目会出现在工作台与
    项目列表里，生成软著截图时会被拍进去。
    """
    with app.app_context():
        from app.models import Project, ReferenceEcosystem
        leftovers = Project.query.filter(Project.code.like("
    WEIGHT-%")).all()
        for p in leftovers:
            for m in p.patches.all():
                uids = [u.id for u in
    MicroUnit.query.filter_by(patch_id=m.id)]
                if uids:
                    IndicatorData.query.filter(
                        IndicatorData.unit_id.in_(uids)).del
    ete(
                        synchronize_session=False)
                EvaluationResult.query.filter_by(patch_id=m
    id).delete()
                ConstraintCheck.query.filter_by(patch_id=m.i
    d).delete()
                ReferenceEcosystem.query.filter_by(patch_id=
    m.id).delete()
                MicroUnit.query.filter_by(patch_id=m.id).del
    ete()
                db.session.delete(m)
            db.session.delete(p)
        db.session.commit()
        return len(leftovers)
def cleanup(pid):
    with app.app_context():
        from app.models import ScoreDetail
        rids = [r.id for r in EvaluationResult.query.filter_
    by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.i
    n_(rids)).delete(
                synchronize_session=False)
        EvaluationResult.query.filter_by(patch_id=pid).delet
    e()
        ConstraintCheck.query.filter_by(patch_id=pid).delete
    ()
        uids = [u.id for u in MicroUnit.query.filter_by(patc
```

```
    h_id=pid)]                                                          13001
        if uids:                                                        13002
            IndicatorData.query.filter(IndicatorData.unit_id        13003
    .in_(uids)).delete(                                                 13004
                synchronize_session=False)                              13005
        MicroUnit.query.filter_by(patch_id=pid).delete()                13006
        m = db.session.get(MinePatch, pid)                              13007
        if m:                                                           13008
            db.session.delete(m)                                        13009
        p = Project.query.filter_by(code="WEIGHT-%s" %                  13010
    TAG).first()                                                        13011
        if p:                                                           13012
            db.session.delete(p)                                        13013
        db.session.commit()                                             13014
def run_eval(pid):                                                      13015
    c.post("/evaluate/%d" % pid, data={},                               13016
    follow_redirects=False)                                             13017
    with app.app_context():                                             13018
        return EvaluationResult.query.filter_by(patch_id=pid        13019
    ) \                                                                 13020
            .order_by(EvaluationResult.id.desc()).first().fi        13021
    nal_score                                                           13022
print(LINE)                                                             13023
print("一、默认应使用系统内置权重")                                          13024
print(LINE)                                                             13025
print("  （开跑前清掉 %d 个上一次残留的自检项目）" % purge_fixtures())          13026
with app.app_context():                                                 13027
    ws.reset_weights(UNIT_TYPE)                                         13028
    w, label, scheme = ws.effective_weights(UNIT_TYPE)                  13029
ok(scheme is None and "系统默认权重" in label,                              13030
   "未配置时权重来源为系统默认权重", label)                                  13031
ok(abs(w["l1"]["土壤指标"] - 0.44) < 1e-9, "一级权重取自系统内置权重",          13032
   "土壤指标 %.2f" % w["l1"]["土壤指标"])                                  13033
pid = build_case()                                                      13034
base_score = run_eval(pid)                                              13035
print("  默认权重下的评价值 F = %.4f" % base_score)                          13036
print()                                                                 13037
print(LINE)                                                             13038
print("二、保存 AHP 算出的权重后，评价应改用新权重")                             13039
print(LINE)                                                             13040
with app.app_context():                                                 13041
    # 模拟 AHP 计算结果：把土壤指标权重提高、植被与地形降低（仍和为 1）          13042
    ws.save_weights(UNIT_TYPE, [("soil", 0.70), ("veg",             13043
    0.20),                                                              13044
                                ("terrain", 0.10)])                     13045
    n_items = WeightItem.query.count()                                  13046
ok(n_items >= 3, "weight_items 表已写入权重条目", "%d 条" % n_items)        13047
with app.app_context():                                                 13048
    w2, label2, scheme2 = ws.effective_weights(UNIT_TYPE)               13049
ok(scheme2 is not None and "自定义" in label2, "权重来源变为自定义方案",        13050
```

```
        label2)                                                    13051
ok(abs(w2["l1"]["土壤指标"] - 0.70) < 1e-9, "新权重已生效",          13052
    "土壤指标 %.2f" % w2["l1"]["土壤指标"])                         13053
ok(abs(w2["l1"]["植被指标"] - 0.20) < 1e-9, "同批保存的其他项也已生效")  13054
ok(abs(w2["l2"]["土壤指标"]["土石比"] - 0.25) < 1e-9,               13055
    "未覆盖的二级权重仍保留系统内置取值",                           13056
    "土石比 %.2f" % w2["l2"]["土壤指标"]["土石比"])                 13057
new_score = run_eval(pid)                                          13058
print("  新权重下的评价值 F = %.4f" % new_score)                   13059
ok(abs(new_score - base_score) > 1e-4,                             13060
    "权重变化确实改变了评价结果",                                   13061
    "%.4f → %.4f" % (base_score, new_score))                       13062
print()                                                            13063
print(LINE)                                                        13064
print("三、结果应记录所用权重来源")                                  13065
print(LINE)                                                        13066
import json                                                        13067
with app.app_context():                                            13068
    rec = EvaluationResult.query.filter_by(patch_id=pid) \         13069
        .order_by(EvaluationResult.id.desc()).first()              13070
    stored = json.loads(rec.unit_scores)                           13071
ok(any(u.get("weight_source") for u in stored),                    13072
    "评价记录中标注了权重来源",                                     13073
    stored[0].get("weight_source") or "")                          13074
print()                                                            13075
print(LINE)                                                        13076
print("四、恢复默认后应回到系统默认权重")                            13077
print(LINE)                                                        13078
with app.app_context():                                            13079
    ws.reset_weights(UNIT_TYPE)                                    13080
    w3, label3, scheme3 = ws.effective_weights(UNIT_TYPE)          13081
ok(scheme3 is None and "系统默认权重" in label3,                    13082
    "已恢复为系统默认权重", label3)                                 13083
back_score = run_eval(pid)                                         13084
print("  恢复后评价值 F = %.4f（默认时 %.4f）" % (back_score,        13085
    base_score))                                                   13086
ok(abs(back_score - base_score) < 1e-9, "评价结果与默认时一致")      13087
print()                                                            13088
print(LINE)                                                        13089
print("五、权重首页概览")                                          13090
print(LINE)                                                        13091
with app.app_context():                                            13092
    rows = ws.summary()                                            13093
ok(len(rows) == 3, "三种微地质单元均有权重来源说明", "%d 行" % len(rows))  13094
r = c.get("/ahp/")                                                 13095
body = r.get_data(as_text=True)                                    13096
ok(r.status_code == 200 and "当前生效权重" in body, "权重首页正常展示")  13097
print()                                                            13098
print(LINE)                                                        13099
print("六、一致性不通过的矩阵不得保存生效")                          13100
```

```
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
n = 3
# 明显矛盾的矩阵：a12=9, a13=1/9, a23=9 —— CR 必然超限
mats = [(0, 1, "9"), (0, 2, "0.1111"), (1, 2, "9")]
data = {"csrf_token": "", "action": "save"}
for i, j, v in mats:
    data["m_%d_%d" % (i, j)] = v
r = c.post("/ahp/matrix?unit_type=%s&level=1" % UNIT_TYPE,
    data=data,
           follow_redirects=True)
with app.app_context():
    _, label4, scheme4 = ws.effective_weights(UNIT_TYPE)
ok(scheme4 is None, "CR 超限的矩阵未被保存生效", label4)
ok("一致性" in r.get_data(as_text=True), "页面给出了未通过一致性的提示")
cleanup(pid)
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("权重生效链路自检全部通过 √")
print(LINE)
# ==== app/services/__init__.py ====
# -*- coding: utf-8 -*-
# ==== app/services/cleanup.py ====
# -*- coding: utf-8 -*-
"""级联清理：删图斑、删项目时，把挂在它下面的数据一并清掉。
为什么单独抽出来：`Project.patches` 与 `MinePatch.units` 上有 ORM 级联，
但图斑下面的**指标数据、限制性判定、评价结果、得分明细、参照生态系统**
都没有级联关系。原先只有「删除图斑」那条路径里手写了清理，
于是「删除项目」会留下一批孤儿行——自检实测：删一个项目后，
上面五张表各残留 1 行（项目与图斑本身都删掉了，数据却还在库里）。
两条删除路径都调用这里，才不会漏。
"""
from app.models import (db, MinePatch, MicroUnit,
    IndicatorData,
                        EvaluationResult, ScoreDetail,
    ConstraintCheck,
                        ReferenceEcosystem)
def delete_patch_children(patch_id):
    """清掉挂在某个图斑下的全部附属数据（不含图斑本身）。
    返回各表清理条数，便于调用方打印或断言。
    """
    counts = {}
    unit_ids = [u.id for u in MicroUnit.query.filter_by(patch_id=patch_id).all()]
```

```
    if unit_ids:                                                    13151
        counts["indicator_data"] = IndicatorData.query.filte        13152
    r(                                                              13153
            IndicatorData.unit_id.in_(unit_ids)).delete(synch       13154
    ronize_session=False)                                           13155
    result_ids = [r.id for r in                                     13156
                  EvaluationResult.query.filter_by(patch_id=        13157
    patch_id).all()]                                                13158
    if result_ids:                                                  13159
        counts["score_details"] = ScoreDetail.query.filter(         13160
            ScoreDetail.result_id.in_(result_ids)).delete(sy        13161
    nchronize_session=False)                                        13162
    counts["evaluation_results"] = EvaluationResult.query.fi        13163
    lter_by(                                                        13164
        patch_id=patch_id).delete(synchronize_session=False)        13165
    counts["constraint_checks"] = ConstraintCheck.query.filt        13166
    er_by(                                                          13167
        patch_id=patch_id).delete(synchronize_session=False)        13168
    counts["micro_units"] = MicroUnit.query.filter_by(              13169
        patch_id=patch_id).delete(synchronize_session=False)        13170
    counts["reference_ecosystems"] = ReferenceEcosystem.quer        13171
    y.filter_by(                                                    13172
        patch_id=patch_id).delete(synchronize_session=False)        13173
    return counts                                                   13174
def delete_patch(patch_id):                                         13175
    """删除一个图斑及其全部附属数据。图斑不存在时返回 (None, {})。"""          13176
    patch = db.session.get(MinePatch, patch_id)                     13177
    if patch is None:                                               13178
        return None, {}                                             13179
    counts = delete_patch_children(patch_id)                        13180
    db.session.delete(patch)                                        13181
    db.session.commit()                                             13182
    return patch, counts                                            13183
def delete_project(project):                                        13184
    """删除一个项目：先清掉每个图斑的附属数据，再删项目本身。                    13185
    项目下的图斑由 `Project.patches` 的 ORM 级联删除，但图斑下面那五张表        13186
    必须先手工清，否则会留孤儿。                                           13187
    """                                                             13188
    cleaned = 0                                                     13189
    for patch in project.patches.all():                             13190
        counts = delete_patch_children(patch.id)                    13191
        cleaned += sum(counts.values())                             13192
    name = project.name                                             13193
    db.session.delete(project)                                      13194
    db.session.commit()                                             13195
    return name, cleaned                                            13196
# ==== app/services/standard_check.py ====                          13197
# -*- coding: utf-8 -*-                                             13198
"""评价结论与地方标准 / 政策文件的对照。                                    13199
标准知识库只作文件储备与比对，**不参与评价因子 F 的计算**：评价仍按内置的          13200
```

```
指标体系与权重算，算完之后再拿结论去对照文件里的判定项，回答
「这个结论是否符合当地文件的要求」。
对照结果有三种，映射不上的项如实标为「待人工判断」，不由系统替文件下结论：
    符合　　——　系统里有对应数据，且满足文件里的判定条件
    不符合　　——　系统里有对应数据，但不满足
    待人工判断　——　文件里的定性要求，系统内没有对应数据（如「地形地貌与周边协调」）
对照用到的数据全部来自系统已有记录：限制性条件判定结果、各微地质单元实测值、
参照生态系统基准、最近一次评价记录。
"""
import re
from app.models import (db, MinePatch, MicroUnit,
    IndicatorData, ConstraintCheck,
                           Standard, StandardRule,
    ReferenceEcosystem, EvaluationResult)
OK = "符合"
BAD = "不符合"
MANUAL = "待人工判断"
# 结论 → 页面上的标签样式（沿用既有的 tag 样式，不新增样式）
TAG_CLASS = {OK: "tag--ok", BAD: "tag--warn", MANUAL:
    "tag--muted"}
def province_of(region):
    """从行政区里认出已收录的省份：「河北省唐山市玉田县」→「河北」。
    库里没有这个省份的标准时返回 None，由调用方给出「未收录」的提示。
    """
    if not region:
        return None
    provinces = {p for (p,) in db.session.query(Standard.pro
    vince).distinct()
                 if p}
    for p in sorted(provinces, key=len, reverse=True):
        if p in region:
            return p
    return None
def _num(text):
    if text is None:
        return None
    m = re.search(r"-?\d+(?:\.\d+)?", str(text))
    return float(m.group(0)) if m else None
def _cmp(value, op, threshold):
    """按比较符判断；阈值不是数字或取值缺失时返回 None（无法判断）。"""
    t = _num(threshold)
    if value is None or t is None:
        return None
    try:
        v = float(value)
    except (TypeError, ValueError):
        return None
    if op == ">=":
        return v >= t
    if op == ">":
```

```
            return v > t
        if op == "<=":
            return v <= t
        if op == "<":
            return v < t
        if op == "==":
            return abs(v - t) < 1e-9
        return None
    def _coverage_stats(patch_id):
        """各单元植被覆盖度的面积加权值与最低值（只统计录了实测值的单元）。"""
        rows = (db.session.query(MicroUnit, IndicatorData)
                .outerjoin(IndicatorData, IndicatorData.unit_id
        == MicroUnit.id)
                .filter(MicroUnit.patch_id == patch_id).all())
        values, num, den = [], 0.0, 0.0
        for unit, data in rows:
            cov = getattr(data, "coverage", None) if data is
        not None else None
            if cov is None:
                continue
            values.append(cov)
            w = unit.ratio or 0.0
            num += cov * w
            den += w
        if not values:
            return None, None
        return (num / den if den else None), min(values)
    def _context(patch):
        """把对照要用到的数据一次性取出来。"""
        check = (ConstraintCheck.query.filter_by(patch_id=patch
        id)
                 .order_by(ConstraintCheck.id.desc()).first())
        ref = ReferenceEcosystem.query.filter_by(patch_id=patch
        id).first()
        result = (EvaluationResult.query.filter_by(patch_id=pat
        h.id)
                  .order_by(EvaluationResult.id.desc()).first())
        weighted, lowest = _coverage_stats(patch.id)
        return {
            "patch": patch,
            "constraint": check,
            "benchmark": getattr(ref, "coverage_benchmark",
        None) if ref else None,
            "has_reference": ref is not None,
            "result": result,
            "grade": getattr(result, "grade", None) if result
        else None,
            "score": getattr(result, "final_score", None) if
        result else None,
            "coverage": weighted,
```

```
        "coverage_min": lowest,
    }
def _judge(item, op, threshold, ctx):
    """把一条文件规则映射到系统数据，返回 （结论， 依据说明）。"""
    name = item or ""
    cov, cov_min = ctx["coverage"], ctx["coverage_min"]
    check = ctx["constraint"]
    def cov_text():
        if cov is None:
            return "尚未录入植被覆盖度"
        return "面积加权覆盖度 %.2f%%（最低单元 %.2f%%）" % (cov,
    cov_min)
    # ---- 植被覆盖度（含趋势）----
    if "趋势" in name:
        return MANUAL, "需连续多年的监测数据，系统内没有时间序列"
    if "植被覆盖" in name:
        if cov is None:
            return MANUAL, cov_text()
        r = _cmp(cov, op, threshold)
        if r is None:
            return MANUAL, cov_text() + "；文件给出的条件为「%s %s」"
    % (op, threshold)
        return (OK if r else BAD), cov_text()
    # ---- 参照生态系统 ----
    if "参照生态系统调查" in name or "参照系调查" in name:
        return ((OK, "已录入参照生态系统的植被覆盖度与生物多样性基准")
                if ctx["has_reference"] else
                (MANUAL, "尚未录入参照生态系统基准"))
    if "参照系" in name:
        if cov is None or ctx["benchmark"] is None:
            return MANUAL, "缺参照生态系统基准或覆盖度实测值"
        gap = (ctx["benchmark"] - cov)
        gap = 0.0 if gap < 0 else gap
        r = _cmp(gap, op, threshold)
        if r is None:
            return MANUAL, "与参照系差距 %.2f 个百分点" % gap
        return (OK if r else BAD), "与参照系差距 %.2f 个百分点（基准
    %.2f%%）" % (
            gap, ctx["benchmark"])
    # ---- 污染与地质灾害 ----
    if "污染" in name:
        if check is None or check.pollution_pass is None:
            return MANUAL, "尚未完成限制性条件判定"
        return ((OK, "限制性判定：土壤污染一项通过") if
    check.pollution_pass else
                (BAD, "限制性判定：土壤污染一项未通过"))
    if "地质灾害" in name:
        if check is None or check.geo_pass is None:
            return MANUAL, "尚未完成限制性条件判定"
        if not check.geo_pass:
```

```
                return BAD, "限制性判定：存在地质灾害"
            if "发育程度" in name:
                return MANUAL, "系统只记录「有无地质灾害」，未记录发育程度分级"
            return OK, "限制性判定：地质灾害一项通过"
        # ---- 责任主体 ----
        if "责任主体" in name:
            missing = ctx["patch"].responsible_entity_missing
            if missing is None:
                return MANUAL, "图斑档案未填写责任主体是否缺失"
            if missing:
                return OK, "图斑档案记录责任主体缺失，按文件属政府主导治理"
            if "政府" in name:
                return MANUAL, "责任主体未缺失，是否符合「政府」要求需人工核实"
            return OK, "图斑档案记录责任主体明确"
        # ---- 适宜性分级 / 指数 ----
        if "适宜性分级" in name or "适宜性评价" in name:
            # 文件只要求「做过自然恢复适宜性分级」，系统已经给出等级
            if ctx["grade"] is None:
                return MANUAL, "尚未执行定量评价"
            return OK, "系统已给出适宜性等级：%s（F=%.4f）" % (ctx["grade"],
        ctx["score"])
        if "适宜性指数" in name:
            # 指数是要跟阈值比的：文件没给具体阈值时不能替它下结论
            if ctx["grade"] is None:
                return MANUAL, "尚未执行定量评价"
            t = _num(threshold)
            if t is None:
                return MANUAL, ("文件未给出具体阈值，系统评价因子
        F=%.4f（%s），需人工比对"
                                % (ctx["score"], ctx["grade"]))
            r = _cmp(ctx["score"], op, t)
            if r is None:
                return MANUAL, "系统评价因子 F=%.4f" % ctx["score"]
            return (OK if r else BAD), ("系统评价因子 F=%.4f，文件要求 %s
        %s"
                                        % (ctx["score"], op,
        threshold))
        # ---- 其它定性要求 ----
        return MANUAL, "系统内无对应数据，需人工核实"
def compare(patch_id):
    """生成某图斑的评价结论与当地标准 / 政策文件的对照结果。"""
    patch = db.session.get(MinePatch, patch_id)
    if patch is None:
        return None
    province = province_of(patch.admin_region)
    out = {
        "patch": patch,
        "region": patch.admin_region or "",
        "province": province,
        "covered": False,
```

```
        "message": "",
        "standards": [],
        "totals": {OK: 0, BAD: 0, MANUAL: 0},
        "necessary": {"符合": 0, "共计": 0},
    }
    if province is None:
        out["message"] = ("未收录该省份的地方标准与政策文件"
                          "（图斑所在行政区：%s）" %
    (patch.admin_region or "未填写"))
        return out
    standards = (Standard.query.filter_by(province=province)
                 .order_by(Standard.category.desc(),
    Standard.name).all())
    if not standards:
        out["message"] = "未收录该省份的地方标准与政策文件（%s）" % province
        return out
    ctx = _context(patch)
    out["covered"] = True
    for s in standards:
        rules = (StandardRule.query.filter_by(standard_id=s.
    id)
                 .order_by(StandardRule.id).all())
        rows = []
        for r in rules:
            verdict, evidence = _judge(r.item, r.operator,
    r.threshold, ctx)
            rows.append({
                "item": r.item, "operator": r.operator,
    "threshold": r.threshold,
                "condition_type": r.condition_type or "",
    "region": r.region,
                "note": r.note, "verdict": verdict, "tag":
    TAG_CLASS[verdict],
                "evidence": evidence,
            })
            out["totals"][verdict] = out["totals"].get(verd
    ct, 0) + 1
            if (r.condition_type or "") == "必要条件":
                out["necessary"]["共计"] += 1
                if verdict == OK:
                    out["necessary"]["符合"] += 1
        out["standards"].append({
            "name": s.name, "doc_number": s.doc_number,
    "category": s.category,
            "summary": s.summary, "rules": rows,
            "counts": {k: sum(1 for x in rows if
    x["verdict"] == k)
                       for k in (OK, BAD, MANUAL)},
        })
    return out
```

```
# ==== app/services/unit_config.py ====
# -*- coding: utf-8 -*-
"""标准化参数与必填项：让界面上能改的东西真正参与计算。
以前有个说不通的地方：图斑的「参照生态系统基准」、指标定义里的「理想取值」
和「启用单元」在界面上都能改、也能落库，**但计算链只读代码常量**
（`app/utils/constants.py`），从不查这两张表。于是报告里写着「基准 60%」，
实际按常量 75% 算，同一份报告两个口径。
现在统一从这里取：
- 基准值（植被覆盖度、Shannon）来自图斑的 `reference_ecosystems`；
- 各指标的理想取值来自 `indicators.ideal_value`（能解析成数字的才覆盖，
  像「5~9」「25° 分界」这种区间/规则型仍按代码里的算法处理）；
- 启用单元来自 `indicators.enabled_units`（岩质边坡无土壤、平台无地形）。
库里没有行、或值不是数字时一律回落到代码常量 —— 所以历史数据与
`验证数据/` 里的 8 个算例结果不会因为这次改动而变化。
"""
from app.models import Indicator, ReferenceEcosystem
from app.utils.constants import (
    UNIT_TYPES, IDEAL_SOIL_RATIO, IDEAL_THICKNESS,
    IDEAL_ORGANIC,
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,
    HEAVY_METALS,
)
# 二级指标编码 → 标准化参数名（同时也是 IndicatorData 的字段名）
NUMERIC_IDEALS = {
    "soil_ratio": ("soil_ratio", IDEAL_SOIL_RATIO),
    "thickness": ("thickness", IDEAL_THICKNESS),
    "organic_matter": ("organic_matter", IDEAL_ORGANIC),
    "coverage": ("coverage", IDEAL_COVERAGE),
    "shannon": ("shannon", IDEAL_SHANNON),
}
# 重金属编码 → 金属名
METAL_CODES = {"hm_" + m: m for m in HEAVY_METALS}
# 必填的实测字段（污染指标按方法约定允许「未检测」，故不在必填之列）
REQUIRED_CODES = ["soil_ratio", "thickness", "ph",
    "organic_matter",
                  "coverage", "shannon", "slope", "aspect"]
# 字段 → 界面上的中文名，用于提示「缺什么」
FIELD_LABELS = {
    "soil_ratio": "土石比", "thickness": "土层厚度", "ph": "pH",
    "organic_matter": "有机质", "coverage": "植被覆盖度",
    "shannon": "生物多样性（Shannon）", "slope": "坡度", "aspect":
    "坡向",
}
def _numeric(text):
    """把理想取值的文本解析成数字；区间、规则型返回 None。"""
    if text is None:
        return None
    try:
        return float(str(text).strip())
    except (TypeError, ValueError):
```

```
        return None
class UnitSettings:
    """一个图斑上下文的标准化参数。空对象表示「全部按代码常量」。"""
    def __init__(self, params=None, enabled=None):
        self.params = params or {}              # 覆盖用的标准值
        self.enabled = enabled or {}            # 单元类型 →
    启用的二级指标编码集合
    def ideal(self, code, fallback):
        """取某指标的理想取值：库里有数字就用库里的，否则用常量。"""
        value = self.params.get(code)
        return fallback if value is None else value
    def group_enabled(self, unit_type, l1_name, l1_code):
        """该单元类型下某个一级指标是否启用（库里没配就按代码常量）。"""
        codes = self.enabled.get(unit_type)
        if codes is None:
            return None                         # 交给调用方按常量判断
        return l1_code in codes
    def enabled_codes(self, unit_type):
        """该单元类型启用的全部二级指标编码；库里没配返回 None。"""
        return self.enabled.get(unit_type)
    def required_codes(self, unit_type):
        """该单元类型必填的实测字段。
        按「启用的二级指标」推导：岩质边坡不启用土壤，平台不启用地形。
        库里没配该项时用代码常量里的结构性规则。
        """
        from app.utils.constants import
    UNIT_ENABLED_INDICATORS
        codes = self.enabled.get(unit_type)
        if codes is None:
            groups = UNIT_ENABLED_INDICATORS.get(unit_type,
    ())
            l2_by_group = {
                "土壤指标": ["soil_ratio", "thickness", "ph",
    "organic_matter"],
                "植被指标": ["coverage", "shannon"],
                "地形指标": ["slope", "aspect"],
            }
            codes = {c for g in groups for c in
    l2_by_group.get(g, [])}
        return [c for c in REQUIRED_CODES if c in codes]
    def missing_fields(self, unit_type, data):
        """返回该单元缺哪些必填项（中文名列表）。"""
        missing = []
        for code in self.required_codes(unit_type):
            value = data.get(code)
            if value is None or (isinstance(value, str) and
    value.strip() == ""):
                missing.append(FIELD_LABELS.get(code, code))
        return missing
    def undetected_metals(self, data):
```

```
        """按方法约定，未检测的重金属按「未受污染」计入；这里列出是哪些。"""
        measured = data.get("heavy_metals") or {}
        return [m for m in HEAVY_METALS if measured.get(m) is None]
def for_patch(patch_id):
    """读某个图斑的标准化参数；库里有什么就用什么，没有就用常量。
    优先级：图斑自己的参照生态系统基准  **高于**  指标定义里的全局理想取值  ——
    前者说的是「这个图斑的参照生态系统是什么样」，后者是「一般情况下的理想值」，
    越具体的越应该说了算。两者都优先于代码常量。
    """
    params = {}
    enabled = {}
    for ind in Indicator.query.all():
        code = ind.code
        if code in NUMERIC_IDEALS:
            value = _numeric(ind.ideal_value)
            if value is not None:
                params[code] = value
        elif code in METAL_CODES:
            # 重金属的「理想值」存的是限值，作为越小越优的极值使用
            value = _numeric(ind.ideal_value)
            if value is not None:
                params[code] = value
        units = (ind.enabled_units or "").strip()
        if units:
            for unit in UNIT_TYPES:
                if unit in units:
                    enabled.setdefault(unit, set()).add(code)
    # 再用该图斑的参照生态系统基准覆盖：它是「这个图斑的参照生态系统」，
    # 比指标定义里的全局理想取值更具体，应该它说了算。
    ref = ReferenceEcosystem.query.filter_by(patch_id=patch_id).first()
    if ref is not None:
        if ref.coverage_benchmark is not None:
            params["coverage"] = float(ref.coverage_benchmark)
        if ref.shannon_benchmark is not None:
            params["shannon"] = float(ref.shannon_benchmark)
    return UnitSettings(params=params, enabled=enabled)
def metal_limits(settings):
    """重金属的极值表：库里有配置的用库里的，其余用常量。"""
    limits = dict(HEAVY_METAL_LIMIT)
    for code, metal in METAL_CODES.items():
        value = settings.params.get(code) if settings else None
        if value is not None:
            limits[metal] = value
    return limits
# ==== app/templates/standard/_check_block.html ====
```

```
{# 结论对照内容块：标准知识库对照页与评价结果页共用同一份渲染 #}
{% if check %}
<h3>与地方标准 / 政策文件的对照</h3>
{% if not check.covered %}
<p class="muted">{{ check.message }}。标准知识库只作文件储备与比对，不参与评价计算；
  如需对照，请先在「标准知识库」中收录该省份的标准或政策文件。</p>
{% else %}
<p class="muted">
  图斑所在行政区：{{ check.region or "未填写" }}（对照 {{ check.province
    }} 的标准与政策文件）。
  对照只是把文件里的判定项逐条映射到系统已有数据：共 {{ check.totals["符合"] +
    check.totals["不符合"] + check.totals["待人工判断"] }} 条，
  符合 <strong>{{ check.totals["符合"] }}</strong> 条、不符合
    <strong>{{ check.totals["不符合"] }}</strong> 条、
  待人工判断 <strong>{{ check.totals["待人工判断"] }}</strong> 条。
  文件里的定性要求（如「地形地貌与周边协调」）系统没有对应数据，一律标为待人工判断，不代为结论。
</p>
{% for s in check.standards %}
<h4 style="margin:14px 0 6px;">{{ s.name }}{% if
    s.doc_number %} <span class="muted mono">{{
    s.doc_number }}</span>{% endif %}</h4>
<table>
  <caption class="muted">{{ s.category }}｜符合 {{
    s.counts["符合"] }} 条、不符合 {{ s.counts["不符合"] }} 条、待人工判断
    {{ s.counts["待人工判断"] }} 条</caption>
  <thead>
    <tr>
      <th scope="col">文件判定项</th>
      <th scope="col">条件</th>
      <th scope="col">类型</th>
      <th scope="col">对照结果</th>
      <th scope="col">依据（系统内的对应数据）</th>
    </tr>
  </thead>
  <tbody>
    {% for r in s.rules %}
    <tr>
      <td style="text-align:left;">{{ r.item }}</td>
      <td class="mono nowrap">{{ r.operator }} {{
    r.threshold or "" }}</td>
      <td>{{ r.condition_type or "—" }}</td>
      <td><span class="tag {{ r.tag }}">{{ r.verdict
    }}</span></td>
      <td style="text-align:left;">{{ r.evidence }}</td>
    </tr>
    {% endfor %}
  </tbody>
</table>
{% endfor %}
<p class="muted">
```

```
    对照结论仅供评价与报送时参考：符合全部必要条件的，说明评价结论与文件要求一致；
    出现「不符合」或「待人工判断」的，需由评价人员结合现场情况核定后再报送。
</p>
{% endif %}
{% endif %}
# ==== app/templates/standard/check.html ====
{% extends "base.html" %}
{% block title %}结论对照 · {{ check.patch.name }}{% endblock %}
{% block content %}
<div class="card">
  <h2>结论对照 · {{ check.patch.name }}</h2>
  {% include "standard/_check_block.html" %}
  <p class="actions" style="margin-top:16px;">
    <a class="btn btn--ghost" href="/standard/check">返回图斑列表</a>
    <a class="btn btn--ghost" href="/standard/">返回标准列表</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/standard/check_index.html ====
{% extends "base.html" %}
{% block title %}结论对照{% endblock %}
{% block content %}
<div class="card">
  <h2>评价结论与地方标准 / 政策文件的对照</h2>
  <p class="muted">
    标准知识库不参与评价计算，只在评价完成后回答一个问题：
    本次评价结论是否符合图斑所在省份的文件要求。选择一个图斑查看逐条对照。
  </p>
  {% if not rows %}
  <p class="muted">暂无矿山图斑。请先在「项目与矿山档案」中建立图斑并录入数据。</p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">矿山图斑</th>
        <th scope="col">所在行政区</th>
        <th scope="col">已收录该省份文件</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for row in rows %}
      <tr>
        <td style="text-align:left;">{{ row.patch.name
    }}</td>
        <td style="text-align:left;">{{
    row.patch.admin_region or "—" }}</td>
        <td>
          {% if row.province %}<span class="tag tag--ok">{{
```

```
      row.province }}</span>
            {% else %}<span class="tag tag--muted">未收录</span>{
      % endif %}
          </td>
          <td><a class="link-btn" href="/standard/check/{{
      row.patch.id }}">查看对照</a></td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% endif %}
    <p class="actions" style="margin-top:16px;">
      <a class="btn btn--ghost" href="/standard/">返回标准列表</a>
    </p>
</div>
{% endblock %}
#  ==== app/utils/__init__.py ====
#  -*- coding: utf-8 -*-
#  ==== app/utils/parsing.py ====
#  -*- coding: utf-8 -*-
"""表单数值的安全解析。
背景：页面上的数字输入框原来写成 `float(request.form.get("area"))`，
用户填「约 3.5」或全角数字就会撞上 ValueError —— 页面上是默认英文 500 页。
真人演示时这是最容易踩到的一类。统一走这里：
没填返回 None，填了但不是数字抛 FieldError，让视图 flash 一句中文再回退。
"""
import math
class FieldError(ValueError):
    """表单里某个字段填得不对，携带给用户看的中文提示。"""
def parse_float(raw, field, *, minimum=None, maximum=None):
    """把表单值解析成 float。
    raw       表单原始值（可能是 None / "" / "abc" / "1e400"）
    field     字段名，用于拼提示
    minimum / maximum   可选的取值范围，越界同样抛 FieldError
    """
    if raw is None:
        return None
    text = str(raw).strip()
    if text == "":
        return None
    try:
        value = float(text)
    except (TypeError, ValueError):
        raise FieldError("「%s」请填数字" % field)
    # nan / inf 能通过 float()，但存进库里会把标准化算成满分，必须挡住
    if not math.isfinite(value):
        raise FieldError("「%s」请填正常的数字" % field)
    if minimum is not None and value < minimum:
        raise FieldError("「%s」不能小于 %g" % (field, minimum))
    if maximum is not None and value > maximum:
```

```
            raise FieldError("「%s」不能大于 %g" % (field, maximum))
        return value
def parse_int(raw, field, *, default=None, minimum=None,
        maximum=None):
    """解析整数，用于查询参数与计数类字段。"""
    if raw is None:
        return default
    text = str(raw).strip()
    if text == "":
        return default
    try:
        value = int(text)
    except (TypeError, ValueError):
        raise FieldError("「%s」请填整数" % field)
    if minimum is not None and value < minimum:
        value = minimum
    if maximum is not None and value > maximum:
        value = maximum
    return value
#  ==== app/utils/timefmt.py ====
#  -*- coding: utf-8 -*-
"""时间显示：库里一律存 UTC，展示时换成服务器本地时间。
为什么不在写库时直接用本地时间：一个系统里时间基准只能有一个。
`app/models.py` 里所有 `created_at` / `checked_at` 都用
    `datetime.utcnow`，
存的是 UTC；如果写库改成本地时间，已经存进库的几千条记录会新旧混着差 8 小时。
所以统一在「展示」这一层换算，老数据不用动。
用法（Jinja）：
    {{ row.created_at | localdt }}                    →
    2026-09-26 23:33
    {{ row.created_at | localdt("%Y-%m-%d %H:%M:%S") }} →
    2026-09-26 23:33:28
"""
from datetime import datetime, timezone
DEFAULT_FORMAT = "%Y-%m-%d %H:%M"
def to_local(value, fmt=DEFAULT_FORMAT, empty="—"):
    """把 UTC 时间换算成服务器本地时间并格式化。
    传进来的值按「无时区即 UTC」处理；带时区的原样换算，便于以后改用
    带时区的时间戳。空值返回占位符，免得页面上出现 None。
    """
    if value is None:
        return empty
    if value.tzinfo is None:
        value = value.replace(tzinfo=timezone.utc)
    return value.astimezone().strftime(fmt)
def now_utc():
    """取当前 UTC 时间（带时区），用于只作展示、不落库的时间戳。"""
    return datetime.now(timezone.utc)
def utcnow():
    """取当前 UTC 时间（无时区），用于数据库默认值。
```

```
    `datetime.utcnow()` 在 Python 3.12 已标记弃用，导入时会刷告警；
    这里用带时区的 now() 再摘掉时区，得到的仍是 naive-UTC，行为一致。
    """
    return datetime.now(timezone.utc).replace(tzinfo=None)
# ==== app/views/__init__.py ====
# -*- coding: utf-8 -*-
```